

Applied Basic Science

Part I · ABS

A full-scope, blueprint-mapped review of every discipline and condition area the Part I ABS examination can draw from, built for retrieval practice rather than passive reading.

FORMAT	Computer-based, Pearson test centers
SCORED ITEMS	350 scored + 20 unscored pre-test
SESSIONS	2 sessions × 185 items · 4 hours each
ITEM STYLES	Single best answer + 10–15% multiple-response
BREAK	Optional, up to 45 minutes between sessions
SCORING	Scaled 100–900 · 300 and above passes
MAJOR DOMAINS	2 condition groups · 17 condition areas
DISCIPLINES TESTED	12, from gross anatomy to pharmacology

Structure, item counts, and condition-area weights follow the NBEO Part I ABS content matrix and current published exam information. Eponyms appear in the non-possessive form used on the exam (Horner syndrome, Descemet membrane, Schlemm canal).

How to use this guide

This guide is built around one finding that dominates the learning-science literature: **you remember what you retrieve, not what you re-read**. Every chapter is therefore structured so that reading is the shortest phase and self-testing is the longest. The layout does the work for you if you follow the conventions below.

The page furniture, and what each element is for

READING CONVENTIONS USED THROUGHOUT

Element	Signal	How to work it
Chapter opener	Objectives + blueprint weight	Read the objectives first and try to answer them cold. Whatever you cannot answer is your actual reading assignment for that chapter.
Blue bold	Key term	Terms the exam names directly. If you cannot define it in one sentence, it is not learned.
12.5 D	Memorable value	Monospaced values are numbers you are expected to reproduce, not estimate.
High yield box	Amber bar	Content with the highest historical item density. Do these first on a short timeline.
Exam trap box	Crimson bar	Look-alike pairs and reversed relationships. Most avoidable losses live here.
Clinical pearl box	Teal bar	The basic-science fact reframed as the clinical consequence, which is how ABS items are usually worded.
Mnemonic box	Violet bar	Use only for arbitrary ordered lists. Do not mnemonic-ise anything you can reason out.
Numbers to know	Amber-topped panel	Cover the right column, recite, then check. Repeat until the panel is boring.
Retrieval practice	Slate-topped panel	Answer out loud or in writing before looking. The answer key sits directly below on purpose: the delay should be seconds, not days.
Recap	Framed blue panel	End-of-chapter checkpoint. If you cannot expand every line into a paragraph, the chapter is not finished.

HIGHEST-RETURN STUDY BEHAVIOURS

- 1. Retrieve before you review.** Close the guide and write what you know about a condition area, then read to fill gaps. This single change outperforms re-reading by a wide margin.
- 2. Interleave disciplines within a condition area.** Study the cornea's anatomy, physiology, pharmacology, and pathology in one sitting. That is exactly how ABS items are written: one tissue, many disciplines.
- 3. Space, do not mass.** Three 40-minute passes over eight days beat one 2-hour pass, even at equal total time.
- 4. Explain the mechanism.** For every fact, ask "so what happens if this fails?" Mechanistic elaboration is what converts a fact into an answerable item.
- 5. Practise in exam conditions.** 185 items in 4 hours is **77 s** per item. Train at that pace or you will lose points you actually knew.

Exam blueprint: how the 350 scored items are distributed

Part I ABS is organised as two major condition groups containing seventeen specific condition areas. Each condition area is then sampled across twelve basic-science disciplines. The ranges below are the minimum and maximum number of scored items per area.

GROUP 1 · REFRACTIVE STATUS / SENSORY PROCESSES / OCULOMOTOR PROCESSES · 122 ITEMS

Condition area	Scored items	Share
Ametropia	29–39	~10%
Ophthalmic optics / spectacles	17–25	~6%
Contact lenses	8–14	~3%
Low vision	5–9	~2%
Accommodation / vergence / oculomotor function	10–18	~4%
Amblyopia / strabismus	9–17	~4%
Perceptual function / color vision	10–18	~4%
Visual and human development	6–10	~2%

GROUP 2 · NORMAL HEALTH / DISEASE / TRAUMA · 228 ITEMS

Condition area	Scored items	Share
Systemic health	55–75	~19%
Conjunctiva / cornea / refractive surgery	34–50	~12%
Lids / lashes / lacrimal system / ocular adnexa / orbit	20–28	~7%
Vitreous / retina / choroid	17–25	~6%
Episclera / sclera / anterior uvea	16–24	~6%
Optic nerve / neuro-ophthalmic pathways	14–22	~5%
Glaucoma	13–21	~5%
Lens / cataract / IOL / pre- and post-operative care	8–14	~3%
Emergencies / trauma	8–14	~3%

READ THE BLUEPRINT CORRECTLY

Emergencies / trauma items are **embedded inside the other condition areas** of Normal health / disease / trauma rather than presented as a standalone block. Do not study trauma as a separate silo; study it as the acute presentation of each tissue you already cover.

Systemic health is the single largest area on the exam at roughly one item in five. Candidates routinely under-prepare it because it feels like general medicine rather than optometry. That misjudgement costs more points than any optics gap.

The twelve disciplines sampled across those areas

1. Anatomy (gross)

2. Anatomy (neuroanatomy)

3. Anatomy (histology)	9. Pathology
4. Anatomy (developmental / embryology)	10. Optics: geometrical, physical, ophthalmic, physiological
5. Biochemistry	11. Pharmacology
6. Physiology	12. Public health context embedded in systemic health
7. Immunology	
8. Microbiology	

This guide is organised by *discipline* because that is how basic science is encoded in memory, and each chapter is cross-tagged to the condition areas it feeds. Use the tag chips under each chapter heading to check your blueprint coverage.

A workable revision schedule

Pick the column that matches the time you actually have, not the time you wish you had.

THREE TIMELINES, SAME PRINCIPLES

Phase	10 weeks	5 weeks	2 weeks (salvage)
Pass 1: build	Weeks 1–5. One chapter per day, objectives cold-tested first. Make no new notes beyond cloze prompts.	Weeks 1–2. Two chapters per day, high-yield and trap boxes only on first read.	Days 1–5. Read only the high-yield boxes, numbers panels, and recaps.
Pass 2: retrieve	Weeks 6–8. All retrieval blocks, closed book. Re-read only what you miss.	Weeks 3–4. All retrieval blocks + the numbers appendix daily.	Days 6–10. Retrieval blocks for the six largest condition areas only.
Pass 3: simulate	Weeks 9–10. Timed 185-item blocks, then error-log by condition area.	Week 5. Two timed blocks, error-log, targeted repair.	Days 11–13. One timed block, repair the top three error clusters.
Day before	Numbers appendix, mnemonic boxes, and the recap panels. Nothing new. Sleep is a study technique.		

ERROR LOG, THE ONE EXTRA DOCUMENT YOU SHOULD KEEP

For every practice item you miss, record four things: the condition area, the discipline, the specific fact you lacked, and *why* you chose the wrong option. The fourth column is the valuable one. Most repeated errors are reasoning habits, not knowledge gaps, and only the fourth column exposes them.

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Orbit, lids, lacrimal system, extraocular muscles

Lids / lashes / lacrimal / adnexa / orbit · 20–28 items

Emergencies / trauma (embedded)

Optic nerve / neuro-ophthalmic pathways

ANSWER THESE COLD BEFORE READING

1. Name the seven orbital bones and state which wall fails first in blunt trauma and why.
2. List every structure passing through the superior orbital fissure and sort them into inside versus outside the annulus of Zinn.
3. Give the seven layers of the upper eyelid from skin to conjunctiva.
4. Trace the parasympathetic pathway from brainstem nucleus to lacrimal gland, naming every ganglion and nerve.
5. State the spiral of Tillaux distances and the origin, insertion, and innervation of all six extraocular muscles.
6. Explain why an eyelid infection can reach the cavernous sinus.

BLUEPRINT WEIGHT

7%

20–28 scored items in the lids / lacrimal / orbit area, plus heavy cross-loading into neuroanatomy and trauma.

The bony orbit

The orbit is a four-walled pyramid whose apex points posteromedially. Its volume is ~30 mL, of which the globe occupies only about 6.5 mL. Orbital depth from rim to apex is 40–45 mm; the entrance measures roughly 35 mm high by 40–45 mm wide, with the widest point about 1 cm behind the anterior rim. The **medial walls are essentially parallel** to one another and to the sagittal plane; the lateral walls diverge at about 45° from the medial wall, so the two orbital axes diverge ~45° from each other and each axis sits ~23° from the sagittal plane.

SEVEN BONES OF THE ORBIT

"Many Friendly Zebras Enjoyed Lazy Sunday Picnics"

Maxillary, Frontal, Zygomatic, Ethmoid, Lacrimal, Sphenoid, Palatine. Only the palatine contributes no part of a named wall boundary you will be asked to draw: it forms a small posteromedial piece of the floor.

THE FOUR WALLS: COMPOSITION AND CLINICAL CONSEQUENCE

Wall	Bones	Key relations and clinical point
Roof	Orbital plate of the frontal bone; lesser wing of sphenoid posteriorly	Separates orbit from anterior cranial fossa and frontal sinus. Contains the lacrimal fossa superotemporally and the trochlear fossa superomedially (4 mm behind the rim). Roof fractures risk CSF leak and pulsatile proptosis.

Wall	Bones	Key relations and clinical point
Lateral	Zygomatic anteriorly; greater wing of sphenoid posteriorly	Thickest and strongest wall. Its anterior edge sits posterior to the globe equator, which is why the temporal field is relatively unobstructed. Whitnall (lateral orbital) tubercle lies ~11 mm below the frontozygomatic suture and anchors the lateral canthal tendon, Lockwood ligament, levator aponeurosis lateral horn, and check ligaments.
Floor	Maxilla (largest part), zygomatic , orbital process of the palatine	Roof of the maxillary sinus. Weakest area is the posteromedial floor medial to the infraorbital groove: the classic blowout site. Does not reach the orbital apex, ending at the pterygopalatine fossa. The floor slopes upward laterally.
Medial	Frontal process of maxilla , lacrimal bone, lamina papyracea of the ethmoid , body of sphenoid	Thinnest wall (lamina papyracea, 0.2–0.4 mm). Ethmoid sinusitis spreads here to cause orbital cellulitis in children. Blowout fractures of the medial wall produce orbital emphysema on nose-blowing.

THINNEST VERSUS WEAKEST

The **medial wall is the thinnest bone**, but the **floor is the wall that most commonly fractures in a blowout**, because the medial wall is buttressed by the ethmoidal air-cell septa while the floor is unsupported over the maxillary sinus. Both statements are true and the exam uses one to distract from the other.

Openings, and exactly what passes through them**ORBITAL APERTURES AND CONTENTS**

Opening	Bone	Contents
Optic canal	Lesser wing of sphenoid	Optic nerve (CN II) in its dural sheath, ophthalmic artery (inferolateral to the nerve), sympathetic fibres from the carotid plexus. Length 8–10 mm, diameter ~6.5 mm. Lies within the annulus of Zinn.
Superior orbital fissure	Between greater and lesser wings of sphenoid	<i>Inside the annulus of Zinn:</i> superior and inferior divisions of CN III, CN VI, nasociliary nerve (V1), sympathetic fibres. <i>Outside (above/lateral to) the annulus:</i> CN IV (trochlear), frontal nerve (V1), lacrimal nerve (V1), superior ophthalmic vein.
Inferior orbital fissure	Between maxilla, sphenoid greater wing, and zygomatic	Infraorbital nerve and zygomatic nerve (both V2), infraorbital artery, inferior ophthalmic vein branch to the pterygoid venous plexus, parasympathetic branches from the pterygopalatine ganglion.
Supraorbital notch / foramen	Frontal	Supraorbital nerve and artery (terminal frontal nerve branch, V1).

Opening	Bone	Contents
Anterior ethmoidal foramen	Frontoethmoidal suture	Anterior ethmoidal nerve and artery. Sits ~20 mm behind the anterior lacrimal crest.
Posterior ethmoidal foramen	Frontoethmoidal suture	Posterior ethmoidal nerve and artery. Sits ~12 mm further back, and only ~6 mm anterior to the optic canal: the surgical danger point.
Nasolacrimal canal	Maxilla, lacrimal, inferior nasal concha	Nasolacrimal duct, opening into the inferior nasal meatus at the valve of Hasner. Runs inferiorly, posteriorly, and slightly laterally.
Infraorbital foramen / groove / canal	Maxilla	Infraorbital nerve (V2) and artery. Foramen is ~4 mm below the inferior orbital rim: numbness of the cheek, upper lip, and upper gum localises a floor fracture.
Zygomaticofacial and zygomaticotemporal foramina	Zygomatic	Corresponding branches of the zygomatic nerve (V2), supplying cheek and temple skin.

THE ANNULUS OF ZINN DISTINCTION IS A FAVOURITE ITEM

The **annulus (anulus) of Zinn** is the common tendinous ring from which the four rectus muscles originate at the apex. Nerves destined for muscles *inside* the muscle cone (III, VI, and the sensory nasociliary) pass inside the ring; the trochlear nerve enters *outside* it, which is consistent with the superior oblique lying outside the cone. Frontal and lacrimal nerves and the superior ophthalmic vein also stay outside.

Orbital apex syndrome = superior orbital fissure syndrome plus optic nerve involvement: ophthalmoplegia (III, IV, VI), V1 anaesthesia, and reduced vision with an afferent defect. **Superior orbital fissure syndrome** spares the optic nerve.

Orbital soft tissue architecture

Periorbita

Periosteum lining the orbital bones, loosely adherent except at sutures, foramina, and the orbital rim. Its loose attachment is why subperiosteal abscess and haematoma can dissect widely.

Orbital septum

Fibrous sheet from the arcus marginalis at the orbital rim to the levator aponeurosis (upper, 2–5 mm above the tarsus) and to the inferior tarsal border (lower). It is the anatomic barrier that separates **preseptal** from **orbital** cellulitis, the single most consequential anatomic boundary in lid disease.

Tenon capsule (fascia bulbi)

Dense fibroelastic envelope from the limbus to the optic nerve, separated from the sclera by the episcleral space. Fuses with the intermuscular septum and is pierced by the EOM tendons ~10 mm behind their insertions. Preserving it prevents fat prolapse and adhesive restriction after strabismus surgery.

Check ligaments

Extensions of the medial rectus sheath to the lacrimal bone and the lateral rectus sheath to Whitnall tubercle; they limit excursion.

Lockwood ligament

Suspensory hammock beneath the globe formed by the fused inferior rectus and inferior oblique sheaths; it prevents the globe sinking after maxillectomy.

Whitnall ligament

Superior transverse ligament of the levator: the fulcrum where the levator's horizontal vector turns vertical. Sits ~10–12 mm above the superior tarsal border.

Orbital fat

Intraconal (central) and extraconal (peripheral) compartments, separated by the intermuscular septum. Intraconal fat expansion drives axial proptosis; the medial extraconal fat pad is the one removed in decompression.

The eyelids**Dimensions and landmarks**

- Palpebral fissure: 8–11 mm vertically, 28–30 mm horizontally in the adult. Upper lid margin normally sits 1–2 mm below the superior limbus; lower lid rests at the inferior limbus.
- Marginal reflex distance 1 (MRD1), pupil centre to upper lid margin: 4–5 mm. MRD2, pupil centre to lower lid margin: 5–5.5 mm.
- Upper lid crease: 8–11 mm above the margin in Caucasian lids, 3–6 mm or absent in East Asian lids because the levator aponeurosis inserts lower and preaponeurotic fat descends further.
- Tarsal plates: dense connective tissue (not cartilage). Upper 10–11 mm vertical height, lower 4–5 mm, both about 1 mm thick and 29 mm long.
- Lashes: ~100–150 in two to three irregular rows on the upper lid, ~50–75 in one to two rows on the lower. No lashes medial to the punctum. Lifespan 3–5 months; they do not grey with age at the same rate as scalp hair.
- **Gray line:** the visible line on the lid margin corresponding to the muscle of Riolan (the most superficial pretarsal orbicularis). It marks the surgical plane separating anterior lamella (skin, orbicularis) from posterior lamella (tarsus, conjunctiva).
- Meibomian orifices lie just anterior to the mucocutaneous junction and posterior to the gray line.

Layers of the upper lid, superficial to deep

SEVEN LAYERS, IN ORDER

- 1 **Skin:** the thinnest in the body, <1 mm, with no subcutaneous fat over the pretarsal region.
- 2 **Subcutaneous areolar tissue:** loose, which is why oedema and blood accumulate dramatically here.
- 3 **Orbicularis oculi** (CN VII): pretarsal, preseptal, and orbital portions. Pretarsal and preseptal heads form the lacrimal pump; the orbital portion produces forced closure.
- 4 **Orbital septum,** with preaponeurotic (postseptal) fat immediately behind it. Fat behind the septum is the surgical landmark for the levator: the levator aponeurosis lies directly deep to the preaponeurotic fat pad.
- 5 **Levator palpebrae superioris aponeurosis,** with its medial and lateral horns.
- 6 **Müller muscle** (superior tarsal muscle): smooth muscle, sympathetically innervated, from the undersurface of the levator to the superior tarsal border, contributing ~2 mm of lid elevation.
- 7 **Palpebral conjunctiva:** firmly adherent to the tarsus, non-keratinised stratified squamous with goblet cells.

The lower lid mirrors this: skin, orbicularis, septum, **capsulopalpebral fascia** (the inferior rectus analogue of the levator aponeurosis), the **inferior tarsal muscle** (sympathetic Müller analogue), tarsus, and conjunctiva. Disinsertion of the capsulopalpebral fascia produces involutional entropion.

PTOSIS LOCALISATION FOLLOWS THE ANATOMY

- **CN III palsy:** levator denervated, ptosis is severe (>4 mm), often with pupil involvement and ophthalmoplegia.
- **Horner syndrome:** Müller denervated, ptosis is mild (1–2 mm), with *reverse ptosis* of the lower lid from inferior tarsal muscle loss, miosis, and anhidrosis. Apparent enophthalmos is an illusion created by the narrowed fissure.
- **Aponeurotic (involutional) ptosis:** levator function is normal, the aponeurosis has stretched. High crease, good excursion, deep superior sulcus.
- **Myasthenia gravis:** variable, fatigable ptosis with Cogan lid twitch; pupil is *always* spared.

Glands of the eyelid and ocular surface

ADNEXAL GLANDS: TYPE, SECRETION, AND THE LESION EACH PRODUCES

Gland	Type	Location	Secretion / role and associated pathology
Meibomian (tarsal)	Holocrine sebaceous	Vertically within the tarsus; ~25 lower, 30–40 upper	Meibum: the outer lipid layer of the tear film, retarding evaporation. Obstruction causes MGD and evaporative dry eye; retained obstruction with granulomatous reaction gives an INTERNAL HORDEOLUM or, when sterile and chronic, a CHALAZION . Sebaceous cell carcinoma arises here.
Zeis	Holocrine sebaceous	Attached to lash follicles	Oils the lash shaft. Infection gives an EXTERNAL HORDEOLUM (STYE) .

Gland	Type	Location	Secretion / role and associated pathology
Moll	Apocrine sweat	Lid margin, near lashes	Modified sweat gland; contributes antimicrobial peptides. Ductal occlusion gives the translucent APOCRINE HIDROCYSTOMA (SUDORIFEROUS CYST) .
Krause	Accessory lacrimal, serous	Fornices: ~40 upper, ~6–8 lower	Basal aqueous tear secretion. Preserved in some cases of lacrimal gland loss, which is why basal tearing can persist.
Wolfring (Ciaccio)	Accessory lacrimal, serous	Along the proximal tarsal border: ~2–5 upper, ~2 lower	Basal aqueous secretion.
Goblet cells	Unicellular mucous	Conjunctival epithelium, densest in the inferonasal fornix and plica	Secrete MUC5AC , the principal gel-forming mucin. Destroyed in cicatrising conjunctivitis (ocular cicatricial pemphigoid, Stevens-Johnson, trachoma) and in vitamin A deficiency, producing Bitot spots and keratinisation.
Crypts of Henle / glands of Manz	Mucous	Henle: superior tarsal conjunctiva; Manz: limbal (present in animals, debated in humans)	Accessory mucin.
Main lacrimal gland	Tubuloacinar serous	Superotemporal lacrimal fossa	Reflex and the bulk of aqueous tear secretion. Target of Sjögren syndrome and of sarcoid/IgG4 dacryoadenitis.

WHICH GLAND, WHICH CYST

Moll is **M**oist (sweat, apocrine, clear cyst = hidrocystoma). **Zeis** is **Z**ebaceous (oil on the lash, styte). Meibomian is **M**eaty and lives in the tarsus (chalazion).

Eyelid vasculature and its clinical danger

The lids receive a double supply, which is why lid flaps survive well.

- **From the internal carotid** via the ophthalmic artery: supraorbital, supratrochlear, dorsal nasal, lacrimal, and the medial palpebral arteries.
- **From the external carotid** via the facial artery (angular artery), superficial temporal artery (lateral palpebral, transverse facial), and infraorbital artery.
- These anastomose in the **marginal arcade** (in the pretarsal plane **2–4 mm** from the margin) and the **peripheral arcade** (at the upper tarsal border, between levator aponeurosis and Müller muscle).

WHY LID CELLULITIS IS NOT A TRIVIAL DIAGNOSIS

Venous drainage runs medially through the **angular vein** to the superior ophthalmic vein and then to the **cavernous sinus**. The orbital and facial veins are **valveless**, so flow can reverse. This is the anatomic basis of the "danger triangle" of the face and of cavernous sinus thrombosis complicating a medial canthal or nasal infection. Bilateral involvement, proptosis, chemosis, and multiple cranial neuropathies (III, IV, V1, V2, VI) signal that the infection has crossed into the sinus.

Eyelid sensory innervation**CUTANEOUS SENSORY MAP**

Region	Nerve	Division
Upper lid, central and lateral	Supraorbital (from frontal nerve)	V1
Upper lid, medial	Supratrochlear (from frontal nerve)	V1
Medial canthus, side of nose	Infratrochlear (from nasociliary)	V1
Lateral upper lid, lacrimal gland, conjunctiva	Lacrimal nerve	V1
Lower lid, central	Infraorbital	V2
Lower lid, lateral / cheek	Zygomaticofacial	V2
Tip of nose (Hutchinson sign)	External nasal branch of anterior ethmoidal, from nasociliary	V1

HUTCHINSON SIGN

Herpes zoster vesicles on the **tip, side, or root of the nose** indicate involvement of the nasociliary nerve. Because the nasociliary also supplies the globe through the long ciliary nerves, this predicts a substantially higher risk of zoster keratitis and iritis. The rest of V1 dermatomal involvement does not carry the same predictive weight.

The lacrimal system**Secretory apparatus**

The **main lacrimal gland** sits in the lacrimal fossa of the frontal bone in the superotemporal orbit. The lateral horn of the levator aponeurosis divides it into a larger **orbital lobe** and a smaller **palpebral lobe**, the latter visible through the superior temporal fornix on lid eversion. About 12 ductules drain the orbital lobe *through* the palpebral lobe into the superior temporal fornix.

SURGICAL COROLLARY

Because the orbital lobe's ducts traverse the palpebral lobe, **biopsy or excision of the palpebral lobe can de-function the entire gland**. Biopsy is therefore taken from the orbital lobe when tissue is needed.

PARASYMPATHETIC PATHWAY TO THE LACRIMAL GLAND: MEMORISE THIS CHAIN

- 1 **Lacrimal (superior salivatory) nucleus** in the pons.
- 2 Fibres leave with the **nervus intermedius** of CN VII.
- 3 Branch off at the geniculate ganglion as the **greater superficial petrosal nerve**.
- 4 Joins the deep petrosal (sympathetic) to become the **nerve of the pterygoid canal (vidian nerve)**.
- 5 Synapse in the **pterygopalatine (sphenopalatine) ganglion**.
- 6 Postganglionic fibres travel with the **zygomatic nerve (V2)**, then its zygomaticotemporal branch, and join the **lacrimal nerve (V1)** to reach the gland.

Sympathetic supply is from the superior cervical ganglion along the internal carotid plexus, reaching the gland via the deep petrosal nerve and the lacrimal artery. Sensory supply is the lacrimal nerve (V1). Arterial supply is the **lacrimal artery** from the ophthalmic artery.

CROCODILE TEARS AND THE VIDIAN NERVE

After a proximal facial nerve injury, regenerating salivatory fibres may misroute into the lacrimal pathway, producing **gustatory lacrimation**: tearing when eating. The shared pterygopalatine relay is the reason this aberrant regeneration is possible.

Excretory apparatus, in order of tear flow**DRAINAGE PATHWAY: DIMENSIONS MATTER HERE**

Structure	Dimension	Detail
Puncta	0.2–0.3 mm	Upper punctum ~6 mm and lower ~6.5 mm from the medial canthus; the lower sits slightly more lateral so they do not appose on closure. Normally directed posteriorly into the tear lake.
Ampulla	2 mm vertical	Vertical segment before the canaliculus turns horizontal. Dilate here before probing to avoid a false passage.
Canaliculi	8–10 mm horizontal	Lined by non-keratinised stratified squamous epithelium. Join to form a common canaliculus in ~90% of people.
Valve of Rosenmüller	–	At the junction of the common canaliculus with the sac; prevents reflux back into the canaliculi. Its competence explains why a mucocele distends the sac rather than refluxing.
Lacrimal sac	10–12 mm tall, 4–8 mm wide	In the lacrimal fossa between the anterior (maxillary) and posterior (lacrimal bone) lacrimal crests. Fundus rises 3–5 mm above the medial canthal tendon. Lined by pseudostratified columnar epithelium with goblet cells.
Nasolacrimal duct	12–18 mm	Intraosseous then membranous; travels inferiorly, posteriorly, and laterally.
Valve of Hasner	–	Mucosal fold at the duct's opening into the inferior nasal meatus . Failure to canalise here causes congenital nasolacrimal duct obstruction , which resolves spontaneously in most infants within the first year.

THE LACRIMAL PUMP (JONES MECHANISM)

Contraction of the **pretarsal orbicularis** shortens the canaliculi and compresses the ampullae while the **preseptal orbicularis (Horner muscle)** pulls the sac's posterolateral wall to create negative pressure; on relaxation, elastic recoil plus gravity and capillary force move tears into the duct. This is why **orbicularis weakness (CN VII palsy) causes epiphora even with a patent system**, and why lower lid laxity or ectropion produces a functional obstruction with normal irrigation.

Roughly **10–25%** of tear volume is lost to evaporation; the remainder drains, with the lower canaliculus historically credited with the larger share, though both contribute substantially.

Extraocular muscles

Origins, insertions, and actions

THE SIX EXTRAOCULAR MUSCLES

Muscle	Origin	Insertion from limbus	CN	Primary / secondary / tertiary action
Medial rectus	Annulus of Zinn	5.5 mm	III (inferior division)	Adduction only. Longest arc of contact, strongest rectus.
Inferior rectus	Annulus of Zinn	6.5 mm	III (inferior division)	Depression / extorsion / adduction. Acts on a 23° angle from the visual axis.
Lateral rectus	Annulus of Zinn + spina recti lateralis	6.9 mm	VI	Abduction only.
Superior rectus	Annulus of Zinn	7.7 mm	III (superior division)	Elevation / intorsion / adduction. 23° angle.
Superior oblique	Body of sphenoid above and medial to the optic canal; functional origin is the trochlea	Posterosuperior temporal quadrant, behind the equator	IV	Intorsion / depression / abduction. Longest muscle and longest tendon (~26 mm tendon; total ~40 mm). Tendon makes a 51° angle with the visual axis.
Inferior oblique	Periosteum of the maxilla, lateral to the nasolacrimal canal opening; the only EOM not originating at the apex	Posteroinferior temporal quadrant, near the macula	III (inferior division)	Extorsion / elevation / abduction. Shortest muscle (~37 mm), essentially no tendon. 51° angle.

TWO MNEMONICS THAT CARRY A LOT OF WEIGHT

LR₆SO₄, all the rest CN III. Lateral rectus is CN VI, superior oblique is CN IV.

Spiral of Tillaux: 5-6-7-8 (approximately) for Medial 5.5, Inferior 6.5, Lateral 6.9, Superior 7.7 mm from the limbus. Read it as "MILS", nasal to superior, increasing distance.

"Obliques do the opposite": the obliques *abduct* while the vertical recti *adduct*; the superior muscles *intort* and the inferior muscles *extort* (Superiors are Intorters: "SIN" and "RAID": Recti ADduct, obliques abduct).

WHY THE FIELD OF GAZE CHANGES MUSCLE ACTION

The vertical recti are most efficient **elevators/depressors in abduction** (their line of pull aligns with the visual axis at 23° of abduction). The obliques are most efficient **elevators/depressors in adduction** (aligning at 51° of adduction). This is the entire logic of the **H-pattern / cardinal positions** test and of the three-step (Parks-Bielschowsky) test.

In torsion the obliques dominate: SO intorts, IO extorts, and they do so maximally in *abduction*, which is why a superior oblique palsy shows worsening hypertropia on *ipsilateral head tilt*.

Muscle structure and blood supply

- EOMs are among the fastest and most fatigue-resistant striated muscles in the body, with the **highest ratio of nerve fibres to muscle fibres** (about 1:3-1:5 versus 1:100+ in limb muscle), permitting exquisitely graded movement.
- Two layers: an **orbital layer** (outer, inserting on the pulley connective tissue, entirely fatigue-resistant) and a **global layer** (inner, inserting on the sclera). Fibre types include singly innervated fast twitch and multiply innervated slow tonic fibres, the latter unusual outside the EOMs.
- Arterial supply**: the superior and inferior muscular branches of the ophthalmic artery. The **lateral rectus is supplied by the lacrimal artery**, and the inferior oblique also receives infraorbital contributions.
- Anterior ciliary arteries** arise from the muscular branches and run in the rectus tendons to the limbus: **two each** from the medial, superior, and inferior recti and **one** from the lateral rectus, for **seven** total. They supply the anterior segment. This is why detaching three or more rectus muscles in one sitting risks **ANTERIOR SEGMENT ISCHAEMIA**.
- The **vortex veins** (four to eight, one to two per quadrant) exit near the equator and drain the uvea into the superior and inferior ophthalmic veins.
- Levator palpebrae superioris, though a lid muscle, originates at the apex above the annulus and is supplied by CN III superior division along with the superior rectus, which is why ptosis and elevation deficit travel together.

NUMBERS TO KNOW: ORBIT, LIDS, LACRIMAL, EOM

Orbital volume	~30 mL	Globe volume	~6.5 mL
Orbital depth (rim to apex)	40-45 mm	Orbital axis divergence	~45° total
Optic canal length	8-10 mm	Lamina papyracea thickness	0.2-0.4 mm

Palpebral fissure height	8–11 mm	Fissure width	28–30 mm
MRD1	4–5 mm	Upper tarsus height	10–11 mm
Lower tarsus height	4–5 mm	Müller muscle contribution	~2 mm
Punctum diameter	0.2–0.3 mm	Canaliculus length	8–10 mm
Lacrimal sac height	10–12 mm	Nasolacrimal duct length	12–18 mm
Spiral of Tillaux MR	5.5 mm	IR	6.5 mm
LR	6.9 mm	SR	7.7 mm
SO tendon angle to visual axis	51°	Vertical recti angle	23°
Anterior ciliary arteries	7 total	Meibomian glands, upper lid	30–40

RETRIEVAL PRACTICE

Answer aloud, then uncover the key. Do not read the key first.

- Which two bones form the orbital roof, and what structure lies immediately superior to it?
- A patient has cheek numbness, diplopia on upgaze, and orbital emphysema after being struck by a ball. Name the two fractures and the nerve involved.
- Which four structures enter the orbit *outside* the annulus of Zinn?
- A patient has 1.5 mm of ptosis, miosis, and elevation of the lower lid. What is the lesion, and which muscle is denervated?
- Name the ganglion where lacrimal parasympathetic fibres synapse and the two nerves that carry the postganglionic fibres to the gland.
- Where does the nasolacrimal duct open, and what is the name of the fold at that opening?
- Why does a seventh nerve palsy cause epiphora when syringing is patent?
- Which extraocular muscle does not originate at the orbital apex, and what is its primary action?
- How many anterior ciliary arteries are there, and which muscle contributes only one?
- In which horizontal position of gaze is the inferior rectus the most efficient depressor, and why?

KEY

- Orbital plate of the frontal bone and lesser wing of the sphenoid; the anterior cranial fossa (and frontal sinus anteriorly).
- Orbital floor blowout (maxillary sinus, entrapment of inferior rectus/inferior oblique on upgaze) and medial wall (lamina papyracea, emphysema); the infraorbital nerve, V2.
- CN IV, frontal nerve, lacrimal nerve, and the superior ophthalmic vein.
- Horner syndrome, an oculosympathetic lesion; Müller (superior tarsal) muscle, with the inferior tarsal muscle producing the reverse ptosis.
- Pterygopalatine (sphenopalatine) ganglion; zygomatic branch of V2 (via zygomaticotemporal) joining the lacrimal nerve of V1.
- Into the inferior nasal meatus; the valve of Hasner.
- The orbicularis-driven lacrimal pump fails, so tears are not actively propelled; the anatomic lumen is open but the pump is not working.
- Inferior oblique, from the maxilla near the nasolacrimal canal; primary action is extorsion.
- Seven; the lateral rectus.
- In abduction of about 23°, where its line of pull aligns with the visual axis so its entire force becomes depression.

CHAPTER RECAP: EXPAND EACH LINE INTO A PARAGRAPH

1. Seven bones, four walls, and one rule: thinnest is medial, weakest is the posteromedial floor, strongest is lateral.
2. The annulus of Zinn sorts the apex: III, VI, nasociliary, sympathetics inside; IV, frontal, lacrimal, superior ophthalmic vein outside.
3. The orbital septum divides preseptal from orbital cellulitis; the gray line divides anterior from posterior lamella.
4. Levator is CN III and gives gross ptosis; Müller is sympathetic and gives 1 to 2 mm with reverse ptosis.
5. Gland to lesion: Meibomian to chalazion, Zeis to sty, Moll to hidrocystoma, goblet cell loss to keratinisation.
6. Valveless facial and orbital veins connect the medial canthus to the cavernous sinus.
7. Lacrimal parasympathetics: pons to VII to greater petrosal to vidian to pterygopalatine to V2 to lacrimal nerve.
8. Tear drainage: punctum, ampulla, canaliculus, Rosenmüller, sac, duct, Hasner, inferior meatus, and an orbicularis pump that must work.
9. Spiral of Tillaux 5.5 / 6.5 / 6.9 / 7.7; LR6SO4; recti adduct and obliques abduct; superiors intort.
10. Seven anterior ciliary arteries mean three-muscle surgery risks anterior segment ischaemia.

Conjunctiva, cornea, sclera, and the limbus

Conjunctiva / cornea / refractive surgery • 34–50 items

Episclera / sclera / anterior uvea • 16–24 items

Ametropia (corneal power)

Contact lenses

ANSWER THESE COLD BEFORE READING

1. Give central and peripheral corneal thickness, both radii of curvature, the refractive index, and the surface powers.
2. Name the corneal layers in order with thickness, cell type, and whether each regenerates.
3. State why the cornea is transparent and the sclera is not, in mechanistic terms.
4. Give normal adult endothelial cell density and the density at which decompensation becomes likely.
5. Describe the conjunctival lymphatic drainage and explain why it matters clinically.
6. State scleral thickness at four named locations and name the three sets of emissaria.

BLUEPRINT WEIGHT

12%

Conjunctiva / cornea / refractive surgery is the largest single non-systemic area at 34–50 scored items.

The cornea: dimensions and optics

CORNEAL QUANTITATIVE ANATOMY

Parameter	Value	Why it is asked
Horizontal diameter	11.5–12.0 mm	Below 10 mm is microcornea; above 13 mm is megalocornea (or buphthalmos if with raised IOP).
Vertical diameter	10.5–11.0 mm	The cornea is a horizontally oval structure viewed from the front; the sclera overlaps it more superiorly and inferiorly.
Central thickness	0.52–0.54 mm	Central corneal thickness corrects Goldmann IOP and is a risk factor in ocular hypertension.
Peripheral thickness	0.65–0.70 mm	The cornea thickens peripherally, which is the opposite of a simple lens and explains peripheral pachymetry values.
Anterior radius of curvature	7.7–7.8 mm	Base curve selection in contact lens fitting starts from this.
Posterior radius of curvature	6.5–6.8 mm	Steeper than the front surface, giving the posterior surface negative power.
Refractive index	1.376	Stroma 1.375–1.376; tears 1.336; aqueous 1.336.
Anterior surface power	+48.8 D	Air (1.000) to cornea (1.376) across 7.7 mm.
Posterior surface power	–5.8 D	Cornea (1.376) to aqueous (1.336) across 6.8 mm.

Parameter	Value	Why it is asked
Net corneal power	~+43 D	About two thirds to 70% of the eye's total ~60 D. Keratometers are calibrated to a single index of 1.3375 to report this net value from the front surface alone.
Optical zone / apex	central 4 mm	The central 4 mm is nearly spherical; the cornea flattens progressively toward the limbus (prolate, positive asphericity, Q value about -0.26).
Corneal hydration	78% water	Stromal deturgescence is actively maintained; loss of that control is the mechanism of every form of corneal oedema.

KERATOMETRY INDEX TRAP

The keratometer measures only the *anterior* surface but reports *total* corneal power by using a fictitious index of 1.3375 instead of the true 1.376. After myopic refractive surgery this assumption fails, because the surgery changes the anterior surface without changing the posterior, so keratometry overestimates corneal power and IOL calculations go wrong. That is the anatomic reason post-LASIK IOL formulas exist.

Corneal layers, from anterior to posterior

TEARS (lipid / aqueous / mucin) n = 1.336

1	EPITHELIUM	50-52 um	~10%	5-7 cell layers	REGENERATES (days)
	superficial squamous	2-3 layers		microplicae, tight junctions = barrier	
	wing cells	2-3 layers		polyhedral	
	basal columnar	1 layer		ONLY mitotic layer, hemidesmosomes	
2	BASAL LAMINA	0.05 um		type IV collagen + laminin	regenerates slowly
3	BOWMAN LAYER	8-12 um		acellular type I/V collagen	DOES NOT REGENERATE
4	STROMA	~500 um	~90%	200-250 lamellae	heals with scar
	type I (+ V) collagen fibrils,	22-35 nm,		regular lattice spacing	
	keratocytes ~2-3% of volume;			keratan sulfate is the dominant GAG	
5	DESCMET MEMBRANE	3 um	birth -> 10-12 um	adult type IV/VIII	REGENERATES
	anterior banded (fetal) + posterior			non-banded (lifelong deposition)	
6	ENDOTHELIUM	5 um		single hexagonal layer	AMITOTIC in humans
	Na+/K+ ATPase pumps; density	2000-3000/mm2		adult	

AQUEOUS n = 1.336

Epithelium

- **5-7 cell layers, 50-52 µm**, about **10%** of total thickness. Turnover **7-10 days**.
- **Basal columnar cells**: the only layer capable of mitosis. Attached to the basal lamina by **hemidesmosomes** anchored by anchoring fibrils (type VII collagen) into the stroma. Disruption of this adhesion complex is the mechanism of **RECURRENT CORNEAL EROSION** and of epithelial basement membrane dystrophy.
- **Wing cells**: transitional, polyhedral, joined by desmosomes and gap junctions.
- **Superficial squamous cells**: bear **microvilli and microplicae** that anchor the mucin layer, and are joined by **zonulae occludentes (tight junctions)**. Those tight junctions are the ocular surface's principal barrier to fluorescein, drugs, and pathogens: break them and the cornea stains.

- Epithelium is **~10×** more permeable to lipophilic than hydrophilic molecules, while the stroma is the reverse. A drug must therefore be **biphasic** to penetrate, which is why prodrug esters (dipivefrin, latanoprost) exist.

Bowman layer versus Descemet membrane

THE TWO LIMITING LAMINAE COMPARED

Feature	Bowman (anterior limiting lamina)	Descemet (posterior limiting lamina)
Nature	Condensed acellular anterior stroma, <i>not</i> a true basement membrane	A true basement membrane, secreted by the endothelium
Collagen	Type I and V, randomly arranged	Type IV and VIII
Thickness	8–12 μm , thins with age	3 μm at birth to 10–12 μm in adults
Regeneration	Never. Injury leaves a permanent scar	Readily, by healthy endothelium
Elasticity	Not elastic; does not curl	Elastic; when torn it curls inward (scrolls) : the classic finding in birth forceps injury and in acute hydrops
Named pathology	Reis-Bücklers dystrophy; loss in keratoconus with breaks producing Vogt striae precursors	Haab striae (congenital glaucoma, horizontal), breaks in keratoconus hydrops, guttae in Fuchs dystrophy, Hassall-Henle bodies peripherally with normal ageing

GUTTAE VERSUS HASSALL-HENLE BODIES

Identical histology, different location. **Peripheral** excrescences of Descemet membrane are **HASSALL-HENLE BODIES** and are a normal age change. **Central** excrescences are **GUTTAE** and define **FUCHS ENDOTHELIAL CORNEAL DYSTROPHY**.

Stroma and the physics of transparency

The stroma is **~90%** of corneal thickness, built from **200–250** lamellae of type I collagen running parallel to the surface and crossing at angles between lamellae. Keratocytes occupy the interlamellar spaces and are joined by gap junctions.

WHY THE CORNEA IS TRANSPARENT

1. **Fibril diameter is small and uniform** (**22–35 nm**), well below the wavelength of visible light.
2. **Fibril spacing is regular and short** (centre to centre **~60 nm**), so scattered wavefronts from adjacent fibrils interfere destructively: the **Maurice lattice theory**.
3. **Avascularity**.
4. **Relative dehydration** to **78%** water, actively maintained. Above roughly **80%** hydration the lattice spacing becomes irregular and the cornea scatters: clinical oedema.
5. **Smooth optical surfaces** maintained by the tear film and the epithelium.

The **sclera** has the same collagen types but *irregular* fibril diameters (**25–230 nm**) and irregular spacing, plus higher hydration. That irregularity is the entire reason it is white and the cornea is clear.

Glycosaminoglycans: **keratan sulfate** is the predominant corneal GAG, with chondroitin and dermatan sulfate also present. GAGs are strongly hydrophilic and generate the stromal swelling pressure that the endothelium must continuously oppose. Keratan sulfate accumulates in

MACULAR CORNEAL DYSTROPHY, which is why that dystrophy stains with Alcian blue and colloidal iron and is the only classic stromal dystrophy that is autosomal *recessive*.

Endothelium

ENDOTHELIAL CELL DENSITY ACROSS LIFE

Stage	Density (cells/mm ²)	Note
Birth	~5000–6000	Cells are small and uniform
Young adult	~3000–3500	Coefficient of variation low, hexagonality high
Typical adult	2000–3000	Physiologic loss ~0.6% per year
At-risk threshold	<1000	Reduced functional reserve; guarded for intraocular surgery
Decompensation	~500	Stromal oedema and bullous keratopathy become likely

Human endothelium is effectively **amitotic**: healing occurs by cell enlargement (**polymegathism**, rising coefficient of variation) and shape change (**pleomorphism**, falling percentage of hexagonal cells) rather than division. Both indices worsen before density does, so they are the early markers of contact-lens-induced hypoxic stress.

Corneal innervation

- **70–80** stromal nerve trunks derived from the **long ciliary nerves** (nasociliary branch of V1) enter radially at the mid-peripheral mid-stroma.
- They lose their myelin sheaths **1–2 mm** inside the limbus, which preserves transparency, then form the subepithelial and **subbasal nerve plexus** before terminating as free nerve endings between epithelial cells.
- The cornea is the **most densely innervated tissue in the body**, roughly **300–600×** the density of skin. Sensitivity is highest centrally.
- Nerves are **trophic as well as sensory**. Denervation (herpes zoster, diabetes, post-LASIK, CN V lesions) produces **NEUROTROPHIC KERATOPATHY** with poor healing despite an adequate tear film.

NUTRITION OF AN AVASCULAR TISSUE

Oxygen comes almost entirely from the atmosphere dissolved in the tear film (from the palpebral conjunctival vessels during sleep), which is why contact lens oxygen transmissibility is a corneal, not a conjunctival, problem. **Glucose** comes from the aqueous humour posteriorly. The peripheral cornea additionally receives from the limbal arcades. Waste and lactate leave via aqueous and tears.

The limbus

A **1–2 mm** transition zone in which corneal structures convert to conjunctival and scleral ones. It contains, in a single small volume, most of the anterior segment's clinically critical anatomy.

- **Palisades of Vogt**: radial ridges of pigmented limbal epithelium housing the **limbal epithelial stem cells**. Densest superiorly and inferiorly. Their loss (chemical burn, Stevens-Johnson,

aniridia, contact lens overwear) produces **LIMBAL STEM CELL DEFICIENCY** with conjunctivalisation and persistent epithelial defects.

- Bowman layer terminates; the epithelium thickens to **10–12** layers.
- Descemet membrane terminates at **Schwalbe line**.
- The internal scleral sulcus houses the **trabecular meshwork** and **Schlemm canal**; the **scleral spur** is the posterior lip.
- Limbal arcades and the terminal anterior ciliary vessels arrive here, along with the marginal corneal arcade: the substrate for limbal flush in iritis and for peripheral corneal infiltrates.
- **Thoft X, Y, Z hypothesis** of epithelial maintenance: X = proliferation of basal cells, Y = centripetal migration from the limbus, Z = epithelial cell loss from the surface. X + Y must equal Z for a stable epithelium.

The conjunctiva

Regional anatomy

CONJUNCTIVAL ZONES

Zone	Features
Palpebral	Marginal (from the mucocutaneous junction to the tarsal edge), tarsal (tightly adherent, so follicles and papillae are best seen here), and orbital (loose, with horizontal folds).
Forniceal	Loose and redundant to permit eye movement. Deepest superiorly (~10 mm from limbus) and shallower inferiorly (~8 mm); there is no true medial fornix . Site of the glands of Krause. Greatest goblet cell density regionally lies inferonasally.
Bulbar	Thin and translucent over the sclera, freely mobile except at the limbus where it fuses with Tenon capsule and episclera.
Plica semilunaris	Vestigial third eyelid at the medial canthus; a crescentic fold of conjunctiva with goblet cells, allowing globe adduction.
Caruncle	Modified <i>skin</i> at the medial canthus. Contains hair follicles, sebaceous glands, sweat glands, and accessory lacrimal tissue. This mixed composition is why oncocytoma, sebaceous lesions, and hidrocystoma all occur here.

Histology

- **Epithelium:** 2–5 layers of non-keratinised stratified squamous over the tarsus and bulbar surface, becoming more columnar in the fornices, with interspersed **goblet cells** (about **1.5 million** total). Also contains melanocytes, Langerhans cells, and intraepithelial lymphocytes.
- **Substantia propria (stroma)**, two layers:
 - **Adenoid (lymphoid) layer**, superficial, containing lymphocytes, plasma cells, mast cells, and the **conjunctiva-associated lymphoid tissue (CALT)** with follicles. **It does not develop until roughly 2 to 3 months of age.**
 - **Fibrous layer**, deeper, carrying vessels and nerves, and fusing with the tarsus.

TWO AGE-DRIVEN DIAGNOSTIC RULES FROM ONE HISTOLOGIC FACT

Because the adenoid layer is absent at birth, **a neonate cannot mount a follicular response**. Neonatal chlamydial conjunctivitis therefore presents as a *papillary*, not follicular, conjunctivitis, whereas in an adult the same organism gives florid follicles. Conversely, membrane formation is prominent in the neonate.

Papilla = a vascular core with fibrous septa tethering the epithelium to the tarsus, so the vessel is *central*. Seen in allergic and bacterial disease.

Follicle = a lymphoid germinal centre, avascular in the middle with vessels *encircling* the base. Seen in viral, chlamydial, and toxic disease.

Vessels, lymphatics, and nerves

- **Arteries:** the palpebral arcades supply the palpebral and forniceal conjunctiva and the posterior conjunctival arteries continue onto the peripheral bulbar conjunctiva; the **anterior ciliary arteries** supply the perilimbal conjunctiva and give the anterior episcleral plexus.
- **Distinguishing the plexuses at the slit lamp:**
 - *Superficial conjunctival vessels:* bright red, move with the conjunctiva on cotton-bud manipulation, blanch with topical phenylephrine 2.5%. Injection here is conjunctival.
 - *Superficial episcleral plexus:* salmon pink, radial, moves less, blanches partially. Injected in episcleritis.
 - *Deep episcleral plexus:* violaceous, does **not** move and does **not** blanch with phenylephrine. Injected in scleritis and in anterior uveitis (ciliary flush).
- **Lymphatics:** the conjunctiva and lids are the only ocular structures with lymphatic drainage (the globe itself has none, which underpins ocular immune privilege). **Lateral two thirds drain to the preauricular (parotid) nodes; medial one third drains to the submandibular nodes.** Preauricular lymphadenopathy is therefore a cardinal sign of adenoviral and chlamydial conjunctivitis.
- **Sensory innervation:** V1 via the lacrimal, infratrochlear, supratrochlear, and long ciliary nerves; the inferior conjunctiva additionally by the infraorbital nerve (V2).

The sclera and episclera**SCLERAL THICKNESS BY LOCATION: A FAVOURITE ITEM**

Location	Thickness	Clinical consequence
Posterior pole / around optic nerve	1.0 mm	Thickest region
At the limbus	0.6–0.8 mm	Site of trabeculectomy and tunnel incisions
Equator	0.4–0.6 mm	Vortex vein exit
Immediately behind the rectus insertions	0.3 mm	Thinnest point of the eye wall. The site of inadvertent perforation in strabismus surgery and of scleral staphyloma in scleritis or high myopia.

- Scleral radius of curvature ~11.5 mm; hydration 65–70%; it becomes translucent when dehydrated, which is why a drying sclera looks bluish at surgery.

- **Layers:** **episclera** (vascular, loose, fused with Tenon capsule), **scleral stroma** (dense irregular type I and III collagen with fibroblasts, sparse vessels), and the **lamina fusca** (inner, pigmented, with melanocytes, blending into the suprachoroidal space).
- **Emissaria** (channels through the sclera):
 - *Anterior* (near the limbus): anterior ciliary arteries and veins, aqueous veins.
 - *Middle* (at the equator): the four to eight **vortex veins**.
 - *Posterior*: long and short posterior ciliary arteries, long and short ciliary nerves, and the optic nerve through the **lamina cribrosa**.
- The sclera is largely avascular and nourished by the episcleral and choroidal circulations; it is richly supplied by **sensory ciliary nerves**, which is why scleritis produces boring, night-waking pain and episcleritis does not.
- **Blue sclera** means thin sclera revealing underlying uvea: osteogenesis imperfecta, Ehlers-Danlos, Marfan, and normal infancy.

Refractive surgery anatomy

WHICH LAYERS EACH PROCEDURE VIOLATES

Procedure	Tissue plane	Consequence that follows from the anatomy
PRK	Epithelium and Bowman layer removed, anterior stroma ablated	Bowman does not regenerate, so haze and subepithelial fibrosis are the characteristic complications. No flap, so no flap complications and no dislocation risk. More pain, slower visual recovery.
LASIK	Lamellar flap through epithelium, Bowman, and anterior stroma; ablation in the stromal bed	Severs the subbasal nerve plexus : transient neurotrophic dry eye is expected. Flap interface permits epithelial ingrowth, diffuse lamellar keratitis, and late traumatic dislocation. Anterior stroma is biomechanically the strongest part, so removing it from load-bearing predisposes to ectasia.
SMILE	Intrastromal lenticule removed through a small incision; Bowman and most anterior stroma preserved	Less nerve transection and no flap, so less dry eye and no flap dislocation.
Phakic IOL / ICL	No corneal tissue removed	Endothelial cell loss and pigment dispersion / pupillary block become the issues instead.

THE STROMAL ARITHMETIC YOU MUST BE ABLE TO DO

Residual stromal bed after LASIK should be at least **250–300 μm** , and total residual corneal thickness at least **400 μm** . Roughly **12–15 μm** of central stroma is ablated per dioptre of myopia in a **6 mm** optical zone (Munnerlyn: ablation depth $\approx$ diopters $\times$ (optical zone diameter)² / 3).

Worked example. CCT **530 μm** , flap **110 μm** , correction **–6.00 D** at a **6 mm** zone. Ablation $\approx 6 \times 36 / 3 = 72 \mu\text{m}$. Residual bed = $530 - 110 - 72 = 348 \mu\text{m}$: acceptable. Repeat with CCT **480 μm** and the bed is **298 μm** : marginal, and PRK or SMILE becomes the better anatomic choice.

NUMBERS TO KNOW: CORNEA, CONJUNCTIVA, SCLERA

Corneal horizontal diameter	11.5–12 mm	Central thickness	0.52–0.54 mm
Peripheral thickness	0.65–0.70 mm	Anterior radius	7.7–7.8 mm
Posterior radius	6.5–6.8 mm	Corneal index	1.376
Anterior surface power	+48.8 D	Posterior surface power	–5.8 D
Net corneal power	~+43 D	Keratometer index	1.3375
Epithelium thickness	50–52 μm	Epithelial turnover	7–10 days
Bowman layer	8–12 μm	Descemet, adult	10–12 μm
Stromal lamellae	200–250	Collagen fibril diameter	22–35 nm
Corneal hydration	78%	Endothelium, adult	2000–3000 /mm ²
Endothelial loss rate	~0.6% /yr	Decompensation density	~500 /mm ²
Stromal nerve trunks	70–80	Goblet cells, total	~1.5 million
Sclera, posterior pole	1.0 mm	Sclera behind rectus insertion	0.3 mm
Scleral fibril diameter	25–230 nm	LASIK residual bed minimum	250–300 μm

RETRIEVAL PRACTICE

Cover the key. Say each answer out loud before checking.

1. Which corneal layer contains the only mitotically active cells, and how is it anchored?
2. Give the two reasons the cornea is transparent that relate specifically to collagen fibrils.
3. A tear in Descemet membrane scrolls inward. Name two conditions in which you would see this.
4. Central endothelial excrescences versus peripheral: name each and state which is pathologic.
5. Which GAG predominates in the cornea, and which dystrophy accumulates it?
6. Why must a topical drug be biphasic to reach the anterior chamber?
7. A 3-week-old infant has a purulent conjunctivitis with membranes but no follicles. What histologic fact explains the absence of follicles?
8. A red eye does not blanch with phenylephrine 2.5%. Which vascular plexus is involved, and name two diagnoses.
9. Where does the lateral two thirds of the conjunctiva drain, and which two infections classically produce that node?
10. Where is the eye wall thinnest, and name two clinical consequences.
11. CCT is 505 μm , planned flap 100 μm , correction -5.00 D at a 6.5 mm zone. Is LASIK anatomically safe?
12. Name the structure that houses limbal stem cells and one disease that destroys it.

KEY

1. Basal columnar layer; hemidesmosomes to the basal lamina with type VII collagen anchoring fibrils into the stroma.
2. Fibril diameter is uniform and smaller than the wavelength of light, and fibril spacing is regular so scattered light interferes destructively (Maurice lattice).
3. Birth forceps injury and acute hydrops in keratoconus (also congenital glaucoma with Haab striae).
4. Central = guttae, pathologic, Fuchs dystrophy. Peripheral = Hassall-Henle bodies, a normal age change.

- | | |
|---|--|
| <ul style="list-style-type: none"> 5. Keratan sulfate; macular corneal dystrophy (autosomal recessive). 6. The epithelium favours lipophilic molecules and the stroma favours hydrophilic ones, so a drug must have both characters to cross both. 7. The adenoid (lymphoid) layer of the substantia propria does not develop until about 2 to 3 months of age, so no germinal centres can form. 8. Deep episcleral plexus; scleritis and anterior uveitis (ciliary flush). | <ul style="list-style-type: none"> 9. Preauricular (parotid) nodes; adenoviral and chlamydial conjunctivitis. 10. Sclera immediately behind the rectus insertions at about 0.3 mm; inadvertent scleral perforation during strabismus surgery, and staphyloma formation in scleritis or high myopia. 11. Ablation $\approx 5 \times 42.25 / 3 \approx 70 \mu\text{m}$. Residual bed = $505 - 100 - 70 = 335 \mu\text{m}$: acceptable, but total residual thickness and topography must also be checked. 12. Palisades of Vogt; chemical burn, Stevens-Johnson syndrome, aniridia, or contact lens overwear. |
|---|--|

CHAPTER RECAP

1. Cornea: 12 mm wide, 540 μm central, 7.8 front / 6.8 back radius, index 1.376, +48.8 and -5.8 giving +43 D, two thirds of ocular power.
2. Six layers, and only three of them regenerate: epithelium, Descemet, and (as scar) stroma. Bowman never does; endothelium never divides.
3. Transparency = uniform small fibrils, regular spacing, avascularity, 78% hydration, smooth surface. The sclera fails all but the first.
4. Endothelium 2000–3000/ mm^2 , losing 0.6% per year, decompensating near 500. Polymegathism and pleomorphism precede density loss.
5. Long ciliary nerves from V1 make the cornea the most innervated tissue in the body and make denervation a healing problem.
6. Limbus = palisades of Vogt, Schwalbe line, trabecular meshwork, Schlemm canal, scleral spur, limbal arcades, in 1 to 2 mm.
7. Conjunctival adenoid layer arrives at 2 to 3 months: no follicles in a neonate. Papilla is central-vessel, follicle is peripheral-vessel.
8. Phenylephrine blanches conjunctival and superficial episcleral vessels, not the deep episcleral plexus.
9. Lateral conjunctiva to preauricular nodes, medial to submandibular. The globe has no lymphatics.
10. Sclera: 1.0 mm posteriorly, 0.3 mm behind rectus insertions; three sets of emissaria; pain because of ciliary nerves.

Uvea, anterior chamber angle, lens, and vitreous

Episclera / sclera / anterior uvea • 16–24 items

Lens / cataract / IOL • 8–14 items

Glaucoma • 13–21 items

Vitreous / retina / choroid • 17–25 items

ANSWER THESE COLD BEFORE READING

1. Name the four layers of the iris and state which embryonic layer each derives from.
2. Name the three parts of the ciliary muscle, their innervation, and what each does to outflow.
3. List the gonioscopic angle structures anterior to posterior and give the trabecular subdivisions with pore sizes.
4. State the percentage of aqueous leaving by trabecular versus uveoscleral routes and what shifts the balance.
5. Give lens diameter, thickness, capsule thickness front and back, and refractive index gradient.
6. Rank the sites of vitreoretinal adhesion from strongest to weakest.

BLUEPRINT WEIGHT

~20%

Combined across anterior uvea, lens, glaucoma, and vitreous: roughly 54 to 84 scored items depend on this chapter.

The iris

Surface anatomy

- Diameter ~12 mm; thickest at the **collarette** (~0.6 mm), thinnest at the root (~0.1 mm), which is why the root is where **IRIDODIALYSIS** and **PERIPHERAL IRIDOTOMY** occur.
- The **collarette** divides the **pupillary zone** (inner third, containing the sphincter) from the **ciliary zone** (outer two thirds, containing the dilator). It marks where the embryonic pupillary membrane attached; persistent pupillary membrane strands arise from it.
- Pupil diameter ranges 1–8 mm, decentred slightly nasally and inferiorly. Physiologic anisocoria of up to 0.4–1.0 mm occurs in about 20% of people and is equal in light and dark.
- **Crypts of Fuchs**: stromal openings near the collarette and root. **Contraction furrows**: circumferential grooves in the ciliary zone. **Fuchs freckles** (naevi) sit on the anterior border layer; **Brushfield spots** (Down syndrome) are areas of relative stromal hyperplasia; **Lisch nodules** (neurofibromatosis 1) are melanocytic hamartomas of the anterior border layer.

Layers, anterior to posterior

IRIS LAYERS AND THEIR EMBRYONIC ORIGIN

Layer	Origin	Composition and clinical relevance
Anterior border layer	Neural crest mesenchyme	Condensed stroma of fibroblasts and melanocytes. There is no endothelium and no epithelium on the anterior iris surface , so the stroma is in direct contact with aqueous. Naevi and Lisch nodules sit here. Loss of this layer in pigment dispersion gives the "moth-eaten" transillumination pattern.

Layer	Origin	Composition and clinical relevance
Stroma	Neural crest mesenchyme	Loose collagen, melanocytes, mast cells, radially oriented vessels with characteristically thick collagenous walls , nerves, and the sphincter pupillae . Iris colour is determined by the <i>amount of melanin</i> in these stromal melanocytes, not by their number.
Anterior (myo)epithelium	Neuroectoderm, outer layer of optic cup	Pigmented cuboidal cells whose <i>basal</i> processes form the dilator pupillae : this is a myoepithelium, an unusual arrangement worth remembering.
Posterior pigment epithelium	Neuroectoderm, inner layer of optic cup	Densely pigmented columnar cells, continuous with the non-pigmented ciliary epithelium and, embryologically, with the neurosensory retina. It curls around the pupil margin to form the pupillary ruff . Its pigment is what disperses in pigmentary glaucoma and what forms Krukenberg spindle.

MUSCLE, NERVE, AND ROUTE: DO NOT MIX THESE UP

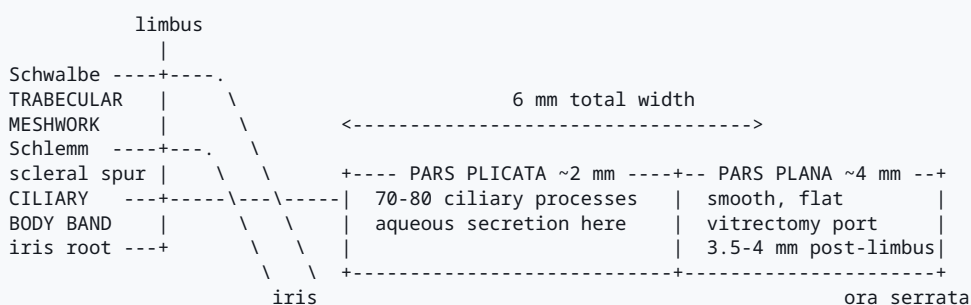
Muscle	Type	Autonomic	Nerve route	Receptor
Sphincter pupillae	Circular smooth muscle in the stroma, 0.75–1 mm wide	Parasympathetic	CN III inferior division → branch to inferior oblique → ciliary ganglion → short ciliary nerves	M ₃
Dilator pupillae	Radial myoepithelium from the anterior epithelium	Sympathetic	Superior cervical ganglion → carotid plexus → nasociliary → long ciliary nerves (passing <i>through</i> but not synapsing in the ciliary ganglion)	α ₁

The sphincter is stronger than the dilator, which is why a pharmacologically or structurally intact sphincter always wins: mydriasis requires either sphincter paralysis or a very potent α-agonist.

Iris vasculature

- The **major arterial circle** lies in the *ciliary body*, not the iris, and is formed by the two **long posterior ciliary arteries** and the **anterior ciliary arteries**. Radial iris arterioles run from it to a **minor arterial circle** near the collarette.
- Iris capillaries are **non-fenestrated with tight junctions**: together with the non-pigmented ciliary epithelium they constitute the **blood-aqueous barrier**. Breakdown of this barrier is what produces cells and flare.
- Normal iris vessels are **radial and regular**. Neovascularisation of the iris is **irregular, arborising, and begins at the pupil margin and in the angle**: that distinction is the observation on which neovascular glaucoma is diagnosed.

The ciliary body



- Total width ~6 mm from the scleral spur to the ora serrata. Triangular in cross section with the base facing forward.
- **Pars plicata** (corona ciliaris), anterior ~2 mm, carrying 70-80 ciliary processes. This is the site of aqueous formation and of zonular origin.
- **Pars plana**, posterior ~4 mm, smooth and relatively avascular. It is **wider temporally than nasally**, which is why the standard vitrectomy sclerotomy is placed 3.5-4.0 mm posterior to the limbus in a phakic adult (3.0-3.5 mm in a pseudophake, less in a child).

Layers, outside to inside

1. **Supraciliary lamina** (continuous with the lamina fusca and the suprachoroidal space). Fluid accumulation here is a *ciliochoroidal detachment*; it is also the potential space that cyclodialysis exploits.
2. **Ciliary muscle** (see below).
3. **Stroma** with the major arterial circle and vessels of the ciliary processes.
4. **Pigmented ciliary epithelium**: continuous with the retinal pigment epithelium.
5. **Non-pigmented ciliary epithelium**: continuous with the neurosensory retina. **Tight junctions between non-pigmented cells form the blood-aqueous barrier**, and these cells are rich in Na^+/K^+ ATPase and **carbonic anhydrase II**: the two pharmacologic targets for suppressing aqueous production.
6. **Internal limiting membrane**, continuous with the retinal ILM.

CILIARY MUSCLE: THREE PARTS, TWO CONSEQUENCES

Portion	Orientation and insertion	Functional effect
Longitudinal (Brücke)	Outermost, meridional, inserts into the scleral spur and trabecular meshwork	Contraction puts traction on the meshwork, widening its spaces and increasing trabecular outflow . This is precisely how pilocarpine lowers IOP, and it simultaneously <i>reduces</i> uveoscleral outflow.
Radial / oblique	Middle, fan-shaped	Intermediate.
Circular (Müller)	Innermost, sphincter-like, near the processes	Contraction moves the ciliary body forward and inward, slackening the zonules and allowing the lens to become more convex: accommodation.

Innervation: parasympathetic from CN III via the ciliary ganglion and short ciliary nerves acting on M_3 receptors to contract; sympathetic β_2 input relaxes it (a minor effect clinically, but the reason β -blockers can slightly shift accommodation).

The choroid

- Thickness **0.22–0.25 mm** at the posterior pole, thinning to **~0.1 mm** anteriorly. Extends from the optic nerve to the ora serrata.
- **Layers, outer to inner:** lamina fusca / suprachoroid → **Haller layer** (large vessels) → **Sattler layer** (medium vessels) → **choriocapillaris** → **Bruch membrane**.
- The **choriocapillaris is fenestrated** and has the **highest blood flow per unit weight of any tissue in the body**, with a low arteriovenous oxygen difference. Its jobs are to nourish the outer retina across Bruch membrane and to act as a heat sink for absorbed light.
- **Bruch membrane**, five layers: RPE basal lamina, inner collagenous zone, elastic layer, outer collagenous zone, choriocapillaris endothelial basal lamina. Thickens and calcifies with age; breaks here permit choroidal neovascularisation and produce *lacquer cracks* in pathologic myopia and *angioid streaks* in pseudoxanthoma elasticum, Paget disease, sickle cell disease, and Ehlers-Danlos.
- **Supply:** short posterior ciliary arteries feed the choriocapillaris in a lobular, segmental pattern (hence **TRIANGULAR AMALRIC INFARCTS** and **ELSCHNIG SPOTS**); long posterior ciliary arteries and anterior ciliary arteries feed the anterior choroid. Drainage is by the **vortex veins**. The **watershed zone** near the disc explains the pattern of ischaemic optic neuropathy.
- **Innervation** by long and short ciliary nerves; the choroid is richly supplied with sympathetic vasoconstrictor and parasympathetic vasodilator fibres.

Anterior chamber, posterior chamber, and the angle

CHAMBER DIMENSIONS

Parameter	Value
Anterior chamber volume	0.20–0.31 mL (~0.25)
Central AC depth	3.0–3.5 mm

Parameter	Value
AC diameter	11.3–12.4 mm
Posterior chamber volume	~0.06 mL
Aqueous formation rate	2.0–3.0 $\mu\text{L}/\text{min}$ (~2.5)
Aqueous turnover	~1–1.5% of AC volume / min

Gonioscopic anatomy

ANGLE STRUCTURES, ANTERIOR TO POSTERIOR

"**I Can See Some Structures**" read *backwards* from the cornea inward, or learn the forward sequence directly:

Schwalbe line → Trabecular meshwork → Scleral spur → Ciliary body band → Iris root

Forward mnemonic: "**Steep TRAIl Sends Cattle Inward**", or simply remember the white line, the meshwork, the white spur, the brown band, the iris.

WHAT EACH ANGLE LANDMARK LOOKS LIKE AND MEANS

Structure	Gonioscopic appearance	Significance
Schwalbe line	Fine translucent ridge; the termination of Descemet membrane. Located by the corneal wedge : the point where the two beams of an optic section meet.	A prominent, anteriorly displaced Schwalbe line is POSTERIOR EMBRYOTOXON , seen in Axenfeld-Rieger. Pigment on it is the SAMPAOLESI LINE of pigment dispersion syndrome and pseudoexfoliation.
Trabecular meshwork	Ground-glass band; anterior (non-filtering) portion is pale, posterior (filtering) portion overlying Schlemm canal is greyer and takes up pigment.	Blood in Schlemm canal is visible here when episcleral venous pressure exceeds IOP (carotid-cavernous fistula, Sturge-Weber, or over-compression with the goniolens).
Scleral spur	The whitest, most sharply defined line in the angle; the posterior lip of the internal scleral sulcus.	The universal reference landmark: the angle is called "open to the spur", "open to the ciliary body band", and so on. Insertion point of the longitudinal ciliary muscle.
Ciliary body band	Grey to dark brown band; the anterior face of the ciliary body.	Abnormally wide band means ANGLE RECESSION from blunt trauma, a cause of late unilateral glaucoma. Also wide in myopia and aphakia.
Iris root / processes	Iris insertion; fine iris processes (of Busacca) may bridge to the meshwork.	Distinguish delicate, lace-like iris processes from broad, tent-like peripheral anterior synechiae , which obliterate the view of underlying structures.

Trabecular meshwork microanatomy and outflow

THREE TRABECULAR LAYERS, INNER TO OUTER

Layer	Pore size	Role
Uveal meshwork	25–75 μm	Innermost, from iris root to Schwalbe line. Largest openings, negligible resistance.
Corneoscleral meshwork	5–50 μm	Middle, sheet-like beams from scleral spur to the lateral wall of the scleral sulcus.
Juxtacanalicular (cribriform) tissue	smallest	The principal site of normal aqueous outflow resistance. Ground substance plus the inner wall endothelium of Schlemm canal. This is where extracellular matrix accumulates in primary open-angle glaucoma.

- **Schlemm canal:** an endothelium-lined circumferential channel, **190–370 μm** in diameter, whose inner wall cells form **giant vacuoles** that transport aqueous transcellularly. **25–35 collector channels** leave it for the deep and intrascleral venous plexuses and then the episcleral veins. Direct connections that carry clear aqueous are the **aqueous veins of Ascher**, visible at the slit lamp as a clear stream laminating with blood.
- **Outflow split:** conventional (trabecular) route **~80–90%**, pressure-dependent; uveoscleral (unconventional) route **~10–20%**, largely pressure-independent, through the ciliary muscle interstitium into the supraciliary space and out through the sclera.

TWO DRUGS, TWO OPPOSITE MUSCLE EFFECTS ON TWO DIFFERENT ROUTES

- **Pilocarpine** (muscarinic agonist): ciliary muscle contracts → spur traction → *increases trabecular outflow*, but simultaneously *decreases uveoscleral outflow* by compacting the muscle bundles.
- **Prostaglandin analogues:** remodel ciliary muscle extracellular matrix via matrix metalloproteinases → *increases uveoscleral outflow* predominantly.
- **Atropine / cycloplegics:** muscle relaxes → uveoscleral outflow increases slightly, trabecular outflow falls. In an anatomically narrow angle this can precipitate angle closure, which is a mechanical, not a pharmacologic, effect.

The crystalline lens

LENS QUANTITATIVE ANATOMY

Parameter	Value	Note
Equatorial diameter	6.5 mm at birth → 9–10 mm adult	Grows throughout life; fibres are never shed.
Axial thickness	3.5–4 mm young → ~5 mm by 70	Thickening plus increased curvature underlies the myopic shift of nuclear sclerosis and the "second sight" phenomenon.
Anterior radius of curvature	10 mm relaxed → ~6 mm accommodated	The anterior surface does nearly all the accommodative work.
Posterior radius of curvature	6 mm → ~5.5 mm	Changes little.

Parameter	Value	Note
Refractive index	1.386 cortex → 1.406 nucleus	A gradient index optic; this is why the lens delivers more power than its surfaces alone predict and why an equivalent index of about 1.42 is used in schematic eyes.
Power in situ	+15 to +20 D (~+19 D)	About one third of ocular power; in air the isolated lens would be roughly +100 D .
Weight	65 mg at birth → ~250 mg at 80	Continuous growth.
Water content	~66%	Lowest water content of any tissue except enamel; protein content ~33% is the highest of any tissue.

Capsule, epithelium, fibres

- **Capsule:** the **thickest basement membrane in the body**, type IV collagen, secreted by the epithelium. **Anterior capsule** ~14–15.5 μm (thickest pre-equatorially), **posterior capsule** ~4 μm (thinnest at the posterior pole). That single asymmetry explains why posterior capsule rupture is the feared complication of phacoemulsification and why posterior lenticonus and posterior subcapsular opacity behave as they do.
- **Epithelium:** a single layer beneath the *anterior* capsule only. **There is no posterior lens epithelium after embryogenesis**. The **germinative (pre-equatorial) zone** divides; new cells migrate to the equator, elongate, lose their organelles, and become fibres. Migration of these cells across the posterior capsule after cataract surgery causes **POSTERIOR CAPSULE OPACIFICATION** (and Elschnig pearls, and Soemmering ring).
- **Sutures:** fibres from opposite sides meet in sutures. The fetal nucleus shows an **upright Y anteriorly and an inverted Y posteriorly**. The embryonic nucleus (from the primary lens fibres) has no sutures.
- **Zones, oldest to newest:** embryonic nucleus → fetal nucleus → infantile nucleus → adult nucleus → cortex. The centre is the oldest tissue in the body.
- **Zonules of Zinn:** fibrillin-rich fibres from the basal lamina of the non-pigmented epithelium in the valleys between ciliary processes (pars plana and plicata) to the lens capsule pre-equatorially and post-equatorially, leaving a zonule-free zone at the equator.
- The lens is **avascular and non-innervated**. All nutrition and waste exchange occur through the aqueous, predominantly anteriorly.

DIRECTION OF ECTOPIA LENTIS LOCALISES THE DISEASE

Displacement	Disease	Mechanism
Superotemporal	Marfan syndrome	<i>FBN1</i> fibrillin defect; zonules stretched but usually intact. Autosomal dominant, with aortic dilatation and arachnodactyly.
Inferonasal	Homocystinuria	Cystathionine β -synthase deficiency; zonules disintegrate. Autosomal recessive, with thromboembolism risk under general anaesthesia.
Superonasal or variable, microspherophakia	Weill-Marchesani	Short stature, brachydactyly, small round lens prone to pupillary block.
Downward, with aniridia or coloboma	Sulfite oxidase deficiency, hyperlysinaemia, trauma	Traumatic dialysis displaces the lens <i>away</i> from the torn quadrant.

The vitreous

- Volume **~4.0 mL**, about **80%** of the globe's volume. **98–99%** water, held as a gel by a scaffold of **type II collagen** filled with **hyaluronan**. Refractive index **1.336**. Hyalocytes populate the cortex.
- **Named spaces and remnants:**
 - **Patellar fossa:** the saucer-shaped depression accommodating the posterior lens.
 - **Berger space:** the retrolental space between posterior capsule and anterior hyaloid.
 - **Wieger (hyaloideocapsular) ligament:** annular attachment of the anterior hyaloid to the posterior capsule; its anterior margin is the **ring of Egger**.
 - **Cloquet canal:** the S-shaped remnant of the primary vitreous running from the posterior lens to the disc. Its funnel-shaped termination at the disc is the **area of Martegiani**.
 - A **persistent hyaloid artery remnant** on the disc is a **BERGMEISTER PAPILLA**; on the lens it is a **MITTENDORF DOT** (typically inferonasal).

VITREORETINAL ADHESION, STRONGEST TO WEAKEST

- 1 **Vitreous base:** straddles the ora serrata, extending ~2 mm anterior and ~3–4 mm posterior to it. Adhesion here is so strong that the base essentially never detaches; traction instead avulses retina, producing **giant retinal tears and dialyses** at its posterior margin.
- 2 **Margins of lattice degeneration and chorioretinal scars:** abnormally firm focal adhesion, which is why lattice generates horseshoe tears at its posterior edge.
- 3 **Posterior lens capsule** (Wieger ligament) in the young eye.
- 4 **Optic disc margin:** separation releases a glial annulus, the **Weiss ring**, the pathognomonic sign of complete posterior vitreous detachment.
- 5 **Perimacular / foveal attachment:** incomplete separation here causes vitreomacular traction, macular hole, and epiretinal membrane.
- 6 **Along major retinal vessels:** traction here tears a vessel and causes vitreous haemorrhage without a retinal break.
- 7 **Rest of the posterior retina:** weakest, so this is where separation begins.

Ageing. Hyaluronan-collagen dissociation produces **syneresis** (liquefaction) and **syneretic cavities**, then collapse and **posterior vitreous detachment**. PVD is the proximate cause of most acute symptomatic floaters, of most retinal tears, and of the single most important history question in an acute floaters presentation.

NUMBERS TO KNOW: UVEA, ANGLE, LENS, VITREOUS

Iris diameter	~12 mm	Pupil range	1–8 mm
Sphincter width	0.75–1 mm	Ciliary body total width	~6 mm
Pars plicata / pars plana	2 mm / 4 mm	Ciliary processes	70–80
Vitrectomy port from limbus	3.5–4 mm	Choroid, posterior pole	0.22–0.25 mm
AC volume	~0.25 mL	Central AC depth	3.0–3.5 mm
Aqueous production	2.0–3.0 µL/min	Trabecular / uveoscleral outflow	80–90% / 10–20%
Schlemm canal diameter	190–370 µm	Collector channels	25–35
Uveal meshwork pores	25–75 µm	Corneoscleral pores	5–50 µm
Lens diameter, adult	9–10 mm	Lens thickness, young	3.5–4 mm
Anterior capsule	~14–15.5 µm	Posterior capsule	~4 µm
Lens index range	1.386–1.406	Lens power in situ	+15 to +20 D
Vitreous volume	~4 mL	Vitreous water content	98–99%
Vitreous base, anterior extent	~2 mm pre-ora	posterior extent	3–4 mm post-ora

RETRIEVAL PRACTICE

Twelve questions. Target: eleven correct with no hesitation.

1. Which iris layer gives rise to the dilator pupillae, and what is unusual about that muscle?
2. Why does the anterior iris surface have direct contact with aqueous, and what does that mean for drug penetration?
3. Name the two arterial sources of the major arterial circle and state where it lies.
4. Which epithelium forms the blood-aqueous barrier, and name its two key enzymes.
5. Explain, mechanically, how pilocarpine lowers IOP and what it does to uveoscleral outflow.
6. Give the gonioscopic sequence from cornea inward and state which landmark is the reference.
7. Which trabecular layer generates most outflow resistance?
8. A wide ciliary body band in one eye only. What is the likely history and the late risk?
9. Anterior versus posterior lens capsule thickness, and the surgical consequence.
10. Why is there no posterior lens epithelium, and what condition results from anterior epithelial cells migrating posteriorly?
11. Superotemporal versus inferonasal lens subluxation: name the disease and the molecular defect for each.
12. Rank the top three sites of vitreoretinal adhesion and give the lesion each produces on traction.

KEY

1. The anterior (myo)epithelium, derived from the outer layer of the optic cup; it is a myoepithelium, unusual for a muscle to arise from an epithelial layer.
2. There is no anterior endothelium or epithelium, only a condensed anterior border layer, so drugs and pigment interact directly with stroma.
3. The two long posterior ciliary arteries and the anterior ciliary arteries; it lies in the ciliary body, not the iris.
4. Non-pigmented ciliary epithelium, joined by tight junctions; Na^+/K^+ ATPase and carbonic anhydrase II.
5. Longitudinal (Brücke) fibres pull on the scleral spur, opening the trabecular spaces and increasing conventional outflow; uveoscleral outflow decreases.
6. Schwalbe line, trabecular meshwork, scleral spur, ciliary body band, iris root; the scleral spur is the reference landmark.
7. Juxtacanalicular (cribriform) tissue with the inner wall of Schlemm canal.
8. Blunt ocular trauma with angle recession; late unilateral open-angle glaucoma.
9. Anterior about 14 to 15.5 μm , posterior about 4 μm ; the posterior capsule is fragile and is the structure at risk in phacoemulsification.
10. Posterior epithelial cells elongate into the primary lens fibres during embryogenesis and are consumed; migration of residual equatorial epithelium onto the posterior capsule causes posterior capsule opacification.
11. Superotemporal = Marfan, *FBN1* fibrillin-1. Inferonasal = homocystinuria, cystathionine β -synthase deficiency.
12. Vitreous base (giant retinal tear or dialysis), lattice margins (horseshoe tear), optic disc (Weiss ring with PVD).

CHAPTER RECAP

1. Iris: four layers, two from neural crest and two from neuroectoderm; sphincter is parasympathetic via short ciliary nerves, dilator sympathetic via long ciliary nerves.
2. Ciliary body 6 mm: pars plicata 2 mm secretes aqueous, pars plana 4 mm is the surgical entry.
3. Brücke to the spur opens the meshwork; Müller slackens zonules for accommodation.
4. Blood-aqueous barrier = non-pigmented ciliary epithelium tight junctions plus non-fenestrated iris capillaries.
5. Choriocapillaris is fenestrated with the highest blood flow per gram in the body; Bruch membrane breaks give CNV, lacquer cracks, and angioid streaks.
6. Angle: Schwalbe, meshwork, spur, band, root. Juxtacanalicular tissue is the resistor. 80 to 90% trabecular, 10 to 20% uveoscleral.
7. Lens grows forever, has no posterior epithelium, a 4 μm posterior capsule, and a gradient index.
8. Lens dislocation direction names the metabolic disease.
9. Vitreous is 4 mL of 99% water on a type II collagen and hyaluronan scaffold.
10. Adhesion order: base, lattice margins, capsule, disc, macula, vessels, posterior retina. That order predicts every traction lesion.

CHAPTER 4 · GROSS ANATOMY AND HISTOLOGY

Retina, RPE, photoreceptors, and retinal circulation

Vitreous / retina / choroid · 17–25 items

Perceptual function / color vision

Systemic health (diabetes, hypertension)

Optic nerve / neuro-ophthalmic pathways

ANSWER THESE COLD BEFORE READING

1. Name the ten retinal layers from vitreous to choroid and state which cell bodies live in each nuclear layer.
2. Give the diameters of macula, fovea, foveola, and the foveal avascular zone.
3. State which retinal layers the retinal circulation supplies and which the choroid supplies, and name the two blood-retinal barriers.
4. Give rod and cone total counts, peak cone density, and the three cone absorption peaks.
5. List six functions of the retinal pigment epithelium.
6. Explain the anatomic basis of the arcuate visual field defect.

BLUEPRINT WEIGHT

6%

17–25 items directly, plus the anatomic substrate for most systemic health and neuro items.

The ten layers

VITREOUS

=====	1	INTERNAL LIMITING MEMBRANE
Muller cell footplates + basal lamina. Peeled in ILM surgery.	2	NERVE FIBRE LAYER
-----	3	GANGLION CELL LAYER
Unmyelinated ganglion cell axons, arcuate bundles.]	4	INNER PLEXIFORM LAYER] CIRCULATION
-----	5	INNER NUCLEAR LAYER
Ganglion cell bodies (~1.0-1.2 million).] RETINAL	6	OUTER PLEXIFORM LAYER <-- WATERSHED
Bipolar-to-ganglion + amacrine synapses. ON and OFF sublaminae.	7	OUTER NUCLEAR LAYER
-----	8	EXTERNAL LIMITING MEMBRANE
Bipolar, horizontal, amacrine, Muller nuclei.]	9	PHOTORECEPTOR LAYER
Photoreceptor-to-bipolar/horizontal synapses.	10	RETINAL PIGMENT EPITHELIUM

Rod and cone nuclei.]		

Zonulae adherentes between Muller cells and photoreceptors.]		

Inner and outer segments.]		
=====		
BRUCH MEMBRANE / CHORIOCAPILLARIS / CHOROID		

THE WATERSHED AT THE OUTER PLEXIFORM LAYER IS THE HIGHEST-YIELD SINGLE FACT HERE

The **retinal circulation** (central retinal artery) supplies the inner retina down to and including the **outer plexiform layer**. The **choriocapillaris** supplies the outer nuclear layer and the photoreceptors by diffusion across Bruch membrane and the RPE.

Therefore: a **central retinal artery occlusion** infarcts the inner retina, which becomes opaque and white, while the intact choroid still perfuses the foveola where the retina is thinnest, producing the **CHERRY RED SPOT**. A **choroidal** occlusion instead damages photoreceptors and RPE. Every ischaemia question in retina descends from this one boundary.

Cell populations

RETINAL CELLS: COUNTS, LAYER, AND FUNCTION

Cell	Nuclei in	Count / role
Rods	ONL	110–125 million . Scotopic, high sensitivity, low resolution, no colour. Absent in the foveola. Peak density $\sim 160,000/\text{mm}^2$ at about $18\text{--}20^\circ$ eccentricity.
Cones	ONL	6–7 million . Photopic, colour, high acuity. Peak density $\sim 199,000/\text{mm}^2$ at the foveola.
Bipolar cells	INL	The vertical relay. ON (depolarised by light, metabotropic mGluR6) and OFF (hyperpolarised, ionotropic) types. Rod bipolars are all ON-type and feed AII amacrine cells.
Horizontal cells	INL	Lateral inhibition in the OPL: creates the antagonistic receptive field surround.
Amacrine cells	INL	Over 30 subtypes; lateral and vertical modulation in the IPL. AII amacrine carry the rod signal into the cone pathway. Dopaminergic and starburst (direction-selective) subtypes are named on exams.
Ganglion cells	GCL	1.0–1.2 million , matching the number of optic nerve axons. Types below.
Müller cells	INL	The principal glia: span the whole retina from ILM to ELM, form both limiting membranes, buffer potassium and glutamate, store glycogen, and generate the b-wave of the ERG indirectly. Their contraction is implicated in macular pucker.
Astrocytes and microglia	NFL / GCL	Astrocytes ensheath vessels and axons in the inner retina; microglia are the resident immune cells.

GANGLION CELL CLASSES: THE THREE PARALLEL CHANNELS

Class	Share	Projection and properties
Midget (P cells)	$\sim 80\%$	→ parvocellular LGN layers 3, 4, 5, 6. Small receptive fields, sustained response, red-green opponency, high spatial and low temporal resolution. Carry central acuity and colour. In the foveola the ratio approaches one cone : one midget bipolar : one midget ganglion cell : the anatomic basis of 20/20 or better.
Parasol (M cells)	$\sim 10\%$	→ magnocellular LGN layers 1, 2. Large receptive fields, transient response, high contrast sensitivity at low contrast, motion and flicker. Preferentially damaged early in glaucoma, which is the rationale for frequency-doubling and motion-based perimetry.
Small bistratified	$\sim 8\text{--}10\%$	→ koniocellular layers between the main LGN laminae. Blue-yellow (S-cone) opponency. Targeted by short-wavelength automated perimetry.
Intrinsically photosensitive (ipRGC)	$\sim 1\text{--}3\%$	Contain melanopsin , peak sensitivity $\sim 480\text{ nm}$. Project to the suprachiasmatic nucleus (circadian entrainment) and olivary pretectal nucleus (pupil light reflex). They explain why the pupil still reacts in some cases of profound photoreceptor loss.

Topography of the posterior pole

MACULAR SUBDIVISIONS: MEMORISE THE DIAMETERS

Region	Diameter	Definition and features
Area centralis / macula lutea	5.5–6.0 mm	Anatomically, the region with more than one ganglion cell layer. Yellow from xanthophyll carotenoids: lutein and zeaxanthin , concentrated in the Henle fibre layer, which filter short wavelengths and quench free radicals.
Perifovea	1.5 mm annulus	Outermost macular ring.

Region	Diameter	Definition and features
Parafovea	0.5 mm annulus	Ring immediately around the fovea; thickest retina in the eye (~0.56 mm).
Fovea (centralis)	1.5 mm (~1 disc diameter, 5°)	The central depression. Inner layers are displaced peripherally, so photoreceptor axons run obliquely as the Henle fibre layer . Retinal thickness ~0.25 mm.
Foveola	0.35 mm (~1.2°)	Rod-free. Contains only cones, with peak density and the thinnest retina in the eye (0.13–0.18 mm). No inner retinal layers, no vessels.
Umbo	0.15–0.2 mm	The very centre of the foveola; source of the foveal reflex whose loss is an early sign of macular disease.
Foveal avascular zone	0.4–0.5 mm	The capillary-free zone. Its enlargement or irregularity on angiography or OCT-A is a quantitative marker of macular ischaemia in diabetes.

WHY THE FOVEA SEES BEST

1. Highest cone density, with the thinnest cones and the smallest inter-receptor spacing (~2–2.5 µm), setting the Nyquist limit near 60 cycles/degree.
 2. Near one-to-one cone-to-ganglion-cell wiring: minimal spatial summation.
 3. Inner retinal layers and vessels are displaced aside so light reaches cones with minimal scatter.
 4. Cones are optically funnelled by the Stiles-Crawford effect, favouring rays through the pupil centre.
- Every one of those advantages is also a vulnerability: the fovea has no capillaries, so it depends entirely on the choriocapillaris and on the perifoveal capillary ring.

The optic disc and the nerve fibre layer

- Optic disc: 1.5 mm horizontally by 1.75–1.9 mm vertically (vertically oval), carrying ~1.2 million axons. Located ~4 mm nasal to and slightly above the fovea, corresponding to a blind spot ~15° temporal to fixation in the visual field.
- **Nerve fibre layer arrangement:**
 - **Papillomacular bundle:** fibres from the macula enter the temporal disc directly. Selective loss gives a **central or centrocaecal scotoma**: toxic, nutritional, and hereditary optic neuropathies.
 - **Arcuate (arciform) bundles:** temporal retinal fibres arch above and below the papillomacular bundle to enter the superior and inferior poles. They respect the **horizontal temporal raphe** and never cross it.
 - Nasal fibres enter radially and directly.

ANATOMY OF THE ARCUATE DEFECT

Because arcuate bundles arch around the macula and stop at the horizontal raphe, damage at the superior or inferior disc pole (the thinnest neuroretinal rim, hence the **ISNT rule**: rim is normally thickest Inferior > Superior > Nasal > Temporal) produces a field defect that:

- arches from the blind spot toward fixation,
- **ends abruptly at the horizontal midline nasally** (a nasal step), because fibres do not cross the raphe,
- and spares central acuity until late.

That is the glaucomatous field defect, and it is a purely anatomic prediction. A defect that crosses the horizontal midline is not arcuate and points elsewhere.

The retinal pigment epithelium

- A single layer of **~3.5 million** hexagonal cells extending from the disc to the ora serrata, where it continues as the pigmented ciliary epithelium. Cells are **tall, narrow, and densely pigmented centrally** (~14 µm tall, 10–14 µm wide) and **flatter and broader peripherally**.
- Photoreceptor-to-RPE ratio: about **30:1** at the macula, **20:1** peripherally. Each RPE cell phagocytoses roughly that many outer segment tips daily.
- **Apical microvilli** interdigitate with photoreceptor outer segments; there is no anatomic attachment, only apposition maintained by the RPE pump, the interphotoreceptor matrix, and vitreous pressure. That is why the retina detaches at this plane and why subretinal fluid resolves when RPE function returns.
- **Basal infoldings** sit on Bruch membrane; the RPE basal lamina is Bruch's innermost layer.
- **Tight junctions (zonulae occludentes)** between RPE cells form the **outer blood-retinal barrier**.

SIX FUNCTIONS OF THE RPE, AND THE DISEASE THAT FOLLOWS EACH FAILURE

Function	Failure produces
Phagocytosis and recycling of shed outer segments	Accumulation of lipofuscin (A2E) and drusen; the basis of age-related macular degeneration and of the dark choroid in Stargardt disease (<i>ABCA4</i>).
The visual cycle: re-isomerisation of all-<i>trans</i> retinol to 11-<i>cis</i>-retinal via RPE65 and LRAT	Leber congenital amaurosis (<i>RPE65</i>) and retinitis punctata albescens; the target of voretigene gene therapy.
Transport of ions, water, and metabolites; maintenance of the subretinal space	Serous detachment: central serous chorioretinopathy, and the pigment epithelial detachments of AMD.
Outer blood-retinal barrier (tight junctions)	Leakage patterns on fluorescein angiography; RPE-level pooling.
Absorption of stray light by melanin	Photophobia and glare in albinism; also loss of the barrier that makes hypopigmented fundi transilluminate.
Secretion of growth factors: VEGF basally toward the choroid, PEDF apically	Imbalance drives choroidal neovascularisation; this is the target of every anti-VEGF injection.

Photoreceptors in detail

RODS VERSUS CONES

Feature	Rods	Cones
Number	110-125 million	6-7 million
Outer segment discs	Detached from the plasma membrane; renewed at the base, shed at the tip, complete turnover in ~10 days. Shedding is at dawn .	Continuous with the plasma membrane (infoldings). Shedding is at dusk .
Photopigment	Rhodopsin , peak ~496-500 nm	Three opsins: S/blue ~419-440 nm, M/green ~531-535 nm, L/red ~558-565 nm
Sensitivity	Single-photon detection; saturate in daylight	Need ~100 photons; do not saturate
Convergence	High (up to 100:1 onto a ganglion cell): great sensitivity, poor resolution	Low, approaching 1:1 at the foveola
Synaptic terminal	Spherule : one invagination, contacts rod bipolar and 2 horizontal processes	Pedicle : many invaginations, each a triad of one bipolar dendrite between two horizontal cell processes
Dark adaptation	Slow (20-30 min), second limb of the curve	Fast (5-7 min), first limb
Distribution	Absent in the foveola; peak at 18-20°	Peak at foveola, sparse in periphery

TWO EASILY REVERSED FACTS

Rod discs are free-floating; cone discs are continuous with the membrane. That is why rods can renew an entire outer segment in ten days and why rod-specific dystrophies present with nyctalopia first.

Blue (S) cones are the rarest, comprise only about 5-10% of cones, and are **absent from the very centre of the foveola** (the foveal tritanopia). Nonetheless the S-cone pathway is the most vulnerable to early acquired disease, which is why acquired colour defects are usually blue-yellow and inherited defects are usually red-green (Köllner rule, with its well-known exceptions).

Retinal circulation

- The **central retinal artery**, the first branch of the ophthalmic artery, enters the optic nerve ~10-12 mm behind the globe and divides into superior and inferior branches, then nasal and temporal: four arcades. Branches are **end arteries with no anastomoses**, which is why occlusion produces a sharply defined wedge of infarction.
- **Capillary beds**: a superficial plexus in the NFL/GCL, a deep plexus at the INL/OPL borders, an intermediate plexus, and the **radial peripapillary capillaries** in the NFL near the disc, which are the ones whose infarction produces a cotton wool spot.
- **Inner blood-retinal barrier**: non-fenestrated endothelium with tight junctions, supported by pericytes at roughly a 1:1 endothelium-to-pericyte ratio (the highest in the body). **Pericyte loss is the earliest histologic lesion of diabetic retinopathy**, and it precedes microaneurysm formation.

- **Cilioretinal artery:** present in **15–32%** of eyes, arising from the choroidal (short posterior ciliary) circulation. It spares a wedge of retina, often including the fovea, in CRAO, and conversely it is the vessel that infarcts in isolation with giant cell arteritis.
- Retinal vessels are **autoregulated and have no autonomic innervation**; the **choroid is not autoregulated and is richly innervated**. Remember this pair.
- Arteriole-to-venule diameter ratio is normally **2:3**. Arterioles and venules share a common adventitial sheath at crossings, which is the anatomic reason hypertensive arteriolar thickening causes venous nicking and branch retinal vein occlusion at arteriovenous crossings.

The peripheral retina

- **Ora serrata:** **~2 mm** wide, located **6–7 mm** behind the limbus nasally and **7–8 mm** temporally. Dentate processes are more pronounced nasally; oral bays lie between them. Retinal thickness here falls to **~0.1 mm**.
- **Long ciliary nerves** run in the horizontal meridians at 3 and 9 o'clock and are the landmark separating superior from inferior on scleral depression.
- **Normal peripheral variants** to distinguish from pathology: pavingstone (cobblestone) degeneration, peripheral cystoid degeneration (universal after age 8, begins at the ora temporally), honeycomb (reticular) degeneration, pars plana cysts, granular tissue and meridional folds.
- **Lattice degeneration:** present in **6–10%** of the population, bilateral in **~50%**, most often in the vertical meridians. Histologically shows inner retinal thinning, overlying vitreous liquefaction, and **firm vitreous adhesion at its margins** with sclerotic vessels. That combination is why lattice causes horseshoe tears at its posterior edge and atrophic holes within it.

NUMBERS TO KNOW: RETINA AND RPE

Rods	110–125 million	Cones	6–7 million
Ganglion cells / ON axons	~1.2 million	RPE cells	~3.5 million
Peak cone density	~199,000/mm ²	Peak rod density	~160,000/mm ²
Macula diameter	5.5–6.0 mm	Fovea	1.5 mm
Foveola	0.35 mm	Foveal avascular zone	0.4–0.5 mm
Retina at parafovea (thickest)	~0.56 mm	Retina at foveola (thinnest)	0.13–0.18 mm
Optic disc, horizontal	1.5 mm	Optic disc, vertical	1.75–1.9 mm
Blind spot position	~15° temporal	1 disc diameter subtends	~5°
Rhodopsin peak	~500 nm	S / M / L cone peaks	~430 / 530 / 560 nm
Rod outer segment turnover	~10 days	Photoreceptors per RPE cell	30 macula, 20 periphery
A:V diameter ratio	2:3	Cilioretinal artery prevalence	15–32%

Ora serrata from limbus, nasal	6–7 mm	temporal	7–8 mm
Lattice degeneration prevalence	6–10%	bilateral in	~50%

RETRIEVAL PRACTICE

Write the answers, then check. Writing beats thinking-you-know.

1. Name the ten retinal layers in order from the vitreous.
2. Which layer is the boundary between retinal and choroidal supply, and what clinical sign follows from it in CRAO?
3. Give the two blood-retinal barriers and the cell type responsible for each.
4. What is the earliest histologic change in diabetic retinopathy?
5. Why does a glaucomatous field defect stop abruptly at the horizontal midline nasally?
6. State the ISNT rule and what it describes.
7. Which ganglion cell class is preferentially lost early in glaucoma, and which perimetric technique exploits that?
8. Which cell contains melanopsin, and where do its axons go?
9. Name the pigments producing macular yellow and state where they concentrate.
10. Give three structural reasons the foveola has the best acuity in the eye.
11. Rod discs versus cone discs: state the difference and one clinical consequence.
12. Which vessels are autoregulated and lack autonomic innervation, and which are the opposite?
13. Why does lattice degeneration cause horseshoe tears at its posterior margin?

KEY

1. ILM, NFL, GCL, IPL, INL, OPL, ONL, ELM, photoreceptor layer, RPE.
2. The outer plexiform layer; the inner retina whitens while the choroid still perfuses the thin foveola, giving a cherry red spot.
3. Inner: retinal capillary endothelium with tight junctions and pericytes. Outer: RPE tight junctions.
4. Pericyte loss, preceding microaneurysms and basement membrane thickening.
5. Arcuate nerve fibre bundles arch around the macula and do not cross the horizontal temporal raphe.
6. Neuroretinal rim thickness normally Inferior > Superior > Nasal > Temporal; violation suggests glaucomatous rim loss.
7. Parasol / magnocellular cells; frequency-doubling technology perimetry (and motion-based tests).
8. Intrinsically photosensitive retinal ganglion cells; to the suprachiasmatic nucleus and the olivary pretectal nucleus.
9. Lutein and zeaxanthin, concentrated in the Henle fibre layer of the macula.
10. Highest cone density with smallest spacing, near one-to-one cone-to-ganglion-cell wiring, and lateral displacement of inner layers and vessels.
11. Rod discs are detached from the plasma membrane and are fully renewed in about 10 days; cone discs are continuous with it. Rod turnover dependence makes rods fail first in RPE phagocytic disease, giving nyctalopia.
12. Retinal vessels are autoregulated with no autonomic supply; choroidal vessels are richly innervated and not autoregulated.
13. The vitreous is liquefied over the lesion but firmly adherent at its margins, so traction concentrates at the posterior edge.

CHAPTER RECAP

1. Ten layers, two limiting membranes both made by Müller cells, and one watershed at the OPL.
2. Retinal circulation inner; choroid outer; two barriers, endothelial tight junctions and RPE tight junctions.
3. Macula 5.5 mm, fovea 1.5 mm, foveola 0.35 mm and rod-free, FAZ 0.4 to 0.5 mm.
4. 1.2 million ganglion cells in four classes: midget/parvo, parasol/magno, bistratified/konio, and melanopsin ipRGC.
5. The arcuate bundle and the horizontal raphe explain the glaucomatous field; the papillomacular bundle explains toxic central scotomas.
6. RPE does six jobs; each failure has a named disease.
7. Rods: 120 million, rhodopsin at 500 nm, free discs, slow dark adaptation. Cones: 6 million, three opsins, membrane-continuous discs.
8. Pericyte loss starts diabetic retinopathy; the 1:1 pericyte ratio is the highest in the body.
9. Retinal arteries are end arteries; a cilioretinal artery in up to a third of eyes changes the CRAO picture.
10. Lattice: 6 to 10% of people, liquefied centre, adherent margins, tears posteriorly.

Optic nerve, visual pathway, cranial nerves, pupil, gaze

Optic nerve / neuro-ophthalmic pathways · 14–22 items

Accommodation / vergence / oculomotor function

Amblyopia / strabismus

Systemic health (stroke, MS, diabetes)

ANSWER THESE COLD BEFORE READING

1. Give the four segments of the optic nerve with lengths, and state where myelination begins.
2. Draw the visual pathway and name the lesion site for eight named field defects.
3. Which LGN layers are magnocellular, and which layers receive crossed fibres?
4. Trace the pupil light reflex in four neurons and explain light-near dissociation.
5. Trace the three-neuron oculosympathetic pathway and give the three pharmacologic tests.
6. Explain why compressive third nerve palsies involve the pupil and ischaemic ones spare it.
7. Define internuclear ophthalmoplegia and one-and-a-half syndrome anatomically.

BLUEPRINT WEIGHT

5%+

14–22 items directly, and neuroanatomy is the discipline most often used to make systemic health items ocular.

The optic nerve: four segments

OPTIC NERVE ANATOMY BY SEGMENT

Segment	Length	Features and clinical consequence
Intraocular	~1 mm	The disc and the lamina cribrosa (fenestrated scleral collagen). Axons are unmyelinated here, which is why the nerve narrows from ~3 mm in diameter behind the sclera to 1.5 mm at the disc. Backward bowing of the lamina under pressure is the structural event in glaucomatous cupping.
Intraorbital	25–30 mm	Runs an S-shaped, slack course in a distance of only about 18 mm: the redundancy permits ~9 mm of proptosis before traction. Surrounded by pia, arachnoid, and dura continuous with the intracranial meninges, so the subarachnoid space communicates with the intracranial CSF space . That single fact produces papilloedema in raised intracranial pressure and permits optic nerve sheath fenestration as a treatment.
Intracanalicular	4–10 mm	Tethered to the periosteum within the optic canal, with the ophthalmic artery inferolateral. Deceleration injury transmits shear here: this is the site of TRAUMATIC OPTIC NEUROPATHY . Also the site compressed by sphenoid ridge meningioma and fibrous dysplasia.
Intracranial	10–16 mm	From the canal to the chiasm, lying above the cavernous sinus and beside the internal carotid and anterior cerebral arteries. Loses the dural sheath, gaining only pia. Compression here by an olfactory groove or tuberculum sellae meningioma gives a junctional scotoma.

THE OPTIC NERVE IS A CENTRAL NERVOUS SYSTEM TRACT, NOT A PERIPHERAL NERVE

- Myelination is by **oligodendrocytes** and begins **immediately posterior to the lamina cribrosa**. There are no Schwann cells and no endoneurium.
- Consequence 1: **it does not regenerate**.
- Consequence 2: it is a target of central demyelination, so optic neuritis is an MS phenomenon and not a Guillain-Barré phenomenon.
- Consequence 3: **myelinated retinal nerve fibres** represent oligodendrocytes that migrated *anterior* to the lamina, appearing as feathery white patches obscuring vessels.
- Consequence 4: the nerve has meninges, so it has a subarachnoid space, so intracranial pressure is transmitted to the disc.

Optic nerve blood supply**BY SEGMENT: THE BASIS OF ISCHAEMIC OPTIC NEUROPATHY**

Level	Supply
Surface nerve fibre layer	Retinal arteriolar branches (central retinal artery)
Prelaminar and laminar	Short posterior ciliary arteries via the circle of Zinn-Haller . This is the watershed territory infarcted in non-arteritic anterior ischaemic optic neuropathy.
Retrolaminar	Pial branches of the short posterior ciliary arteries plus centrifugal branches of the central retinal artery
Intracanalicular and intracranial	Ophthalmic artery, internal carotid, anterior cerebral and superior hypophyseal branches. Poorly collateralised: the site of posterior ischaemic optic neuropathy.

The visual pathway

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RETINA          nasal fibres CROSS (~53%)      temporal fibres stay
|               \                               /
OPTIC NERVE  ----- CHIASM -----          pituitary BELOW
|               /                               \          carotids LATERAL
OPTIC TRACT  -----                      --- 3rd ventricle ABOVE

+--> ~10% leave before the LGN:
|   brachium of superior colliculus --> PRETECTAL OLIVARY N. (pupil)
|   superior colliculus (saccades), SCN (circadian), pulvinar
|
LATERAL GENICULATE NUCLEUS (6 layers)
|   layers 1,2 = MAGNOcellular      layers 3,4,5,6 = PARVOcellular
|   CONTRAlateral (crossed) input --> layers 1, 4, 6
|   IPSIlateral (uncrossed) input --> layers 2, 3, 5
|   koniocellular = interlaminar zones
|
OPTIC RADIATIONS
|   inferior fibres --> MEYER LOOP, temporal lobe --> SUPERIOR field
|   superior fibres --> parietal lobe --> INFERIOR field
|
V1 STRIATE CORTEX (Brodmann 17), banks of the CALCARINE fissure
|   upper bank (cuneus) = INFERIOR field
|   lower bank (lingual) = SUPERIOR field
|   occipital POLE = MACULA (dual MCA + PCA supply)

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Chiasm relationships

- Inferior: the **pituitary gland** and sella turcica, separated by the diaphragma sellae. A pituitary adenoma grows upward and compresses the **inferior chiasmal fibres first**, which carry the

superior visual field, so the classic evolution is a **bitemporal superior quadrantanopia progressing to bitemporal hemianopia**.

- Superior: the third ventricle floor and hypothalamus. A craniopharyngioma compresses from above, taking the **inferior** field first.
- Lateral: the internal carotid arteries and the cavernous sinuses.
- Anterior: the anterior cerebral arteries and anterior communicating artery.
- Position relative to the sella is variable: pre-fixed (anterior, ~15%), normal (~80%), post-fixed (~5%). This variability is why identical tumours produce different fields.

VISUAL FIELD DEFECT LOCALISATION: LEARN THIS TABLE COLD

Defect	Lesion site	Typical cause and distinguishing feature
Monocular field loss with a relative afferent pupillary defect	Optic nerve, anterior to the chiasm	Optic neuritis, AION, compression, glaucoma. Any purely monocular defect is prechiasmal.
Central or centrocaecal scotoma	Papillomacular bundle / optic nerve	Toxic (ethambutol, methanol), nutritional (B ₁₂ , thiamine), hereditary (Leber, dominant optic atrophy), optic neuritis.
Altitudinal defect	Anterior optic nerve, watershed of the short posterior ciliary arteries	Non-arteritic AION, typically inferior altitudinal, with a crowded disc and disc-at-risk fellow eye.
Arcuate defect / nasal step	Arcuate nerve fibre bundle at the disc pole	Glaucoma. Respects the horizontal raphe.
Junctional scotoma	Optic nerve at its junction with the chiasm	Ipsilateral central scotoma plus a contralateral superotemporal defect. Classically explained by Wilbrand knee; the localisation remains valid clinically.
Bitemporal hemianopia	Chiasm, crossing nasal fibres	Pituitary adenoma (superior first), craniopharyngioma (inferior first), meningioma, aneurysm. Look for a hemifield slide phenomenon and loss of stereopsis.
Binasal hemianopia	Lateral chiasmal compression, or bilateral optic nerve/retinal disease	Rare and usually not chiasmal: think tilted discs, bilateral glaucoma, or bilateral optic nerve drusen before accepting a chiasmal cause.
Incongruous homonymous hemianopia	Optic tract or LGN	Tract lesions may show contralateral optic atrophy and a mild RAPD. LGN lesions can give a wedge (sectoranopia) from anterior choroidal or lateral posterior choroidal artery infarction.
Superior homonymous quadrantanopia ("pie in the sky")	Meyer loop , temporal lobe	Temporal lobectomy, tumour. Often with formed visual hallucinations or complex partial seizures.
Inferior homonymous quadrantanopia ("pie on the floor")	Parietal optic radiations	Often with an asymmetric optokinetic response toward the lesion side and, if the non-dominant parietal lobe, hemineglect.

Defect	Lesion site	Typical cause and distinguishing feature
Complete congruous homonymous hemianopia with macular sparing	Occipital cortex, PCA territory	Macula spared because the occipital pole receives collateral supply from the middle cerebral artery . Pupils normal, optic discs normal.
Bilateral homonymous defects with normal pupils and normal discs, patient denies blindness	Bilateral occipital cortex	Cortical blindness; with denial it is ANTON SYNDROME .
Temporal crescent loss or sparing	Most anterior calcarine cortex	The monocular temporal crescent (60–90°) is represented only in the most anterior striate cortex, so it may be lost alone or spared alone.

CONGRUITY IS A LOCALISING SIGN

The further posterior the lesion, the more **congruous** (identical in the two eyes) the homonymous defect, because corresponding fibres from the two eyes converge progressively toward the cortex. Optic tract lesions are markedly incongruous; occipital lesions are exquisitely congruous. **Congruity is worth more than any other single feature in localising a homonymous defect.**

Cranial nerves of the orbit

CN III: oculomotor

- **Nuclei** in the midbrain at the level of the **superior colliculus**: a paired somatic motor complex plus the **Edinger-Westphal nucleus** supplying preganglionic parasympathetics.
- Fibres pass ventrally through the red nucleus and cerebral peduncle. Combined lesions here are named: **NOTHNAGEL** (III + cerebellar ataxia), **BENEDIKT** (III + red nucleus, contralateral tremor), **WEBER** (III + corticospinal tract, contralateral hemiparesis), **CLAUDE** (Benedikt plus Nothnagel features).
- Exits into the interpeduncular fossa between the **posterior cerebral** and **superior cerebellar** arteries, then runs beside the **posterior communicating artery**: a PCom aneurysm is the classic cause of a painful, pupil-involving palsy.
- Passes through the **lateral wall of the cavernous sinus**, then the superior orbital fissure inside the annulus of Zinn, dividing into:
 - **Superior division**: superior rectus and levator palpebrae superioris.
 - **Inferior division**: medial rectus, inferior rectus, inferior oblique, and the **preganglionic parasympathetic fibres** to the ciliary ganglion (branching from the nerve to the inferior oblique).

THE RULE OF THE PUPIL, AND ITS ANATOMIC REASON

Parasympathetic pupillomotor fibres travel **superficially and dorsomedially** in the nerve, and they are supplied by pial vessels. Somatic motor fibres run in the core, supplied by intraneural vasa nervorum.

- **Compression** (aneurysm, uncal herniation, tumour) squeezes the outside first, so the **pupil is involved**, often before or with the motility deficit. This is a **neurosurgical emergency** requiring same-day vascular imaging.
- **Microvascular ischaemia** (diabetes, hypertension) infarcts the core, so the **pupil is spared** and the palsy is painful but benign, usually recovering in **8–12 weeks**.
- **Aberrant regeneration** (lid elevation on adduction or downgaze, pupil constriction on adduction) implies a slowly compressive lesion: cavernous meningioma, aneurysm, or trauma. **It never follows an ischaemic palsy.**

CN IV: trochlear

- Nucleus in the midbrain at the level of the **inferior colliculus**.
- **The only cranial nerve to decussate completely and the only one to exit the brainstem dorsally**, just below the inferior colliculus. Therefore a *nuclear* fourth nerve lesion causes a **contralateral** superior oblique palsy while a lesion anywhere along the nerve causes an ipsilateral palsy.
- Longest intracranial course (**~75 mm**) and thinnest nerve, wrapping around the brainstem, passing between PCA and SCA, then along the lateral cavernous sinus wall and through the superior orbital fissure **outside** the annulus of Zinn.
- Its dorsal exit and long course make it **the nerve most often injured in closed head trauma**, frequently bilaterally.

CN VI: abducens

- Nucleus in the **pons**, beneath the **facial colliculus**, because the motor root of CN VII loops around it. Therefore a **nuclear** sixth nerve lesion gives an **ipsilateral horizontal gaze palsy** (not an isolated abduction deficit, because the nucleus also contains interneurons projecting via the MLF to the contralateral medial rectus subnucleus) **plus an ipsilateral lower motor neuron facial palsy**. That pairing localises the lesion to the pons with near certainty.
- Long subarachnoid course, ascends the petrous ridge and passes through **Dorello canal** under the petroclinoid (Gruber) ligament: this fixation makes it the classic **false localising sign** in raised intracranial pressure and the nerve affected in **GRADENIGO SYNDROME** (petrous apicitis, with VI palsy, facial pain, and otorrhoea).
- **Runs inside the cavernous sinus**, adjacent to the internal carotid artery, not in the lateral wall. It is therefore the nerve most often affected first by an intracavernous carotid aneurysm or carotid-cavernous fistula.
- **MILLARD-GUBLER SYNDROME**: ventral pontine lesion with VI and VII palsies plus contralateral hemiplegia. **FOVILLE SYNDROME**: dorsal pontine, with gaze palsy, VII palsy, and Horner syndrome.

CN V: trigeminal, and CN VII: facial**TRIGEMINAL DIVISIONS RELEVANT TO THE EYE**

Division	Branches	Territory
V1 ophthalmic	Frontal (→ supraorbital, supratrochlear), lacrimal, nasociliary (→ long ciliary, infratrochlear; posterior/anterior ethmoidal, external nasal)	Cornea, iris, ciliary body, sclera, upper lid, forehead, scalp to vertex, nasal tip. Sensory limb of the corneal reflex.
V2 maxillary	Infraorbital, zygomatic (zygomaticofacial, zygomaticotemporal), superior alveolar	Lower lid, cheek, upper lip, upper teeth, lateral nose. Carries lacrimal postganglionic parasympathetics.
V3 mandibular	Auriculotemporal, lingual, inferior alveolar; motor to muscles of mastication	Not orbital, but its motor function distinguishes trigeminal from other neuropathies.

- **CN VII** has a motor root to the muscles of facial expression, including **orbicularis oculi** (efferent limb of the blink and corneal reflex), and the **nervus intermedius** carrying taste and the parasympathetic secretomotor fibres to the lacrimal, submandibular, and sublingual glands.
- Upper facial muscles receive **bilateral** cortical input. A **supranuclear (upper motor neuron) lesion therefore spares the forehead and permits lid closure**; a **nuclear or peripheral lesion causes complete hemifacial weakness with lagophthalmos**. This is the single most useful clinical discriminator in facial weakness, and it determines whether corneal exposure is a risk.

The ciliary ganglion

THREE ROOTS, ONE SYNAPSE, AND A 30-TO-1 RATIO

The ciliary ganglion lies in the posterior orbit, roughly **1 cm** anterior to the superior orbital fissure, lateral to the optic nerve and medial to the lateral rectus.

Root	Origin	Synapse?
Parasympathetic (motor)	Preganglionic from Edinger-Westphal via CN III inferior division	Yes. The only root that synapses.
Sensory	Nasociliary nerve (V1)	No, passes through.
Sympathetic	Internal carotid plexus, from the superior cervical ganglion	No, passes through.

Postganglionic fibres leave as **6-10 short ciliary nerves**. Crucially, about **97%** supply the **ciliary muscle** and only **3%** supply the **iris sphincter**. That ratio explains two clinical phenomena: after ganglion damage, aberrantly regenerating accommodative fibres readily reinnervate the sphincter, giving **light-near dissociation**; and it explains the sectoral, vermiform movements of a tonic (Adie) pupil.

The pupil

LIGHT REFLEX: FOUR NEURONS

- 1 **Retina to pretectum.** Photoreceptors and intrinsically photosensitive ganglion cells → optic nerve → chiasm (nasal fibres cross) → optic tract → fibres **leave before the LGN** via the brachium of the superior colliculus → **pretectal olivary nucleus**.
- 2 **Pretectum to both Edinger-Westphal nuclei**, crossing in the **posterior commissure**. This double decussation (chiasm and posterior commissure) is why the reflex is consensual and symmetric.
- 3 **Edinger-Westphal to ciliary ganglion** via CN III inferior division.
- 4 **Ciliary ganglion to sphincter** via the short ciliary nerves.

The **near reflex** (accommodation, convergence, miosis) is driven from the occipital association cortex to the Edinger-Westphal nucleus **without a pretectal relay**, and the fibres approach the nucleus more ventrally.

LIGHT-NEAR DISSOCIATION: NEAR RESPONSE PRESERVED, LIGHT RESPONSE LOST

Cause	Localisation and clue
Dorsal midbrain (Parinaud) syndrome	Pretectum and posterior commissure: upgaze palsy, convergence-retraction nystagmus, lid retraction (Collier sign), mid-dilated pupils. Pinealoma in a young patient; stroke in an older one.
Argyll Robertson pupils	Neurosyphilis. Small, irregular, bilateral pupils that accommodate but do not react, with poor dilation to mydriatics.
Adie (tonic) pupil	Postganglionic ciliary ganglion damage. Large pupil, sector palsy with vermiform movements, tonic near response, slow redilation, supersensitivity to pilocarpine 0.125% . With areflexia it is Holmes-Adie.
Severe afferent disease	Any profound bilateral optic neuropathy or retinopathy: the pupil cannot respond to light but the near pathway is intact.
Aberrant regeneration of CN III	Constriction on adduction; implies compression, not ischaemia.
Diabetic autonomic neuropathy	Bilateral, symmetric, with other autonomic features.

The oculosympathetic pathway and Horner syndrome

THREE NEURONS, THREE LESION LEVELS

- 1 **First order (central):** posterolateral hypothalamus → descends uncrossed through the brainstem and cervical cord to the **cilio-spinal centre of Budge-Waller** at **C8 to T2**. Lesions: brainstem stroke (lateral medullary/Wallenberg), demyelination, syringomyelia. Look for anhidrosis of the whole ipsilateral body and other brainstem signs.
- 2 **Second order (preganglionic):** exits the cord, crosses the **apex of the lung** and the subclavian artery, ascends the sympathetic chain to the **superior cervical ganglion**. Lesions: **Pancoast tumour**, thyroid or neck surgery, brachial plexus trauma, aortic or subclavian aneurysm, and in children **neuroblastoma**. Anhidrosis of the face and neck.
- 3 **Third order (postganglionic):** travels on the **internal carotid artery** plexus, enters the cavernous sinus, joins **V1**, then the **nasociliary** and **long ciliary nerves** to the dilator pupillae and Müller muscle. Lesions: **carotid dissection** (painful Horner is a dissection until proven otherwise), cavernous sinus disease, cluster headache (Raeder paratrigeminal syndrome). Little or no anhidrosis, since sudomotor fibres leave with the external carotid.

PHARMACOLOGIC PUPIL TESTING: WHAT EACH DRUG PROVES

Agent	Mechanism	Interpretation
Cocaine 4–10%	Blocks noradrenaline reuptake, so dilation requires intact tonic release	Confirms Horner syndrome at any level. The Horner pupil <i>fails</i> to dilate; anisocoria increases. Positive if post-drop anisocoria is ≥ 0.8 – 1.0 mm.
Apraclonidine 0.5–1%	Weak α_1 agonist acting on denervation-supersensitive receptors	Practical replacement for cocaine. Reversal of anisocoria: the Horner pupil dilates and its ptosis lifts, while the normal pupil constricts slightly. Can be falsely negative in the first days.
Hydroxyamphetamine 1% (Paredrine)	Releases stored noradrenaline from an intact postganglionic terminal	Localises the lesion. Dilates in first- and second-order (preganglionic) lesions because the third neuron is intact; fails to dilate in a third-order (postganglionic) lesion.
Pilocarpine 0.125%	Cholinergic supersensitivity of a denervated sphincter	Constricts a tonic (Adie) pupil ; no effect on a normal pupil or on pharmacologic mydriasis.
Pilocarpine 1–2%	Full-strength muscarinic agonist	Constricts a pupil in third nerve palsy or in mechanical (traumatic) mydriasis; fails to constrict a pharmacologically blocked pupil (atropine, scopolamine patch, plant alkaloid exposure).

Supranuclear control of eye movements

GAZE SYSTEMS AND THEIR LESIONS

System	Pathway	Lesion effect
Horizontal saccades	Frontal eye field (Brodmann area 8) → crosses → contralateral paramedian pontine reticular formation (PPRF) → ipsilateral CN VI nucleus → via the medial longitudinal fasciculus to the contralateral CN III medial rectus subnucleus	<i>FEF lesion</i> : cannot make saccades to the opposite side; eyes deviate toward the lesion (transiently). <i>PPRF lesion</i> : cannot look toward the lesion; eyes deviate away . Cortical (FEF) gaze palsies can be overcome by the oculocephalic reflex; pontine ones cannot.
Smooth pursuit	Parieto-occipito-temporal junction (MT/MST) → ipsilateral pursuit; relays through dorsolateral pontine nuclei and cerebellum	Ipsilateral pursuit defect. Parietal lesions produce an asymmetric optokinetic response with the drum rotated toward the lesioned side.
Vertical saccades	riMLF (rostral interstitial nucleus of the MLF) in the midbrain, with the interstitial nucleus of Cajal for gaze holding; upgaze fibres cross the posterior commissure	Posterior commissure lesion → upgaze palsy (Parinaud). Bilateral riMLF → loss of vertical saccades, as in progressive supranuclear palsy.
Vergence	Midbrain near the CN III nuclei, driven from occipital cortex	Convergence insufficiency or spasm; convergence-retraction nystagmus with dorsal midbrain lesions.
Vestibulo-ocular reflex	Semicircular canals → vestibular nuclei → ocular motor nuclei; three-neuron arc	Preserved in supranuclear gaze palsy, which is how doll's head testing distinguishes supranuclear from nuclear or infranuclear lesions.

INTERNUCLEAR OPHTHALMOPLEGIA AND ONE-AND-A-HALF SYNDROME

Internuclear ophthalmoplegia (INO) = a lesion of the **medial longitudinal fasciculus**. On attempted gaze away from the lesion, the **ipsilateral eye fails to adduct** while the **contralateral abducting eye shows nystagmus**. Convergence is preserved if the lesion is caudal to the CN III nucleus. **Bilateral INO in a young adult means multiple sclerosis** until proven otherwise; in an older patient, pontine infarct. Bilateral INO with exotropia is **WEBINO**.

One-and-a-half syndrome = ipsilateral **PPRF (or CN VI nucleus) plus the adjacent MLF**. The patient has a complete ipsilateral horizontal gaze palsy (the "one") and, on gaze to the other side, only abduction of the contralateral eye (the "half"), because the ipsilateral eye cannot adduct. The only remaining horizontal movement is abduction of the contralateral eye.

The cavernous sinus

----- CAVERNOUS SINUS (coronal) -----	
LATERAL WALL	INSIDE THE SINUS
CN III (oculomotor)	CN VI (abducens)
CN IV (trochlear)	INTERNAL CAROTID ARTERY
CN V1 (ophthalmic)	sympathetic plexus
CN V2 (maxillary)	

Above: optic nerve, chiasm. Below: sphenoid sinus. Medial: pituitary.

Cavernous sinus syndrome: any combination of III, IV, V1, V2, VI, and a Horner syndrome, with proptosis and chemosis. Because both sympathetic and parasympathetic pupillomotor supply can be interrupted, the **pupil may be mid-sized and unreactive**. A **painful** multiple ophthalmoplegia

with V1 numbness and normal imaging suggests **TOLOSA-HUNT SYNDROME**. Pulsatile proptosis with an orbital bruit, dilated "corkscrew" episcleral vessels, and raised IOP suggests a **CAROTID-CAVERNOUS FISTULA**.

Ophthalmic artery branches, in order of origin

ORDER OF OPHTHALMIC ARTERY BRANCHES

"Cranial Lacerations Prompt Many Serious Problems, And Might Send Doctors Nuts"

Central retinal, Lacrimal, Posterior ciliary (long and short), Muscular, Supraorbital, Posterior ethmoidal, Anterior ethmoidal, Medial palpebral, Supratrochlear, Dorsal Nasal.

The central retinal artery is the first branch: that is why an embolus lodging early produces isolated retinal infarction, and why the ophthalmic artery is the first branch of the internal carotid after the cavernous segment (making the eye a window on carotid disease).

NUMBERS TO KNOW: NEUROANATOMY

Optic nerve axons	~1.2 million	Intraorbital ON length	25–30 mm
Intracranial ON	4–10 mm	Intracranial ON	10–16 mm
ON diameter, at disc / behind sclera	1.5 / 3.0 mm	Nasal fibres crossing at chiasm	~53%
LGN layers	6	Magnocellular layers	1, 2
Crossed input to LGN layers	1, 4, 6	Uncrossed input	2, 3, 5
Striate cortex	Brodman 17	Frontal eye field	Brodman 8
Ciliospinal centre	C8–T2	Short ciliary nerves	6–10
Short ciliary fibres to ciliary muscle	~97%	to iris sphincter	~3%
CN IV intracranial course	~75 mm	Ischaemic III palsy recovery	8–12 weeks
Monocular temporal crescent	60–90°	Physiologic anisocoria prevalence	~20%

RETRIEVAL PRACTICE

Fourteen questions. These are the highest-yield items in the chapter.

1. Where does optic nerve myelination begin, and give two clinical consequences.
2. A pituitary adenoma first produces which field defect, and why that quadrant?
3. Distinguish a temporal-lobe from a parietal-lobe homonymous quadrantanopia by name and by one additional sign.
4. Why is the macula spared in a PCA occipital infarct?
5. Which LGN layers are magnocellular, and which layers receive crossed fibres?
6. Explain in anatomic terms why compressive third nerve palsies involve the pupil.
7. What does aberrant regeneration of CN III tell you about aetiology?

8. A patient has an ipsilateral horizontal gaze palsy plus an ipsilateral complete facial palsy. Localise precisely.
9. Which cranial nerve decussates completely and exits dorsally, and what does a nuclear lesion cause?
10. Which nerve runs inside the cavernous sinus rather than in its wall?
11. List the three roots of the ciliary ganglion and state which synapses.
12. Give the three-drug sequence for evaluating a suspected Horner syndrome and what each proves.
13. Define one-and-a-half syndrome and name the two structures involved.
14. Name four causes of light-near dissociation.

KEY

1. Immediately posterior to the lamina cribrosa, by oligodendrocytes. Consequences: no regeneration; it is a CNS tract subject to central demyelination; and myelinated nerve fibres represent anterior oligodendrocyte migration.
2. Bitemporal **superior** quadrantanopia, because the tumour grows upward from the sella and compresses the inferior chiasmal fibres, which carry the superior field.
3. Temporal = "pie in the sky" superior quadrantanopia from Meyer loop, often with complex partial seizures or formed hallucinations. Parietal = "pie on the floor" inferior quadrantanopia, often with hemineglect and an asymmetric optokinetic response toward the lesion.
4. The occipital pole receives collateral supply from the middle cerebral artery.
5. Layers 1 and 2 are magnocellular; crossed (contralateral) fibres go to layers 1, 4, and 6.
6. Pupillomotor fibres run superficially and dorsomedially and are pially supplied, so external compression reaches them first; motor fibres in the core are spared until later.
7. It indicates a slowly compressive or traumatic lesion, and effectively excludes microvascular ischaemia.
8. The pons, at the CN VI nucleus beneath the facial colliculus, where the facial motor root loops around it.
9. CN IV; a nuclear lesion causes a **contralateral** superior oblique palsy.
10. CN VI, adjacent to the internal carotid artery.
11. Parasympathetic (from CN III inferior division) synapses; sensory (nasociliary) and sympathetic (carotid plexus) pass through without synapsing.
12. Cocaine or apraclonidine confirms Horner at any level; hydroxyamphetamine then localises, dilating in first- and second-order lesions and failing in third-order.
13. Ipsilateral complete horizontal gaze palsy plus loss of adduction of the ipsilateral eye on contralateral gaze; the PPRF (or CN VI nucleus) and the adjacent MLF.
14. Dorsal midbrain (Parinaud) syndrome, Argyll Robertson pupils, Adie tonic pupil, severe bilateral afferent disease (also aberrant CN III regeneration and diabetic autonomic neuropathy).

CHAPTER RECAP

1. Four ON segments; 1 mm intraocular, 25 to 30 mm orbital with slack, 4 to 10 mm tethered in the canal, 10 to 16 mm intracranial.
2. It is a CNS tract with meninges: no regeneration, central demyelination, papilloedema.
3. Inferior chiasmal fibres carry the superior field; that is why pituitary tumours start superotemporally.
4. Inferior radiations loop through the temporal lobe (Meyer) for the superior field; congruity increases posteriorly.
5. LGN: 1 and 2 magno; crossed to 1, 4, 6.
6. Pupil fibres are superficial: compression involves the pupil, ischaemia spares it, aberrant regeneration means compression.
7. Nuclear IV is contralateral; nuclear VI gives gaze palsy plus facial palsy; VI runs inside the cavernous sinus.
8. Ciliary ganglion: only the parasympathetic root synapses, and 97% of output drives accommodation.
9. Horner: three neurons, three drugs, and painful Horner means dissection.
10. INO is an MLF lesion; add the PPRF and it becomes one-and-a-half.

CHAPTER 6 · DEVELOPMENTAL ANATOMY

Ocular embryology, visual and human development

Visual and human development · 6–10 items

Developmental anatomy across all tissue areas

Amblyopia / strabismus

Ametropia (emmetropisation)

ANSWER THESE COLD BEFORE READING

1. Assign every ocular structure to one of four embryonic sources: surface ectoderm, neuroectoderm, neural crest, mesoderm.
2. State when the optic vesicle, lens placode, and lens vesicle appear, and when the optic fissure closes.
3. Explain why a typical coloboma is inferonasal.
4. Distinguish primary, secondary, and tertiary vitreous and name three remnants of the hyaloid system.
5. Give axial length, corneal power, and lens power at birth versus adulthood.
6. State visual acuity, stereopsis, and accommodation milestones by age.

BLUEPRINT WEIGHT

2%

6–10 direct items, but developmental origin is the single most reliably asked framework in ABS anatomy.

Embryonic origins: the table that answers a dozen items

EVERY OCULAR STRUCTURE BY GERM LAYER

Source	Structures
Surface ectoderm	Lens (entire), corneal epithelium , conjunctival epithelium, epidermis of the eyelids, lacrimal gland and accessory lacrimal glands, the whole lacrimal drainage epithelium, meibomian and other adnexal glands, lash follicles.
Neuroectoderm (optic cup, from diencephalon)	Inner layer of the optic cup → neurosensory retina, non-pigmented ciliary epithelium, posterior iris pigment epithelium. Outer layer → retinal pigment epithelium , pigmented ciliary epithelium, anterior iris epithelium and hence the sphincter and dilator pupillae . Also: optic nerve axons and glia (astrocytes, oligodendrocytes, Müller cells), optic nerve, part of the primary vitreous.
Neural crest (the largest contributor)	Corneal stroma and endothelium , Descemet membrane, Bowman layer, trabecular meshwork , sclera (most of it), choroidal stroma, iris stroma , ciliary muscle , all uveal and epithelial melanocytes , orbital bones and cartilage, orbital connective tissue and fat, meninges of the optic nerve, ciliary ganglion, Schwann cells, and the connective tissue of the lids and EOM sheaths.
Mesoderm	Extraocular muscles , vascular endothelium of every ocular vessel (including the endothelium of Schlemm canal), and the temporal portion of the sclera in some accounts.

TWO RULES THAT RESOLVE MOST ORIGIN ITEMS

"If it is transparent and epithelial at the front, it is surface ectoderm. If it is pigmented or neural, it is neuroectoderm. If it is connective tissue, it is neural crest. Only muscle-that-moves-the-eye and blood vessel linings are mesoderm."

The cornea is the trap: epithelium is surface ectoderm; stroma and endothelium are neural crest. Similarly the **iris** is split: stroma from neural crest, epithelia and both muscles from neuroectoderm.

Timeline of ocular morphogenesis

KEY EVENTS WITH GESTATIONAL TIMING

Timing	Event
Day 22 (wk 3)	Optic sulci (pits) appear in the neural folds of the forebrain.
Day 25 (wk 4)	Optic vesicles evaginate from the diencephalon, still connected by the optic stalk.
Day 27	Optic vesicle induces the overlying surface ectoderm to thicken into the lens placode . This induction is the classic example of reciprocal induction: without contact, no lens forms.
Day 28–32	Vesicle invaginates to form the bilayered optic cup ; the lens placode invaginates as the lens pit . The inferonasal groove that admits vascular mesenchyme is the optic (embryonic, choroid) fissure .
Day 33 (wk 5)	Lens vesicle separates from the surface ectoderm. Failure of separation is the mechanism of PETERS ANOMALY and anterior lenticonus. Hyaloid artery enters through the fissure.
Wk 5–7 (day 33–44)	Optic fissure closes , from the middle outward. Failure produces a typical coloboma , which is therefore inferonasal and may involve iris, ciliary body, choroid, retina, and optic disc in continuity.
Wk 5–6	Primary lens fibres from the posterior lens vesicle cells form the embryonic nucleus . Extraocular muscles and lids begin.
Wk 6–7	Optic nerve axons grow from ganglion cells into the optic stalk. Eyelid folds meet and begin to fuse.
Wk 8–9	Eyelids fuse . Cornea has recognisable layers. Ciliary body begins. Anterior chamber appears.
Mo 3 (wk 9–12)	Secondary lens fibres from the equator create the Y sutures and the fetal nucleus . Secondary (definitive, avascular) vitreous is laid down. Retinal vessels begin to grow from the disc.
Mo 4–5	Ciliary processes and zonules (tertiary vitreous) form. Iris sphincter differentiates (dilator follows around month 6). Sclera condenses. Lacrimal drainage canalises from above downward.
Mo 5–7 (wk 20–26)	Eyelids reopen . Photoreceptor outer segments appear. Pupillary membrane begins to regress.
Mo 7	Hyaloid system regresses (from posterior to anterior). Optic nerve myelination begins at the chiasm and proceeds anteriorly.

Timing	Event
Mo 8–9	Retinal vascularisation reaches the nasal ora at about 36 weeks and the temporal ora at or shortly after term (40 weeks) : the incomplete temporal periphery is exactly why RETINOPATHY OF PREMATURITY is a temporal disease of preterm infants.
Birth to 1 mo	Myelination reaches the lamina cribrosa and stops. Advancing past it later gives myelinated retinal nerve fibres.
Birth to 4 mo	Foveal maturation : inner layers migrate away, cones elongate and pack. Not complete until about 4 months, and cone packing continues to about 4 years.
Birth to 1 yr	Trabecular meshwork and angle continue to mature. Maldevelopment here produces PRIMARY CONGENITAL GLAUCOMA (CYP1B1) . Valve of Hasner canalises: failure gives congenital nasolacrimal duct obstruction, which resolves spontaneously in most infants by 12 months.

VITREOUS IN THREE STAGES, AND ITS CLINICAL REMNANTS

Stage	Timing / nature	Remnants and disease
Primary	Wk 4–6. Vascular : hyaloid artery, vasa hyaloidea propria, and the tunica vasculosa lentis	Persists as Cloquet canal . Failure of regression: PERSISTENT FETAL VASCULATURE (PHPV) , MITTENDORF DOT (on the posterior lens, usually inferonasal), BERGMEISTER PAPILLA (on the disc), persistent pupillary membrane .
Secondary	Wk 6–12. Avascular , becomes the definitive adult vitreous	Type II collagen and hyaluronan gel; undergoes syneresis in adulthood.
Tertiary	Mo 3–4. Condenses between lens equator and ciliary epithelium	Becomes the zonules of Zinn .

Named developmental anomalies and their genes

CONGENITAL ANOMALIES MAPPED TO THE FAILED DEVELOPMENTAL STEP

Anomaly	Failed step	Features and gene
Anophthalmia / microphthalmia	Optic vesicle formation or growth	<i>SOX2</i> , <i>RAX</i> , <i>OTX2</i> , <i>CHX10</i> . Also maternal rubella and vitamin A deficiency.
Typical coloboma	Optic fissure closure, wk 5–7	Inferonasal , may involve iris to disc. Part of CHARGE association (<i>CHD7</i>) and cat-eye syndrome.
Aniridia	Neuroectodermal and mesenchymal failure of iris and angle	<i>PAX6</i> at 11p13 . Screen for Wilms tumour : the WAGR contiguous gene deletion (Wilms, Aniridia, Genitourinary, Retardation) involves the adjacent <i>WT1</i> . Also foveal hypoplasia, nystagmus, keratopathy, cataract, glaucoma.
Peters anomaly	Incomplete separation of the lens vesicle / neural crest migration	Central corneal opacity with iridocorneal or keratolenticular adhesion and a Descemet defect. <i>PAX6</i> , <i>PITX2</i> , <i>CYP1B1</i> , <i>FOXE3</i> .
Axenveld-Rieger spectrum	Neural crest maldevelopment of the anterior segment	Posterior embryotoxon, iris strands to a prominent Schwalbe line, iris hypoplasia, corectopia, polycoria; ~50% develop glaucoma. <i>PITX2</i> , <i>FOXC1</i> . Systemic: dental hypoplasia, umbilical skin, maxillary hypoplasia.

Anomaly	Failed step	Features and gene
Primary congenital glaucoma	Trabecular meshwork and angle maturation	<i>CYP11B1</i> . Buphthalmos, Haab striae, epiphora, photophobia, blepharospasm.
Optic nerve hypoplasia	Ganglion cell axon growth / retinal apoptosis regulation	Small disc with a double ring sign . Look for DE MORSIER SEPTO-OPTIC DYSPLASIA : absent septum pellucidum with pituitary dysfunction; screen endocrine function urgently.
Morning glory disc anomaly	Abnormal closure of the posterior fissure / mesenchymal defect	Funnel-shaped excavation, central glial tuft, radiating vessels. Associated with basal encephalocele and moyamoya.
Persistent fetal vasculature	Hyaloid regression, mo 7 onward	Unilateral leukocoria, microphthalmia, retrolental membrane, elongated ciliary processes, cataract. A key differential for retinoblastoma.
Congenital cataract	Lens fibre organisation, any stage	Timing of the insult determines the layer opacified. Causes: rubella (nuclear, plus "salt and pepper" retinopathy), galactosaemia (oil-droplet), Lowe syndrome, trisomy 21, Fabry (spoke-like), myotonic dystrophy (Christmas tree).
Retinopathy of prematurity	Retinal vascularisation incomplete at birth	Temporal periphery avascular; risk driven by gestational age, birth weight, and supplemental oxygen. Zones I to III and stages 1 to 5 with plus disease.

Postnatal ocular growth and emmetropisation

THE GROWING EYE: NEWBORN TO ADULT

Parameter	Newborn	Adult	Consequence
Axial length	16–17 mm	23–24 mm	Rapid phase to age 3 (to about 23 mm), then slow to about age 14. 1 mm of axial length is worth roughly 3 D.
Corneal power	~51 D	43–44 D	Cornea flattens steeply over the first year, offsetting axial growth.
Corneal diameter	9.5–10.5 mm	11.5–12 mm	Above 12 mm in an infant suggests buphthalmos.
Lens power	~34 D	~19 D	Lens flattening is the other half of emmetropisation.
Refraction	+2 to +3 D	~plano	Distribution at birth is broad; emmetropisation narrows it markedly by age 6. Astigmatism is common in infancy (often against the rule) and largely resolves by age 4.
Orbital volume	~ 1/3 adult	30 mL	Reaches adult size by adolescence; enucleation in infancy arrests orbital growth, hence conformers and implants.
Interpupillary distance	~40 mm	60–65 mm	Relevant to prism and to pediatric spectacle fitting.

EMMETROPISATION IS ACTIVE, NOT PASSIVE

The retina detects defocus and modulates axial elongation through a signalling cascade involving the retina, RPE, choroid, and scleral remodelling. **Hyperopic defocus (image behind the retina) accelerates axial growth; myopic defocus slows it.** That single mechanism is the rationale for every modern myopia control strategy: peripheral myopic defocus with orthokeratology or multifocal soft lenses, defocus incorporated spectacle lenses, and low-dose atropine (which acts on a non-accommodative retinal or scleral pathway, since it works in chicks with no accommodative mechanism).

Risk factors for myopia progression: two myopic parents, early age of onset, less than about **1 hour** of daily outdoor time, near work with sustained short working distance, and an accommodative lag with esophoria at near.

Visual development milestones

FUNCTIONAL MILESTONES: KNOW THE AGES

Function	Timeline
Pupil light reflex	Present by 29–31 weeks gestation; a useful marker of gestational maturity.
Blink to bright light	~ 30 weeks gestation.
Blink to threat	2–5 months postnatally.
Fixation and following	Brief fixation at birth; steady fix-and-follow by 6–8 weeks . Absence of fix-and-follow by 3 months requires investigation.
Optokinetic nystagmus	Present at birth (monocular OKN asymmetry is normal until 3–6 months and persists abnormally in early-onset strabismus).
Accurate saccades	Hypometric at birth; adult-like by 2–3 months .
Smooth pursuit	Saccadic ("cogwheel") until 2–3 months .
Accommodation	Inaccurate at birth; adult-like accuracy by 3–4 months .
Ocular alignment	Variable in the first weeks; a constant deviation at any age is abnormal, and any deviation persisting past 3 to 4 months is abnormal.
Fusion and stereopsis	Onset 3–6 months , developing rapidly; near-adult stereoacuity by 2–3 years .
Colour discrimination	Present in rudimentary form by 2–3 months .
Contrast sensitivity	Peak shifts to higher spatial frequencies through the first year; adult-like around 5–7 years .
Visual acuity (behavioural)	Roughly 20/400 at birth, 20/200 at 2–3 months, 20/100 at 6 months, 20/50 at 1 year, 20/30 at 2–3 years, 20/20 by 3–5 years .
Visual acuity (VEP / preferential looking)	Matures faster than recognition acuity: VEP acuity approaches 20/20 by 6–12 months . Expect a discrepancy and know which test you are quoting.

CRITICAL PERIODS, AND WHY THEY MATTER FOR AMBLYOPIA

- **Period of greatest susceptibility to deprivation:** birth to about **2 years**, peaking in the first **6 months**. A congenital unilateral cataract must be removed within weeks, not months.
- **Period of emmetropisation and visual maturation:** to about **6–8 years**.
- **Period during which amblyopia can be reversed:** classically to about **7–9 years**, but treatment response has been demonstrated into the teens and beyond, so age alone should not withhold treatment.
- **Amblyopia severity by type:** deprivation > anisometropic > strabismic > isoametropic / meridional. Deprivation is the most damaging because it removes pattern input entirely.

Human development beyond the eye

ABS items on human development also test general developmental and ageing physiology in the framework of the visual system.

AGEING CHANGES WITH DIRECT CLINICAL CONSEQUENCES

System	Change and consequence
Accommodation	Amplitude falls roughly linearly (Hofstetter: mean amplitude = $18.5 - 0.30 \times \text{age}$); functionally presbyopic near 40–45 . Mechanism: lens hardening and capsule/zonule geometry, not ciliary muscle failure.
Pupil	Senile miosis: resting diameter falls about 0.5 mm per decade after 20, reducing retinal illuminance and increasing depth of focus.
Lens	Yellowing (chromophore accumulation) reduces short-wavelength transmission, producing acquired tritan colour errors; nuclear sclerosis causes myopic shift.
Tear film	Reduced aqueous secretion and meibomian dropout; dry eye prevalence rises steeply with age and with female sex after menopause.
Vitreous	Syneresis and PVD, with peak incidence in the sixth and seventh decades, earlier in myopes and after cataract surgery.
Retina and RPE	Lipofuscin accumulation, drusen, reduced rod density (rods are lost faster than cones with age), slower dark adaptation.
Cornea	Endothelial density falls, guttae increase peripherally (Hassall-Henle), arcus senilis appears, corneal sensitivity declines.
Contrast and glare	Contrast sensitivity and disability glare worsen faster than Snellen acuity, which is why acuity alone under-describes functional loss in the elderly.

NUMBERS TO KNOW: DEVELOPMENT

Optic vesicle forms	day 25 (wk 4)	Lens placode induced	day 27
Lens vesicle separates	day 33 (wk 5)	Optic fissure closes	wk 5–7
Lids fuse / reopen	wk 8–9 / mo 5–7	Hyaloid regression begins	mo 7
Retinal vessels reach nasal ora	~36 wk	temporal ora	~40 wk / term
Myelination reaches lamina cribrosa	~1 mo postnatal	Foveal maturation complete	~4 mo
Axial length at birth / adult	16–17 / 23–24 mm	Corneal power birth / adult	~51 / 43–44 D

Lens power birth / adult	~34 / ~19 D	Newborn refraction	+2 to +3 D
Acuity at birth / 1 yr / 3–5 yr	20/400 / 20/50 / 20/20	Stereopsis onset	3–6 mo
Adult-like accommodation	3–4 mo	Amblyopia critical period	to ~7–9 yr
Senile miosis rate	~0.5 mm/decade	Hofstetter mean amplitude	18.5 – 0.30(age)

RETRIEVAL PRACTICE

1. Corneal epithelium, corneal stroma, corneal endothelium: give the embryonic source of each.
2. Iris stroma versus iris epithelium and muscles: source of each?
3. Which structures come from mesoderm?
4. Why is a typical coloboma inferonasal?
5. A newborn has a unilateral small eye with leukocoria and a retrolental membrane. Name the condition and the failed developmental process.
6. Name the gene and chromosome for aniridia, and the tumour you must screen for.
7. Why is retinopathy of prematurity a temporal disease?
8. Give axial length and corneal power at birth and in adulthood, and explain how the eye stays roughly emmetropic.
9. At what age should absence of steady fix-and-follow prompt investigation?
10. Rank amblyopia types by severity.
11. Which vitreous stage becomes the zonules, and when?
12. A small disc with a double ring sign: what must you exclude systemically?

KEY

1. Epithelium: surface ectoderm. Stroma and endothelium: neural crest.
2. Stroma: neural crest. Both epithelia and both the sphincter and dilator: neuroectoderm.
3. Extraocular muscles and all vascular endothelium, including the endothelium of Schlemm canal.
4. Because the optic (embryonic) fissure lies inferonasally and closes from the middle outward during weeks 5 to 7; failure leaves a defect along that line.
5. Persistent fetal vasculature (PHPV); failure of regression of the primary vitreous and hyaloid system beginning at month 7.
6. *PAX6* on 11p13; Wilms tumour, via contiguous deletion of *WT1* in the WAGR syndrome.
7. Retinal vascularisation reaches the nasal ora at about 36 weeks but the temporal ora only at term, so the temporal periphery is the region left avascular in a preterm infant.
8. Axial length 16 to 17 mm to 23 to 24 mm; corneal power about 51 D to 43 to 44 D. Corneal and lens flattening offset axial elongation, and the process is actively guided by retinal defocus.
9. By 3 months of age.
10. Deprivation > anisometropic > strabismic > isoametropic or meridional.
11. Tertiary vitreous, at months 3 to 4.
12. Optic nerve hypoplasia with possible septo-optic dysplasia (de Morsier): absent septum pellucidum and pituitary or hypothalamic dysfunction, which can be life-threatening if unrecognised.

CHAPTER RECAP

1. Four sources: surface ectoderm for lens and surface epithelia, neuroectoderm for retina and pigmented layers and iris muscles, neural crest for all connective tissue, mesoderm for EOM and vessel linings.
2. Optic vesicle day 25, lens placode day 27, lens vesicle day 33, fissure closed by week 7.
3. Coloboma is inferonasal because that is where the fissure is.
4. Primary vitreous is vascular and leaves Cloquet canal, Mittendorf dot, and Bergmeister papilla; secondary is the definitive gel; tertiary becomes zonules.
5. Lids fuse at 8 weeks and reopen at 5 to 7 months; myelination stops at the lamina cribrosa about a month after birth.
6. Temporal retina vascularises last, which is the whole geography of ROP.
7. 16 to 24 mm, 51 to 43 D, 34 to 19 D: emmetropisation is a coordinated, defocus-guided process.
8. Milestones: fix and follow by 6 to 8 weeks, accommodation by 4 months, stereopsis from 3 to 6 months, 20/20 by 3 to 5 years.
9. Deprivation amblyopia in the first 6 months is the most damaging visual event in a human lifetime.
10. Named anomalies map one-to-one onto failed developmental steps and to specific genes.

Ocular and systemic biochemistry

Systemic health · 55–75 items

Lens / cataract

Conjunctiva / cornea

Vitreous / retina / choroid

ANSWER THESE COLD BEFORE READING

1. Give the percentage of lens glucose handled by glycolysis, the TCA cycle, the pentose phosphate shunt, and the polyol pathway.
2. Explain sugar cataract formation in two mechanistic steps, and why galactose is worse than glucose.
3. Name the crystallin that acts as a molecular chaperone.
4. Write out the phototransduction cascade from photon to hyperpolarisation.
5. Name four antibacterial proteins in tears and give tear osmolarity.
6. State ascorbate and protein concentrations in aqueous relative to plasma.

BLUEPRINT WEIGHT

High

Biochemistry is sampled through the systemic health area, the largest on the exam, plus lens and retina.

Metabolic pathways you must be able to place

CORE PATHWAYS: LOCATION, YIELD, AND OCULAR RELEVANCE

Pathway	Location	Yield / purpose and ocular relevance
Glycolysis (Embden-Meyerhof)	Cytosol	Glucose → 2 pyruvate, net 2 ATP and 2 NADH anaerobically. Rate-limiting enzymes: hexokinase (glucokinase in liver) and phosphofructokinase-1 . Dominant in the lens and the corneal epithelium, both of which are relatively hypoxic.
TCA (Krebs) cycle and oxidative phosphorylation	Mitochondria	~ 36–38 ATP per glucose. Requires oxygen. Dominant in the corneal endothelium, ciliary epithelium, retinal inner segments, and RPE. Lens fibres have no mitochondria , so they cannot use it.
Pentose phosphate (hexose monophosphate) shunt	Cytosol	Generates NADPH and ribose-5-phosphate. NADPH is required to regenerate reduced glutathione and to drive aldose reductase. Unusually active in the corneal epithelium (~ 35% of glucose) and the lens (~ 5%). Rate-limiting enzyme: glucose-6-phosphate dehydrogenase ; its deficiency causes haemolysis with sulfonamides and with some topical agents.
Polyol (sorbitol) pathway	Cytosol	Aldose reductase (uses NADPH) converts glucose → sorbitol; sorbitol dehydrogenase converts sorbitol → fructose. Normally minor (~ 5%), but aldose reductase has a high K_m , so it takes over in hyperglycaemia. The pathway of diabetic cataract, and implicated in diabetic neuropathy and retinopathy.
Glycogen metabolism	Cytosol	The corneal epithelium and Müller cells store glycogen as a local energy buffer, which is why the epithelium can survive brief hypoxia and why Müller cells sustain retinal neurons.
Glutathione cycle	Cytosol	GSH (a tripeptide: glutamate, cysteine, glycine) is the main intracellular reducing agent. Glutathione peroxidase detoxifies H_2O_2 ; glutathione reductase regenerates GSH using NADPH . Lens GSH concentration is very high and falls with age and with cataract.

Lens biochemistry

GLUCOSE DISPOSAL IN THE LENS

Route	Share	Note
Anaerobic glycolysis	~85%	Produces almost all lens ATP; lactate diffuses into aqueous.
TCA cycle	~3%	Confined to the epithelium and superficial cortex, the only regions with mitochondria; nevertheless supplies about 20–30% of lens ATP because of its high yield.
Pentose phosphate shunt	~5%	Supplies NADPH for glutathione regeneration.
Polyol pathway	~5%	The route that becomes pathologic in hyperglycaemia.

SUGAR CATARACT IN TWO STEPS, AND WHY GALACTOSE IS WORSE

- Osmotic step.** Hyperglycaemia saturates hexokinase, so excess glucose enters the polyol pathway. **Aldose reductase** makes **sorbitol**, which is *impermeant* and cannot leave the fibre. Sorbitol dehydrogenase is comparatively slow, so sorbitol accumulates, water follows osmotically, fibres swell and rupture, and the lens opacifies. This is the mechanism of the acute, sometimes reversible, refractive shift and cataract of poorly controlled diabetes.
- Oxidative step.** Aldose reductase consumes **NADPH**, competing with glutathione reductase. GSH falls, thiol groups on crystallins oxidise and cross-link, and protein aggregates scatter light irreversibly.

Galactose is worse because aldose reductase converts it to **galactitol (dulcitol)**, and there is *no* dehydrogenase to metabolise galactitol further. It therefore accumulates without limit, which is why classic galactosaemia produces a rapid **oil-droplet cataract** in infancy.

Lens proteins

- Protein content ~33% by weight, the highest of any tissue. Water ~66%.
- Crystallins**, about 90% of lens protein:
 - α-crystallin**: the largest, and a **small heat-shock protein that functions as a molecular chaperone**, binding partially denatured proteins and keeping them soluble. Loss of chaperone capacity with age is a central event in cataractogenesis.
 - β-crystallin**: most abundant by number, oligomeric.
 - γ-crystallin**: smallest, monomeric, highest in the nucleus. Mutations cause several congenital cataracts.
- Albuminoid** is the water-insoluble fraction; it rises with age and correlates with nuclear opacity.
- Post-translational damage that accumulates** because lens proteins are never replaced: non-enzymatic **glycation**, deamidation, **racemisation** of amino acids, disulfide cross-linking, carbamylation (in uraemia, hence uraemic cataract), and photo-oxidation by UV-A. These are the molecular reasons the lens centre is the oldest and most vulnerable protein pool in the body.

- **Ion homeostasis:** a **pump-leak** system maintains high intracellular potassium and low sodium, driven by **Na⁺/K⁺ ATPase** concentrated in the anterior epithelium. Pump failure produces cortical hydration and cortical cataract. This is why the anterior epithelium is described as the metabolic engine of an otherwise inert tissue.

Corneal biochemistry

- **Substrate sources:** glucose from the *aqueous*, oxygen from the *tear film* (and from palpebral conjunctival vessels during lid closure).
- **Epithelium:** glycogen stores, high pentose phosphate activity (~35%), and anaerobic glycolysis producing lactate. Under contact lens hypoxia lactate accumulates in the stroma, raises osmotic load, and causes stromal oedema and striae.
- **Endothelium:** densely packed with mitochondria (second only to the photoreceptor inner segment among ocular tissues) and with **Na⁺/K⁺ ATPase** and **carbonic anhydrase**: the machinery of deturgescence.
- **Antioxidants:** high ascorbate (protecting against UV-B, which the cornea absorbs almost entirely), glutathione, superoxide dismutase, catalase.
- **Collagen:** type I predominates in stroma with type V regulating fibril diameter; type IV in basal lamina and Descemet; type VII in anchoring fibrils; type VIII in Descemet.
- **Corneal wound healing biochemistry:** epithelial defects heal by sliding then mitosis, requiring fibronectin and an intact basement membrane; stromal wounds recruit keratocytes that transform into fibroblasts and myofibroblasts (TGF-β driven), depositing type III collagen and irregular matrix, hence haze and scar. Matrix metalloproteinases (MMP-9 particularly) mediate melting in dry eye, alkali burns, and rosacea, which is the rationale for doxycycline.

Retinal biochemistry: phototransduction and the visual cycle

PHOTOTRANSDUCTION: PHOTON TO HYPERPOLARISATION

- 1 In darkness cGMP is high, **cyclic-nucleotide-gated (CNG) channels are open**, Na⁺ and Ca²⁺ flow inward (the **dark current**), the photoreceptor sits depolarised near **-40 mV**, and it **tonically releases glutamate**.
- 2 A photon isomerises **11-cis-retinal to all-trans-retinal** within rhodopsin, producing the active intermediate **metarhodopsin II**.
- 3 Metarhodopsin II activates the G protein **transducin** (exchanging GDP for GTP).
- 4 Transducin activates **cGMP phosphodiesterase (PDE6)**.
- 5 PDE6 hydrolyses cGMP, so **cGMP falls**.
- 6 **CNG channels close**, cation influx stops, and the cell **hyperpolarises** toward **-70 mV**.
- 7 **Glutamate release decreases**. ON bipolar cells (sign-inverting, mGluR6) depolarise; OFF bipolar cells (sign-conserving) hyperpolarise.
- 8 **Recovery:** falling intracellular Ca²⁺ activates guanylate cyclase (via GCAP) to remake cGMP; rhodopsin kinase and arrestin quench metarhodopsin II; the RGS9 complex deactivates transducin.

THE COUNTER-INTUITIVE CORE OF RETINAL PHYSIOLOGY

Light hyperpolarises the photoreceptor and reduces neurotransmitter release. The photoreceptor is depolarised and busy in the dark. Sildenafil inhibits PDE5 but cross-inhibits **PDE6**, which is precisely why it can cause transient blue-tinged vision and why it is cautioned in retinitis pigmentosa.

THE VISUAL (RETINOID) CYCLE

- 1 All-*trans*-retinal is released from opsin and reduced to **all-*trans*-retinol** by retinol dehydrogenase in the photoreceptor.
 - 2 Retinol crosses to the **RPE**, carried by interphotoreceptor retinoid-binding protein (IRBP).
 - 3 In the RPE, **LRAT** esterifies it to all-*trans*-retinyl ester.
 - 4 **RPE65 isomerohydrolase** converts the ester to **11-*cis*-retinol**.
 - 5 11-*cis*-retinol dehydrogenase oxidises it to **11-*cis*-retinal**.
 - 6 11-*cis*-retinal returns to the photoreceptor and recombines with opsin to regenerate rhodopsin.
- **Vitamin A deficiency** therefore causes **nyctalopia** first (rhodopsin cannot be regenerated), then conjunctival xerosis, **Bitot spots** (keratinised patches of conjunctival epithelium), keratomalacia, and blindness. It remains the leading cause of preventable childhood blindness worldwide.
 - **Mutations:** *RPE65* causes Leber congenital amaurosis (the target of voretigene neparvovec); *ABCA4* failure to clear **all-*trans*-retinal adducts** leads to **A2E lipofuscin** accumulation and Stargardt disease with a dark choroid on angiography.
 - **Nutrients concentrated in the retina:** **taurine** is the most abundant free amino acid in the retina and its deficiency causes photoreceptor degeneration; **docosahexaenoic acid (DHA, 22:6n-3)** makes up a large fraction of outer segment phospholipid; **zinc** is a cofactor for retinol dehydrogenase and is highly concentrated in RPE and choroid; **lutein and zeaxanthin** filter blue light and quench singlet oxygen at the macula.

Tears, aqueous, and vitreous composition**FLUID BIOCHEMISTRY: THE NUMBERS THAT GET ASKED**

Fluid	Composition and key values
Tears	pH 7.4 (range 6.5–7.6); osmolarity 295–310 mOsm/L, with >308 supporting dry eye; protein 6–10 mg/mL; glucose ~5 mg/dL (too low and too variable for reliable diabetes monitoring). Antibacterial proteins: lysozyme (most abundant, cleaves peptidoglycan), lactoferrin (iron chelation, also antioxidant), lipocalin (tear-specific prealbumin) (lipid binding, and the main protein by concentration in some assays), secretory IgA (the principal immunoglobulin), and defensins . Mucins: gel-forming MUC5AC from goblet cells and membrane-bound MUC1, MUC4, MUC16 from the epithelium.

Fluid	Composition and key values
Aqueous humour	Formed at 2.0–3.0 $\mu\text{L}/\text{min}$. Compared with plasma: ascorbate about 20× higher (UV protection); protein about 200× lower ($\sim 0.02\%$ versus plasma $\sim 7\%$), which is the basis of the optically empty anterior chamber and of the Tyndall effect when the barrier breaks; lactate and hydrogen ion higher; glucose about 80% of plasma; amino acids actively transported and <i>above</i> plasma levels; bicarbonate and chloride shifted; TGF-β2 present and immunosuppressive, contributing to anterior chamber associated immune deviation.
Vitreous	98–99% water . Type II collagen fibrils in a hyaluronan matrix, with type IX collagen and opticin regulating spacing. Contains ascorbate, hyaluronidase-resistant proteoglycans, and very little protein. Its high viscosity and slow diffusion govern intravitreal drug half-lives.

Collagen types: a compact reference

WHERE EACH COLLAGEN TYPE APPEARS

Type	Location	Disease link
I	Corneal and scleral stroma, dermis, bone	Osteogenesis imperfecta: blue sclera
II	Vitreous , cartilage	Stickler syndrome (<i>COL2A1</i>): optically empty vitreous, high myopia, retinal detachment, cleft palate, deafness, arthropathy
III	Reticular fibres, scar tissue, blood vessels	Vascular Ehlers-Danlos
IV	All basement membranes: lens capsule, Descemet, corneal basal lamina, Bruch, ILM	Alport syndrome : anterior lenticonus, dot-and-fleck retinopathy, nephritis, deafness
V	With type I in corneal stroma, regulating fibril diameter	Classic Ehlers-Danlos; thin cornea and keratoconus association
VII	Anchoring fibrils under the corneal epithelium	Epidermolysis bullosa; recurrent erosion
VIII	Descemet membrane	<i>COL8A2</i> in early-onset Fuchs dystrophy
IX, XI	Vitreous and cartilage minor collagens	Marshall and Stickler variants

Systemic metabolic disease with ocular findings

INBORN ERRORS AND METABOLIC DISEASE: ENZYME, METABOLITE, AND EYE SIGN

Disease	Defect	Ocular finding
Classic galactosaemia	Galactose-1-phosphate uridylyltransferase	Oil-droplet cataract in infancy; reversible with dietary galactose restriction if caught early.
Homocystinuria	Cystathionine β -synthase	Inferonasal ectopia lentis ; thromboembolism risk with anaesthesia; treat with pyridoxine, betaine, methionine restriction.
Wilson disease	ATP7B copper transport	Kayser-Fleischer ring in peripheral Descemet (copper), sunflower cataract; neuropsychiatric and hepatic disease.
Cystinosis	Lysosomal cystine transporter (<i>CTNS</i>)	Refractile corneal cystine crystals , photophobia, retinopathy; renal Fanconi syndrome. Treated with cysteamine drops.

Disease	Defect	Ocular finding
Fabry disease	α -galactosidase A (X-linked)	Cornea verticillata (whorl-like epithelial deposits), spoke-like cataract, conjunctival and retinal vessel tortuosity.
Gyrate atrophy	Ornithine aminotransferase	Scalloped chorioretinal atrophy, high myopia, cataract; hyperornithinaemia; treat with arginine restriction and pyridoxine.
Refsum disease	Phytanoyl-CoA hydroxylase (phytanic acid accumulation)	Retinitis pigmentosa, cataract, night blindness, with polyneuropathy, deafness, ataxia, ichthyosis.
Abetalipoproteinaemia (Bassen-Kornzweig)	Microsomal triglyceride transfer protein; fat-soluble vitamin malabsorption	Pigmentary retinopathy responsive to vitamin A and E supplementation; acanthocytosis, ataxia.
Tay-Sachs / Niemann-Pick A	Hexosaminidase A / sphingomyelinase	Cherry red spot from ganglioside-laden ganglion cells surrounding the thin foveola.
Mucopolysaccharidoses	Various lysosomal hydrolases (Hurler, Scheie, Hunter, Morquio)	Diffuse corneal clouding (except Hunter and Sanfilippo, which typically spare the cornea), pigmentary retinopathy, optic nerve swelling, glaucoma.
Diabetes mellitus	Insulin deficiency or resistance	Polyol pathway cataract, refractive shifts with glycaemic swings, retinopathy, cranial neuropathies, and a diagnostic HgbA1c threshold of 6.5% .

Vitamins and cofactors

VITAMINS WITH OCULAR CONSEQUENCES

Vitamin	Role	Deficiency or excess
A (retinol)	Retinal chromophore; epithelial differentiation	Deficiency: nyctalopia, xerosis, Bitot spots, keratomalacia. Excess: pseudotumour cerebri with papilloedema.
B₁ thiamine	Pyruvate dehydrogenase and transketolase cofactor	Deficiency: Wernicke encephalopathy with nystagmus, CN VI palsy, and ophthalmoplegia; also nutritional optic neuropathy.
B₂ riboflavin	FAD precursor; used as the photosensitiser in corneal cross-linking	Deficiency: angular stomatitis, corneal vascularisation.
B₃ niacin	NAD precursor	High-dose niacin can cause a reversible cystoid maculopathy that does not leak on fluorescein .
B₁₂ and folate	Methylation and myelin maintenance	Deficiency: bilateral centrocaecal scotomas and optic neuropathy; tobacco-alcohol amblyopia.
C (ascorbate)	Collagen hydroxylation; major aqueous and corneal antioxidant	Deficiency: scurvy with subconjunctival and orbital haemorrhage, poor wound healing.
D	Calcium homeostasis, immunomodulation	Deficiency associated with dry eye and with myopia progression in observational data.

Vitamin	Role	Deficiency or excess
E (α-tocopherol)	Lipid-phase antioxidant, protects outer segment membranes	Deficiency: pigmentary retinopathy, especially in fat malabsorption.
Zinc, copper	Enzyme cofactors; zinc for retinol dehydrogenase	High-dose zinc depletes copper, which is why AREDS formulations add copper.

AREDS2 FORMULATION: KNOW THE EXACT CONTENTS

Vitamin C **500 mg**, vitamin E **400 IU**, **lutein 10 mg and zeaxanthin 2 mg** (replacing β-carotene, which increased lung cancer risk in smokers), zinc oxide **80 mg** (a **25 mg** arm was non-inferior), and cupric oxide **2 mg**. Indicated for **intermediate AMD in one or both eyes, or advanced AMD in one eye**; it reduces progression risk by roughly **25%** over five years and does not prevent AMD in people without it.

NUMBERS TO KNOW: BIOCHEMISTRY

Lens glycolysis / TCA / PPP / polyol	85 / 3 / 5 / 5 %	Lens protein content	~33%
Corneal epithelium PPP share	~35%	Glycolysis ATP yield	2 ATP
Oxidative phosphorylation yield	~36–38 ATP	Tear pH	7.4
Tear osmolarity, normal	295–310 mOsm/L	Dry eye threshold	>308 mOsm/L
Tear protein	6–10 mg/mL	Aqueous protein vs plasma	~1/200
Aqueous ascorbate vs plasma	~20×	Aqueous glucose vs plasma	~80%
Rhodopsin chromophore	11-cis-retinal	Dark membrane potential	~-40 mV
Light membrane potential	~-70 mV	HgbA1c diabetes threshold	6.5%
AREDS2 lutein / zeaxanthin	10 / 2 mg	AREDS2 zinc	80 mg (25 mg ok)

RETRIEVAL PRACTICE

1. Which enzyme starts the polyol pathway, what cofactor does it consume, and why does that matter beyond osmosis?
2. Why does galactosaemia cause cataract faster than diabetes?
3. Which crystallin is a chaperone, and what happens when it fails?
4. In darkness, is the photoreceptor depolarised or hyperpolarised, and is it releasing glutamate?
5. Name the G protein and the effector enzyme of phototransduction, and the drug that cross-inhibits the effector.
6. Which RPE enzyme performs the isomerisation step of the visual cycle, and which disease results from its loss?
7. Name four antibacterial proteins in tears and the principal immunoglobulin.

8. Give two ways aqueous differs from plasma and the clinical sign each explains.
9. Type II collagen defect: name the syndrome and three features.
10. A child has anterior lenticonus, nephritis, and deafness. Which collagen type and which syndrome?
11. Refractile corneal crystals with renal Fanconi syndrome: diagnosis and treatment?
12. State the exact AREDS2 contents and the indication.

KEY

1. Aldose reductase; it consumes NADPH, which depletes reduced glutathione and produces oxidative cross-linking of crystallins in addition to osmotic swelling.
2. Aldose reductase converts galactose to galactitol, for which there is no further dehydrogenase, so it accumulates without limit.
3. α -crystallin; partially denatured proteins aggregate, become insoluble, and scatter light.
4. Depolarised (about -40 mV) and tonically releasing glutamate.
5. Transducin and cGMP phosphodiesterase 6; sildenafil (a PDE5 inhibitor that cross-inhibits PDE6).
6. RPE65; Leber congenital amaurosis.
7. Lysozyme, lactoferrin, lipocalin, defensins; secretory IgA.
8. Protein about 200 times lower (optically clear chamber; and Tyndall flare when the barrier fails) and ascorbate about 20 times higher (UV protection).
9. Stickler syndrome: optically empty vitreous, high myopia with retinal detachment, cleft palate, deafness, arthropathy.
10. Type IV collagen; Alport syndrome.
11. Cystinosis; cysteamine eye drops and systemic cysteamine.
12. Vitamin C 500 mg, vitamin E 400 IU, lutein 10 mg, zeaxanthin 2 mg, zinc oxide 80 mg, cupric oxide 2 mg; for intermediate AMD in one or both eyes or advanced AMD in one eye.

CHAPTER RECAP

1. Lens: 85% glycolysis, no mitochondria in fibres, pump-leak run from the anterior epithelium.
2. Polyol pathway causes sugar cataract twice: osmotically through sorbitol and oxidatively through NADPH theft.
3. α -crystallin is the chaperone; ageing damage is glycation, racemisation, and disulfide cross-linking of proteins that are never replaced.
4. Cornea gets glucose from aqueous and oxygen from tears; the endothelium is the pump, the epithelium is the barrier.
5. Light closes CNG channels and reduces glutamate. Transducin then PDE6 then cGMP fall.
6. Visual cycle turns on RPE65; failure gives Leber congenital amaurosis, and A2E accumulation gives Stargardt.
7. Tears: lysozyme, lactoferrin, lipocalin, defensins, secretory IgA, MUC5AC; osmolarity above 308 supports dry eye.
8. Aqueous: ascorbate high, protein 200-fold low, TGF- β 2 immunosuppressive.
9. Collagen II vitreous (Stickler), IV basement membrane (Alport), I stroma (osteogenesis imperfecta), VIII Descemet (Fuchs).
10. Every classic inborn error has a signature eye finding; learn the enzyme, the metabolite, and the sign as a triple.

Tear film, cornea, aqueous dynamics, IOP, retina, electrodiagnostics

Glaucoma • 13–21 items

Conjunctiva / cornea • 34–50 items

Contact lenses

Vitreous / retina / choroid

Accommodation / vergence

ANSWER THESE COLD BEFORE READING

1. Write the Goldmann equation and define every term with units.
2. Explain corneal deturgescence and the pump-leak hypothesis, and give the Dk/t criteria for daily and extended wear.
3. Give normal IOP mean and distribution, diurnal variation, and the Imbert-Fick assumptions behind Goldmann tonometry.
4. Name the three tear layers with thicknesses and the three mechanisms of aqueous formation with percentages.
5. Describe the ERG waveform and state the cellular origin of each component.
6. State Hering and Sherrington laws and give one clinical application of each.

BLUEPRINT WEIGHT

High

Physiology is the discipline that carries glaucoma, contact lens, and retinal function items.

The tear film

TEAR FILM STRUCTURE: THE CLASSICAL THREE-LAYER MODEL

Layer	Thickness	Source and function
Lipid	~0.1 μm	Meibomian glands. Retards evaporation (by roughly 4–10 $\times$), lowers surface tension, and provides a smooth optical surface. Composed of wax and cholesterol esters, triglycerides, and polar lipids. Failure here causes evaporative dry eye and a rapid tear break-up time.
Aqueous	~7 μm	Main and accessory lacrimal glands. Carries electrolytes, glucose, oxygen, antibacterial proteins, immunoglobulins, and growth factors. Failure causes aqueous-deficient dry eye.
Mucin	~0.02–0.05 μm plus a glycocalyx	Goblet cells (MUC5AC) and epithelial membrane-bound mucins (MUC1, 4, 16). Converts the hydrophobic epithelial surface to a wettable one and traps debris. Failure causes surface non-wetting and staining despite normal volume.

The modern view treats the film as a **two-phase structure**: a lipid layer over a mucoaqueous gel whose mucin concentration rises toward the epithelium. Either model answers exam items; know the classical thicknesses.

TEAR DYNAMICS

Total film thickness	3–10 μm (~7)	Resting tear volume	6–9 μL
Maximum comfortable volume	~25–30 μL	Basal secretion rate	1.2 $\mu\text{L}/\text{min}$
Turnover rate	~16% /min	Blink rate	12–15 /min
Tear break-up time, normal	>10 s	Abnormal TBUT	<5–10 s
Schirmer I, normal	>10 mm / 5 min	Phenol red thread	>10 mm / 15 s
Tear meniscus height	0.2–0.5 mm	Interblink evaporative loss	~10–25% of volume

Corneal physiology

Deturgescence and the pump-leak hypothesis

WHY THE CORNEA STAYS CLEAR

Stromal glycosaminoglycans generate a **swelling pressure of about 50–60 mmHg**, so the stroma continuously imbibes water. Two barriers and one pump oppose it:

1. **Epithelial tight junctions:** the strongest barrier to water and solute entry from tears.
2. **Endothelial barrier:** leaky by design (a "pump-leak" system), allowing nutrient entry from aqueous.
3. **Endothelial Na^+/K^+ ATPase and bicarbonate-dependent transport**, supported by **carbonic anhydrase**, actively pumps ions (and osmotically obligated water) from stroma into aqueous.

Net stromal hydration settles at **78% water**. Endothelial dysfunction or epithelial barrier loss shifts the balance and the cornea swells; because collagen lamellae are tethered laterally, **swelling occurs almost entirely in the anteroposterior direction**, which is why oedema shows as increased thickness and posterior striae rather than lateral expansion.

- **Temperature reversal:** cool the cornea and the pump slows while the leak continues, so the cornea swells; rewarm it and thickness recovers. This experiment established that the endothelium is an active pump.
- **Physiologic overnight oedema:** closed-lid hypoxia produces **2–4%** corneal thickening in a normal eye, resolving within an hour or two of waking. Contact lens wear that exceeds this is pathologic.
- **Critical oxygen requirements (Holden-Mertz):** to avoid more than **4%** oedema, a lens needs $\text{Dk}/t \geq 24 \times 10^{-9}$ units for **daily wear** and $\text{Dk}/t \geq 87$ for **extended (overnight) wear**. Harvitt and Bonanno later proposed higher targets (~35 and ~125) to avoid anoxia in any corneal layer.
- **Hypoxic sequelae in order of appearance:** epithelial microcysts and vacuoles → striae (at about **5%** oedema) → folds (at about **8%**) → polymegathism and pleomorphism → stromal thinning → neovascularisation.
- **Corneal sensitivity** is highest centrally, decreases with contact lens wear, with age, in diabetes, after LASIK, and in herpetic disease. It rises through the day and is lowest on waking.

Aqueous humour dynamics

Formation: three mechanisms

HOW AQUEOUS IS MADE

Mechanism	Share	Detail
Active secretion	80–90%	Energy-dependent transport across the non-pigmented ciliary epithelium , driven by Na^+/K^+ ATPase with carbonic anhydrase II supplying bicarbonate. Independent of IOP . This is the step blocked by carbonic anhydrase inhibitors and by β -blockers (which reduce cAMP-driven secretion).
Ultrafiltration	~10–20%	Hydrostatic pressure gradient across ciliary capillaries. IOP-dependent, so it falls as IOP rises (pseudofacility).
Simple diffusion	small	Lipid-soluble substances move down concentration gradients.

Flow is **2.0–3.0 $\mu\text{L}/\text{min}$** (about **2.5**) with a marked **diurnal rhythm: production falls by roughly half during sleep**. That single fact explains why nocturnal IOP does not simply track daytime IOP and why aqueous suppressants have limited nocturnal effect while prostaglandins retain it.

The Goldmann equation

GOLDMANN EQUATION: KNOW EVERY TERM

$$P_o = (F - U) / C + P_v$$

Term	Meaning	Normal value
P_o	Intraocular pressure	~15.5 mmHg
F	Rate of aqueous formation	2.0–3.0 $\mu\text{L}/\text{min}$
U	Uveoscleral (pressure-independent) outflow	~0.2–0.5 $\mu\text{L}/\text{min}$, i.e. 10–20% of flow
C	Facility of trabecular outflow	0.2–0.3 $\mu\text{L}/\text{min}/\text{mmHg}$
P_v	Episcleral venous pressure	8–10 mmHg

Outflow *resistance* $R = 1/C$. Every IOP-lowering intervention acts on one of these four variables, and identifying which one is a recurring item stem:

- **Reduce F:** β -blockers, α_2 -agonists, carbonic anhydrase inhibitors.
- **Increase C:** pilocarpine, laser trabeculoplasty, trabeculotomy, MIGS stents, Rho kinase inhibitors (netarsudil also lowers P_v).
- **Increase U:** prostaglandin analogues, α_2 -agonists (partly), cyclodialysis.
- **Reduce P_v :** netarsudil; conversely a **raised P_v** from carotid-cavernous fistula, Sturge-Weber, thyroid eye disease, or superior vena cava obstruction raises IOP and shows **blood in Schlemm canal** on gonioscopy.

Intraocular pressure and tonometry

- Population mean **15.5 mmHg**, standard deviation **2.6 mmHg**. The distribution is **not Gaussian: it is skewed to the right**, which is why **21 mmHg** (mean + 2 SD) is a statistical convention rather than a physiologic threshold.
- **Diurnal variation** **3–5 mmHg** in normal eyes and often more in glaucoma; typically highest in the early morning. Supine posture raises IOP by **2–6 mmHg**.
- **Factors that raise IOP**: supine position, Valsalva, breath-holding, tight collar, eyelid squeezing, water drinking, caffeine (small), corticosteroids, and inverted body positions. **Factors that lower it**: aerobic exercise, general anaesthesia, alcohol, metabolic acidosis, and pregnancy.
- IOP rises modestly with age in most populations, and is higher with higher blood pressure and body mass index.

IMBERT-FICK AND THE FOUR ASSUMPTIONS GOLDMANN TONOMETRY MAKES

Imbert-Fick principle: $P = F / A$, valid only for a sphere that is (1) *infinitely thin*, (2) *perfectly elastic*, (3) *perfectly spherical*, and (4) *dry*. The cornea satisfies none of these, so Goldmann engineered a compromise:

- The applanated diameter is fixed at **3.06 mm** (area **7.35 mm²**), the diameter at which the force resisting applanation (corneal rigidity) is exactly balanced by the force assisting it (tear film surface tension, capillary attraction).
- At that diameter, grams of force $\times 10 = \text{mmHg}$, which is why the dial is calibrated the way it is.
- Calibration assumes a **central corneal thickness of about 520 μm** .

Consequences. A **thicker** cornea reads falsely **high**; a **thinner** cornea reads falsely **low** (roughly **0.5–0.7 mmHg per 10 μm** , though the relationship is not linear and correction nomograms are approximations). Thin corneas are also an *independent* risk factor for glaucoma progression in the Ocular Hypertension Treatment Study, so do not simply "correct away" a low reading. **Corneal astigmatism above 3 D** requires rotating the prism so the red mark aligns with the *minus cylinder axis*, or averaging two readings 90° apart. Excess fluorescein gives wide mires and overestimates; too little gives thin mires and underestimates.

TONOMETRY METHODS COMPARED

Method	Principle and behaviour
Goldmann applanation	Variable force, fixed area. The clinical reference standard. CCT-sensitive.
Perkins	Hand-held Goldmann; same physics, usable supine.
Tono-Pen	Mackay-Marg principle; small probe separates corneal rigidity from pressure. Useful on irregular or oedematous corneas; tends to overestimate at low IOP and underestimate at high IOP.
Non-contact (air puff)	Measures time to applanation. More CCT-sensitive than Goldmann and more variable; good for screening.
Rebound (iCare)	Induction-based probe rebound. No anaesthetic needed; excellent in children. Slightly CCT-sensitive.
Dynamic contour tonometry (Pascal)	Contour-matching sensor; largely independent of CCT and gives ocular pulse amplitude.
Ocular response analyser	Gives corneal-compensated IOP, corneal hysteresis, and corneal resistance factor. Low hysteresis is an independent risk factor for glaucoma progression.

Method	Principle and behaviour
Schiötz indentation	Fixed weight, variable indentation. Highly dependent on ocular rigidity : reads low in high myopia and after scleral buckle, high in hyperopia. Largely historical but conceptually examined.
Tonography	Sustained weight on the cornea; the rate of IOP decay estimates outflow facility C. Conceptually important for the Goldmann equation.

Accommodation

HELMHOLTZ MECHANISM, AND THE SIX CHANGES OF THE NEAR TRIAD

Helmholtz (capsular) theory: parasympathetic stimulation contracts the ciliary muscle, which moves the ciliary body forward and inward, **relaxing zonular tension**; the elastic lens capsule then moulds the lens into a more convex shape. Accommodation is therefore an *active* process producing a *passive* lens deformation.

What changes on accommodating:

1. Anterior lens surface steepens markedly (radius **10** → **~6 mm**); posterior surface steepens slightly.
2. Lens axial thickness increases; equatorial diameter decreases.
3. Anterior lens pole moves forward, so anterior chamber depth decreases.
4. Lens nucleus density redistributes; the gradient index steepens.
5. **Pupil constricts** (part of the near triad, increasing depth of focus).
6. **Convergence** occurs (the third element of the near triad).

Presbyopia is primarily *lenticular*: the lens and capsule stiffen and the lens grows, so the same ciliary effort produces less shape change. The ciliary muscle itself remains functional, which is why accommodating intraocular lenses remain an active field.

- **Hofstetter formulae** for amplitude of accommodation in dioptres: maximum = **25 - 0.4 × age**; mean = **18.5 - 0.30 × age**; minimum = **15 - 0.25 × age**.
- **Types of accommodation:** reflex (blur-driven, up to about **2 D**), vergence (driven by disparity, via the AC/A relationship), proximal (awareness of nearness), and tonic (the resting state, about **1 D** in the dark: "dark focus").
- **Accommodative lag** of about **+0.25 to +0.75 D** is normal on monocular estimate method retinoscopy; a high lag with esophoria at near is the classic accommodative-insufficiency pattern and a putative myopia risk factor.

Retinal physiology and electrodiagnostics

Adaptation

- **Dark adaptation curve** is biphasic: a rapid **cone limb** plateauing at **5-7 minutes**, then the **rod-cone break**, then a slow **rod limb** reaching maximum sensitivity at **20-40 minutes**. Total sensitivity gain is about **5 log units**.
- **Purkinje shift:** peak spectral sensitivity moves from **555 nm** (photopic, cones) to **507 nm** (scotopic, rods), which is why red objects look darker and blue objects relatively brighter at dusk.
- **Photopic** above about **3 cd/m²**, **mesopic** between, **scotopic** below about **0.003 cd/m²**.

- **Stiles-Crawford effect:** cones are most sensitive to light entering through the pupil centre; this waveguide directionality reduces the visual impact of scattered and peripheral rays.

Receptive fields and lateral inhibition

- Ganglion cells have **concentric centre-surround antagonistic receptive fields**: ON-centre cells depolarise to light in the centre and are inhibited by surround light; OFF-centre cells do the reverse. The surround is created by **horizontal cell lateral inhibition** in the outer plexiform layer and by amacrine cells in the inner.
- Consequences: the retina encodes **contrast, not absolute luminance**; simultaneous contrast, Mach bands, and the Hermann grid illusion are direct behavioural read-outs of lateral inhibition.
- Receptive field size increases with eccentricity, which is the physiologic basis of the acuity gradient across the field.
- **Cortex**: V1 **simple cells** respond to oriented edges at a specific retinal position; **complex cells** are position-invariant; **hypercomplex** cells are end-stopped. Cells are organised into **orientation columns, ocular dominance columns** (the anatomic substrate modified in amblyopia), and cytochrome oxidase blobs for colour. Beyond V1, the **dorsal stream** (magno, to parietal) handles motion and spatial relations ("where") and the **ventral stream** (parvo, to temporal) handles form and colour ("what").

Electrodiagnostic tests: what each one localises

ERG, EOG, VEP: COMPONENTS AND ORIGINS

Test	Component	Cellular origin and clinical use
Full-field ERG	a-wave (negative)	Photoreceptors (outer segment hyperpolarisation). Reduced in photoreceptor disease.
	b-wave (positive)	ON bipolar cells , recorded via Müller cell potassium currents. A selectively reduced b-wave with a normal a-wave (negative ERG) localises to the inner retina: congenital stationary night blindness, X-linked retinoschisis, CRAO, quinine toxicity, melanoma-associated retinopathy.
	c-wave	RPE (and photoreceptor contribution). Slow, requires DC recording.
	Oscillatory potentials	Amacrine and inner plexiform circuitry; among the earliest abnormalities in diabetic retinopathy.
Pattern ERG	P50 / N95	P50 largely macular; N95 ganglion cell . Distinguishes macular from optic nerve disease when the VEP is abnormal.
Multifocal ERG	Local responses	Topographic map of central retinal function; localises macular versus diffuse loss and detects hydroxychloroquine toxicity.
EOG	Arden ratio = light peak / dark trough	Requires a healthy RPE and photoreceptor interface . Normal >1.8 ; borderline $1.65-1.8$; subnormal <1.65 . A markedly abnormal EOG with a normal ERG is the signature of Best vitelliform dystrophy (bestrophin, <i>BEST1</i>).

Test	Component	Cellular origin and clinical use
VEP	P100 latency and amplitude	Occipital cortical response dominated by the central $\sim 10^\circ$. Delayed latency with relatively preserved amplitude means demyelination (optic neuritis, MS); reduced amplitude with normal latency means axonal loss (compression, ischaemia, glaucoma). Sweep VEP estimates infant acuity.

THREE ELECTRODIAGNOSTIC PATTERNS WORTH MEMORISING AS PAIRS

- **Normal ERG, abnormal EOG** → Best disease.
- **Reduced a- and b-wave** → widespread photoreceptor disease (retinitis pigmentosa: *rod responses affected first and most*).
- **Normal a-wave, reduced b-wave** → inner retinal or bipolar dysfunction.
- **Normal ERG, abnormal VEP** → optic nerve or cortex, not retina.
- **Normal ERG and VEP with poor vision** → consider non-organic visual loss.

Extraocular muscle and binocular physiology

THE TWO INNERVATIONAL LAWS AND THEIR CLINICAL USE

Law	Statement	Clinical application
Sherrington law of reciprocal innervation	Contraction of a muscle is accompanied by relaxation of its ipsilateral antagonist.	Explains why forced duction testing distinguishes restriction (mechanical) from paresis, and why the antagonist does not resist a normal saccade.
Hering law of equal innervation	Yoke muscles in the two eyes receive equal and simultaneous innervation.	Explains secondary deviation exceeding primary deviation in a paretic strabismus, and enhancement of ptosis in the fellow lid when the ptotic eye is made to fixate in myasthenia. Also underlies the three-step test logic.

- **Blood-ocular barriers:** blood-aqueous (non-pigmented ciliary epithelium tight junctions plus non-fenestrated iris capillaries) and blood-retinal (retinal capillary endothelium inner, RPE outer). Their breakdown produces, respectively, cells and flare, and macular oedema.
- **Autoregulation:** retinal and optic nerve head vessels autoregulate over a wide perfusion pressure range and have **no autonomic innervation**; the **choroid does not autoregulate** and is **richly innervated**. Ocular perfusion pressure = mean arterial pressure - IOP, which is why nocturnal systemic hypotension is a glaucoma risk factor.
- **Aqueous flow direction:** posterior chamber → pupil → anterior chamber → angle. Obstruction at the pupil gives **iris bombé**; obstruction behind the iris or lens gives **malignant (aqueous misdirection) glaucoma**, which does not respond to iridotomy.

NUMBERS TO KNOW: PHYSIOLOGY

Tear film thickness	3–10 μm	Tear volume	6–9 μL
Tear turnover	$\sim 16\%$ /min	Normal TBUT	>10 s
Stromal swelling pressure	50–60 mmHg	Normal stromal hydration	78%

Overnight physiologic oedema	2–4%	Dk/t daily / extended wear	≥ 24 / ≥ 87
Aqueous formation	2.0–3.0 $\mu\text{L}/\text{min}$	Nocturnal reduction	~50%
Outflow facility C	0.2–0.3 $\mu\text{L}/\text{min}/\text{mmHg}$	Episcleral venous pressure	8–10 mmHg
Mean IOP $\pm$ SD	15.5 $\pm$ 2.6 mmHg	Diurnal variation	3–5 mmHg
GAT applanation diameter	3.06 mm	GAT assumed CCT	~520 μm
Cone dark adaptation plateau	5–7 min	Rod plateau	20–40 min
Photopic peak sensitivity	555 nm	Scotopic peak	507 nm
Normal EOG Arden ratio	>1.8	VEP major component	P100
Hofstetter mean amplitude	18.5 – 0.30 (age)	Tonic (dark) focus	~1 D

RETRIEVAL PRACTICE

1. Write the Goldmann equation and state which term each drug class modifies.
2. A patient has raised IOP and blood in Schlemm canal. Which Goldmann term is abnormal and name two causes.
3. Explain temperature reversal and what it proved.
4. Give the Dk/t thresholds for daily and extended wear and the percentage oedema they were designed to prevent.
5. Why is 3.06 mm the chosen applanation diameter?
6. CCT is 610 μm and Goldmann reads 24 mmHg. What are the two competing interpretations?
7. Which tonometer is least affected by corneal thickness?
8. List the six changes that occur during accommodation.
9. Give the Hofstetter mean amplitude formula and compute the expected amplitude at age 45.
10. What produces the ERG b-wave, and what does a normal a-wave with a reduced b-wave mean?
11. Normal ERG with a grossly abnormal EOG: diagnosis and gene?
12. Delayed VEP latency with preserved amplitude versus reduced amplitude with normal latency: interpret each.
13. State Hering law and explain enhancement of ptosis in myasthenia.
14. Which ocular vessels autoregulate and lack autonomic supply?

KEY

1. $P_o = (F - U)/C + P_v$. β -blockers, α_2 -agonists, and CAIs reduce F; pilocarpine, SLT, and MIGS raise C; prostaglandins raise U; netarsudil raises C and lowers P_v .
2. P_v , episcleral venous pressure; carotid-cavernous fistula and Sturge-Weber (also thyroid eye disease, SVC obstruction).
3. Cooling the cornea causes swelling because the active pump slows while passive leak continues; rewarming

reverses it. It proved the endothelium is an active, energy-dependent pump.

4. Dk/t of at least 24 for daily wear and 87 for extended wear, designed to keep oedema under about 4%.
5. At that diameter the force of corneal rigidity resisting applanation is balanced by the tear surface tension assisting it, so the reading reflects IOP alone.
6. Either the true IOP is lower than measured because of the thick cornea, or the thick cornea is a real biomechanical feature and the eye genuinely has elevated pressure. Thickness correction is

approximate, so treat trends, discs, and fields rather than a corrected number alone.

7. Dynamic contour tonometry (Pascal).
8. Anterior surface steepens, posterior steepens slightly, lens thickens axially and narrows equatorially, anterior pole moves forward reducing chamber depth, pupil constricts, and the eyes converge.
9. $18.5 - 0.30 \times \text{age}$; at 45 that is $18.5 - 13.5 = 5.0$ D.
10. ON bipolar cells, recorded through Müller cell potassium currents; a normal a-wave with a reduced b-wave (negative ERG) localises to the inner retina, as in congenital stationary night blindness, X-linked retinoschisis, CRAO, or quinine toxicity.
11. Best vitelliform macular dystrophy; *BEST1* encoding bestrophin.
12. Delayed latency with preserved amplitude indicates demyelination; reduced amplitude with normal latency indicates axonal loss.
13. Yoke muscles receive equal innervation. When the more ptotic eye fixates, extra innervation goes to both levators; occluding it removes that drive and the fellow lid falls.
14. Retinal and optic nerve head vessels.

CHAPTER RECAP

1. Three tear layers, 7 μm total, 16% turnover per minute, TBUT above 10 seconds.
2. Deturgescence = epithelial barrier plus a leaky endothelium plus an active pump against a 50 to 60 mmHg swelling pressure.
3. Dk/t 24 daily, 87 extended; hypoxia gives microcysts, then striae at 5%, then folds at 8%.
4. Aqueous: 80 to 90% active secretion, 2.5 $\mu\text{L}/\text{min}$, halved at night.
5. Goldmann equation with five terms, and every drug maps to one of them.
6. IOP 15.5 ± 2.6 with a right-skewed distribution; 21 is a convention.
7. Imbert-Fick needs a thin, elastic, spherical, dry sphere; 3.06 mm is the engineered compromise.
8. Helmholtz: contraction slackens zonules and the capsule moulds the lens; presbyopia is lenticular.
9. ERG a-wave photoreceptors, b-wave ON bipolars via Müller, c-wave RPE; EOG tests RPE; VEP latency means myelin, amplitude means axons.
10. Sherrington for antagonists, Hering for yokes; retinal vessels autoregulate, choroidal vessels do not.

Ocular immunology and infectious agents

Systemic health • 55–75 items

Conjunctiva / cornea • 34–50 items

Episclera / sclera / anterior uvea

Vitreous / retina / choroid

ANSWER THESE COLD BEFORE READING

1. Name the four Gell and Coombs hypersensitivity types with mediator, timing, and one ocular example each.
2. Define anterior chamber associated immune deviation and name three components of ocular immune privilege.
3. List every organism that can penetrate an intact corneal epithelium.
4. Give the classic HLA associations in uveitis.
5. Distinguish the five clinical patterns of adenoviral eye disease.
6. Name the organisms behind each of the four classic causes of infectious keratitis and their risk factors.

BLUEPRINT WEIGHT

High

Immunology and microbiology feed the systemic health, cornea, uvea, and retina areas simultaneously.

Immunology fundamentals

INNATE VERSUS ADAPTIVE IMMUNITY

Feature	Innate	Adaptive
Speed	Minutes to hours	Days on first exposure, hours on re-exposure
Specificity	Pattern recognition (Toll-like receptors recognising PAMPs)	Antigen-specific receptors generated by somatic recombination
Memory	None (some trained immunity)	Yes
Cells	Neutrophils, macrophages, dendritic cells, mast cells, natural killer cells, eosinophils	T lymphocytes (CD4 helper, CD8 cytotoxic, regulatory) and B lymphocytes / plasma cells
Soluble mediators	Complement, lysozyme, lactoferrin, defensins, interferons, acute phase proteins	Immunoglobulins, cytokines
Ocular representation	Tear lysozyme, lactoferrin, lipocalin, defensins; corneal and conjunctival Langerhans cells; complement in tears	Secretory IgA in tears; CALT; regional lymph node priming

IMMUNOGLOBULINS

Class	Key properties and ocular relevance
IgG	Most abundant in serum; four subclasses; the only one crossing the placenta; fixes complement; predominates in the secondary response. IgG4 is central to IgG4-related orbital disease and dacryoadenitis.
IgA	Dimeric secretory IgA is the principal immunoglobulin of tears and all mucosal surfaces; does not fix complement classically; neutralises without inflammation.
IgM	Pentameric, first antibody produced, best complement fixer. A positive IgM titre implies recent infection (relevant to toxoplasmosis and syphilis serology).

Class	Key properties and ocular relevance
IgE	Binds mast cell FcεRI; mediates type I hypersensitivity: allergic and vernal conjunctivitis.
IgD	B cell surface receptor; minimal free serum role.

Hypersensitivity: the four types with ocular examples

GELL AND COOMBS CLASSIFICATION

Type	Mechanism	Timing	Ocular examples
I	IgE on mast cells → degranulation, histamine, tryptase, then leukotrienes and prostaglandins	Minutes (immediate), with a late phase at 4–8 h	Seasonal and perennial allergic conjunctivitis, acute anaphylactic reactions to fluorescein or anaesthetic, the immediate component of vernal and atopic keratoconjunctivitis.
II	Antibody against cell-surface or matrix antigen → complement and cytotoxicity	Hours to days	Ocular cicatricial pemphigoid (antibodies to basement membrane antigens), myasthenia gravis (acetylcholine receptor antibodies), Mooren ulcer (partly), Graves disease (stimulating TSH receptor antibodies).
III	Immune complex deposition → complement activation, neutrophil influx	Hours to days	Stevens-Johnson syndrome , peripheral ulcerative keratitis and scleritis in rheumatoid arthritis and granulomatosis with polyangiitis, marginal (catarrhal) infiltrates from staphylococcal exotoxin, lupus retinal vasculitis, Wessely immune ring .
IV	T cell mediated , delayed; CD4 Th1 with macrophages, or CD8 cytotoxic	24–72 h and beyond	Contact dermatoblepharitis (atropine, neomycin, preservatives), phlyctenulosis , corneal graft rejection , granulomatous uveitis in sarcoid and tuberculosis, the tuberculin skin test, giant cell arteritis.

HYPERSENSITIVITY TYPES

"ACID": type I Allergic (anaphylactic), type II Cytotoxic, type III Immune complex, type IV Delayed.

Ocular immune privilege

WHY A CORNEAL GRAFT SUCCEEDS WITHOUT HLA MATCHING

1. **No lymphatics and no blood vessels in the cornea, lens, or vitreous.** Antigens cannot reach a draining node by the usual route.
2. **Blood-ocular barriers** restrict lymphocyte and antibody entry.
3. **An immunosuppressive microenvironment:** aqueous contains **TGF- β 2**, α -melanocyte stimulating hormone, VIP, calcitonin gene-related peptide, somatostatin, and complement regulatory proteins.
4. **Fas ligand expression** on corneal endothelium and other ocular tissues induces apoptosis of infiltrating activated T cells.
5. **Low MHC class II expression** and few resident antigen-presenting cells in the central cornea (Langerhans cells are limbal).
6. **Anterior chamber associated immune deviation (ACAID):** antigen introduced into the anterior chamber is carried by F4/80-positive antigen-presenting cells through the trabecular meshwork into the blood, reaching the **spleen**, where it induces regulatory T cells that *suppress* the delayed hypersensitivity response to that antigen systemically.

The price of privilege: once these mechanisms are breached, as when a graft becomes vascularised or a herpetic cornea becomes inflamed, rejection risk rises sharply. And privilege explains **sympathetic ophthalmia:** penetrating trauma exposing uveal antigen through a lymphatic-bearing wound bypasses privilege and sensitises the host against its own uvea, producing bilateral granulomatous panuveitis with **Dalen-Fuchs nodules** weeks to years later.

HLA ASSOCIATIONS YOU SHOULD RECOGNISE INSTANTLY

HLA	Association
HLA-B27	Acute anterior uveitis, ankylosing spondylitis, reactive arthritis, psoriatic arthritis, inflammatory bowel disease. Recurrent, unilateral-alternating, non-granulomatous, often with fibrinous exudate and hypopyon.
HLA-B51	Behçet disease: occlusive retinal vasculitis, recurrent hypopyon uveitis, oral and genital ulcers.
HLA-A29	Birdshot chorioretinopathy: one of the strongest HLA associations in medicine.
HLA-DR4 / DRB1*0405	Vogt-Koyanagi-Harada and sympathetic ophthalmia; also rheumatoid arthritis (DR4).
HLA-B7 and DR2	Presumed ocular histoplasmosis syndrome.
HLA-DR15 / DR2	Multiple sclerosis and intermediate uveitis / pars planitis.
HLA-B12 / DR4	Ocular cicatricial pemphigoid.

Microbiology: organisms and the eye

Bacteria

OCULAR BACTERIAL PATHOGENS BY CLASS

Organism	Morphology / stain	Ocular disease and distinguishing feature
Staphylococcus aureus	Gram-positive cocci in clusters, catalase and coagulase positive	Blepharitis, hordeolum, keratitis, endophthalmitis, and marginal infiltrates from exotoxin (a type III reaction, sterile, with a clear zone from the limbus). MRSA is now common in lid and orbital disease.

Organism	Morphology / stain	Ocular disease and distinguishing feature
Staphylococcus epidermidis	Gram-positive cocci, coagulase negative	The commonest organism in acute postoperative endophthalmitis and in normal lid flora; a low-grade, later-onset, better-prognosis endophthalmitis than streptococcal.
Streptococcus pneumoniae	Gram-positive lancet-shaped diplococci, α -haemolytic, optochin sensitive, capsulated	Aggressive central keratitis with a serpiginous edge, dacryocystitis, and a severe endophthalmitis. Historically the classic cause of ulcer serpens.
Streptococcus pyogenes	Group A, β -haemolytic	Preseptal and orbital cellulitis, necrotising fasciitis of the lids, severe conjunctivitis with membranes.
Pseudomonas aeruginosa	Gram-negative rod, oxidase positive, non-lactose fermenting, produces pyocyanin	The classic contact-lens-related keratitis. Rapidly progressive, with a dense stromal infiltrate, thick mucopurulent adherent discharge, ring infiltrate, and rapid melting from proteases. Can penetrate an <i>intact</i> epithelium.
Neisseria gonorrhoeae	Gram-negative intracellular diplococci	Hyperacute purulent conjunctivitis with marked chemosis, lid oedema, preauricular adenopathy, and the ability to penetrate an intact cornea and perforate within days. Requires systemic ceftriaxone. In the neonate it presents on days 2–5.
Haemophilus influenzae	Gram-negative pleomorphic rod, requires factors X and V (chocolate agar)	Pediatric conjunctivitis-otitis syndrome, preseptal cellulitis with a violaceous lid; incidence fell dramatically after Hib vaccination.
Moraxella	Gram-negative diplobacilli ("boxcar")	Angular blepharoconjunctivitis and an indolent, deep, oval keratitis in debilitated or alcoholic patients.

Organism	Morphology / stain	Ocular disease and distinguishing feature
Chlamydia trachomatis	Obligate intracellular; Giemsa shows basophilic intracytoplasmic inclusions ; nucleic acid amplification is the modern test	Serotypes A, B, Ba, C: trachoma , with the WHO SAFE strategy and the FISTO grading (Follicles, Intense inflammation, Scarring, Trichiasis, Opacity); Arlt line and Herbert pits . Serotypes D to K: adult inclusion conjunctivitis , chronic follicular with superior limbal infiltrates, treated with oral azithromycin 1 g and partner treatment. Serotypes L1–L3: lymphogranuloma venereum .
Mycobacterium tuberculosis	Acid-fast bacillus, Ziehl-Neelsen; Lowenstein-Jensen medium	Granulomatous uveitis, serpiginous-like choroiditis, choroidal tubercles, phlyctenulosis (a type IV reaction), and interstitial keratitis.
Treponema pallidum	Spirochaete, darkfield or immunofluorescence; not culturable	"The great imitator": any uveitis pattern, interstitial keratitis with ghost vessels, salt-and-pepper fundus, Argyll Robertson pupils , acute syphilitic posterior placoid chorioretinitis, optic neuritis. Screen with treponemal-specific tests plus RPR/VDRL for activity.
Borrelia burgdorferi	Spirochaete, <i>Ixodes</i> tick vector	Lyme disease: follicular conjunctivitis early, then keratitis, uveitis, and bilateral facial palsy ; erythema migrans rash.
Bartonella henselae	Gram-negative bacillus, cat scratch	Neuroretinitis with a macular star , Parinaud oculoglandular syndrome (conjunctival granuloma with regional adenopathy).
Propionibacterium (Cutibacterium) acnes	Gram-positive anaerobic rod	Delayed, chronic postoperative endophthalmitis with a white capsular plaque, granulomatous uveitis, and initial steroid responsiveness. Requires prolonged anaerobic culture.
Bacillus cereus	Gram-positive spore-forming rod	Fulminant post-traumatic endophthalmitis, especially with intravenous drug use or soil-contaminated injury.

THE ORGANISMS THAT PENETRATE AN INTACT CORNEAL EPITHELIUM

"Shigella, Corynebacterium diphtheriae, Neisseria gonorrhoeae, Neisseria meningitidis, Haemophilus (aegyptius / influenzae biotype III), Listeria"

The high-yield trio to remember is **Neisseria gonorrhoeae, Corynebacterium diphtheriae, and Shigella**. *Pseudomonas* is also capable of rapid invasion, especially in the presence of a contact lens biofilm. All other bacteria require an epithelial break.

Viruses

OCULAR VIRAL PATHOGENS

Virus	Disease patterns and key facts
Adenovirus (double-stranded DNA, non-enveloped)	Five patterns: (1) non-specific follicular conjunctivitis; (2) pharyngoconjunctival fever (serotypes 3, 4, 7) with fever and pharyngitis in children; (3) epidemic keratoconjunctivitis (serotypes 8, 19, 37) with subepithelial infiltrates at 7–14 days, membranes, and preauricular adenopathy; (4) chronic; (5) acute haemorrhagic. Extremely contagious for 10 to 14 days ; non-enveloped, so it survives on surfaces and resists alcohol. Treat supportively; steroids relieve but prolong shedding.
Herpes simplex virus 1 and 2	Primary: blepharoconjunctivitis with vesicles and follicles. Recurrent: dendritic epithelial keratitis with terminal bulbs (staining centrally with fluorescein and at the edges with rose bengal or lissamine), geographic ulcer, stromal (immune) keratitis with a Wessely ring, disciform endotheliitis , neurotrophic keratopathy , iritis with sectoral iris atrophy and raised IOP, and acute retinal necrosis. Latency in the trigeminal ganglion . HEDS trials: topical steroids help stromal keratitis (with antiviral cover), oral aciclovir 400 mg twice daily halves recurrence.
Varicella zoster virus	Chickenpox, then latency in the trigeminal ganglion. Herpes zoster ophthalmicus : dermatomal V1 vesicles, pseudodendrites (elevated, mucous, no terminal bulbs), stromal keratitis, uveitis with sectoral atrophy, scleritis, cranial nerve palsies, post-herpetic neuralgia. Hutchinson sign predicts ocular involvement. Acute retinal necrosis in the immunocompetent; progressive outer retinal necrosis in AIDS. Ramsay Hunt when the geniculate ganglion is involved.
Cytomegalovirus	Retinitis with a "pizza pie" or "brushfire" appearance : confluent necrosis, haemorrhage, and mild vitritis, classically when CD4 is below 50 cells/μL . Also causes corneal endotheliitis with linear keratic precipitates and hypertensive anterior uveitis in immunocompetent adults, and congenital infection with chorioretinitis and hearing loss.
Epstein-Barr virus	Dacryoadenitis, follicular conjunctivitis, nummular keratitis, and an association with ocular adnexal lymphoma.
Molluscum contagiosum (poxvirus)	Umbilicated lid margin nodule causing a chronic follicular conjunctivitis by viral shedding; multiple lesions suggest immunosuppression.
Human papillomavirus	Conjunctival and lid papilloma (types 6, 11); ocular surface squamous neoplasia association (types 16, 18).
Rubella (congenital)	Nuclear cataract, salt-and-pepper retinopathy , microphthalmia, glaucoma, plus deafness and cardiac defects.
Rubeola (measles)	Keratitis, and it precipitates blindness in vitamin A deficient children.
HIV	Immunosuppression enabling CMV retinitis, toxoplasmosis, PORN, cryptococcal choroiditis; direct effects include HIV retinopathy with cotton wool spots and conjunctival microvasculopathy.

Fungi, parasites, and prions

NON-BACTERIAL, NON-VIRAL OCULAR PATHOGENS

Organism	Presentation and diagnostic clue
Filamentous fungi: <i>Fusarium, Aspergillus</i>	Keratitis after vegetative trauma or contaminated contact lens solution. Grey-white infiltrate with feathery margins , satellite lesions, an endothelial plaque, and a hypopyon with a relatively intact epithelium. Grows on Sabouraud agar; visible with calcofluor white, Giemsa, or Gomori methenamine silver. Treat with topical natamycin 5% (best evidence for <i>Fusarium</i>) or voriconazole. Do not give steroids.
Yeast: <i>Candida albicans</i>	Keratitis in a compromised surface (chronic steroid use, dry eye, penetrating keratoplasty), and endogenous endophthalmitis with a fluffy white "string of pearls" in the vitreous in intravenous drug users and patients with indwelling lines.
<i>Acanthamoeba</i>	Contact lens wearer, tap water or hot tub exposure . Pain grossly out of proportion to signs , radial perineuritis early, then a ring infiltrate ; frequently misdiagnosed as HSV. Cysts are double-walled; culture on non-nutrient agar with an <i>E. coli</i> overlay; confocal microscopy is useful. Treat with biguanides (PHMB or chlorhexidine) plus a diamidine, for months.
<i>Toxoplasma gondii</i>	The commonest cause of infectious posterior uveitis worldwide. Focal necrotising retinochoroiditis adjacent to a pigmented scar ("headlight in the fog" with vitritis). Cat definitive host; undercooked meat and oocysts are the routes. Congenital form is bilateral and macular. Treat significant lesions with pyrimethamine, sulfadiazine, and folinic acid, or trimethoprim-sulfamethoxazole, plus steroid cover.
<i>Toxocara canis</i>	Unilateral, in a child with pica and puppy exposure. Peripheral or posterior pole granuloma with a traction band to the disc, or a diffuse endophthalmitis. No systemic eosinophilia in ocular larva migrans , unlike visceral disease. A major leukocoria differential.
<i>Onchocerca volvulus</i>	River blindness, blackfly (<i>Simulium</i>) vector. Sclerosing keratitis, iritis, chorioretinal atrophy, optic atrophy. Treated with ivermectin; do not use diethylcarbamazine , which causes a severe inflammatory reaction.
<i>Loa loa</i>, cysticercosis, <i>Taenia solium</i>	Subconjunctival migrating worm; intraocular cysticercus cyst.
<i>Demodex folliculorum</i> and <i>brevis</i>	Mite in lash follicles and meibomian glands; cylindrical dandruff (collarettes) at the lash base. Treated with tea tree oil derivatives or lotilaner.
<i>Phthirus pubis</i>	Lash infestation with nits; a marker of sexual abuse in children.
Prions	Creutzfeldt-Jakob disease: Heidenhain variant with cortical visual loss; a theoretical transmission risk in corneal transplantation.

INFECTIOUS KERATITIS: MATCHING ORGANISM TO HISTORY

History	First-guess organism	Signature sign
Soft contact lens, overnight wear	Pseudomonas aeruginosa	Rapid, dense infiltrate, adherent mucopurulent discharge, ring infiltrate, melting
Vegetative or organic trauma	Filamentous fungus (<i>Fusarium</i> , <i>Aspergillus</i>)	Feathery margins, satellites, endothelial plaque, relatively intact epithelium
Tap water, hot tub, poor lens hygiene	Acanthamoeba	Pain out of proportion, radial perineuritis, then ring infiltrate
Recurrent, unilateral, reduced sensation	Herpes simplex	Dendrite with terminal bulbs; reduced corneal sensation
Elderly, dermatomal rash, nasal tip vesicles	Varicella zoster	Pseudodendrite without terminal bulbs; sectoral iris atrophy
Chronic steroid use or ocular surface disease	Candida or atypical mycobacteria	Indolent, cracked-windshield or feathery infiltrate; post-LASIK interface keratitis
Sterile, at 2, 4, 8, or 10 o'clock, clear limbal zone	Staphylococcal marginal infiltrate (immune, not infective)	Minimal pain, clear zone from the limbus, associated blepharitis

Diagnostic microbiology**STAINS AND MEDIA**

Test	Detects
Gram stain	Bacteria (positive purple, negative pink) and fungi.
Giemsa	Chlamydial inclusions, cell morphology, <i>Acanthamoeba</i> cysts, and cytology (eosinophils in allergy, multinucleated giant cells in herpes).
Ziehl-Neelsen / Kinyoun	Acid-fast organisms: mycobacteria, <i>Nocardia</i> (partially acid fast).
Calcofluor white / Gomori methenamine silver / PAS	Fungi.
Blood agar	Most aerobic bacteria.
Chocolate agar	<i>Haemophilus</i> , <i>Neisseria</i> (factors X and V).
Thayer-Martin	<i>Neisseria</i> selectively.
Sabouraud dextrose agar	Fungi.
Non-nutrient agar with <i>E. coli</i>	<i>Acanthamoeba</i> .
Thioglycollate broth / anaerobic culture	Anaerobes including <i>C. acnes</i> .
PCR / nucleic acid amplification	Chlamydia, herpesviruses, <i>Toxoplasma</i> , and now the standard for most viral identification.

WHEN TO CULTURE A CORNEAL ULCER

Culture before treating if the infiltrate is **central, larger than about 1–2 mm, involves the deep stroma, has an anterior chamber reaction, is unresponsive to initial therapy, or occurs in an atypical host** (post-surgical, contact lens associated, immunocompromised). Scrape the *edge and base* of the ulcer, not the centre of the slough, and plate directly. Small peripheral non-vision-threatening infiltrates can be treated empirically.

RETRIEVAL PRACTICE

1. Give the hypersensitivity type for allergic conjunctivitis, ocular cicatricial pemphigoid, Stevens-Johnson syndrome, and corneal graft rejection.
2. Name six mechanisms of ocular immune privilege.
3. What is ACAID, and which organ is essential to it?
4. HLA-A29, HLA-B51, HLA-B27: give the disease for each.
5. Which bacteria can penetrate an intact corneal epithelium?
6. Distinguish the corneal lesion of HSV from that of VZV.
7. Which adenovirus serotypes cause epidemic keratoconjunctivitis, and when do subepithelial infiltrates appear?
8. A contact lens wearer has pain far exceeding the visible signs and a ring infiltrate. Diagnosis, culture medium, and two drug classes?
9. Which organism causes a chronic postoperative endophthalmitis with a capsular plaque?
10. Give the CD4 count below which CMV retinitis becomes likely, and describe the fundus.
11. A child with a puppy has unilateral leukocoria and a peripheral granuloma with a traction band. Diagnosis, and what laboratory finding is characteristically absent?
12. Name the two most abundant antibacterial proteins of tears and the principal immunoglobulin.

KEY

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| <ol style="list-style-type: none"> 1. Type I; type II; type III; type IV. 2. No lymphatics or vessels in cornea/lens/vitreous, blood-ocular barriers, immunosuppressive aqueous factors (TGF-β2 and neuropeptides), Fas ligand expression, low MHC class II with few central antigen-presenting cells, and ACAID. 3. Anterior chamber associated immune deviation: antigen from the anterior chamber induces systemic regulatory T cells that suppress delayed hypersensitivity; the spleen is essential. 4. A29 birdshot chorioretinopathy; B51 Behçet disease; B27 acute anterior uveitis with the spondyloarthropathies. 5. <i>Neisseria gonorrhoeae</i>, <i>Neisseria meningitidis</i>, <i>Corynebacterium diphtheriae</i>, <i>Shigella</i>, <i>Haemophilus</i> (aegyptius / influenzae biotype III), and <i>Listeria</i>; | <p><i>Pseudomonas</i> can invade rapidly in the presence of a lens biofilm.</p> <ol style="list-style-type: none"> 6. HSV gives a true dendrite with terminal bulbs and an ulcerated base; VZV gives an elevated mucous pseudodendrite without terminal bulbs. 7. Serotypes 8, 19, and 37; subepithelial infiltrates appear at about 7 to 14 days. 8. <i>Acanthamoeba</i>; non-nutrient agar with an <i>E. coli</i> overlay; a biguanide (PHMB or chlorhexidine) plus a diamidine. 9. <i>Cutibacterium</i> (<i>Propionibacterium</i>) <i>acnes</i>. 10. Below about 50 cells/μL; confluent necrotising retinitis with haemorrhage and mild vitritis, described as pizza pie or brushfire. 11. Ocular toxocariasis; systemic eosinophilia is characteristically absent in the ocular form. 12. Lysozyme and lactoferrin; secretory IgA. |
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CHAPTER RECAP

1. ACID: allergic, cytotoxic, immune complex, delayed. Assign every inflammatory eye disease to one.
2. Immune privilege has six pillars, and breaking them explains both graft rejection and sympathetic ophthalmia.
3. HLA: B27 anterior uveitis, B51 Behçet, A29 birdshot, DR4 VKH.
4. Only a short list of bacteria cross an intact epithelium; everything else needs a break.
5. Contact lens plus water equals *Acanthamoeba*; contact lens plus overnight wear equals *Pseudomonas*; vegetation equals fungus.
6. Adenovirus 8, 19, 37 for EKC; infiltrates at 7 to 14 days; contagious for two weeks.
7. HSV dendrite has terminal bulbs; VZV pseudodendrite does not.
8. CMV retinitis below CD4 of 50; toxoplasmosis sits next to an old scar.
9. *C. acnes* for chronic postoperative inflammation with a capsular plaque; *S. epidermidis* for acute.
10. Culture central, large, deep, or atypical ulcers; scrape the edge, not the slough.

General pathology and ocular pathology by tissue

All Normal health / disease / trauma areas · 228 items

Emergencies / trauma (embedded)

Systemic health

ANSWER THESE COLD BEFORE READING

1. Define the five cardinal cellular adaptations and give an ocular example of each.
2. Distinguish acute from chronic and granulomatous from non-granulomatous inflammation histologically.
3. Name the histologic hallmark of each of the four common corneal dystrophies by layer.
4. Give the histology and the two most feared prognostic features of choroidal melanoma.
5. Describe Flexner-Wintersteiner rosettes and where they occur.
6. List the histologic sequence of diabetic retinopathy.

BLUEPRINT WEIGHT

Very high

Pathology is sampled in every one of the nine Normal health / disease / trauma condition areas.

General pathology framework

CELLULAR ADAPTATION AND INJURY, WITH OCULAR EXAMPLES

Process	Definition	Ocular example
Hypertrophy	Increase in cell <i>size</i>	Retinal pigment epithelial cell hypertrophy overlying a naevus or after trauma.
Hyperplasia	Increase in cell <i>number</i>	Reactive RPE hyperplasia at the edge of a chorioretinal scar; conjunctival goblet cell hyperplasia in early allergy.
Atrophy	Decrease in size or number	Optic atrophy after axonal loss; iris stromal atrophy after herpetic iritis; geographic atrophy in AMD.
Metaplasia	Replacement of one differentiated cell type by another	Squamous metaplasia and keratinisation of conjunctival epithelium in vitamin A deficiency and cicatrising conjunctivitis.
Dysplasia	Disordered growth with cytologic atypia, still confined by basement membrane	Conjunctival intraepithelial neoplasia; the precursor to ocular surface squamous cell carcinoma.
Necrosis	Cell death with membrane rupture and inflammation	Coagulative necrosis in CRAO; caseous necrosis in a tubercle; liquefactive necrosis in a bacterial abscess.
Apoptosis	Programmed death, no inflammation, cell shrinkage with nuclear fragmentation	Ganglion cell death in glaucoma; photoreceptor death in retinitis pigmentosa; the mechanism of Fas ligand mediated immune privilege.

INFLAMMATION: RECOGNISING THE PATTERN

Pattern	Dominant cell	Ocular correlate
Acute	Neutrophils , oedema, fibrin	Bacterial keratitis, hypopyon, acute dacryocystitis, endophthalmitis.

Pattern	Dominant cell	Ocular correlate
Chronic non-granulomatous	Lymphocytes and plasma cells	HLA-B27 anterior uveitis, chronic blepharitis, pterygium stroma.
Chronic granulomatous	Epithelioid histiocytes and multinucleated giant cells , sometimes with caseation	Sarcoidosis, tuberculosis, sympathetic ophthalmia (with Dalen-Fuchs nodules), chalazion (lipogranuloma), Langhans giant cells in TB, Touton giant cells in juvenile xanthogranuloma.
Zonal granulomatous	Palisaded histiocytes around a nidus	Chalazion around extravasated meibum; phacoantigenic (phacoanaphylactic) uveitis around lens material.
Eosinophil-rich	Eosinophils	Vernal keratoconjunctivitis (eosinophils in conjunctival scrapings are diagnostic), parasitic infection.
Granulomatous uveitis, clinically	–	Mutton-fat keratic precipitates , Koeppe (pupil margin) and Busacca (stromal) iris nodules, and posterior involvement. Contrast with fine, non-pigmented precipitates in non-granulomatous disease.

Lids and adnexa

LID LESIONS: HISTOLOGY DRIVES THE DIAGNOSIS

Lesion	Histology	Clinical key
Chalazion	Lipogranuloma: zonal granulomatous inflammation around extravasated meibum with lipid vacuoles and giant cells	Painless nodule; recurrent or atypical lesions in the elderly must be biopsied.
Seborrheic keratosis	Acanthosis, hyperkeratosis, and pseudo-horn cysts of keratin	Greasy, stuck-on, benign.
Actinic keratosis	Epidermal dysplasia with parakeratosis and solar elastosis	Premalignant; scaly, sun-exposed skin.
Keratoacanthoma	Rapidly growing cup-shaped lesion with a keratin-filled crater; well-differentiated squamous proliferation	Considered a low-grade squamous cell carcinoma variant; grows over weeks.
Basal cell carcinoma	Nests of basaloid cells with peripheral palisading and retraction artefact; nodular, morpheaform, or superficial	85–90% of lid malignancies; commonest on the lower lid and medial canthus . Pearly rolled edge with telangiectasia and central ulceration. Locally invasive, rarely metastasises; medial canthal morpheaform lesions are the most dangerous for orbital invasion.
Squamous cell carcinoma	Atypical keratinocytes with keratin pearls, invading the dermis; may show perineural invasion	Second commonest; more aggressive with true metastatic potential and perineural spread to the orbit and cavernous sinus.

Lesion	Histology	Clinical key
Sebaceous cell carcinoma	Lobules of atypical cells with foamy, lipid-laden cytoplasm; oil-red-O positive on fresh tissue ; pagetoid spread through the epithelium; multicentric	The great masquerader. Presents as a recurrent chalazion, or as a unilateral chronic "blepharoconjunctivitis" that fails to respond. Favours the upper lid (more meibomian glands). Loss of lashes (madarosis) is a red flag. Associated with Muir-Torre syndrome. High mortality if missed.
Melanoma	Atypical melanocytes; Breslow depth and mitotic rate drive prognosis	Rare on the lid; consider in any pigmented lesion with ABCDE features. Lentigo maligna in sun-damaged skin.
Merkel cell carcinoma	Small blue cells with neuroendocrine markers (CK20 positive, dot-like perinuclear pattern)	Violaceous, rapidly growing; aggressive with early nodal metastasis; Merkel cell polyomavirus association.
Xanthelasma	Dermal foamy lipid-laden histiocytes	Medial lids; check the lipid profile, especially in the young.
Molluscum contagiosum	Lobular epidermal hyperplasia with Henderson-Paterson intracytoplasmic inclusion bodies	Umbilicated nodule; multiple lesions suggest HIV.

TWO LID MALIGNANCY RULES

Any "recurrent chalazion", especially in the upper lid of a patient over 50, must be biopsied to exclude sebaceous cell carcinoma. Similarly, a unilateral chronic blepharoconjunctivitis unresponsive to treatment is a sebaceous carcinoma until proven otherwise. Warn the pathologist to keep fresh tissue for lipid stains, because routine processing dissolves the lipid.

Madarosis, loss of lid margin architecture, telangiectasia, ulceration, induration, and distortion of the punctum are the six features that convert a benign-looking lesion into a biopsy.

Conjunctiva and cornea

CONJUNCTIVAL PATHOLOGY

Lesion	Histology and significance
Pinguecula	Subepithelial elastotic degeneration of collagen with basophilic hyaline change; no corneal involvement.
Pterygium	Same elastotic degeneration plus destruction of Bowman layer and fibrovascular ingrowth onto the cornea; a Stocker line of iron marks the advancing edge.
Conjunctival naevus	Nests of benign melanocytes, often with intralesional cysts ; almost never involves the palpebral conjunctiva or fornix (a lesion there is suspicious).
Primary acquired melanosis	Intraepithelial melanocytic proliferation; with atypia it progresses to melanoma in a substantial minority ; flat, patchy, acquired in middle age.
Conjunctival melanoma	Invasive atypical melanocytes; arises from PAM with atypia, naevus, or de novo. Spreads to preauricular and submandibular nodes because the conjunctiva <i>has</i> lymphatics.
Ocular surface squamous neoplasia	Dysplasia to invasive squamous carcinoma; gelatinous, leukoplakic, or papilliform with "hairpin" vessels; UV and HPV related. Rose bengal stains it.

Lesion	Histology and significance
Lymphoid tumour	Salmon-pink patch; monomorphic small lymphocytes suggest MALT lymphoma, polymorphic suggests reactive hyperplasia; flow cytometry and molecular clonality distinguish them.
Amyloidosis	Amorphous eosinophilic deposits, Congo red positive with apple-green birefringence under polarised light; presents as waxy salmon nodules that bleed easily.

CORNEAL DYSTROPHIES: LAYER, INHERITANCE, AND HISTOLOGIC HALLMARK

Dystrophy	Layer	Inheritance	Hallmark and stain
Epithelial basement membrane (map-dot-fingerprint, Cogan)	Epithelium / basement membrane	AD (often sporadic)	Thickened, multilaminar basement membrane with intraepithelial extension and microcysts. The commonest dystrophy; causes recurrent erosions and irregular astigmatism. <i>TGFBI</i> in some families.
Meesmann	Epithelium	AD	Myriad tiny intraepithelial cysts containing "peculiar substance"; keratin genes <i>KRT3</i> , <i>KRT12</i> .
Reis-Bücklers / Thiel-Behnke	Bowman layer	AD	Replacement of Bowman by fibrocellular tissue; honeycomb (Thiel-Behnke) or geographic (Reis-Bücklers) opacity. <i>TGFBI</i> .
Lattice	Stroma	AD	Amyloid : Congo red positive, apple-green birefringence, metachromatic with crystal violet. Refractile branching lines; recurs in grafts; <i>TGFBI</i> . Gelsolin variant is systemic (Meretoja).
Granular	Stroma	AD	Hyaline : Masson trichrome positive, PAS negative. Discrete crumb-like opacities with clear intervening stroma and a clear periphery. <i>TGFBI</i> .
Macular	Stroma	AR	Keratan sulfate mucopolysaccharide : Alcian blue and colloidal iron positive. Dense grey opacities with hazy intervening stroma extending to the limbus , and a thin cornea . The only classic stromal dystrophy that is autosomal recessive ; <i>CHST6</i> .
Schnyder	Stroma	AD	Cholesterol and phospholipid crystals; central crystalline ring with arcus. Check serum lipids (<i>UBIAD1</i>).
Fuchs endothelial	Endothelium / Descemet	AD (late), <i>COL8A2</i> early	Central guttae (posterior Descemet excrescences), endothelial pleomorphism and loss, thickened Descemet, then stromal oedema and epithelial bullae. Female predominance; <i>TCF4</i> trinucleotide repeat expansion is the major risk locus.
Posterior polymorphous	Endothelium	AD	Endothelium undergoes epithelial metaplasia with multilayering and microvilli; vesicular and band lesions, iris adhesions, glaucoma. <i>ZEB1</i> , <i>OVOL2</i> .
Congenital hereditary endothelial dystrophy	Endothelium	AR (<i>SLC4A11</i>)	Diffuse ground-glass corneal oedema from birth; thickened Descemet; a leukocoria and congenital-glaucoma differential.

THREE STROMAL DYSTROPHIES, THREE STAINS**"Marilyn Monroe Always Gets Her Man in LA County"**

Macular = Mucopolysaccharide = Alcian blue. Granular = Hyaline = Masson trichrome. Lattice = Amyloid = Congo red.

Corneal degenerations and depositions

DEGENERATIONS (ACQUIRED, OFTEN PERIPHERAL AND ASYMMETRIC) VERSUS DYSTROPHIES (INHERITED, CENTRAL, SYMMETRIC)

Entity	Nature
Arcus senilis	Peripheral stromal lipid with a clear zone at the limbus ; a lipid profile is warranted under age 40, and unilateral arcus suggests contralateral carotid disease.
Vogt limbal girdle	Chalk-like peripheral elastotic band; benign.
Band (calcific) keratopathy	Calcium hydroxyapatite in Bowman layer, in the interpalpebral zone with a clear limbal interval and Swiss-cheese holes ; follows chronic uveitis, silicone oil, phosphate exposure, or hypercalcaemia.
Salzmann nodular degeneration	Bluish-white subepithelial hyaline nodules after chronic surface disease.
Terrien marginal degeneration	Slowly progressive superior peripheral thinning with lipid at the central edge, vascularisation, intact epithelium and no pain ; against-the-rule astigmatism.
Pellucid marginal degeneration	Inferior peripheral thinning with the cornea protruding <i>above</i> the thinnest zone; "crab claw" topography, high against-the-rule astigmatism.
Keratoconus	Breaks in Bowman layer , stromal thinning, iron deposition (Fleischer ring), Vogt striae, and folds in Descemet with acute hydrops on rupture. Munson and Rizzuti signs; scissors reflex on retinoscopy; associated with atopy, eye rubbing, Down syndrome, Leber congenital amaurosis, and connective tissue disease.
Iron lines	Hudson-Stähli (age, inferior interpalpebral), Fleischer (keratoconus cone base), Stocker (pterygium), Ferry (filtering bleb), Bergmeister -unrelated. All are epithelial iron.
Metal and drug deposits	Kayser-Fleischer copper in peripheral Descemet (Wilson); chalcosis from intraocular copper; siderosis from iron with heterochromia and mydriasis; cornea verticillata from amiodarone, chloroquine, indomethacin, or Fabry disease; argyrosis from silver; chrysiasis from gold; ciprofloxacin white precipitate in an epithelial defect.

Lens, uvea, and intraocular tumours**LENS PATHOLOGY**

Entity	Histology and consequence
Nuclear cataract	Crystallin aggregation and chromophore accumulation; myopic shift and reduced blue transmission.
Cortical cataract	Fibre hydration, Morgagnian globules (liquefied cortex), and clefts; glare and monocular diplopia.
Posterior subcapsular cataract	Posterior migration of aberrant epithelial bladder (Wedl) cells ; classic with steroids, radiation, diabetes, and uveitis; disproportionate near and glare loss.
Phacolytic glaucoma	Leaked high-molecular-weight lens protein from a hypermature cataract engulfed by macrophages that obstruct the meshwork. Open angle, no synechiae, heavy cell-and-flare.
Phacomorphic glaucoma	An intumescent lens producing mechanical angle closure. Closed angle.

Entity	Histology and consequence
Phacoantigenic (phacoanaphylactic) uveitis	Zonal granulomatous reaction to exposed lens protein after capsule rupture.
Posterior capsule opacification	Residual equatorial epithelium migrating onto the posterior capsule; Elschnig pearls and a Soemmering ring .
Pseudoexfoliation	Fibrillar basement-membrane-like material on the anterior capsule, zonules, iris, and meshwork; target pattern with a clear intermediate zone, Sampaolesi line, weak zonules, poor dilation. Systemic elastosis (<i>LOXL1</i>) with vascular associations.

UVEAL MELANOMA: THE HISTOLOGY THAT PREDICTS SURVIVAL

- **Cell type (modified Callender):** **spindle** cells have the best prognosis; **mixed** intermediate; **epithelioid** the worst.
- **Size** is the strongest clinical predictor: largest basal diameter and thickness. Thickness above **2 mm**, subretinal fluid, symptoms, orange pigment (lipofuscin), margin within **3 mm** of the disc, and absence of drusen or a halo distinguish a **small melanoma from a naevus** ("To Find Small Ocular Melanoma Doing IMaging").
- **Extrascleral extension, vortex vein invasion, and high mitotic rate** worsen prognosis.
- **Monosomy 3** and **8q gain**, and **class 2 gene expression profile**, predict metastasis. *GNAQ/GNA11* mutations are near-universal drivers; *BAP1* loss is a strong adverse marker; *EIF1AX* and *SF3B1* mutations are more favourable.
- **Metastasis is haematogenous and overwhelmingly to the liver**, because the eye has no lymphatics. Screen with liver imaging and enzymes.
- **Mushroom (collar-stud) shape** on B-scan with low internal reflectivity and choroidal excavation is characteristic; the shape results from rupture through Bruch membrane.

RETINOBLASTOMA AND OTHER INTRAOCULAR TUMOURS

Tumour	Histology and features
Retinoblastoma	Small round blue cells with scant cytoplasm, high mitotic rate, necrosis with calcification (the finding that makes CT and ultrasound diagnostic), and Flexner-Wintersteiner rosettes (a single layer of cells around a true lumen, indicating photoreceptor differentiation). Homer Wright rosettes (no lumen) and fleurettes also occur. <i>RB1</i> tumour suppressor at 13q14 , following the two-hit hypothesis ; heritable disease is bilateral, multifocal, earlier, and carries a risk of trilateral retinoblastoma (pineoblastoma) and second malignancies, especially osteosarcoma. Leukocoria and strabismus are the presenting signs. Optic nerve invasion beyond the lamina, choroidal invasion, and extrascleral extension are the key adverse pathologic features.
Choroidal metastasis	The commonest intraocular malignancy overall. Creamy, plateau-shaped, often multifocal and bilateral, with high internal reflectivity. Breast in women, lung in men.
Choroidal haemangioma	Cavernous vascular channels; orange-red, high internal reflectivity , early intense fluorescence with late washout. Diffuse form in Sturge-Weber.
Choroidal osteoma	Mature bone in the choroid; juxtapapillary, orange-yellow, dense acoustic shadowing on B-scan.
Astrocytic hamartoma	Glial proliferation; mulberry calcified lesion in tuberous sclerosis (<i>TSC1</i> , <i>TSC2</i>).
Retinal capillary haemangioblastoma	Von Hippel-Lindau (<i>VHL</i> , 3p25); red globe with dilated feeder vessels and exudation; screen for renal cell carcinoma and pheochromocytoma.

Tumour	Histology and features
Combined hamartoma of retina and RPE	Glial, vascular, and pigmented proliferation with retinal traction; associated with neurofibromatosis 2.
Primary vitreoretinal lymphoma	Large atypical B cells, subretinal pigment epithelial infiltration; vitreous IL-10 exceeding IL-6, and <i>MYD88</i> mutation. A masquerade for chronic uveitis in the elderly; requires CNS imaging.

Retina, optic nerve, and glaucoma

HISTOLOGIC SEQUENCE OF DIABETIC RETINOPATHY

- 1 Capillary basement membrane thickening** and loss of **pericytes**.
- 2 Microaneurysms** (saccular outpouchings, best seen on angiography), and increased permeability with leakage of plasma into the retina.
- 3 Intraretinal haemorrhage**: dot-and-blot in the deeper layers, flame-shaped in the nerve fibre layer.
- 4 Hard exudates**: lipoprotein and lipid-laden macrophages, deposited at the junction of oedematous and normal retina, hence their circinate arrangement.
- 5 Capillary closure** with acellular capillaries, cotton wool spots (nerve fibre layer infarcts with cytoid bodies representing swollen axonal end bulbs), **intraretinal microvascular abnormalities**, and venous beading.
- 6 Ischaemia-driven VEGF release** → **neovascularisation** at the disc and elsewhere, growing on the retinal surface and along the posterior hyaloid.
- 7 Fibrovascular proliferation**, vitreous haemorrhage, **tractional retinal detachment**, and finally iris neovascularisation with neovascular glaucoma.

RETINAL AND VASCULAR PATHOLOGY PATTERNS

Entity	Pathology
Cotton wool spot	Focal ischaemic infarct of the nerve fibre layer with cytoid bodies : swollen axonal end bulbs with pseudonuclei.
Cystoid macular oedema	Fluid accumulation predominantly in the inner nuclear and outer plexiform (Henle) layers , producing a petalloid angiographic pattern. Causes: post-cataract surgery (Irvine-Gass), diabetes, vein occlusion, uveitis, epinephrine and prostaglandin analogues, retinitis pigmentosa, nicotinic acid (which does <i>not</i> leak).
Drusen	Focal deposits between the RPE basal lamina and Bruch membrane. Hard (small, discrete) versus soft (large, confluent, higher AMD risk). Distinguish from optic disc drusen , which are calcified hyaline bodies anterior to the lamina cribrosa causing pseudopapilloedema and autofluorescence.
Retinitis pigmentosa	Progressive rod then cone loss with RPE migration into the retina ("bone spicule" pigment), arteriolar attenuation, waxy disc pallor, and posterior subcapsular cataract. Rod-cone ERG reduced early.
Age-related macular degeneration	Non-exudative: drusen, RPE change, and geographic atrophy with photoreceptor and choriocapillaris loss. Exudative: choroidal neovascularisation through Bruch membrane , with pigment epithelial detachment, subretinal haemorrhage, and eventual disciform fibrovascular scar .

Entity	Pathology
Retinal detachment	Separation at the <i>subretinal space</i> , that is between photoreceptors and RPE, not between retina and choroid. Rhegmatogenous (break plus liquefied vitreous plus traction), tractional, or exudative. Chronic detachments show demarcation lines, intraretinal cysts, and subretinal precipitates.
Retinal artery occlusion	Coagulative necrosis of the inner retina with inner nuclear layer and ganglion cell loss; the outer retina survives on choroidal supply.
Glaucomatous optic neuropathy	Loss of retinal ganglion cell axons with posterior bowing and remodelling of the lamina cribrosa , enlargement of the cup, thinning of the neuroretinal rim (worst at the inferior and superior poles), laminar dot sign , bayoneting and nasalisation of vessels, and Schnabel cavernous degeneration retrolaminarly. Ganglion cells die by apoptosis , and there is no inflammation.

Trauma pathology

MECHANISMS AND NAMED FINDINGS IN OCULAR TRAUMA

Injury	Pathology
Blunt (contusion) injury	Anteroposterior compression with equatorial expansion. Produces, from front to back: corneal abrasion, hyphaema, iridodialysis, sphincter tears (traumatic mydriasis), angle recession , cyclodialysis, lens subluxation, Vossius ring (imprinted iris pigment on the anterior capsule), rosette cataract, retinal dialysis (classically superonasal or inferotemporal), commotio retinae (photoreceptor outer segment disruption, not oedema), choroidal rupture, and sclopetaria .
Chemical burn	Alkali saponifies membrane lipids and penetrates rapidly, causing deep and progressive damage with limbal ischaemia and late glaucoma; acid coagulates proteins, which limits penetration (hydrofluoric acid excepted). Limbal blanching extent is the prognostic sign (Roper-Hall and Dua classifications). Copious irrigation until pH normalises comes before everything else.
Intraocular foreign body	Siderosis (iron: heterochromia, mydriasis, cataract, pigmentary retinopathy, reduced ERG) and chalcosis (copper: Kayser-Fleischer-like ring, sunflower cataract; pure copper causes suppurative endophthalmitis). Inert materials include glass, plastic, and gold.
Sympathetic ophthalmia	Bilateral diffuse granulomatous panuveitis sparing the choriocapillaris and retina, with Dalen-Fuchs nodules (epithelioid histiocytes between RPE and Bruch). Latency two weeks to decades, most within three months. Prevention is prompt repair or, rarely, enucleation of a blind injured eye.
Shaken baby syndrome	Multilayered retinal haemorrhages extending to the periphery, retinoschisis with a perimacular fold, and vitreous haemorrhage, with subdural haematoma and no external signs. A mandatory-report finding.
Purtscher retinopathy	Multiple polygonal areas of retinal whitening (Purtscher flecken) after chest compression, long bone fracture, or acute pancreatitis; complement-mediated leucoembolisation.
Solar and laser retinopathy	Photochemical (solar) or photothermal/photodisruptive (laser) foveal damage with a characteristic outer retinal defect on OCT.

RETRIEVAL PRACTICE

1. Give the histologic hallmark and special stain for lattice, granular, and macular dystrophy.
2. Which stromal dystrophy is autosomal recessive, and what else is unusual about the cornea in it?
3. A recurrent upper lid "chalazion" in a 68-year-old with madarosis. Diagnosis, stain required, and what you must tell the pathologist.
4. Name the two giant cell types associated with tuberculosis and juvenile xanthogranuloma.

5. Which uveal melanoma cell type has the worst prognosis, and which cytogenetic findings predict metastasis?
6. Where does uveal melanoma metastasise, and why not to lymph nodes?
7. Describe a Flexner-Wintersteiner rosette and state what it signifies.
8. Give the gene, locus, and inheritance mechanism of retinoblastoma, plus two associated malignancies.
9. List the first three histologic changes of diabetic retinopathy in order.
10. What is a cystoid body and in which lesion is it found?
11. In which retinal layers does cystoid macular oedema accumulate, and which drug causes it without angiographic leakage?
12. Alkali versus acid burn: state the mechanism of each and the single best prognostic sign.
13. Name the histologic hallmark of sympathetic ophthalmia and its typical latency.
14. What is commotio retinae at the cellular level?

KEY

1. Lattice: amyloid, Congo red with apple-green birefringence. Granular: hyaline, Masson trichrome. Macular: keratan sulfate mucopolysaccharide, Alcian blue and colloidal iron.
2. Macular dystrophy; the intervening stroma is hazy and the opacity extends to the limbus, and the cornea is thin.
3. Sebaceous cell carcinoma; oil-red-O on fresh (unfixed) tissue; warn the pathologist to preserve fresh tissue because routine processing dissolves lipid, and request evaluation for pagetoid spread.
4. Langhans giant cells in tuberculosis; Touton giant cells in juvenile xanthogranuloma.
5. Epithelioid; monosomy 3, 8q gain, class 2 gene expression profile, and *BAP1* loss.
6. To the liver; haematogenously, because the eye has no lymphatic drainage.
7. A single layer of tumour cells arranged around a true central lumen, indicating photoreceptor differentiation; characteristic of retinoblastoma.
8. *RB1* at 13q14; two-hit hypothesis with a germline first hit in heritable disease; pineoblastoma (trilateral retinoblastoma) and osteosarcoma.
9. Capillary basement membrane thickening and pericyte loss, then microaneurysms with increased permeability, then intraretinal haemorrhage.
10. A swollen axonal end bulb with a pseudonucleus, found in a cotton wool spot.
11. Inner nuclear and outer plexiform (Henle) layers; nicotinic acid (niacin) maculopathy does not leak on fluorescein.
12. Alkali saponifies lipids and penetrates deeply and progressively; acid coagulates protein, which limits penetration. The extent of limbal (perilimbal) blanching is the best prognostic sign.
13. Dalen-Fuchs nodules, epithelioid histiocytes between the RPE and Bruch membrane; latency two weeks to decades, most within three months.
14. Disruption of photoreceptor outer segments, not true retinal oedema.

CHAPTER RECAP

1. Five adaptations, two death modes; apoptosis is the mode in glaucoma and retinitis pigmentosa.
2. Granulomatous means epithelioid histiocytes and giant cells, and clinically means mutton-fat precipitates and iris nodules.
3. BCC is the commonest lid malignancy, lower lid and medial canthus; sebaceous carcinoma masquerades as chalazion or blepharitis on the upper lid.
4. Marilyn Monroe: macular-mucopolysaccharide-Alcian, granular-hyaline-Masson, lattice-amyloid-Congo.
5. Macular dystrophy is the recessive one, with hazy intervening stroma and a thin cornea.
6. Melanoma: epithelioid, large, monosomy 3, BAP1 loss, class 2 profile; liver metastasis.
7. Retinoblastoma: RB1 13q14, two hits, calcification, Flexner-Wintersteiner rosettes, trilateral disease.
8. Diabetic retinopathy starts with pericyte loss and ends with VEGF-driven fibrovascular traction.
9. Glaucoma is apoptotic ganglion cell loss with laminar remodelling and no inflammation.
10. Blunt trauma has a named finding at every anteroposterior level; alkali burns are prognosticated by limbal blanching.

Geometrical and physical optics

Ametropia · 29–39 items

Ophthalmic optics / spectacles · 17–25 items

Contact lenses

Low vision

ANSWER THESE COLD BEFORE READING

1. Write the vergence equation, the surface power equation, and the thick lens equivalent power equation.
2. Give Prentice rule and convert 1 prism dioptre to degrees.
3. State the power of a lens in an oblique meridian and transpose a prescription in both directions.
4. Define Abbe number and rank common lens materials by it.
5. State the Rayleigh criterion and compute the diffraction limit for a 3 mm pupil.
6. Give the wavelength and tissue interaction for six ophthalmic lasers.

BLUEPRINT WEIGHT

~19%

Optics across all four sub-disciplines is the second largest block on the exam after systemic health.

Core equations

THE EQUATION SET YOU MUST BE ABLE TO WRITE FROM MEMORY

Relationship	Equation	Notes
Reduced vergence	$L = n / l$	l in metres; n is the index of the medium the light is in. Dioptres.
Vergence (paraxial) equation	$L' = L + F$	The single most useful equation in optics. Track light left to right; incoming vergence plus surface power equals outgoing vergence.
Power of a refracting surface	$F = (n' - n) / r$	r in metres, positive if the centre of curvature is to the right.
Thin lens power	$F = 1 / f'$	f' is the secondary focal length in metres, in air.
Lensmaker (thin, in air)	$F = (n - 1)(1/r_1 - 1/r_2)$	Explains why higher index gives the same power with flatter curves and thinner edges.
Thick lens equivalent power	$F_e = F_1 + F_2 - (t/n)F_1F_2$	t is centre thickness in metres, n the lens index. The subtracted term is why a thick plus lens is weaker than the sum of its surfaces.
Back vertex power	$F_v = F_e / (1 - (t/n)F_1)$	The power a lensmeter reads and the number written on a spectacle prescription.
Effective power at a new vertex distance	$F_{\text{new}} = F / (1 - dF)$	d in metres, positive if the lens moves <i>toward</i> the eye. Matters above about ± 4 D.
Transverse magnification	$m = L / L' = -(n l') / (n' l)$	Negative m means inverted.

Relationship	Equation	Notes
Mirror focal length	$f = r / 2, F = -2n / r$	A convex mirror (e.g. the cornea) forms a virtual, minified, upright image: the basis of keratometry and of Hirschberg testing.
Snell law	$n \sin\theta = n' \sin\theta'$	
Critical angle	$\sin\theta_c = n' / n$	Cornea to air: $\sim 46^\circ$. This is why the angle cannot be seen without a gonioscope and why fibre optics work.
Prism deviation (thin prism)	$\Delta = 100 (n - 1) \tan A$	A is the apical angle. Light deviates toward the base ; the image displaces toward the apex .
Prentice rule	$\Delta = c \text{ (cm)} \times F \text{ (D)}$	Decentration in centimetres times lens power gives induced prism. Base direction: for a plus lens, base is toward the direction of decentration relative to the optical centre.
Prism dioptre to degrees	$1 \Delta = 0.57^\circ; 1^\circ \approx 1.75 \Delta$	1 Δ displaces the image 1 cm at 1 m.
Power in an oblique meridian	$F_\theta = F_{\text{sph}} + F_{\text{cyl}} \sin^2\theta$	θ is measured from the cylinder axis.
Spherical equivalent	$SE = \text{sphere} + \text{cyl}/2$	The power at the circle of least confusion.
Simple magnifier	$M = F / 4$	Conventional (trade) magnification, referenced to a 25 cm working distance.
Telescope magnification	$M = -F_{\text{eyepiece}} / F_{\text{objective}}$	Galilean uses a minus eyepiece (upright image, short, small field). Keplerian (astronomical) uses a plus eyepiece with prisms to re-invert (brighter, wider field, longer).
Abbe (constringence) number	$V = (n_d - 1) / (n_F - n_C)$	Higher V means less chromatic dispersion.
Rayleigh resolution criterion	$\theta = 1.22 \lambda / d$	θ in radians, d = pupil diameter. Gives about 1 arcmin at a 2–2.5 mm pupil, which corresponds to 20/20.

Refractive indices to memorise

INDICES OF REFRACTION

Medium	n	Medium	n
Air (vacuum)	1.000	Tears	1.336
Water	1.333	Aqueous and vitreous	1.336
Cornea	1.376	Lens (cortex to nucleus)	1.386–1.406
CR-39 plastic	1.498	Crown glass	1.523
Trivex	1.53	Polycarbonate	1.586
High index plastic	1.60 / 1.67 / 1.74	PMMA (contact lens)	1.49

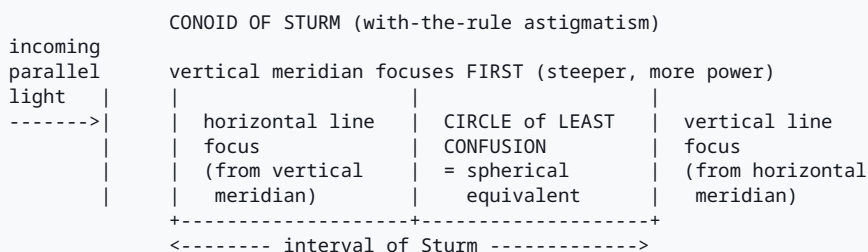
Medium	n	Medium	n
Keratometer assumed index	1.3375	Silicone oil	1.405

ABBE NUMBER, AND THE POLYCARBONATE TRADE-OFF

Crown glass $V \approx 59$; CR-39 $V \approx 58$; Trivex $V \approx 43-45$; polycarbonate $V \approx 30$; 1.67 high index $V \approx 32$; 1.74 $V \approx 33$.

Polycarbonate has the best impact resistance and the worst chromatic dispersion of the common materials. That is the trade-off in a high-prescription patient who notices colour fringing at the lens periphery: Trivex gives comparable impact safety with a better Abbe number and lower specific gravity, at slightly greater thickness. Polycarbonate remains mandatory in monocular patients, children, and sports eyewear.

Astigmatism, the conoid of Sturm, and transposition



- The **circle of least confusion** sits at the dioptric midpoint of the two focal lines, which is why the **spherical equivalent = sphere + cyl/2**.
- With-the-rule:** steeper (more powerful) meridian is vertical, so minus cylinder axis is near **180°**. Typical in youth. **Against-the-rule:** minus cyl axis near **90°**, typical with age and after superior-incision cataract surgery. **Oblique:** axis between **30-60°** or **120-150°**.
- Regular astigmatism** has two perpendicular principal meridians and is correctable with a cylinder. **Irregular astigmatism** does not, and requires a rigid contact lens or wavefront-guided correction.
- Lenticular versus corneal:** compare refractive cylinder with keratometric cylinder. Residual (internal) astigmatism averages about **0.50 D against the rule**, the basis of the **Javal rule**: total astigmatism $\approx 1.25 \times \text{corneal astigmatism} - 0.50 \times 90$.

TRANSPPOSITION, IN THREE STEPS

- Add** the sphere and the cylinder algebraically to get the new sphere.
- Change the sign** of the cylinder.
- Rotate the axis by 90°** (add 90 if the axis is 90 or less, subtract 90 if greater).

Worked example. $-2.00 -1.50 \times 180$ becomes $-3.50 +1.50 \times 090$. And $+1.25 +0.75 \times 135$ becomes $+2.00 -0.75 \times 045$. Check by confirming the two principal meridian powers are unchanged: in the first example the **180** meridian is **-2.00** and the **090** meridian is **-3.50** in both forms.

Aberrations

MONOCHROMATIC AND CHROMATIC ABERRATIONS

Aberration	Mechanism	Ophthalmic manifestation and management
Spherical aberration	Peripheral rays refract more strongly than paraxial rays	Night myopia and halos with a large pupil; reduced by an aspheric lens surface, a smaller pupil, or aspheric IOL design. The normal cornea has positive spherical aberration partly offset by the negative aberration of the young lens.
Coma	Off-axis point images become comet-shaped; asymmetric aberration	Prominent in keratoconus and after decentred refractive surgery; a leading cause of monocular ghosting that spectacles cannot correct.
Oblique (radial or marginal) astigmatism	Off-axis rays meet a tilted surface	Blur in the spectacle periphery; controlled by corrected-curve (best-form) lens design using Tscherning ellipse base curves.
Curvature of field	Image surface is curved, not flat	Also managed by best-form design.
Distortion	Magnification varies with field angle	Pincushion in plus (aphakic) lenses and barrel in minus lenses. Aphakic spectacles also produce the ring scotoma (jack-in-the-box phenomenon) from prismatic effect at the lens edge.
Longitudinal (axial) chromatic aberration	Dispersion: short wavelengths refract more	The eye has about 2 D of chromatic aberration between the ends of the visible spectrum. This is the basis of the duochrome (bichrome) test : with red and green targets, a myope (or over-minused patient) sees green clearer and a hyperope sees red clearer. "RAM-GAP": Red Add Minus, Green Add Plus .
Transverse (lateral) chromatic aberration	Dispersion off-axis, producing coloured fringes	Worse with low Abbe number and with looking away from the optical centre; the practical complaint in polycarbonate.
Higher-order aberrations (Zernike)	Trefoil, quadrafoil, secondary astigmatism, spherical aberration terms	Quantified by wavefront aberrometry as RMS error ; the target of wavefront-guided ablation and custom lenses. Only about 10–15% of ocular blur in a typical eye is higher order.

Aberration	Mechanism	Ophthalmic manifestation and management
Diffraction	Fundamental wave limit at any aperture	Sets the theoretical acuity ceiling. Combined with aberrations, optimum optical performance occurs at a pupil of about 2–3 mm: smaller pupils lose to diffraction, larger pupils lose to aberration.

The schematic and reduced eye

GULLSTRAND AND REDUCED EYE VALUES WORTH KNOWING

Parameter	Value
Total dioptric power (unaccommodated)	+58.6 to +60 D
Corneal contribution	~+43 D (two thirds to 70%)
Lens contribution	~+19 D
Reduced (single-surface) eye power	+60 D, $n = 1.336$
Anterior focal point of the eye	~17 mm in front of the cornea
Posterior focal length	~22.6 mm
Nodal point to retina	~17 mm
Retinal image size per degree	~0.29 mm / degree
Axial length	~24 mm (23–24)
1 mm axial length change	~3 D of refractive change
1 D of spectacle change at the eye's far point	~0.3 mm axial length

WHY THE FAR POINT CONCEPT SOLVES REFRACTION PROBLEMS

The **far point** is the object point conjugate to the retina with accommodation relaxed. A correcting lens must place its **secondary focal point at the eye's far point**. For a **−5.00 D** myope, the far point is **20 cm** in front of the eye; for a **+5.00 D** hyperope it is **20 cm behind** the eye (virtual). Every vertex distance, effective power, and contact-lens-versus-spectacle question is answered by asking where the far point is and where the lens's focal point falls.

Physical optics

Wave properties

- Speed of light in vacuum 3×10^8 m/s; in a medium, $v = c/n$. Frequency does not change with medium; **wavelength does**.
- Visible spectrum **380–760 nm**. Violet-blue short, red long. **Photon energy** $E = h\nu = hc/\lambda$, so shorter wavelengths carry more energy: the reason UV is phototoxic and infrared is thermally damaging.
- **Interference**: coherent waves add. **Anti-reflection coating** requires a layer of **optical thickness** $\lambda/4$ and refractive index equal to $\sqrt{n_{\text{lens}}}$ so the two reflections cancel. High index

materials reflect more light (about 4% per surface at $n = 1.5$, more at 1.74) so AR coating benefits them most.

- **Diffraction:** an aperture spreads light into an **Airy disc**. Resolution $\theta = 1.22\lambda/d$. With $\lambda = 555 \text{ nm}$ and $d = 3 \text{ mm}$, $\theta \approx 2.3 \times 10^{-4} \text{ radians} \approx 0.78 \text{ arcmin}$, slightly better than 20/20. Diffractive multifocal IOLs use concentric steps to split light between two orders.
- **Polarisation:** unpolarised light becomes linearly polarised by reflection at the **Brewster angle** ($\tan\theta_B = n'/n$; about 53° for air to glass). Because glare reflected from horizontal surfaces is horizontally polarised, sunglasses use a **vertical** transmission axis. Crossed polarisers are the basis of the **Titmus and Randot stereotests**, vectographic acuity charts, and of the **Haidinger brush** entoptic phenomenon used to test macular fixation.
- **Scatter:** **Rayleigh** scatter varies as $1/\lambda^4$ (blue sky, blue irides with little stromal melanin, and the faint blue of a normal cornea). **Mie** scatter occurs from particles comparable to the wavelength and is relatively wavelength-independent (the white of a cataract or a scarred cornea). The **Tyndall effect** is the visualisation of scatter by a narrow beam, which is exactly how the slit lamp reveals aqueous cells and flare.

Photometry

RADIOMETRIC AND PHOTOMETRIC UNITS

Quantity	Unit	Meaning
Luminous intensity	candela (cd)	Source output per solid angle. Base SI unit.
Luminous flux	lumen (lm)	Total output; $1 \text{ lm} = 1 \text{ cd} \cdot \text{sr}$.
Illuminance	lux (lm/m^2) or foot-candle	Light <i>falling on</i> a surface. Follows the inverse square law . 1 foot-candle $\approx 10.76 \text{ lux}$. Recommended chart illuminance is 80–320 cd/m^2 for a projected or printed chart.
Luminance	cd/m^2 (nit) or apostilb	Light <i>leaving</i> a surface: the physical correlate of perceived brightness. Perimetric backgrounds are set at 31.5 asb (about $10 \text{ cd}/\text{m}^2$), and stimulus intensity is expressed in decibels relative to the maximum, so higher dB means a dimmer stimulus .
Troland	td	Retinal illuminance: luminance $\times$ pupil area, accounting for pupil size.

Lasers

OPHTHALMIC LASERS: WAVELENGTH, INTERACTION, AND USE

Laser	Wavelength	Interaction	Clinical use
Argon blue-green / green	488 / 514 nm	Photocoagulation (thermal)	Panretinal photocoagulation, focal macular treatment, laser trabeculoplasty, iridotomy in dark irides. Blue is now avoided because xanthophyll absorbs it.
Frequency-doubled Nd:YAG (KTP)	532 nm	Photocoagulation	Modern replacement for argon green; well absorbed by haemoglobin and melanin.
Yellow dye / micropulse	577 nm	Photocoagulation, subthreshold	Macular treatment with less collateral damage; central serous chorioretinopathy.
Krypton red / diode	647 / 810 nm	Photocoagulation, deeper penetration	Treatment through vitreous haemorrhage or media opacity; transscleral cyclophotocoagulation (810 nm diode); retinopathy of prematurity.

Laser	Wavelength	Interaction	Clinical use
Nd:YAG (Q-switched)	1064 nm	Photodisruption: plasma formation and shock wave	Posterior capsulotomy, peripheral iridotomy, vitreolysis, membranectomy. Non-thermal.
Excimer (argon fluoride)	193 nm	Photoablation: breaks molecular bonds directly	PRK, LASIK, PTK. The short wavelength is why there is minimal thermal spread to adjacent tissue.
Femtosecond (Nd:glass)	1053 nm	Photodisruption at ultrashort pulse	LASIK flaps, SMILE lenticules, intrastromal ring channels, capsulotomy and lens fragmentation in FLACS.
Verteporfin PDT	689 nm	Photochemical: activates a photosensitiser	Chronic central serous chorioretinopathy, polypoidal choroidal vasculopathy, circumscribed choroidal haemangioma.
CO₂	10 600 nm	Photovaporisation	Lid and adnexal surgery; strongly absorbed by water.

Laser light has three defining properties: **monochromaticity, coherence (temporal and spatial), and collimation**. Production requires **population inversion** in a gain medium, pumped by an external energy source, within an optical cavity. Energy delivered depends on power, spot size, and duration; **irradiance is inversely proportional to the square of the spot diameter**, which is why halving spot size quadruples the effect at the same power.

Fluorescence and imaging optics

DIAGNOSTIC IMAGING PHYSICS

Modality	Physics and key numbers
Fluorescein angiography	Sodium fluorescein absorbs blue at 465–490 nm and emits green at 520–530 nm ; an exciter filter and a barrier filter isolate the emission. About 70–80% is protein-bound, so the free fraction leaks from incompetent vessels. Phases: choroidal flush at ~10–12 s , arterial, arteriovenous, venous, then recirculation and late phase. Fluoresces yellow-green through a cobalt filter at the slit lamp, which is the same physics used for tear break-up and Seidel testing.
Indocyanine green angiography	Absorbs ~805 nm , emits ~835 nm (infrared). Because it is 98% protein bound and infrared penetrates blood, exudate, and pigment, it images the choroidal circulation : polypoidal vasculopathy, occult CNV, central serous chorioretinopathy, and white dot syndromes. Contains iodine: contraindicated in iodine or shellfish allergy and in significant hepatic disease.
Optical coherence tomography	Low-coherence interferometry using near-infrared light around 840 nm (or 1050 nm for swept-source, giving deeper choroidal penetration). Axial resolution ~5 μm , determined by the light source bandwidth, not the pupil. Spectral-domain uses a spectrometer and a fixed reference arm; swept-source uses a tunable laser. OCT angiography detects motion contrast from flowing erythrocytes and requires no dye.
Ultrasound	Sound velocity 1532 m/s in aqueous and vitreous, 1641 m/s in the lens, ~1550 m/s average for a phakic eye. A-scan (8–10 MHz) for biometry and tissue reflectivity; B-scan (10 MHz) for two-dimensional structure; ultrasound biomicroscopy (50–80 MHz) for angle and ciliary body at the cost of penetration. Higher frequency means better resolution and less penetration, always. Immersion A-scan avoids the corneal compression error of contact technique.

Modality	Physics and key numbers
Optical biometry (IOLMaster, Lenstar)	Partial coherence interferometry or swept-source; measures to the RPE rather than the internal limiting membrane, so values are calibrated to be comparable to immersion ultrasound. More reproducible, and it fails in dense cataract or macular pathology preventing fixation.
Corneal topography and tomography	Placido disc reflection maps the <i>anterior</i> surface only. Scheimpflug photography and OCT-based tomography image both surfaces plus pachymetry, which is essential for detecting posterior elevation in early ectasia.

Ophthalmic instrument optics

HOW EACH INSTRUMENT WORKS OPTICALLY

Instrument	Optical principle
Direct ophthalmoscope	Forms an erect, virtual , ~15× magnified image; field of view only about 7–10°; magnification varies with the patient's refractive error (larger in myopia, smaller in hyperopia).
Binocular indirect ophthalmoscopy	The condensing lens forms a real, inverted, laterally reversed aerial image . Magnification $\approx 60 / F_{\text{lens}}$: a 20 D lens gives about 3× with a wide field; a 30 D lens gives about 2× with a wider field and shorter working distance. Higher lens power means lower magnification and a larger field.
Fundus biomicroscopy lenses	Non-contact 78 D and 90 D lenses give an inverted, reversed image; the 78 D gives more magnification, the 90 D a wider field. Contact lenses (Goldmann three-mirror, Hruby) neutralise corneal power.
Slit lamp	An illumination system in Koehler configuration plus a Galilean telescope in each ocular, with a convergent objective; magnification typically 6–40×. The illumination and observation systems are coupled to a common pivot.
Gonioscopy lens	Overcomes the critical angle at the cornea-air interface (~46°) . Indirect lenses (Goldmann, Zeiss four-mirror, Posner, Sussman) use a mirror so you view the <i>opposite</i> quadrant; the direct Koepple lens gives an upright view.
Keratometer	Treats the cornea as a convex mirror and measures the size of the reflected image of a target of known size at a known distance, converting radius to power with the fictitious index 1.3375. Measures a paracentral ~3 mm zone, which is why it misses central irregularity.
Retinoscope	A streak (or spot) projected into the eye; the reflex from the retina is observed. With motion means the far point is behind the peephole (need plus); against motion means it is between the eye and the peephole (need minus). At neutrality the far point coincides with the peephole, so you subtract the working distance dioptric value (1.50 D at 67 cm, 2.00 D at 50 cm).
Lensmeter (focimeter)	Measures back vertex power by a Badal optometer arrangement; the target moves until the emerging bundle is collimated.
Applanation tonometer	Two prisms split the fluorescein ring into semicircles displaced by exactly half the ring diameter, so the inner edges touch when the applanated area equals 7.35 mm ² .
Specular microscope / pachymeter	Specular reflection from the endothelium-aqueous interface; optical pachymetry uses the split-image doubling principle, ultrasonic uses time-of-flight.

RETRIEVAL PRACTICE: CALCULATIONS

Do these on paper. Numerical fluency is what separates a pass from a struggle in this section.

1. An object sits 25 cm in front of a +8.00 D thin lens in air. Where is the image and what is the magnification?

2. A **-9.00 D** spectacle lens is worn at a **14 mm** vertex distance. What contact lens power is equivalent?
3. A patient wears **+6.00 D** and is decentred **4 mm** nasally from the optical centre in each eye. How much prism is induced, and in which direction?
4. Transpose **-3.25 +2.00 × 065**.
5. What is the power of **-1.00 -3.00 × 180** in the 60° meridian?
6. Give the spherical equivalent of **+2.50 -1.50 × 090**.
7. A biconvex lens has surface powers of **+10.00 D** and **+6.00 D**, centre thickness **6 mm**, index 1.5. What is the equivalent power?
8. Compute the critical angle for the cornea ($n = 1.376$) to air.
9. Using the Rayleigh criterion at 555 nm, what pupil size gives 1 arcmin of resolution?
10. A patient reports the red letters are clearer on duochrome. What is the refractive status of the current lens, and what do you do?
11. What is the magnification of a **+16.00 D** hand magnifier in conventional terms?
12. A Galilean telescope has a **+30 D** objective and a **-90 D** eyepiece. What is its magnification and image orientation?

KEY

1. $L = 1/(-0.25) = -4.00 \text{ D}$; $L' = -4 + 8 = +4.00 \text{ D}$, so $l' = +25 \text{ cm}$ (real, right of lens). $m = L/L' = -4/+4 = -1.00$, i.e. same size, inverted.
2. $F_{\text{new}} = F / (1 - dF) = -9 / (1 - (0.014)(-9)) = -9 / 1.126 = -7.99$, so **-8.00 D**. Moving a minus lens closer requires less minus.
3. $\Delta = 0.4 \text{ cm} \times 6.00 \text{ D} = 2.4 \Delta$ **per eye**. Decentring a plus lens nasally moves the optical centre nasally, giving **base-in** prism in each eye, so **4.8 Δ base-in** total.
4. **-1.25 -2.00 × 155**.
5. θ is measured from the cylinder axis (180), so $\theta = 60^\circ$.
 $F = -1.00 + (-3.00)(\sin^2 60) = -1.00 + (-3.00)(0.75) = -3.25 \text{ D}$.
6. $+2.50 + (-1.50/2) = +1.75 \text{ D}$.
7. $F_e = 10 + 6 - (0.006/1.5)(10)(6) = 16 - (0.004)(60) = 16 - 0.24 = +15.76 \text{ D}$.
8. $\sin \theta_c = 1.000/1.376 = 0.7267$, $\theta_c = 46.6^\circ$.
9. $1 \text{ arcmin} = 2.909 \times 10^{-4} \text{ rad}$. $d = 1.22\lambda/\theta = 1.22(555 \times 10^{-9})/2.909 \times 10^{-4} = 2.33 \text{ mm}$.
10. Red clearer means the eye is effectively under-minused or over-plussed: the current lens leaves the patient **myopic relative to the target**. Add minus (RAM: Red Add Minus) until they equalise or green is just barely clearer.
11. $M = F/4 = 16/4 = 4\times$.
12. $M = -F_e/F_o = -(-90)/(+30) = 3\times$, **and the image is upright** (a Galilean telescope gives an erect image).

CHAPTER RECAP

1. $L' = L + F$ solves most problems; $F = (n' - n)/r$ for surfaces; $F_e = F_1 + F_2 - (t/n)F_1F_2$ for thick lenses.
2. Prentice: $\Delta = c(\text{cm}) \times F$; $1 \Delta = 0.57^\circ$; light bends to the base, images shift to the apex.
3. Transposition: add algebraically, flip the sign, rotate 90° . Spherical equivalent is sphere plus half the cylinder.
4. Abbe number is dispersion resistance: polycarbonate is the toughest and the most dispersive.
5. Aberrations: spherical for halos, coma for keratoconus, oblique astigmatism for best-form design, chromatic for duochrome (RAM-GAP).
6. Reduced eye: $+60 \text{ D}$, $n = 1.336$, nodal point 17 mm from retina, 0.29 mm per degree, 1 mm equals 3 D.
7. Rayleigh $1.22\lambda/d$ gives 1 arcmin at about 2.3 mm; optimum pupil is 2 to 3 mm.
8. Rayleigh scatter for blue, Mie for white, Tyndall for the slit lamp beam.
9. Laser interactions: photocoagulation thermal, photodisruption at 1064 nm, photoablation at 193 nm, photochemical at 689 nm.
10. BIO magnification $\approx 60/F$: higher lens power gives less magnification and more field.

Ametropia, ophthalmic lenses, and spectacle dispensing

Ametropia • 29–39 items (the largest single area)

Ophthalmic optics / spectacles • 17–25 items

Accommodation / vergence

ANSWER THESE COLD BEFORE READING

1. Classify myopia and hyperopia by mechanism, giving one cause for each mechanism.
2. Define total, manifest, latent, facultative, and absolute hyperopia and show how they relate.
3. State Knapp law and explain when it fails in practice.
4. Write the spectacle magnification equation and separate the power factor from the shape factor.
5. Compute image jump for a flat-top and a round-top bifocal with a +2.50 add.
6. Give the ANSI dress eyewear minimum thicknesses and the axis tolerance for a 0.75 D cylinder.

BLUEPRINT WEIGHT

~14%

Ametropia plus ophthalmic optics is 46–64 scored items: more than any other pair of areas.

Classifying ametropia

MYOPIA AND HYPEROPIA BY MECHANISM

Mechanism	Myopia	Hyperopia
Axial	Long eye. The commonest form. Each 1 mm of excess length is about 3 D. Pathologic myopia above 26–26.5 mm.	Short eye: nanophthalmos (high hyperopia, thick sclera, crowded angle, uveal effusion risk), microphthalmos, retinopathy of prematurity in some cases.
Curvature	Steep cornea or lens: keratoconus , keratoglobus, congenital microcornea with steep curvature, post-keratoplasty steepening.	Flat cornea: cornea plana , post-hyperopic overcorrection, flat corneal grafts.
Index	Nuclear sclerosis (the classic myopic shift and "second sight"), acute hyperglycaemia, sorbitol accumulation.	Cortical cataract in some cases; treated diabetes with falling glucose can give a transient hyperopic shift.
Positional	Anterior lens displacement (anterior lens subluxation, ciliary body swelling, plateau iris configuration, drug-induced ciliary oedema).	Posterior lens displacement, and aphakia (about +10 to +12 D at the spectacle plane).
Retinal / other	–	Anything that elevates the retina forward : central serous chorioretinopathy, choroidal tumour, macular oedema, posterior scleritis.

Mechanism	Myopia	Hyperopia
Pharmacologic and functional	Pseudomyopia from accommodative spasm; miotics; sulfonamides, topiramate , acetazolamide, and tetracyclines cause acute myopia with ciliochoroidal effusion and angle narrowing. Night myopia from tonic accommodation plus spherical aberration and the Purkinje shift.	Cycloplegics reveal latent hyperopia; systemic anticholinergics reduce accommodation and unmask it symptomatically.

HYPEROPIA TERMINOLOGY, IN ONE RELATIONSHIP

Total hyperopia = latent + manifest, and manifest = facultative + absolute.

- **Total:** revealed only under full cycloplegia.
- **Latent:** masked by ciliary tonus; the difference between the cycloplegic and the manifest refraction.
- **Manifest:** what you find on a non-cycloplegic maximum-plus refraction.
- **Facultative:** the part the patient can overcome with accommodation (vision is normal with or without the plus lens).
- **Absolute:** the part that cannot be overcome (vision is subnormal without correction).

Clinically this matters because a young hyperope's *symptoms* track the facultative component and their *acuity* tracks the absolute component. It is also why a child with accommodative esotropia needs a **cycloplegic** refraction: the latent component is what drives the deviation.

Anisometropia and aniseikonia

- **Anisometropia:** a difference in refractive error between eyes. **Aniseikonia:** a difference in perceived retinal image size. As a rough rule, **1 D** of spectacle anisometropia produces about **1-2% aniseikonia**, and most patients tolerate up to about **3-5%**.
- **Knapp law:** if the ametropia is purely **axial**, correcting it with a lens whose *secondary focal point coincides with the eye's anterior focal point* (about **15.7 mm** in front of the cornea, i.e. approximately the spectacle plane) yields **equal retinal image sizes** in the two eyes. For purely **refractive** anisometropia, a **contact lens** minimises the size difference.
- **Why Knapp law often fails clinically:** in axial myopia the retina is also *stretched*, so photoreceptor density per degree falls. The theoretically equal image now falls on fewer receptors, and many axial anisometropes actually do better with contact lenses despite the prediction. Treat Knapp law as a starting point, not a rule to be applied blindly.
- **Unequal accommodative demand:** correcting a myope with spectacles requires *less* accommodation at near than a contact lens does; correcting a hyperope with spectacles requires *more*. A **-10 D** myope switching from glasses to contact lenses can suddenly become symptomatic at near for exactly this reason, and this is a common presbyopia-onset complaint.
- **Unequal prismatic effect at near:** in vertical anisometropia, reading below the optical centres induces unequal vertical prism (Prentice rule). Above about **1.5-2 Δ** of vertical imbalance, options are **slab-off** (bicentric grinding of the *more minus / less plus* lens to add base-up at the reading level), **reverse slab** on the other lens, contact lenses, dissimilar segment heights, or a single-vision near pair.

Spectacle magnification

SPECTACLE MAGNIFICATION: TWO FACTORS

$$SM = [1 / (1 - h F_v)] \times [1 / (1 - (t/n) F_1)]$$

Factor	Expression	What changes it
Power factor	$1 / (1 - h F_v)$	h is the distance from the back vertex to the entrance pupil (about 3 mm behind the cornea, so $h \approx$ vertex distance + 3 mm). Plus lenses magnify, minus lenses minify, and reducing vertex distance reduces the magnification effect .
Shape factor	$1 / (1 - (t/n) F_1)$	t is centre thickness, F_1 the front surface power. Flatter front curves and thinner centres reduce magnification. This is the lever the optician controls without changing the prescription.

Practical consequences. A myope in spectacles sees a *smaller* world; a hyperope a *larger* one. Aphakic spectacles magnify by roughly **20–35%**, which is why unilateral aphakia cannot be corrected with spectacles: the aniseikonia is intolerable and fusion is impossible. To reduce aniseikonia in a fixed prescription: **reduce the thickness, flatten the base curve, and shorten the vertex distance of the more plus lens** (or do the reverse on the other lens).

Ophthalmic lens design

LENS FORMS AND WHEN TO USE THEM

Form	Description and indication
Meniscus / periscopic	One convex and one concave surface; the standard modern form.
Best form (corrected curve)	Base curves chosen from the Tscherning ellipse to eliminate oblique astigmatism or power error. Two solutions exist for each power; the flatter (Ostwalt) branch is used because the steep (Wollaston) branch is cosmetically unacceptable.
Aspheric	Surface flattens (or steepens) away from the centre. Allows a flatter, thinner, lighter lens while controlling oblique astigmatism. Standard in high plus, and requires precise optical centre placement because decentration produces unwanted astigmatism.
Lenticular	Central optical portion carried on a flatter carrier. In high plus this reduces weight and thickness; in high minus the myodisc reduces edge thickness.
Atoric	Aspheric in both principal meridians independently; used in high cylinder.
Vogel rule of thumb	Front base curve $\approx$ spherical equivalent / 2 + 6 for plus lenses; for minus lenses, base curve $\approx$ spherical equivalent / 2 + 6 as well, treated as the flattest reasonable base. Base curve is measured on the front surface with a lens clock (a spherometer using the sag principle).

THE SAG FORMULA, WHICH UNDERLIES THICKNESS, BASE CURVES, AND CONTACT LENS FITTING

$$s = y^2 / 2r \text{ or, in terms of surface power, } s = y^2 F / [2000 (n - 1)]$$

where s is sagittal depth, y the semi-diameter, and r the radius of curvature (consistent units).

Steeper radius or larger diameter means greater sag. This one relationship explains lens thickness calculations, lens clock readings, why a larger contact lens diameter behaves as though it were fitted steeper, and the sagittal relationship in every soft lens fitting decision.

Centre thickness of a plus lens $\approx$ edge thickness + sag of front surface - sag of back surface. *Minimum blank size* = effective diameter + 2 $\times$ (total decentration) + 2 mm allowance. Total decentration per lens = (frame PD - patient PD) / 2, where frame PD = eye size + DBL.

Multifocals**BIFOCAL TYPES: JUMP, DISPLACEMENT, AND USE**

Type	Optical centre of segment	Image jump and clinical note
Round top (Kryptok, 22 mm)	At the segment centre, well below the top	Maximum image jump. Jump = (distance from seg top to seg optical centre in cm) $\times$ add. For a 22 mm round seg the centre is 11 mm below the top, so a +2.50 add gives $1.1 \times 2.50 = 2.75 \Delta$ of jump. Base-down jump; the image appears to leap upward as the eye crosses the line.
Flat top (D segment, straight top; 28, 35, 45 mm)	About 4–5 mm below the segment top	Minimal jump: a +2.50 add gives about $0.5 \times 2.50 = 1.25 \Delta$. The default modern bifocal.
Executive (Franklin, one-piece)	On the dividing line	Zero image jump. Full-width near zone. Heavy, thick, cosmetically obvious, and fragile at the ledge.
Blended (invisible) round	At the segment centre	Same jump as round top plus a blurred blend zone; chosen purely for cosmesis.
Trifocal (7$\times$28, 8$\times$35)	Intermediate segment carries half the add (occasionally 40% or 70%)	Two lines, two jumps. Indicated when the add exceeds about +1.75 to +2.00 D and intermediate range matters.
Progressive addition lens	Continuous power gradient in a corridor	No jump and no line , at the cost of unwanted peripheral cylinder and skew distortion in the lower lateral zones. Hard designs have shorter corridors with more peripheral distortion; soft designs the reverse.

JUMP VERSUS DISPLACEMENT: TWO DIFFERENT QUANTITIES

Image jump is the *sudden* prismatic change as the line of sight crosses the segment top, and depends only on the **segment design and add power**, not on the distance prescription.

Total prismatic displacement at the reading level is the *sum* of the distance lens prism (Prentice rule at that point) and the segment prism. It **does** depend on the distance prescription. A myope reading below the optical centre gets base-up from the distance lens; a hyperope gets base-down. To minimise *total* displacement, pair a **round-top segment with a myopic distance prescription** and a **flat-top segment with a hyperopic one**, even though this is the opposite of what minimising jump alone would suggest.

Progressive lens fitting

- **Fitting cross** is aligned with the **pupil centre** in habitual posture, not with the lower lid.
- **Minimum fitting height** is design-dependent: short corridors need about **14 mm**, standard designs **18–22 mm** below the pupil centre. Choose the frame after choosing the design, not before.
- **Vertex distance** as short as lashes allow, **pantoscopic tilt 8–12°**, and modest **face form (wrap)** so the lens plane follows the eye's rotation.
- **Martin rule of tilt**: for every **2°** of pantoscopic tilt, lower the optical centre **1 mm** so the visual axis passes through the optical centre in primary gaze.
- **Non-adapt troubleshooting**: verify monocular PDs and fitting heights first (the two commonest errors), then check for excessive add, unequal seg heights, wrong base curve, or an uncorrected vertical imbalance. Give a two-week adaptation window with explicit instruction to point the nose at the target.

Materials, tints, and safety**LENS MATERIALS COMPARED**

Material	n	Abbe	Sp. gravity	Notes
CR-39	1.498	58	1.32	Excellent optics, tints easily, scratches easily; the plastic benchmark. Minimum 2.0 mm centre thickness for dress use.
Crown glass	1.523	59	2.54	Best scratch resistance and optics; heavy; must be heat or chemically tempered; minimum 3.0 mm .
Trivex	1.53	43–45	1.11	Lightest common material , excellent impact resistance, better Abbe than polycarbonate, good for drilled rimless.
Polycarbonate	1.586	30	1.20	Best impact resistance , inherent UV block, thinnest of the safety materials; poorest Abbe; scratches readily so it is always hard-coated. Mandatory for children, monocular patients, and sports.
High index 1.67 / 1.74	1.67 / 1.74	32 / 33	1.35 / 1.46	Thinnest for a given power; higher surface reflectance so AR coating is effectively required; more chromatic aberration.

- **UV protection**: CR-39 blocks to about **355 nm** untreated; polycarbonate, Trivex, and most high index materials block to **380–400 nm** inherently. Glass requires an additive.
- **Photochromic**: silver halide (glass) or organic naphthopyran (plastic). Activation is **UV-driven**, so most photochromics **do not darken behind a car windscreen**, which is a common patient

complaint; some newer designs respond to visible light. Darkening is faster than fading, and both are temperature-dependent (less dark when hot).

- **Polarised:** a stretched dichroic film oriented to absorb **horizontally** polarised light. Best for water, snow, and road glare. Warn about reduced visibility of LCD dashboards and of the strain patterns in tempered glass.
- **Standards:** **ANSI Z80.1** governs dress ophthalmic prescription tolerances; **ANSI Z87.1** governs occupational and educational safety eyewear (marked "Z87" or "Z87+" for high impact, with a minimum 3 mm centre thickness for most materials and a frame that also meets the standard). The **FDA impact-resistance requirement** (21 CFR 801.410) applies to all dress lenses and is verified by the **drop-ball test:** a 5/8 inch, 15.9 g steel ball dropped from 50 inches. Polycarbonate and Trivex are exempt from batch testing.

ANSI Z80.1 PRESCRIPTION TOLERANCES (SINGLE VISION, DISTANCE)

Parameter	Tolerance
Sphere power	± 0.13 D for meridional powers up to ± 6.50 D; $\pm 2\%$ above that.
Cylinder power	± 0.13 D up to 2.00 D cyl; ± 0.15 D for 2.12–4.50 D; $\pm 4\%$ above 4.50 D.
Cylinder axis	$\pm 14^\circ$ for cyl up to 0.25 D; $\pm 7^\circ$ for 0.37–0.50 D; $\pm 5^\circ$ for 0.62–0.75 D; $\pm 3^\circ$ for 0.87–1.50 D; $\pm 2^\circ$ above 1.50 D.
Add power	± 0.13 D up to 4.00 D; ± 0.18 D above.
Horizontal prism / OC placement	± 0.67 Δ or ± 2.5 mm horizontal decentration (per lens limits apply).
Vertical prism imbalance	± 0.33 Δ or ± 1.0 mm vertical OC difference.

Frame measurement and the boxing system

- **Eye size (A):** horizontal box width of the lens. **B:** vertical box height. **DBL:** distance between lenses (bridge). **ED:** effective diameter, the longest diameter through the box centre (twice the longest radius).
- **Frame PD** (geometric centre distance) = **A + DBL**.
- **Total decentration per lens** = (frame PD - patient PD) / 2. If positive, the optical centre must move nasally.
- **Datum line:** the horizontal midline of the box, used as the reference for segment heights in the datum system; the boxing system references the **lowest point of the lens** instead. Modern practice uses the boxing system.
- Frames are marked **A □ DBL – temple length**, for example **52 □ 18 – 140**.

Presbyopia and the add

DETERMINING THE NEAR ADD: USE AT LEAST TWO METHODS AND CROSS-CHECK

- 1 Age-expected value** as a sanity check: about **+1.00** at 45, **+1.50** at 50, **+2.00** at 55, **+2.50** at 60 and beyond.
- 2 Half the remaining amplitude rule:** leave half the measured amplitude of accommodation in reserve. Add = near demand - (amplitude / 2).
- 3 Balance NRA and PRA:** the correct add is the tentative add plus (NRA + PRA)/2. Equal NRA and PRA indicates a balanced add.
- 4 Fused cross cylinder** at 40 cm with a dissociating grid: the endpoint balances horizontal and vertical line clarity.
- 5 Dynamic (binocular) crossed cylinder or NRA/PRA-refined trial frame check** in the patient's actual working distance and posture, with the actual reading material.
- 6 Verify the range of clear vision** at the chosen add and confirm it brackets the working distance. Range is bounded by the add's far point (1/add) and the near point given add plus half the remaining amplitude.

Worked example. A 52-year-old bookkeeper works at **40 cm** (**2.50 D** demand) with an amplitude of **2.00 D**. Half-amplitude reserve leaves **1.00 D** usable, so add = $2.50 - 1.00 = +1.50 \text{ D}$. At **+1.50** the far point of the add is **67 cm** and, using the remaining **1.00 D** of reserve, the near point is **40 cm**: the range is **40–67 cm**, which brackets the task. If she also uses a monitor at **65 cm**, that sits at the very edge of the range and an occupational progressive or a separate intermediate pair is the better answer.

NUMBERS TO KNOW: AMETROPIA AND SPECTACLES

Aphakic spectacle power	+10 to +12 D	Aphakic spectacle magnification	20–35%
Aniseikonia per dioptre	~1–2%	Typical tolerance	~3–5%
Knapp law plane	~15.7 mm	Entrance pupil behind cornea	~3 mm
Vertical imbalance threshold	~1.5–2 Δ	Flat-top seg OC below top	4–5 mm
Round 22 mm seg OC below top	11 mm	Executive seg OC	on the line
Trifocal intermediate	50% of add	PAL minimum fitting height	14–22 mm
Pantoscopic tilt	8–12°	Martin rule of tilt	1 mm per 2°
CR-39 min thickness	2.0 mm	Glass min thickness	3.0 mm
Drop-ball	15.9 g from 50 in	Sphere tolerance to 6.50 D	±0.13 D
Axis tolerance, 0.75 D cyl	±5°	Axis tolerance, >1.50 D cyl	±2°

Vertical prism imbalance tolerance	$\pm 0.33 \Delta$	Frame PD	A + DBL
Pathologic myopia axial length	$> 26.5 \text{ mm}$	Ocular chromatic aberration	$\sim 2 \text{ D}$

RETRIEVAL PRACTICE

1. A 6-year-old has a manifest refraction of $+1.00$ and a cycloplegic of $+4.50$. Name each component of the hyperopia with its value.
2. State Knapp law and explain why an axial myope may still prefer contact lenses.
3. Write the spectacle magnification equation and name the two levers an optician can pull without changing power.
4. Compute image jump for a 22 mm round-top and a flat-top 28 with a $+2.00$ add.
5. Which segment style pairs best with a myopic distance prescription to minimise total displacement, and why does that differ from the jump answer?
6. A patient has R $+4.00$, L $+1.00$ and reads 10 mm below the optical centres. Compute the vertical imbalance and state the management.
7. Frame is 54 □ 16 and the patient PD is 62 mm. What is the decentration per lens and in which direction?
8. Give the two solutions on the Tscherning ellipse and state which is used.
9. Write the sag formula and explain why a larger-diameter soft contact lens behaves as if steeper.
10. Why do most photochromic lenses fail to darken in a car?
11. Give the axis tolerance for a 1.25 D cylinder and the vertical prism imbalance tolerance.
12. A 50-year-old has an amplitude of 2.50 D and works at 33 cm. Compute the add by the half-amplitude rule and give the resulting range of clear vision.

KEY

1. Total $+4.50$; manifest $+1.00$; latent $+3.50$. Of the manifest, the part that can be overcome with normal acuity is facultative and any part leaving reduced acuity is absolute.
2. In purely axial ametropia, correction at the eye's anterior focal point (about 15.7 mm, roughly the spectacle plane) equalises retinal image size. It can fail because axial myopia stretches the retina, reducing receptor density per degree, so the equal-sized image is sampled more coarsely.
3. $SM = [1/(1 - hf_v)] \times [1/(1 - (t/n)F_1)]$; the levers are **centre thickness** and **front base curve** (plus vertex distance).
4. Round 22: OC is 11 mm below the top, jump = $1.1 \times 2.00 = 2.20 \Delta$. Flat top 28: OC about 5 mm below, jump = $0.5 \times 2.00 = 1.00 \Delta$.
5. A **round-top** segment. A myope reading below the distance optical centre gets base-up prism; the round segment's base-down jump partially cancels it, minimising total displacement even though the jump itself is larger.
6. Right: $1.0 \text{ cm} \times 4.00 = 4.0 \Delta$ base-down (reading below OC of a plus lens gives base-down). Left: $1.0 \times 1.00 = 1.0 \Delta$ base-down. **Imbalance 3.0Δ** , well above tolerance. Options: slab-off on the more plus lens (or the appropriate bicentric grind), dissimilar segment heights, contact lenses, or separate single-vision near glasses.
7. Frame PD = $54 + 16 = 70 \text{ mm}$. Decentration = $(70 - 62)/2 = 4 \text{ mm}$ **nasally per lens**.
8. The steep (Wollaston) and flat (Ostwalt) branches; the **flat Ostwalt branch** is used for cosmetic and weight reasons.
9. $s = y^2/2r$. Increasing the diameter increases y , which increases the sag for the same radius, so the lens vaults more and behaves as though fitted steeper.
10. Activation is driven by **ultraviolet** light, and modern windscreens block most UV.
11. Axis $\pm 3^\circ$ for a 1.25 D cylinder; vertical prism imbalance $\pm 0.33 \Delta$.
12. Demand at 33 cm = 3.00 D; reserve half of 2.50, so 1.25 D usable; add = $3.00 - 1.25 = +1.75 \text{ D}$. Far point of the add = $1/1.75 = 57 \text{ cm}$; near point using the 1.25 D reserve = $1/(1.75 + 1.25) = 33 \text{ cm}$. Range **33 to 57 cm**.

CHAPTER RECAP

1. Ametropia has five mechanisms: axial, curvature, index, positional, and retinal. Drugs cause acute myopia by ciliochoroidal effusion.
2. Total = latent + manifest; manifest = facultative + absolute. Cycloplege every child with esotropia.
3. 1 D of anisometropia gives 1 to 2% aniseikonia; Knapp law says correct axial error at the spectacle plane, but retinal stretch complicates it.
4. Spectacle magnification = power factor $\times$ shape factor; thickness and base curve are what the optician can change.
5. Jump depends on segment design and add only; total displacement adds the distance lens prism.
6. Executive has zero jump, flat top minimal, round top maximal; pair round with myopes for least displacement.
7. Sag formula $s = y^2/2r$ governs thickness, base curves, and contact lens vault.
8. Polycarbonate for safety, Trivex for weight and Abbe, high index for thinness with AR.
9. ANSI axis tolerance tightens as cylinder power rises, down to $\pm 2^\circ$ above 1.50 D; vertical imbalance limit 0.33 Δ .
10. Determine the add by at least two methods, then verify the range of clear vision brackets the task.

Contact lens optics and low vision optics

Contact lenses • 8–14 items

Low vision • 5–9 items

Ametropia

Conjunctiva / cornea

ANSWER THESE COLD BEFORE READING

1. Define the tear lens and give the rule for its sign.
2. Calculate residual astigmatism and state what a spherical rigid lens does and does not correct.
3. State the LARS rule and apply it to a mislocating toric soft lens.
4. Give the FDA hydrogel groups and the Dk/t targets for daily and extended wear.
5. State Kestenbaum rule and compute the predicted add for 20/250 acuity.
6. Give the definition of legal blindness in the United States.

BLUEPRINT WEIGHT

4%

13–23 combined items, and contact lens optics is repeatedly used to test vergence and vertex reasoning.

The tear (lacrimal) lens

THE SINGLE MOST EXAMINED CONCEPT IN RIGID LENS OPTICS

Between a rigid lens and the cornea sits a tear layer whose power is determined by the mismatch between the lens base curve and the corneal curvature:

$$\text{Tear lens power} = \text{base curve (in D)} - \text{flattest K (in D)}$$

- Fitting **steeper** than K creates a **plus** tear lens, so you must **add minus** to the lens power.
- Fitting **flatter** than K creates a **minus** tear lens, so you must **add plus**.

SAM / FAP: Steeper Add Minus, Flatter Add Plus.

Each **0.05 mm** of base curve change is approximately **0.25 D**, and each **0.1 mm** is approximately **0.50 D**, in the physiologic corneal range.

Worked example. Spectacle Rx **-5.00 DS** at **12 mm** vertex; K readings **43.00 / 43.00**. Fit a rigid lens on **42.50** (flatter than K by **0.50 D**).

Step 1, vertex conversion: $F_{\text{corneal}} = -5.00 / (1 - 0.012 \times -5.00) = -5.00 / 1.06 = -4.72$, so **-4.75 D**.

Step 2, tear lens: fitted flatter, so the tear lens is **-0.50 D**. FAP: add plus.

Step 3: lens power = $-4.75 + 0.50 = -4.25$ D.

Residual astigmatism

WHAT A SPHERICAL RIGID LENS CORRECTS, AND WHAT IT LEAVES BEHIND

Residual astigmatism = total refractive astigmatism - corneal astigmatism

A spherical rigid gas permeable lens replaces the anterior corneal surface optically, so it **neutralises corneal (anterior surface) astigmatism completely**. Anything left is **internal (lenticular and posterior corneal) astigmatism**, and it will persist.

- **Expected** residual astigmatism is calculated as above and averages about **0.50 D against the rule** in the population (this is what the **Javal rule** encodes).
- **Induced** residual astigmatism arises when a *toric back surface* rigid lens is used: the tear lens no longer neutralises the toricity evenly, and a front-surface toric compensation (bitoric lens) is needed. The classic solution is the **Mandell-Moore** or CPE (cylinder power effect) bitoric.
- Soft lenses **drrape** over the cornea and therefore do *not* correct corneal astigmatism; a toric soft lens is required for anything above about **0.75–1.00 D**.

Example. Refraction $-3.00 -2.50 \times 180$, K readings $42.00 @ 180 / 45.00 @ 090$ (corneal astigmatism 3.00 D with the rule, i.e. -3.00×180 at the corneal plane). Residual = $-2.50 - (-3.00) = +0.50 \times 180$, **equivalent to -0.50×090** : half a dioptre against the rule will remain with a spherical rigid lens. That is usually acceptable.

Toric soft lens axis correction

LARS

Left Add, Right Subtract. If the lens markings rotate to the patient's **left** (which is clockwise from your view), **add** the amount of rotation to the ordered axis; if they rotate to the patient's **right**, **subtract** it.

Example. Ordered $-2.00 -1.25 \times 180$; the marking sits 15° rotated to the patient's left. New axis = $180 + 15 = 195$, which is 015 . Re-order $-2.00 -1.25 \times 015$. Rotation must be *stable and repeatable* before you compensate; unstable rotation calls for a different stabilisation design instead.

Contact lens materials and oxygen

FDA HYDROGEL CLASSIFICATION AND THE SILICONE HYDROGEL GROUP

Group	Water content	Ionic?	Behaviour
I	Low (<50%)	Non-ionic	Most deposit-resistant; low Dk; tolerant of all care systems. Example: polymacon.
II	High (≥50%)	Non-ionic	Higher Dk; prone to lipid and protein deposit; sensitive to dehydration.
III	Low (<50%)	Ionic	Intermediate; attracts positively charged deposits (lysozyme).
IV	High (≥50%)	Ionic	Most deposit-prone and most reactive to preservatives and to hydrogen peroxide residue. Example: etafilcon A.

Group	Water content	Ionic?	Behaviour
V	Silicone hydrogel (subgroups A, B, C)	Varies	Oxygen is carried by the silicone , not the water, so Dk rises as water content falls : the opposite of conventional hydrogels. Higher modulus (stiffer), which improves handling but predisposes to SEALs , mucin balls, and papillary responses.

- **Dk** is a material property: D is diffusivity and k is solubility, in barrers. **Dk/t** is transmissibility of a specific lens of thickness t, in barrer/cm.
- **Holden-Mertz criteria**: **Dk/t** ≥ 24 to keep daily wear oedema below 4%; **Dk/t** ≥ 87 to keep overnight oedema at the physiologic 4% level. **Harvitt-Bonanno** revised these upward to about 35 and 125 to avoid anoxia in *any* corneal layer.
- In **conventional hydrogels**, oxygen moves through water, so **Dk increases with water content** and decreases with thickness. In **silicone hydrogels**, the relationship inverts.
- **Rigid gas permeable** materials gain Dk from silicone and fluorine content; fluorine improves wettability and deposit resistance. PMMA has a Dk of essentially 0, which is why it is obsolete.
- **Modulus**: higher modulus lenses mask astigmatism better and handle more easily but cause more mechanical complications. **Wettability** is measured by contact angle, lower being better.

Fitting principles

RIGID LENS FLUORESCEIN PATTERNS

Relationship	Fluorescein appearance	Consequence
Alignment (on-K)	Thin, even fluorescein across the optic zone with a narrow band of peripheral pooling at the edge lift	The target fit for a spherical cornea: good tear exchange, centration, and comfort.
Steep (apical clearance)	Central pooling with a ring of mid-peripheral bearing	Poor tear exchange, lens adherence, central corneal oedema and distortion, air bubble trapping. Add plus? No: fitting steep created a plus tear lens, so the lens power needs more minus (SAM).
Flat (apical bearing)	Central touch (dark) with peripheral pooling	Central corneal abrasion, excessive movement, low-riding lens, 3 and 9 o'clock staining. Needs more plus (FAP).
With-the-rule toric cornea	Dumbbell (bowtie) pattern with bearing along the flat horizontal meridian	Consider a back-surface toric or a bitoric lens above about 2.00 D of corneal toricity.
Keratoconus	Target three-point touch : light apical touch with two areas of mid-peripheral bearing	Apical clearance causes bubbles and poor optics; apical bearing causes scarring. Alternatives: scleral, hybrid, or piggyback.

- **Base curve and diameter interact through sagittal depth**: increasing the diameter increases the sag, which behaves like a steeper fit, so a common empirical rule is that a 0.5 mm increase in diameter requires flattening the base curve by about 0.05 mm to maintain the same relationship.

- **Soft lens assessment:** **0.2–0.5 mm** of movement on blink, full corneal coverage with **0.5–1.0 mm** of limbal overlap, moderate resistance and smooth recovery on the push-up test, and stable acuity through the blink. A lens that is too tight shows no movement, conjunctival indentation, and initially good then deteriorating vision; too loose shows excessive movement, decentration, and fluctuating vision that clears momentarily after a blink.
- **Scleral lenses:** aim for **200–400 µm** of central clearance after settling (settling itself is about **100 µm** over the first hours), **~50–80 µm** of limbal clearance, and an aligned haptic with no blanching or impingement. Watch for midday fogging, conjunctival prolapse, and hypoxic oedema when clearance is excessive.
- **Orthokeratology:** a reverse-geometry lens flattens the central cornea by **redistributing epithelium** (central thinning, mid-peripheral thickening) rather than by reshaping stroma. Corrects up to about **–4.00 to –5.00 D** and produces peripheral myopic defocus, which is why it slows axial elongation in children by roughly **40–50%**. The material risk is microbial keratitis with overnight wear.
- **Presbyopic options:** simultaneous vision designs (centre-near or centre-distance aspheric, concentric) present both images at once and rely on neural selection; **translating (alternating) designs** are rigid, prism-ballasted and truncated, and depend on the lower lid to shift the lens on downgaze; **monovision** sacrifices stereopsis and is a legitimate but compromise option (with implications for occupational driving).

CONTACT LENS COMPLICATIONS: MECHANISM TO SIGN

Complication	Key features
Microbial keratitis	Pain, discharge, epithelial defect over an infiltrate, anterior chamber reaction. <i>Pseudomonas</i> with overnight wear, <i>Acanthamoeba</i> with water exposure. Discontinue lenses, culture if central or large, treat aggressively.
Contact lens acute red eye (CLARE)	Sudden unilateral overnight pain and redness with peripheral infiltrates, intact epithelium , from gram-negative endotoxin under a tight lens.
Contact lens peripheral ulcer (CLPU)	Small, round, peripheral, full-thickness epithelial loss over a dense infiltrate; sterile, staphylococcal exotoxin-driven; leaves a circular scar.
Infiltrative keratitis	Multiple small subepithelial infiltrates with variable staining; a spectrum with the above.
Giant papillary conjunctivitis	Upper tarsal papillae above 0.3 mm , mucus, itching, lens awareness and decentration. Mechanical plus immunologic (type I and IV); managed by replacing lenses more frequently, changing material or modality, and mast cell stabilisers.
Superior epithelial arcuate lesion (SEAL)	Arcuate superior epithelial split from mechanical shear, associated with high-modulus silicone hydrogels; often asymptomatic.
3 and 9 o'clock staining	Peripheral desiccation with rigid lenses from poor blink and inadequate edge lift; leads to vascularised limbal keratitis if ignored.
Corneal warpage	Distorted topography and spectacle blur after lens removal; requires weeks out of lenses before refractive surgery planning.
Neovascularisation and microcysts	Chronic hypoxia. Microcysts appear at 2 to 6 months of low-Dk wear and paradoxically increase transiently on switching to a higher-Dk lens.
Solution toxicity / hypersensitivity	Diffuse punctate staining, especially with hydrogen peroxide that has not neutralised, or preservative sensitivity in group IV materials.

NON-NEGOTIABLE PATIENT INSTRUCTIONS

No water, ever: no rinsing, storing, or swimming in lenses without appropriate protection, because tap water carries *Acanthamoeba*. **Rub and rinse** even with "no-rub" solutions. Replace cases every three months. Never top off solution. Overnight wear multiplies microbial keratitis risk several-fold, and smoking multiplies it again.

Low vision optics

Definitions

- **Legal blindness (United States):** best-corrected visual acuity of **20/200** or worse in the better eye, or a visual field diameter of **20°** or less in the better eye.
- **WHO categories (ICD-11):** mild impairment **<6/12**; moderate **<6/18**; severe **<6/60**; blindness **<3/60** or a field under **10°**.
- Leading causes in developed countries: **age-related macular degeneration**, glaucoma, diabetic retinopathy, and cataract; globally, uncorrected refractive error and cataract dominate.

Four ways to make the image bigger

TYPES OF MAGNIFICATION

Type	Method	Example
Relative distance	Move closer to the object; magnification = old distance / new distance	Halving the working distance from 40 cm to 20 cm doubles the angular size, but doubles the accommodative or add demand.
Relative size	Make the object physically larger	Large-print books, enlarged photocopies, big-button phones.
Angular	An optical system increases the visual angle	Telescopes for distance, telemicroscopes for near.
Projection (electronic)	Camera and display	CCTV / video magnifier, smartphone camera, screen reader with zoom; also permits contrast reversal , which is often more valuable than the magnification itself.

KESTENBAUM RULE AND PREDICTING THE ADD

The required add in dioptres is the reciprocal of the best-corrected distance acuity expressed as a fraction. For **20/200**, the add is $200/20 = +10.00$ D. For **20/80** it is $80/20 = +4.00$ D. This predicts the add needed to read **1 M print** (roughly newsprint).

Required magnification = target acuity / current acuity. If a patient sees **20/200** and needs **20/40** to read the target, $M = 200/40 = 5\times$. Then apply a **reserve (acuity reserve) factor of about 2×** for sustained fluent reading rather than spot reading, because threshold acuity does not support continuous reading.

Equivalent viewing power unifies the devices: $EVP = M \times 4$ (for a 25 cm reference), or $EVP = \text{magnifier power} \times (25 / \text{working distance in cm})$. Two devices with the same EVP deliver the same retinal image size regardless of whether the magnification comes from a stronger lens or a shorter distance.

LOW VISION DEVICES: OPTICS, ADVANTAGES, LIMITATIONS

Device	Optics	Trade-offs
Spectacle-mounted plus lens (microscope)	High-plus single vision or aspheric lens; the eye is at the lens	Widest field of any near device and hands-free, but the working distance equals roughly $1/F$, so a +20 D lens forces a 5 cm working distance. Base-in prism is added for binocular use up to about +12 D ; above that, correction becomes monocular.
Hand magnifier	Plus lens held at a variable distance; $M = F/4$ conventionally	Familiar, portable, cheap; requires steady hands and a free hand; field shrinks as the lens moves away from the eye. Field of view is greatest when the lens is held <i>close to the eye</i> .
Stand magnifier	Fixed lens-to-object distance, usually giving a slightly divergent or convergent emergent bundle	No hand tremor issue; excellent for arthritis or tremor. The image is often not at infinity, so the patient must use an add or accommodate . Rated magnification is not comparable across brands; use EVP.
Telescope	Galilean (minus eyepiece, erect, compact, small field) or Keplerian (plus eyepiece plus prisms, brighter, wider field, longer)	The only optical option for distance. Field of view and depth of field both shrink with magnification, and apparent motion is magnified, so hand-held spotting is more practical than continuous wear. Bioptic designs allow driving where legally permitted.
Telemicroscope (reading telescope)	Telescope plus a reading cap	Gives a longer working distance than an equivalent microscope, at the cost of field and depth of focus. Useful for music and computer distances.
Electronic magnifier / CCTV	Camera and display	Very high magnification with a preserved working distance, adjustable contrast and reverse polarity , and no optical aberration. Cost, portability, and set-up are the limits. Often produces the best reading rate of any device.
Field-expansion devices	Fresnel prisms (peripheral field loss, hemianopia), reverse telescopes (minification), mirrors	Prisms for homonymous hemianopia are placed with the base <i>toward</i> the field loss on the affected side. Minifiers shrink the whole scene, reducing acuity, so they suit ambulation in tunnel vision (retinitis pigmentosa, advanced glaucoma) rather than reading.

Device	Optics	Trade-offs
Non-optical aids	–	Task lighting (often the single highest-yield intervention), typoscopes and line guides, high-contrast marking, tinted filters for glare and contrast (yellow and amber for scatter, deeper filters for photophobia in albinism and cone dystrophy), large-print and audio material, and screen-reading software.

THE FUNCTIONAL INTERVENTIONS THAT MATTER AS MUCH AS THE OPTICS

1. **Illumination.** Doubling task luminance often improves reading more than adding magnification, especially in AMD. Position the light close, behind and to the side of the better eye.
2. **Contrast.** Measure it (Pelli-Robson or Mars) and treat it: reverse polarity, bold print, typoscopes, and dark markings on light backgrounds.
3. **Glare control.** Absorptive filters, brimmed hats, side shields, and anti-reflective coatings; distinguish discomfort glare from disability glare.
4. **Eccentric viewing training.** Teach the patient to place the image on a **preferred retinal locus** outside a central scotoma, usually by looking slightly away from the target in a consistent direction, then to build a stable oculomotor habit.
5. **Steady-eye technique.** Move the material rather than the eyes, which reduces the demand on eccentric fixation stability.
6. **Set realistic goals with the patient.** "Read my medication bottles" and "see my grandchild's face" produce better outcomes than "read again", and they determine which device you prescribe.
7. **Refer.** Vision rehabilitation, occupational therapy, orientation and mobility training, state services for the blind, and driving assessment.

NUMBERS TO KNOW: CONTACT LENSES AND LOW VISION

0.05 mm base curve change	~0.25 D	0.1 mm base curve change	~0.50 D
Dk/t daily wear (Holden-Mertz)	≥24	Extended wear	≥87
Harvitt-Bonanno targets	~35 / ~125	PMMA Dk	~0
Soft lens movement	0.2–0.5 mm	Limbal overlap	0.5–1.0 mm
Scleral central clearance	200–400 μm	Scleral settling	~100 μm
Ortho-k correction range	up to ~-4 to -5 D	Ortho-k myopia control	~40–50% slowing
GPC papilla size threshold	>0.3 mm	Mean residual astigmatism	~0.50 D ATR
Legal blindness, acuity	20/200 or worse	Legal blindness, field	≤20° diameter
Kestenbaum add for 20/200	+10.00 D	Acuity reserve factor	~2×
Binocular microscope limit	~+12 D with base-in	Equivalent viewing power	M × 4

RETRIEVAL PRACTICE

1. Spectacle Rx -8.00 DS at 12 mm; K $44.00/44.00$; rigid lens fitted 0.75 D steeper than K. Give the lens power.
2. Refraction $-1.00 -1.75 \times 180$; K $43.00 @ 180 / 44.00 @ 090$. Compute residual astigmatism and comment.
3. A toric soft lens ordered at axis 170 rotates 10° to the patient's right. What axis do you re-order?
4. Which FDA hydrogel group is most deposit-prone, and why?
5. In silicone hydrogels, does Dk rise or fall with water content, and why?
6. Central fluorescein pooling with mid-peripheral bearing. Name the fit and the power adjustment needed.
7. Distinguish CLARE from CLPU from microbial keratitis in one line each.
8. Explain the mechanism of orthokeratology and its effect on axial elongation.
9. Give the US definition of legal blindness, both criteria.
10. A patient sees $20/250$. Give the Kestenbaum add and the working distance it implies.
11. A patient needs $20/50$ to read a label and currently sees $20/300$. What magnification, before and after applying a reserve factor?
12. Which near device gives the widest field, and what is its main drawback?
13. Where do you place the base of a Fresnel prism for a right homonymous hemianopia?

KEY

1. Vertex: $-8.00 / (1 - 0.012 \times -8) = -8.00/1.096 = -7.30$, so -7.25 D at the cornea. Fitted steeper by 0.75 D creates a $+0.75$ tear lens, so SAM: add minus. Lens power = $-7.25 - 0.75 = -8.00$ D.
2. Corneal astigmatism is 1.00 D with the rule, i.e. -1.00×180 . Residual = $-1.75 - (-1.00) = -0.75 \times 180$: three quarters of a dioptre with-the-rule astigmatism will remain, which is enough to blur, so consider a front-surface toric rigid lens or a toric soft lens.
3. Rotation to the patient's right means subtract: $170 - 10 = \text{axis } 160$.
4. Group IV, high water content and ionic: the negative surface charge attracts positively charged proteins such as lysozyme, and high water content admits more material into the matrix.
5. It **falls** as water content rises, because oxygen is transported through the silicone phase, so more water means proportionally less silicone.
6. A **steep** (apical clearance) fit; the plus tear lens means the lens power needs **more minus**, and the fit itself should be flattened.
7. CLARE: overnight, unilateral, peripheral infiltrates with intact epithelium, endotoxin-driven. CLPU: small round peripheral full-thickness epithelial loss over a dense sterile infiltrate, staphylococcal exotoxin. Microbial keratitis: pain, discharge, epithelial defect over the infiltrate with an anterior chamber reaction, and it requires culture and aggressive treatment.
8. A reverse-geometry lens redistributes epithelium, thinning it centrally and thickening it mid-peripherally, which flattens the central cornea and produces relative peripheral myopic defocus; this slows axial elongation by roughly 40 to 50% in children.
9. Best-corrected acuity of $20/200$ or worse in the better eye, or a visual field of 20 degrees or less in diameter in the better eye.
10. $250/20 = +12.50$ D, implying a working distance of about 8 cm.
11. $300/50 = 6\times$; with a $2\times$ acuity reserve for fluent reading, prescribe for about $12\times$ equivalent, which in practice means an electronic device.
12. The **spectacle-mounted high-plus microscope**; its drawback is the very short working distance, roughly $1/\text{F}$, and the need for base-in prism or monocular use.
13. Base **toward the field loss**, so **base-out on the right eye** (and base-in on the left if used binocularly) to shift images from the blind right field into the seeing field.

CHAPTER RECAP

1. Tear lens = base curve minus flattest K. SAM and FAP. 0.05 mm equals 0.25 D.
2. Residual astigmatism = total minus corneal; a spherical rigid lens corrects only the corneal part.
3. LARS for toric soft lens axis, but only when rotation is stable.
4. FDA groups I to IV by water content and ionicity; group IV deposits most; group V silicone hydrogel inverts the water-Dk relationship.
5. Dk/t 24 daily, 87 overnight, revised to 35 and 125.
6. Fluorescein: even is aligned, central pool is steep, central touch is flat, bowtie is toric, three-point touch is the keratoconus target.
7. Never water, always rub and rinse, never sleep in lenses casually.
8. Legal blindness is 20/200 or a 20 degree field.
9. Kestenbaum add is the reciprocal of the acuity fraction; then double the magnification for fluent reading.
10. Light, contrast, glare control, eccentric viewing, and realistic goals often outperform stronger optics.

Accommodation, vergence, amblyopia, strabismus

Accommodation / vergence / oculomotor · 10–18 items

Amblyopia / strabismus · 9–17 items

Visual and human development

ANSWER THESE COLD BEFORE READING

1. Write the calculated AC/A formula and compute it for a given case.
2. Reproduce the Morgan normative table for phorias, vergence ranges, and accommodative findings.
3. State Sheard criterion, Percival criterion, and the 1:1 rule, and say which applies to which deviation.
4. Classify esotropia and exotropia by type with the distinguishing near-distance relationship.
5. Perform the three-step test on paper for a right hypertropia worse in left gaze and on right head tilt.
6. Summarise the PEDIG evidence on patching dose and atropine.

BLUEPRINT WEIGHT

~8%

19–35 combined scored items across accommodation, vergence, amblyopia, and strabismus.

Accommodation and vergence: the measured quantities

MORGAN NORMS: MEMORISE THIS TABLE

Test	Mean	SD
Distance lateral phoria	1 exo	±2 Δ
Near lateral phoria (40 cm)	3 exo	±3 Δ
Distance base-out (blur / break / recovery)	9 / 19 / 10	±4 / 8 / 4
Distance base-in (break / recovery)	x / 7 / 4	±3 / 2
Near base-out	17 / 21 / 11	±5 / 6 / 7
Near base-in	13 / 21 / 13	±4 / 4 / 5
Negative relative accommodation (NRA)	+2.00 D	±0.50
Positive relative accommodation (PRA)	−2.37 D	±1.00
Amplitude of accommodation	18.5 − 0.30 × age	±2.00
Monocular estimate method (accommodative lag)	+0.50 D	±0.25
Accommodative facility (±2.00 flippers, 40 cm)	≥11 cpm binocular, ≥11 cpm monocular	–
Vergence facility (12 BO / 3 BI)	~15 cpm	±3
Near point of convergence (break / recovery)	5 cm / 7 cm	±2.5 / 3
Calculated AC/A	4:1	±2
Gradient AC/A	3–4:1	±2

Test	Mean	SD
Stereoacuity	20 arcsec (Randot), 40 arcsec (Titmus)	-

THE AC/A RATIO, BOTH WAYS

Calculated AC/A (in prism dioptres per dioptre):

$$AC/A = PD(\text{cm}) + m \times (\Delta_{\text{near}} - \Delta_{\text{distance}})$$

where m is the near working distance in metres, and deviations use the convention **eso positive, exo negative**.

Gradient AC/A: change in near deviation per dioptre of added lens power, measured at a fixed distance:

$$AC/A = (\Delta_{\text{with lens}} - \Delta_{\text{without lens}}) / \text{lens power}$$

The gradient method is the more valid because it holds proximal convergence constant; the calculated method includes proximal convergence and therefore reads higher.

Worked example. PD 60 mm, distance phoria 2 exo (-2), near phoria at 40 cm 10 eso (+10). $AC/A = 6 + 0.4 \times (10 - (-2)) = 6 + 0.4(12) = 6 + 4.8 = 10.8:1$. A high AC/A with near esophoria: this is the pattern of **convergence excess**, and a plus add is the primary treatment.

Vergence: components and criteria

- **Maddox components of vergence**: **tonic** (baseline innervation from the physiologic position of rest to the distance phoria position), **proximal** (awareness of nearness), **accommodative** (via the AC/A relationship), and **fusional** (disparity-driven, the only one under blur-free feedback control).
- **Fusional vergence has fast and slow components**: the fast component responds within ~200 ms; the slow component adapts over minutes and is what shifts the measured phoria after sustained near work.

CLINICAL CRITERIA FOR PRESCRIBING PRISM

Criterion	Statement	Applies to
Sheard	The compensating fusional vergence should be at least twice the phoria. Prism = $(2 \times \text{phoria} - \text{compensating fusional reserve}) / 3$.	Exophoria , and generally patients whose symptoms track fusional demand.
Percival	The demand point should lie in the middle third of the total vergence range. Prism = $(G - 2g)/3$ where G is the greater and g the lesser limit.	Any deviation, and it does not require a phoria measurement; useful when the phoria is unreliable.
1:1 rule	Base-in prism such that the near exophoria equals the base-in break. Prism = $(\text{phoria} - \text{BI break}) / 2$.	Esophoria at near is not the target; the 1:1 rule is for exophoria , typically divergence excess and basic exo.
Vertical: associated phoria	Prescribe the full associated phoria (fixation disparity measured with a Mallett unit or Wesson card).	Vertical deviations , where fusional reserves are small and prism is the primary treatment.

TREATMENT HIERARCHY DIFFERS BY DIAGNOSIS, AND THE EXAM TESTS EXACTLY THIS

Condition	Signature findings	First-line treatment
Convergence insufficiency	Exo greater at near, receded NPC, low base-out at near, high NRA relative to PRA, low AC/A	Office-based vergence and accommodative therapy with home reinforcement. The CITT randomised trial showed this outperformed home pencil push-ups and placebo therapy. Base-in prism has limited support here.
Convergence excess	Eso greater at near, high AC/A , low PRA, low base-in at near, high lag	Plus add at near , guided by the gradient AC/A; then vergence therapy for negative fusional vergence.
Divergence insufficiency	Eso greater at distance , low base-in at distance, low AC/A	Base-out prism at distance. Exclude a sixth nerve palsy, raised intracranial pressure, and sagging eye syndrome first.
Divergence excess	Exo greater at distance by more than 10 Δ , high AC/A	Distinguish true from <i>simulated</i> divergence excess with a one-hour patch or a +3.00 add; treat with therapy, minus overcorrection, or surgery.
Basic exo / eso	Roughly equal deviation at distance and near, normal AC/A	Prism by Sheard (exo) or Percival, plus vergence therapy.
Fusional vergence dysfunction	Normal phorias, reduced ranges in both directions , reduced vergence facility	Vergence therapy. Prism is unhelpful because there is no significant phoria.
Accommodative insufficiency	Amplitude more than 2 D below the Hofstetter minimum, high lag, reduced PRA, fails minus on facility	Plus add plus accommodative therapy. Exclude systemic causes: medication, concussion, Adie pupil, diabetes, Graves disease, and mononucleosis in adolescents.
Accommodative excess / spasm	Variable pseudomyopia, reduced NRA, fails plus on facility, often with headache and photophobia	Cycloplegia if severe, remove precipitants, accommodative therapy, plus add for near, and evaluate for functional and neurologic causes (spasm of the near reflex).
Accommodative infacility	Normal amplitude, reduced facility both directions	Accommodative therapy with flippers.

Strabismus classification

ESODEVIATIONS

Type	Features
Infantile (congenital) esotropia	Onset before 6 months , large and stable angle (usually >30 Δ), minimal hyperopia, cross-fixation with apparent abduction deficit, latent nystagmus, inferior oblique overaction, dissociated vertical deviation, and asymmetric monocular optokinetic nystagmus. Early surgical alignment (before 2 years) improves the chance of gross stereopsis.

Type	Features
Refractive accommodative esotropia	Onset 2-3 years, significant hyperopia (average +4.00 D), normal AC/A . Deviation equal at distance and near and fully corrected by full cycloplegic plus .
Non-refractive accommodative esotropia (high AC/A)	Deviation much greater at near, normal or low hyperopia, high AC/A . Treated with a bifocal add (flat-top segment bisecting the pupil) or, in selected cases, surgery.
Partially accommodative esotropia	Some but not all of the deviation corrected by plus; the residual requires surgery.
Sensory esotropia	Unilateral vision loss in a young child (cataract, corneal opacity, retinoblastoma, optic nerve disease). Always dilate and examine the fundus in any new unilateral strabismus in a child.
Divergence insufficiency / sagging eye syndrome	Acquired, comitant, eso greater at distance; in older adults from degeneration of the lateral rectus and superior rectus pulley band. Distinguish from a sixth nerve palsy (incomitant, abduction limited).
Nystagmus blockage syndrome	Esotropia that damps the nystagmus; the eye is adducted to use convergence to suppress nystagmus.
Cyclic and acute acquired comitant esotropia	Acute comitant esotropia in an older child or adult mandates neuroimaging to exclude a posterior fossa lesion or Chiari malformation, even though many cases are benign or digital-device associated.

EXODEVIATIONS AND VERTICAL OR PATTERN DEVIATIONS

Type	Features
Intermittent exotropia	The commonest childhood exodeviation. Subtypes: basic (equal), divergence excess (greater at distance), and convergence insufficiency type (greater at near). Grade control with the Newcastle or LACTOSE scale. Manage with refractive correction, therapy, overminus, prism, or surgery depending on control and size.
Sensory exotropia	Unilateral vision loss, typically in an older child or adult.
Consecutive exotropia	Follows esotropia surgery or spontaneous drift.
A pattern	Deviation more <i>divergent</i> in downgaze (or more convergent in upgaze) by $\geq 10^\circ$; associated with superior oblique overaction .
V pattern	More divergent in upgaze by $\geq 15^\circ$; associated with inferior oblique overaction and with Brown syndrome or craniofacial anomalies.
Dissociated vertical deviation	Slow upward drift of the covered eye with extorsion, then a return with intorsion, violating Hering law. Associated with infantile esotropia and latent nystagmus.
Brown syndrome	Restricted elevation in adduction with a normal or minimal superior oblique overaction, a positive forced duction, and often a down-shoot in adduction. Congenital (tight superior oblique tendon sheath) or acquired (trochleitis in rheumatoid disease, after surgery).
Duane retraction syndrome	Absent abducens nucleus with aberrant oculomotor innervation of the lateral rectus: limited abduction, globe retraction and palpebral fissure narrowing on adduction , with up- or down-shoots. Type I (limited abduction), II (limited adduction), III (both). Part of the congenital cranial dysinnervation disorders.

Type	Features
Möbius syndrome	Congenital CN VI and VII palsies, mask-like face, feeding difficulty.
Congenital fibrosis of the extraocular muscles	Restrictive, non-progressive, bilateral ptosis with the eyes fixed in downgaze; <i>KIF21A</i> , <i>PHOX2A</i> .
Thyroid eye disease and orbital floor fracture	The two great restrictive mimics. Both show positive forced duction, raised IOP on attempted gaze into the restricted field, and a saccadic velocity that is normal rather than slowed.

The Parks-Bielschowsky three-step test

THREE STEPS, EIGHT POSSIBLE ANSWERS, ONE MUSCLE

- 1 Which eye is higher in primary gaze?** A right hypertropia implicates the *depressors of the right eye* (RIR, RSO) or the *elevators of the left eye* (LSR, LIO). That is four muscles.
- 2 Is the deviation worse in right or left gaze?** Worse in left gaze keeps the muscles that act vertically in *left gaze*: RSO (depressor in adduction) and LSR (elevator in abduction). That is two muscles.
- 3 Bielschowsky head tilt: worse on right or left tilt?** On *right* head tilt, the right eye intorts, which requires the RSO and RSR. If the RSO is weak, the RSR's elevating action is unopposed and the hypertropia worsens. Worse on right tilt therefore indicates the **right superior oblique**.

SUPERIOR OBLIQUE PALSY: THE PATTERN TO RECOGNISE INSTANTLY

Hypertropia on the affected side that is **worse in contralateral gaze, worse on ipsilateral head tilt, and worse in downgaze**, with excyclotorsion on double Maddox rod, a compensatory head tilt *away* from the affected side, and often a V pattern. It is the commonest isolated cyclovertical palsy and the commonest cause of acquired vertical diplopia. Congenital cases have large vertical fusional amplitudes ($>3 \Delta$), a long-standing head tilt, and facial asymmetry.

Bilateral superior oblique palsy shows a hypertropia that reverses in lateral gaze, a V pattern, and excyclotorsion greater than 10° : think head trauma.

Beware the mimics of a three-step test: skew deviation (which shows a *conjugate ocular tilt reaction* with head tilt toward the lower eye and incyclotorsion of that eye), restrictive disease, myasthenia, and previous strabismus surgery all break the assumptions of the test.

Measurement and sensory testing

TESTS, WHAT THEY MEASURE, AND THEIR PITFALLS

Test	Measures	Pitfall
Cover-uncover	Distinguishes tropia from phoria	Requires steady fixation; misses intermittent deviations if too fast.
Alternate cover with prism	Total deviation	Breaks fusion, so it overestimates what the patient actually manifests.
Simultaneous prism cover	The manifest portion only	Technically harder; essential in intermittent exotropia and in monofixation syndrome.
Hirschberg	Gross angle from corneal reflex displacement	1 mm of decentration $\approx 7^\circ \approx 15 \Delta$ (values from 15 to 22 Δ/mm are quoted). Confounded by a large angle kappa.
Krimsky	Angle by neutralising the reflex with prism	Useful when one eye cannot fixate.

Test	Measures	Pitfall
Brückner	Compares fundus reflex brightness between eyes	The brighter reflex is the deviating or amblyopic eye; a useful screen for anisometropia, media opacity, and strabismus.
Maddox rod	Subjective phoria including cyclo-deviation with a double Maddox rod	Completely dissociating, so it measures the total deviation; the patient must report accurately.
von Graefe	Phoria by prism dissociation in the phoropter	Prone to proximal convergence and prism adaptation; less repeatable than Thorington.
Modified Thorington	Phoria with a Maddox rod and a numbered card	The most repeatable subjective phoria measurement.
Worth four-dot	Suppression and gross fusion	Dissociating; a small central suppression scotoma can be missed at distance and revealed at near, or the reverse.
Bagolini striated lenses	Fusion status under near-natural conditions	Least dissociating, so best for detecting anomalous retinal correspondence.
4Δ base-out test	Central suppression scotoma (monofixation)	Difficult to interpret; a normal response requires a version movement followed by a fusional recovery.
Stereotests	Titmus (has monocular cues in the fly and circles), Randot and Randot Preschool (random dot, no monocular cues), Frisby, TNO (anaglyphic)	Random-dot tests require bifoveal fusion ; a patient with monofixation syndrome will typically fail random-dot at <67 arcsec but pass contour tests.
Forced duction (traction) test	Restriction versus paresis	The definitive discriminator: resistance means restriction.

Sensory adaptations and amblyopia

- **Suppression:** active cortical inhibition of one image. Central suppression scotoma in monofixation syndrome, hemiretinal suppression in exotropia.
- **Anomalous retinal correspondence:** a shifted relationship between the two retinas producing a shifted horopter; diagnosed when the objective angle differs from the subjective angle (the *angle of anomaly*).
- **Eccentric fixation:** a non-foveal retinal point is used for monocular fixation; measured with a visuoscope. Distinguish from eccentric *viewing*, which is a deliberate strategy in low vision.
- **Monofixation syndrome:** a small (usually $\leq 8^\circ$) deviation with peripheral fusion, a central suppression scotoma, gross but not fine stereopsis, and often mild amblyopia. Frequently the best achievable outcome after infantile strabismus surgery.
- **Panum fusional area** permits fusion of slightly disparate images; it is narrowest at the fovea (about $6\text{--}10^\circ$) and widens with eccentricity, which is why peripheral fusion survives when central fusion is lost.
- **Physiologic diplopia** is normal and is the basis of much vergence therapy.

AMBLYOPIA: TYPES, SEVERITY, AND TREATMENT EVIDENCE

Type	Cause	Notes
Deprivation (amblyopia ex anopsia)	Congenital cataract, corneal opacity, complete ptosis, prolonged patching	The most severe form. A dense unilateral congenital cataract must be removed within the first weeks of life.
Anisometropic	Unequal refractive error, generally ≥ 1.00 D spherical or ≥ 1.50 D cylindrical difference	Often undetected until screening because the child appears normal. Responds well to refractive correction alone in many cases.
Strabismic	Constant unilateral deviation, most often esotropia	Crowding phenomenon is prominent: single-letter acuity exceeds line acuity.
Isoametropic	High bilateral symmetrical error (hyperopia $>+5.00$, myopia >-8.00 , astigmatism >2.50)	Bilateral and often improves slowly over many months with spectacles alone.
Meridional	Uncorrected astigmatism ≥ 1.50 D in early childhood	Acuity deficit is orientation-specific.

PEDIG AND ATS EVIDENCE: THE TRIAL RESULTS THAT GET EXAMINED

- **Refractive adaptation first.** Spectacles alone improve amblyopia in a large fraction of children, and improvement continues for up to **30 weeks**. Do not start patching before giving glasses an adequate trial.
- **Patching dose.** **2 hours** per day is as effective as **6 hours** for **moderate** amblyopia (20/40 to 20/80), and **6 hours** is as effective as full-time for **severe** amblyopia (20/100 to 20/400).
- **Atropine.** 1% daily is equivalent to patching for moderate amblyopia, and **weekend-only** atropine is equivalent to daily. It is a legitimate first-line alternative, with better adherence in many families.
- **Age.** Children aged **7–12** still respond, and **13–17**-year-olds respond if previously untreated. Age alone is not a reason to withhold treatment.
- **Near activities** during patching add little measurable benefit over patching alone in the trials, though they aid compliance.
- **Recurrence** occurs in about a quarter of children after stopping; taper rather than stop abruptly, and follow for at least a year.
- **Binocular (dichoptic) treatments**, including digital therapeutics, are an active area and show benefit in some trials, particularly in older children.

Nystagmus, in brief

NYSTAGMUS PATTERNS AND LOCALISATION

Type	Features and localisation
Infantile (congenital) nystagmus syndrome	Onset in the first months, conjugate, horizontal in all positions, damps with convergence , has a null zone and often a head turn, and shows a paradoxically inverted optokinetic response . No oscillopsia.
Latent (fusion maldevelopment) nystagmus	Appears or worsens on occluding one eye, with the fast phase toward the <i>fixing</i> eye. Associated with infantile esotropia and DVD.
Spasmus nutans	Fine, rapid, asymmetric pendular nystagmus with head nodding and torticollis in infancy. Requires imaging to exclude a chiasmal glioma.

Type	Features and localisation
Gaze-evoked	Jerk nystagmus on eccentric gaze; most commonly drug-induced (anticonvulsants, alcohol, sedatives) or cerebellar.
Downbeat	Cervicomedullary junction: Arnold-Chiari malformation, foramen magnum lesion, lithium or anticonvulsant toxicity, Wernicke encephalopathy.
Upbeat	Medulla or pontomesencephalic junction.
Periodic alternating	Cycles direction every one to two minutes; cerebellar nodulus and uvula; responds to baclofen.
Seesaw	Parasellar or diencephalic lesion; also with achiasma.
Convergence-retraction	Dorsal midbrain (Parinaud) syndrome.
Opsoclonus	Chaotic multidirectional saccades; paraneoplastic (neuroblastoma in children, small cell lung cancer in adults) or post-viral.

RETRIEVAL PRACTICE

1. PD 64 mm, distance orthophoria, near phoria 12 exo at 40 cm. Compute the calculated AC/A and name the likely diagnosis.
2. Give Morgan norms for near base-in and near base-out ranges.
3. State Sheard criterion and compute the prism for a 10 Δ exophoria with a base-out blur/break/recovery of 8/14/6.
4. Which criterion applies to esophoria, and which to exophoria?
5. A 3-year-old with +5.00 D hyperopia and a 25 Δ esotropia equal at distance and near that is fully eliminated by full cycloplegic plus. Diagnosis?
6. A child has esotropia much greater at near with only +1.00 hyperopia. Diagnosis and treatment?
7. Right hypertropia, worse in left gaze, worse on right head tilt. Which muscle?
8. Give three features that distinguish skew deviation from a superior oblique palsy.
9. Distinguish restrictive from paretic limitation using three findings.
10. What does the Brückner test show, and which reflex is abnormal?
11. Which stereotests require bifoveal fusion, and what does a monofixator typically achieve?
12. Give the PEDIG patching dose for moderate and for severe amblyopia, and the atropine equivalence result.
13. A 6-month-old has fine, rapid, asymmetric pendular nystagmus with head nodding. What must you exclude?
14. Downbeat nystagmus: name two structural and one toxic cause.

KEY

1. $AC/A = 6.4 + 0.4(-12 - 0) = 6.4 - 4.8 = 1.6:1$, a low AC/A with near exophoria: **convergence insufficiency**.
2. Base-in x/13/21/13 (blur/break/recovery reported as 13/21/13) and base-out 17/21/11.
3. Sheard: the compensating reserve should be twice the phoria. Prism = $(2 \times 10 - 14)/3 = 6/3 = 2 \Delta$ **base-in**.
4. Sheard for **exophoria**; Percival is general and is often preferred for **esophoria**.
5. **Refractive accommodative esotropia** with a normal AC/A.
6. **Non-refractive accommodative esotropia (high AC/A)**; treat with a plus add at near; typically a flat-top bifocal set to bisect the pupil.
7. **Right superior oblique.**
8. Skew deviation shows a conjugate ocular tilt reaction with head tilt toward the *lower* eye, **incyclotorsion of the lower eye** (whereas SO palsy gives excyclotorsion of the *higher* eye), and accompanying brainstem or cerebellar signs; the hypertropia is often comitant.
9. Restriction: positive forced duction, IOP rise on gaze into the field of limitation, and normal saccadic velocity with an abrupt stop. Paresis: negative forced duction, no IOP change, and slowed saccadic velocity.
10. It compares the brightness of the two fundus reflexes; the **brighter** reflex belongs to the abnormal (deviating, amblyopic, or more ametropic) eye.

- | | |
|---|---|
| <p>11. Random-dot tests (Randot, TNO, Frisby) require bifoveal fusion; a monofixator typically achieves gross stereopsis of about 70–3000 arcsec on contour tests and fails fine random-dot levels.</p> <p>12. Two hours daily for moderate amblyopia (equivalent to six), six hours for severe (equivalent to full-time);</p> | <p>atropine 1% is equivalent to patching for moderate amblyopia, and weekend-only dosing equals daily.</p> <p>13. Spasmus nutans; exclude a chiasmal or hypothalamic glioma with neuroimaging.</p> <p>14. Arnold-Chiari malformation and a foramen magnum or cervicomedullary lesion; lithium (or anticonvulsant) toxicity.</p> |
|---|---|

CHAPTER RECAP

1. Morgan norms are a memorisation task; the exam expects the numbers, not approximations.
2. AC/A calculated = $PD + m(\Delta_{\text{near}} - \Delta_{\text{distance}})$ with eso positive; the gradient method is more valid.
3. Sheard for exo, Percival broadly, 1:1 for exo at near, full associated phoria for verticals.
4. High AC/A with near eso means plus add; low AC/A with near exo means vergence therapy (CITT).
5. Esotropia types are separated by hyperopia and the near-distance relationship; sensory esotropia demands a dilated fundus exam.
6. A pattern goes with superior oblique overaction, V pattern with inferior oblique overaction.
7. Three-step test: higher eye, worse gaze, worse tilt; but check for skew, restriction, and myasthenia.
8. Random-dot stereo needs bifoveal fusion; monofixators pass contour tests only.
9. Glasses first, then 2 hours for moderate and 6 for severe, or atropine; older children still respond.
10. Nystagmus type localises: downbeat cervicomedullary, upbeat medullary, seesaw parasellar, opsoclonus paraneoplastic.

Colour vision, perception, and visual fields

Perceptual function / color vision · 10–18 items

Optic nerve / neuro-ophthalmic pathways

Glaucoma (perimetry)

Low vision

ANSWER THESE COLD BEFORE READING

1. State the trichromatic and opponent-process theories and explain why both are needed.
2. Give the chromosome for each cone opsin gene and the inheritance and prevalence of each congenital defect.
3. State the Köllner rule and give three exceptions.
4. Match six colour vision tests to their purpose.
5. Define the contrast sensitivity function and state where its peak lies.
6. Give the normal extent of the visual field in four directions and define MD, PSD, and GHT.

BLUEPRINT WEIGHT

4%

10–18 scored items, and perimetric interpretation cross-loads into glaucoma and neuro-ophthalmic items.

Theories of colour vision

THREE LEVELS OF EXPLANATION, ALL CORRECT AT THEIR OWN LEVEL

Theory	Claim	Evidence and level of processing
Trichromatic (Young-Helmholtz)	Three cone classes with overlapping broad spectral sensitivities; colour is coded by the ratio of their responses	Correct at the receptor level. Explains why three primaries can match any colour and why metamers exist (physically different spectra producing identical cone ratios).
Opponent process (Hering)	Colour is coded in antagonistic pairs: red-green , blue-yellow , and black-white	Correct at the post-receptoral level (horizontal cells, bipolar cells, ganglion cells, LGN). Explains why there is no reddish-green, why negative afterimages are complementary, and why simultaneous colour contrast occurs.
Zone (stage) theory	Trichromatic at the receptors, then recoded into opponent channels	The synthesis, and the answer to any item asking which theory is "correct".

- Cone peak sensitivities: **S (blue)** ~419–440 nm, **M (green)** ~531–535 nm, **L (red)** ~558–565 nm. Rod (rhodopsin) peak ~496–500 nm.
- **Photopic luminous efficiency** peaks at 555 nm; **scotopic** at 507 nm (the **Purkinje shift**).
- Cone proportions: roughly **L : M : S = 10 : 5 : 1**, with S cones only about **5–10%** and **absent from the very centre of the foveola** (small-field tritanopia).
- **Colour appearance phenomena**: the **Bezold-Brücke** shift (hue changes with intensity), the **Abney effect** (hue changes with desaturation), the **Helmholtz-Kohlrausch effect** (saturated colours look brighter), and **colour constancy** (perceived colour remains stable across illuminants).

Congenital colour vision defects

GENETICS, NOMENCLATURE, AND PREVALENCE

Defect	Photopigment	Inheritance	Prevalence and features
Protanomaly	L cone shifted	X-linked recessive (Xq28)	~1% of males. Reduced red brightness is characteristic of the protan defects.
Protanopia	L cone absent	X-linked recessive	~1% of males. Dichromat; red appears dark.
Deuteranomaly	M cone shifted	X-linked recessive	~5% of males: the commonest colour defect of all . Anomalous trichromat.
Deuteranopia	M cone absent	X-linked recessive	~1% of males. Dichromat with normal luminosity.
Tritanomaly / tritanopia	S cone	Autosomal dominant, chromosome 7	~1 in 15,000–50,000; equal in the sexes. Incomplete penetrance.
Blue cone monochromacy	Both L and M absent	X-linked recessive	Very rare. Acuity ~20/200, nystagmus, photophobia; only S cones and rods function.
Rod monochromacy (complete achromatopsia)	All cones non-functional (CNGB3, CNGA3, GNAT2)	Autosomal recessive	~1 in 30,000. Acuity 20/200 or worse, pendular nystagmus, severe photophobia and paradoxical pupils , normal fundus, flat photopic ERG with a normal scotopic ERG .

PREVALENCE NUMBERS AND THE NOMENCLATURE SYSTEM

Red-green defects affect about 8% of males and 0.5% of females. Because the L and M opsin genes lie in a tandem array on **Xq28**, unequal crossing over between the highly homologous genes is the mechanism, and it explains both the X-linked pattern and why anomalous (shifted) pigments are commoner than absent ones. The S opsin gene is on **chromosome 7**, hence autosomal dominant tritan defects.

Suffix rules: *-anomaly* = anomalous trichromat (three pigments, one shifted); *-anopia* = dichromat (one pigment missing); *monochromacy* = one or no functioning cone class. **Prot-** refers to the first (long-wavelength, L) pigment, **deuter-** the second (M), **trit-** the third (S).

Acquired colour vision defects

KÖLLNER RULE, AND THE EXCEPTIONS THAT GET TESTED

Outer retinal (photoreceptor and RPE) disease tends to cause blue-yellow defects; optic nerve disease tends to cause red-green defects.

Exception	Direction
Glaucoma	Optic nerve disease, but produces an early blue-yellow loss (the basis of short-wavelength automated perimetry).
Autosomal dominant optic atrophy (OPA1)	Optic nerve, but classically blue-yellow (tritan).
Stargardt disease and cone dystrophies	Outer retina, but often red-green .
Chloroquine and hydroxychloroquine maculopathy	Outer retina, and can present with red-green errors.

Other characteristics of acquired defects: they are frequently *asymmetric between the eyes*, are often accompanied by reduced acuity, may be described in qualitative terms ("colours look washed out"), and change over time. Congenital defects are symmetric, stable, and usually have normal acuity. **Asymmetry between eyes is the single best clue that a colour defect is acquired.**

Ocular and systemic causes to know: optic neuritis and other optic neuropathies (red desaturation is an early and sensitive sign), nutritional and toxic optic neuropathy (**ethambutol**, methanol, tobacco-alcohol), **digoxin** (xanthopsia, yellow vision), **sildenafil** (transient blue tinge via PDE6), **amiodarone**, and jaundice or cataract producing a generalised blue-yellow shift by absorption.

COLOUR VISION TESTS: CHOOSE BY PURPOSE

Test	Type	What it does and does not do
Ishihara	Pseudoisochromatic plates	Screening for congenital red-green defects only. It does <i>not</i> detect blue-yellow defects and does not reliably grade severity. Designed for literate adults; a numeral-free tracing version exists for children.
Hardy-Rand-Rittler (HRR)	Pseudoisochromatic plates	Detects and classifies both red-green and blue-yellow defects, and grades severity. The better choice when acquired disease is suspected.
Farnsworth D-15 (and Lanthony desaturated D-15)	Arrangement (hue discrimination)	Sorts 15 caps into a hue circle; the axis of the error pattern indicates the defect type. Quick and useful for occupational screening; the desaturated version is far more sensitive to mild acquired loss.
Farnsworth-Munsell 100-hue	Arrangement, 85 caps in four boxes	Quantitative with a total error score; excellent for monitoring acquired defects over time; slow to administer.

Test	Type	What it does and does not do
Nagel anomaloscope	Colour matching	The reference standard. Distinguishes anomalous trichromats from dichromats, and measures the matching range and midpoint. Not widely available.
Farnsworth lantern (FALANT) and equivalents	Occupational	Signal-light recognition for aviation, maritime, and military standards.
Red cap desaturation and red-green comparison	Bedside	A rapid, sensitive screen for optic nerve disease; ask the patient to compare the saturation of a red target between the eyes.

Perceptual function

Visual acuity: four kinds

ACUITY TYPES AND THEIR THRESHOLDS

Type	Threshold	Task
Detection (minimum visible)	~0.5 arcsec for a bright line on a dark field	Is anything there? Depends on contrast, not on receptor spacing.
Resolution (minimum separable)	~30-60 arcsec (0.5-1 arcmin)	Are these two things separate? Limited by cone spacing (the Nyquist limit) and by diffraction. Grating and Landolt C acuity.
Recognition	~1 arcmin per stroke width, so a 20/20 letter subtends 5 arcmin	What is it? Snellen and logMAR charts. Involves cognitive identification, so it is affected by crowding and literacy.
Vernier (hyperacuity)	~2-5 arcsec	Are these lines aligned? Far finer than cone spacing , because it relies on comparing the centroid of a distribution across many receptors: hence "hyperacuity". Also underlies stereoacuity.

- **MAR and logMAR:** MAR is the minimum angle of resolution in arcmin; **logMAR = $\log_{10}(\text{MAR})$** .
 $20/20 = \text{MAR } 1 = \text{logMAR } 0.00$. $20/200 = \text{MAR } 10 = \text{logMAR } 1.00$. $20/40 = \text{MAR } 2 = \text{logMAR } 0.30$.
 Each line on a logMAR chart is **0.1** log units, with **5** letters per line so each letter is worth **0.02**.
- **Advantages of logMAR (Bailey-Lovie, ETDRS) over Snellen:** equal letter count per line, logarithmic progression of size, constant inter-letter and inter-line spacing proportional to letter size, and equal legibility of letters. This makes acuity a genuine interval scale suitable for statistics, which is why every clinical trial uses it.
- **Pinhole acuity** improves if blur is refractive or from irregular optics; it does not improve, and often worsens, with retinal or neural disease and in a small pupil or media opacity.

Contrast sensitivity

- **Contrast sensitivity = $1 / \text{contrast threshold}$** . Plotted against spatial frequency it gives the **contrast sensitivity function**, an inverted U peaking at **3-6 cycles per degree** and falling to

zero at about **30–60 cpd** (the acuity limit). The low-frequency fall-off is caused by lateral inhibition.

- **Acuity measures only the high-frequency cut-off.** Disease can devastate mid-frequency sensitivity while leaving acuity intact: this is the pattern in **optic neuritis, glaucoma, amblyopia, early cataract, and after refractive surgery**, and it explains patients with 20/20 acuity who complain that vision is not right.
- **Tests:** Pelli-Robson (letter, at a fixed spatial frequency), Vistech and FACT (sine-wave gratings at several frequencies), Mars letter chart, and low-contrast logMAR charts (**2.5%** and **1.25%**).
- **Glare:** *discomfort* glare is a sensation; *disability* glare is measurable loss of contrast sensitivity in the presence of a glare source, quantified by brightness acuity testing. Both are prominent in cataract and in corneal oedema and scatter.

Psychophysical laws worth naming

NAMED RELATIONSHIPS IN VISUAL PSYCHOPHYSICS

Law	Statement
Weber law	The just-noticeable difference is a constant fraction of the background: $\Delta I / I = k$. Underlies contrast constancy across illumination levels.
Fechner law	Perceived magnitude is proportional to the logarithm of stimulus intensity. Why perimetric sensitivity is expressed in decibels.
Ricco law	For small stimuli, threshold intensity $\times$ area = constant: complete spatial summation. Applies below a critical area that grows with eccentricity.
Piper law	For intermediate areas, threshold $\times \sqrt{\text{area}}$ = constant: partial summation.
Bloch law	For brief stimuli, threshold intensity $\times$ duration = constant: complete temporal summation, up to a critical duration of about 100 ms .
Ferry-Porter law	Critical flicker fusion frequency rises linearly with the logarithm of retinal illuminance. CFF is about 50–60 Hz photopically and much lower scotopically, and it is reduced in optic nerve disease.
Talbot-Plateau law	Above the fusion frequency, a flickering light appears as steady light of the time-averaged luminance.
Broca-Sulzer effect	A brief flash of intermediate duration appears brighter than a longer one of the same luminance.
Stiles-Crawford effect	Cones are most sensitive to light entering through the pupil centre; a waveguide directional effect.
Troxler fading	A steadily fixated peripheral stimulus fades from awareness; the basis of the disappearance of stabilised images and of some entoptic phenomena.

Entoptic phenomena and illusions

- **Haidinger brushes:** a yellow propeller pattern seen in polarised blue light, generated by the radially arranged macular xanthophyll and Henle fibres. Used to demonstrate and train **foveal fixation**.
- **Maxwell spot:** a dark central spot seen against certain colours, again from macular pigment absorption.
- **Purkinje tree:** shadows of retinal vessels seen with a moving light source; used to assess retinal function through dense media opacity.

- **Blue field entoptoscope:** white cells moving in perifoveal capillaries, used to assess macular perfusion.
- **Phosphenes:** pressure or mechanically induced light perception, and the basis of the Amsler pressure-phosphenes test in dense media opacity.
- **Mach bands and the Hermann grid:** direct behavioural evidence of lateral inhibition.
- **Gestalt grouping principles:** proximity, similarity, continuity, closure, common fate, figure-ground. These explain contour completion, why a scotoma can be "filled in", and the perceptual completion in hemianopia.

The visual field

- **Normal monocular extent:** temporally **90–100°**, nasally **~60°**, superiorly **~60°**, inferiorly **~70–75°**. The binocular field spans about **200°** horizontally with a central **120°** of overlap; the monocular **temporal crescent** occupies the outer **60–90°**.
- **Physiologic blind spot:** centred about **15°** temporal to fixation and **1.5–2°** below the horizontal, measuring roughly **5°** wide by **7°** tall.
- **Traquair island of vision:** the classic metaphor of a hill of vision in a sea of blindness. Sensitivity is highest at fixation and declines with eccentricity, and the whole hill sinks with age and with media opacity.

PERIMETRIC INDICES AND RELIABILITY

Index	Meaning and interpretation
Mean deviation (MD)	Average departure from age-matched normal across the field. Sensitive to diffuse loss, so it is dragged down by cataract and small pupils as well as by disease.
Pattern standard deviation (PSD)	Irregularity of the hill of vision. Sensitive to localised loss. It paradoxically <i>improves</i> in very advanced, uniformly depressed fields, so a low PSD with a very negative MD is not reassurance.
Visual field index (VFI)	A percentage of age-normal function weighted toward central points, designed for rate-of-progression analysis and less affected by cataract than MD.
Glaucoma hemifield test (GHT)	Compares mirror-image zones above and below the horizontal, exploiting the fact that glaucomatous loss respects the raphe. Reported as within normal limits, borderline, or outside normal limits.
Fixation losses	Blind spot catch trials. >20% is unreliable; may also indicate a mis-mapped blind spot.
False positives	Response with no stimulus ("trigger happy"). >15% is unreliable, and it produces spuriously <i>good</i> , sometimes supranormal, sensitivities with a white scotoma pattern.
False negatives	Failure to respond to a much brighter stimulus in a seeing area. >33% suggests inattention, but is also common in genuinely advanced disease, where it should not be read as unreliability.
Short-term fluctuation	Repeat testing of selected points; rises in early disease and in inexperienced patients.

ARTEFACTS THAT MIMIC DISEASE

- **Lens rim artefact:** a dense, sharply demarcated peripheral arc, usually superior or all around; caused by the trial lens being too far from the eye or decentred.
- **Ptosis or brow:** superior arcuate depression that respects the lid position rather than the raphe; tape the lid and repeat.
- **Learning effect:** the first field is often worse than the second; never diagnose progression on two fields alone.
- **Cloverleaf pattern:** sensitivity preserved only in the four zones tested first; indicates the patient stopped responding.
- **Small pupil or miotic:** generalised depression with a worsened MD.
- **Uncorrected refractive error:** generalised depression, worse centrally, and a high-plus trial lens can also create a rim artefact.
- **Spiral or tunnel fields** with normal structure suggest non-organic visual loss; a true field defect enlarges appropriately with distance, whereas a functional tunnel does not.

NUMBERS TO KNOW: COLOUR, PERCEPTION, FIELDS

S / M / L cone peaks	~430 / 530 / 560 nm	Rod peak	~500 nm
Photopic / scotopic luminosity peak	555 / 507 nm	L:M:S proportion	~10:5:1
Red-green defect, males / females	~8% / 0.5%	Deuteranomaly in males	~5%
L and M opsin locus	Xq28	S opsin locus	chromosome 7
Rod monochromacy prevalence	~1 in 30,000	Tritanopia prevalence	~1 in 15,000–50,000
20/20 letter subtends	5 arcmin	Stroke width at 20/20	1 arcmin
Vernier acuity	2–5 arcsec	logMAR per letter	0.02
CSF peak	3–6 cpd	CSF high cut-off	30–60 cpd
Critical flicker fusion, photopic	50–60 Hz	Bloch critical duration	~100 ms
Field: temporal / nasal	90–100° / 60°	superior / inferior	60° / 70–75°
Blind spot position / size	15° temporal, 5×7°	Perimetric background	31.5 asb
Unreliable fixation losses	>20%	Unreliable false positives	>15%

RETRIEVAL PRACTICE

1. Which theory explains metamers, and which explains negative afterimages?
2. Give the locus and inheritance for red-green and for blue-yellow congenital defects.
3. Name the commonest congenital colour defect and its prevalence in males.
4. A 30-year-old man has 20/20 acuity, a stable symmetric red-green defect, and a normal fundus. Congenital or acquired, and what single feature decided it?
5. State Köllner rule and give three exceptions.

6. Which test would you choose to quantify progression of an acquired defect, and which to classify the axis quickly?
7. Why does Ishihara testing miss tritan defects?
8. Convert 20/50 to MAR and logMAR.
9. Why does a patient with optic neuritis and 20/20 acuity still complain about vision?
10. State Ricco law and Bloch law.
11. What are Haidinger brushes and what are they used for?
12. MD is **-18 dB** and PSD is low. Is the field improving?
13. A sharply demarcated superior arc across the entire superior field with normal discs. Two artefacts to consider.
14. Give normal field extent in all four directions and the blind spot location.

KEY

1. Trichromatic (Young-Helmholtz) explains metamers; opponent process (Hering) explains complementary negative afterimages.
2. Red-green: L and M opsins in a tandem array at Xq28, X-linked recessive. Blue-yellow: S opsin on chromosome 7, autosomal dominant.
3. Deuteranomaly, about 5% of males.
4. **Congenital**; the decisive feature is **symmetry between the eyes** (with stability and normal acuity supporting it).
5. Outer retinal disease gives blue-yellow, optic nerve disease gives red-green. Exceptions: glaucoma (blue-yellow despite being optic nerve), dominant optic atrophy (blue-yellow), and Stargardt or cone dystrophy (red-green despite being outer retinal).
6. Farnsworth-Munsell 100-hue for quantification; Farnsworth D-15 or desaturated Lanthony D-15 for rapid axis classification (HRR also classifies).
7. The plate designs use confusion lines along the red-green axes only; there are no tritan confusion plates in the standard set.
8. $MAR = 50/20 = 2.5 \text{ arcmin}$; $\log MAR = \log_{10}(2.5) = 0.40$.
9. Snellen acuity measures only the high spatial frequency cut-off; optic neuritis characteristically depresses mid-frequency contrast sensitivity, which carries most real-world visual information.
10. Ricco: below a critical area, threshold intensity $\times$ area is constant (complete spatial summation). Bloch: below a critical duration of about 100 ms, threshold intensity $\times$ duration is constant (complete temporal summation).
11. A yellow propeller-shaped entoptic pattern seen in rotating polarised blue light, generated by macular xanthophyll along the radially arranged Henle fibres; used to demonstrate and train foveal fixation.
12. **No**. PSD falls in advanced, uniformly depressed fields because there is little remaining *irregularity*. Interpret with MD and the VFI, and look at the pattern deviation plot.
13. Lens rim artefact and ptosis or brow overhang. Reposition the lens, tape the lid, and repeat.
14. Temporal 90 to 100°, nasal about 60°, superior about 60°, inferior about 70 to 75°; blind spot about 15° temporal and 1.5 to 2° below the horizontal, roughly 5 by 7 degrees.

CHAPTER RECAP

1. Zone theory: trichromatic receptors feeding opponent channels.
2. L and M at Xq28 X-linked; S on chromosome 7 autosomal dominant. 8% of males, 0.5% of females.
3. Deuteranomaly is commonest; rod monochromacy gives a flat photopic ERG with a normal scotopic ERG.
4. Köllner rule with its four exceptions; asymmetry between eyes means acquired.
5. Ishihara screens congenital red-green only; HRR classifies both; 100-hue quantifies; anomaloscope is the reference.
6. Four acuities: detection, resolution, recognition, and vernier hyperacuity at 2 to 5 arcsec.
7. $\log\text{MAR} = \log(\text{MAR})$; 20/20 is $\log\text{MAR} 0$, and each letter is 0.02.
8. The CSF peaks at 3 to 6 cpd; acuity samples only its high-frequency edge.
9. Weber, Fechner, Ricco, Piper, Bloch, Ferry-Porter, Talbot-Plateau: know the statement, not the derivation.
10. MD for diffuse loss, PSD for focal loss, GHT for hemifield asymmetry, and always rule out lens rim and ptosis artefacts.

Ocular and systemic pharmacology

Every Normal health / disease / trauma area

Glaucoma • 13–21 items

Systemic health • 55–75 items

Accommodation / vergence

Contact lenses

ANSWER THESE COLD BEFORE READING

1. Build the autonomic receptor table for the eye: receptor, tissue, effect, agonist, antagonist.
2. Give onset, peak, and duration for six mydriatic and cycloplegic agents.
3. For every glaucoma drug class, state the mechanism in Goldmann-equation terms, the percentage IOP reduction, and the contraindications.
4. Rank the topical corticosteroids by potency and penetration and state why the vehicle matters.
5. Give the mechanism and one ocular toxicity for each major antibiotic class.
6. List twelve systemic drugs with named ocular toxicities and the monitoring required for each.

BLUEPRINT WEIGHT

Very high

Pharmacology is sampled in every condition area and is the discipline most heavily used in Treatment and Management items.

Principles and ocular drug delivery

PHARMACOLOGIC VOCABULARY THAT APPEARS IN STEMS

Term	Definition
Potency	The dose required for a given effect (position of the dose-response curve on the x-axis). Quantified by ED_{50} .
Efficacy	The <i>maximum</i> achievable effect (height of the curve). A less potent drug can be more efficacious.
Therapeutic index	LD_{50} / ED_{50} . A narrow index means small dosing errors matter (digoxin, warfarin, lithium, aminoglycosides).
Competitive antagonist	Shifts the dose-response curve right, surmountable by more agonist; reduces potency, not efficacy.
Non-competitive antagonist	Reduces maximum response; not surmountable.
Partial agonist	Submaximal effect even at receptor saturation; acts as an antagonist in the presence of a full agonist (carteolol has intrinsic sympathomimetic activity on this basis).
Tachyphylaxis	Rapid loss of response with repeated dosing (apraclonidine is the classic ocular example).
Prodrug	Inactive until metabolised: dipivefrin to epinephrine, latanoprost to the free acid, valacyclovir to aciclovir. Used to improve corneal or gastrointestinal penetration.
First-pass effect	Hepatic metabolism of an orally absorbed drug before systemic circulation; bypassed by topical, sublingual, and nasal routes, which is why topical timolol can produce systemic β -blockade.

WHY SO LITTLE TOPICAL DRUG GETS IN, AND WHAT TO DO ABOUT IT

- A commercial drop is **35–50 μL** ; the tear film holds **7–9 μL** and the cul-de-sac can accept at most **25–30 μL** . **Most of the drop is lost immediately.**
- Tear turnover of **$\sim 16\%/min$** clears the rest within minutes. Net corneal bioavailability is typically **$<5\%$** , and often **1–2%**.
- The remainder drains through the nasolacrimal system and is absorbed across **highly vascular nasal mucosa, bypassing hepatic first pass**. This is the route of systemic β -blocker toxicity, of tachycardia from phenylephrine, and of CNS depression from brimonidine in infants.
- **To increase ocular effect and reduce systemic effect:** instil one drop, not two; **occlude the puncta or close the lids gently for 2 to 3 minutes**; space different drugs by **5 minutes**; use gels, ointments, or viscosity agents to prolong contact; and use the lowest effective concentration.
- **Corneal penetration requires biphasic solubility** because the epithelium is lipophilic and the stroma hydrophilic. **Benzalkonium chloride** increases penetration by disrupting epithelial tight junctions, which is also why it causes surface toxicity with chronic use and why preservative-free formulations exist for glaucoma and dry eye patients on multiple drops.
- **Inflamed and abraded eyes absorb far more drug:** the barrier is broken. Dose accordingly and expect greater systemic exposure.

Autonomic pharmacology of the eye

RECEPTORS, TISSUES, EFFECTS, AND THE DRUGS AT EACH

Receptor	Ocular tissue	Effect of stimulation	Agonists / antagonists
M_3	Iris sphincter	Miosis	Agonists: pilocarpine, carbachol, acetylcholine (intracameral), methacholine. Antagonists: atropine, homatropine, scopolamine, cyclopentolate, tropicamide.
M_3	Ciliary muscle	Accommodation ; trabecular outflow increases	As above. Antagonists cause cycloplegia.
M_3	Lacrimal gland	Secretion	Pilocarpine and cevimeline orally for Sjögren dry mouth and dry eye; systemic anticholinergics cause dry eye.
α_1	Iris dilator; Müller muscle; conjunctival vessels	Mydriasis; lid elevation; vasoconstriction	Agonists: phenylephrine, naphazoline, tetrahydrozoline, oxymetazoline (used therapeutically for acquired ptosis). Antagonists: dapiprazole, thymoxamine; systemic tamsulosin (causing intraoperative floppy iris syndrome).
α_2	Ciliary epithelium; presynaptic terminals	Reduced aqueous formation and increased uveoscleral outflow	Agonists: brimonidine, apraclonidine, clonidine. Also brimonidine 0.025% for conjunctival redness.

Receptor	Ocular tissue	Effect of stimulation	Agonists / antagonists
β_1 and β_2	Ciliary epithelium	Increased aqueous formation (via cAMP)	Antagonists reduce IOP: timolol and levobunolol (non-selective), carteolol (non-selective with intrinsic sympathomimetic activity), betaxolol (β_1-selective) , metipranolol.
β_2	Ciliary muscle	Relaxation	Minor clinical relevance, but the reason β -blockers can slightly shift accommodation.
Prostaglandin FP	Ciliary muscle extracellular matrix	Matrix remodelling, increased uveoscleral outflow	Latanoprost, travoprost, bimatoprost, tafluprost, latanoprostene bunod.
Rho kinase	Trabecular meshwork and episcleral veins	Cytoskeletal relaxation, increased trabecular outflow and reduced episcleral venous pressure	Netarsudil.
Carbonic anhydrase II	Non-pigmented ciliary epithelium	Bicarbonate generation for secretion	Inhibitors: dorzolamide, brinzolamide (topical); acetazolamide, methazolamide (oral).

MYDRIATICS AND CYCLOPLEGICS: THE TIMING TABLE

Agent	Mydriasis onset	Cycloplegia peak	Duration	Clinical role
Tropicamide 0.5–1%	15–20 min	20–35 min	4–6 h	Routine dilation. A weak cycloplegic: inadequate for a young hyperope's cycloplegic refraction.
Phenylephrine 2.5%	20–30 min	none	3–5 h	Adjunct mydriatic with no cycloplegia; also used to differentiate conjunctival from deep episcleral injection and to test Müller muscle in ptosis. Avoid 10% in cardiovascular disease and in infants.
Cyclopentolate 0.5–2%	20–30 min	30–60 min	6–24 h (cyclo 6–24 h)	The workhorse cycloplegic refraction agent. Use 0.5% in infants. Can cause CNS effects (ataxia, hallucinations, disorientation), especially at 2% and in children.
Homatropine 2–5%	30–60 min	30–90 min	1–3 days	Uveitis: comfort and synechiae prevention with a moderate duration.
Scopolamine 0.25%	20–30 min	30–60 min	3–7 days	Uveitis. More CNS penetration than atropine; can cause confusion in the elderly.

Agent	Mydriasis onset	Cycloplegia peak	Duration	Clinical role
Atropine 0.5–1%	30–40 min	60–180 min	7–14 days	Strongest cycloplegic; used for severe uveitis, amblyopia penalisation, and cycloplegic refraction in dark irides or accommodative esotropia. Systemic toxicity is a real risk in children: "hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter". Treat with physostigmine.
Atropine 0.01–0.05%	minimal	minimal	-	Myopia control. 0.05% is more effective than 0.01% but with more photophobia and near blur; rebound is greater at higher concentrations.

DILATION RISK, AND THE TWO QUESTIONS THAT ACTUALLY MATTER

The risk of precipitating angle closure with routine dilation is very low (well under **1 in 1000**). The **relevant assessments** are the van Herick or gonioscopic angle grade and the presence of prior symptoms, *not* a history of "glaucoma", since open-angle glaucoma is not a contraindication. Never withhold a dilated fundus examination from a diabetic or a patient with flashes and floaters because of unfounded closure anxiety; assess the angle, then dilate.

Genuine cautions: a very narrow or occludable angle without a patent iridotomy; known plateau iris; an anteriorly subluxated lens; an iris-supported anterior chamber IOL; and cardiac disease with phenylephrine **10%**.

Topical anaesthetics

LOCAL ANAESTHETICS

Agent	Class	Onset	Duration	Notes
Proparacaine 0.5%	Ester	13–30 s	15–20 min	Least stinging; the routine choice. Refrigerate.
Tetracaine 0.5%	Ester	10–20 s	15–20 min	Stings more, slightly longer and deeper effect; more epithelial toxicity.
Benoxinate 0.4% with fluorescein	Ester	~15 s	15–20 min	Combination product for applanation tonometry.
Lidocaine 2–4% (gel or injection)	Amide	1–3 min	30–60 min	Used for lid procedures and intracameral anaesthesia; amides do not cross-react with ester allergy.
Bupivacaine	Amide	5–10 min	4–8 h	Long-acting; cardiotoxic in overdose.

- **Mechanism:** reversible blockade of voltage-gated **sodium channels** from the cytoplasmic side, preventing depolarisation. The unionised form crosses the membrane, the ionised form blocks, which is why anaesthetics work poorly in acidic inflamed tissue.
- **Esters** are metabolised by plasma pseudocholinesterase to **PABA**, the allergen in the rare true anaesthetic allergy; **amides** are metabolised hepatically and do not produce PABA. Remember the mnemonic that **amides contain an "i" in the prefix** (lidocaine, bupivacaine, mepivacaine).

- **Never dispense topical anaesthetics.** Repeated use produces epithelial toxicity, delayed healing, neurotrophic ulceration, ring infiltrates, and permanent scarring. This is an examinable professional standard, not merely a caution.

Glaucoma pharmacology

EVERY CLASS: MECHANISM, MAGNITUDE, DOSING, ADVERSE EFFECTS, CONTRAINDICATIONS

Class	Mechanism (Goldmann term)	IOP drop	Adverse effects and cautions
Prostaglandin analogues latanoprost, travoprost, tafluprost 0.0015–0.004%; bimatoprost 0.01–0.03%	Increase uveoscleral outflow (U) by matrix metalloproteinase-mediated remodelling; bimatoprost also increases trabecular outflow	25–33%	First-line for most patients. Once nightly. Conjunctival hyperaemia, iris and periocular skin hyperpigmentation, eyelash growth (hypertrichosis), prostaglandin-associated periorbitopathy (deepened sulcus, upper lid retraction, relative enophthalmos), reactivation of herpetic keratitis, cystoid macular oedema and anterior uveitis in predisposed eyes (aphakia, ruptured capsule). Retain nocturnal efficacy. Latanoprostene bunod adds nitric oxide donation to increase trabecular outflow.
β-blockers timolol 0.25–0.5%, levobunolol, carteolol, betaxolol	Reduce aqueous formation (F) via reduced cAMP in ciliary epithelium	20–25%	Systemic: bronchospasm (contraindicated in asthma and COPD), bradycardia and heart block, masking of hypoglycaemic warning signs in insulin-treated diabetes , fatigue, depression, reduced exercise tolerance, impotence, and worsened lipid profile. Ocular: reduced tear production, corneal anaesthesia, allergy. Betaxolol is β_1-selective and safer in reactive airway disease, though not risk-free, and may have better ocular blood flow effects. Little nocturnal effect because aqueous production already falls at night. Tachyphylaxis and short-term escape occur.
α_2-agonists brimonidine 0.1–0.2%, apraclonidine 0.5–1%	Reduce formation (F) and increase uveoscleral outflow (U)	20–25%	Ocular: follicular allergic conjunctivitis (up to 15% with brimonidine, higher with apraclonidine), dry mouth, conjunctival blanching then rebound hyperaemia, lid retraction. Systemic: sedation, hypotension, and in infants and small children apnoea, bradycardia, and coma. Contraindicated under age 2 and avoided in young children generally. Contraindicated with monoamine oxidase inhibitors. Apraclonidine shows marked tachyphylaxis , so it is used for short-term prophylaxis around laser procedures.
Carbonic anhydrase inhibitors, topical dorzolamide 2%, brinzolamide 1%	Reduce formation (F) by inhibiting carbonic anhydrase II	15–20%	Bitter taste, stinging (dorzolamide is acidic; brinzolamide is a better-tolerated suspension), transient blur, superficial punctate keratopathy. Sulfonamide moiety: avoid in true sulfa allergy, and use caution with corneal endothelial compromise, where they may precipitate oedema.

Class	Mechanism (Goldmann term)	IOP drop	Adverse effects and cautions
Carbonic anhydrase inhibitors, oral acetazolamide 250–500 mg, methazolamide 25–50 mg	Reduce formation (F) substantially	30–40%	Paraesthesia of fingers and toes, malaise and fatigue, metallic taste (especially with carbonated drinks), anorexia and weight loss, metabolic acidosis, hypokalaemia, renal stones , and rarely aplastic anaemia and Stevens-Johnson syndrome. Avoid in sickle cell disease (acidosis promotes sickling), renal or hepatic failure, and with high-dose aspirin (salicylate toxicity) . Methazolamide is less renally cleared. Also used for idiopathic intracranial hypertension and macular oedema in retinitis pigmentosa.
Cholinergic agonists pilocarpine 1–4%, carbachol 0.75–3%	Increase trabecular outflow (raise C) by ciliary muscle traction on the scleral spur	15–25%	Brow ache, accommodative spasm and induced myopia , miosis with dim vision and reduced field, increased risk of retinal detachment and of posterior synechiae, cataract acceleration. Retains a specific role in acute angle closure (once IOP is lowered enough for the sphincter to respond), in pilocarpine 0.125% for Adie pupil testing, and as pilocarpine 1.25% for presbyopia. Contraindicated in uveitic and neovascular glaucoma.
Rho kinase inhibitor netarsudil 0.02%	Raises C and lowers P_v ; relaxes the meshwork cytoskeleton	~20%	Conjunctival hyperaemia in a majority, conjunctival haemorrhage, cornea verticillata (reversible, non-visually significant), and corneal oedema in compromised endothelium.
Hyperosmotics oral glycerin, isosorbide; IV mannitol 20%	Draw water from the vitreous, reducing volume	acute	Acute angle closure and pre-operative use. Glycerin raises blood glucose (avoid in diabetes); mannitol causes volume overload (avoid in heart or renal failure) .
Fixed combinations	–	–	Dorzolamide-timolol, brinzolamide-brimonidine, brimonidine-timolol, latanoprost-timolol, netarsudil-latanoprost. Improve adherence and reduce preservative and washout exposure.

ACUTE PRIMARY ANGLE CLOSURE: THE SEQUENCE

- 1 **Topical aqueous suppressants immediately:** a β -blocker, an α_2 -agonist, and a topical carbonic anhydrase inhibitor, one drop each.
- 2 **Oral or intravenous acetazolamide (500 mg)**, assuming no sickle cell disease or sulfa allergy.
- 3 **Topical steroid** to reduce inflammation and improve corneal clarity.
- 4 **Pilocarpine 1-2% only once IOP has fallen below roughly 40-50 mmHg**, because a severely ischaemic iris sphincter will not respond and pilocarpine can worsen the situation by increasing pupillary block through anterior lens movement.
- 5 **Corneal indentation** (with a four-mirror gonioscopes) to force aqueous into the angle, and supine positioning to let the lens fall back.
- 6 **Hyperosmotic agent** if still uncontrolled.
- 7 **Definitive treatment: laser peripheral iridotomy** in the affected eye and **prophylactically in the fellow eye**, because the fellow eye has roughly a **50%** risk over five years.

Anti-infective agents**ANTIBACTERIALS: CLASS, MECHANISM, SPECTRUM, OCULAR USE**

Class / agent	Mechanism	Spectrum and ocular role; toxicity
Fluoroquinolones ciprofloxacin, ofloxacin (2nd); levofloxacin (3rd); moxifloxacin, gatifloxacin, besifloxacin (4th)	Inhibit DNA gyrase (topoisomerase II) and topoisomerase IV; bactericidal	Broad gram-negative with good gram-positive coverage in the fourth generation, which also penetrates the cornea best. First-line for bacterial keratitis monotherapy and for prophylaxis. Ciprofloxacin can precipitate as a white crystalline deposit in an epithelial defect. Systemic fluoroquinolones carry tendon rupture, aortic dissection, peripheral neuropathy, and QT prolongation warnings; avoid in children and pregnancy where alternatives exist. Rising resistance, especially MRSA.
Aminoglycosides tobramycin, gentamicin 0.3%; fortified 9–14 mg/mL	Bind the 30S ribosomal subunit, irreversibly; bactericidal; require oxygen for uptake	Excellent gram-negative including <i>Pseudomonas</i> ; poor streptococcal and anaerobic coverage. Marked epithelial toxicity with frequent or fortified dosing; systemically nephrotoxic and ototoxic (vestibular for gentamicin, cochlear for amikacin). Gentamicin is more toxic to the surface than tobramycin.
Macrolides erythromycin ointment, azithromycin 1% drops or oral	Bind the 50S subunit, inhibiting translocation; bacteriostatic	Gram-positive, atypicals, and <i>Chlamydia</i> . Erythromycin ointment for neonatal prophylaxis and blepharitis; azithromycin has anti-inflammatory effects on meibomian glands and is the treatment of choice for adult inclusion conjunctivitis as a single oral 1 g dose (with partner treatment). Watch for QT prolongation.

Class / agent	Mechanism	Spectrum and ocular role; toxicity
Tetracyclines doxycycline 50–100 mg, minocycline	Bind the 30S subunit; bacteriostatic. Also inhibit matrix metalloproteinases and lipase independently of any antibacterial effect	The MMP effect is why doxycycline is used in meibomian gland dysfunction, ocular rosacea, recurrent erosion, and corneal melting , often at sub-antimicrobial doses. Contraindicated under age 8 and in pregnancy (tooth staining and enamel hypoplasia). Causes photosensitivity, oesophagitis, and, notably, pseudotumour cerebri (minocycline in particular). Chelated by dairy, antacids, and iron.
β-lactams penicillins, cephalosporins (cefazolin fortified 50 mg/mL), ceftriaxone	Inhibit peptidoglycan cross-linking (transpeptidase); bactericidal	Fortified cefazolin covers gram-positives including streptococci; ceftriaxone 1 g IM or IV is the treatment for gonococcal conjunctivitis and keratitis (with azithromycin for co-infection). Amoxicillin-clavulanate for preseptal cellulitis. Cross-reactivity between penicillins and cephalosporins is lower than historically believed.
Vancomycin fortified 25–50 mg/mL topical; intravitreal 1 mg	Inhibits cell wall synthesis at a different site from β-lactams	Gram-positives including MRSA and enterococci; no gram-negative activity , so it is paired with ceftazidime or amikacin for endophthalmitis. Not used for prophylaxis, to preserve it.
Polymyxin B / bacitracin / neomycin combinations	Polymyxin disrupts the outer membrane; bacitracin blocks cell wall synthesis; neomycin is an aminoglycoside	Broad topical cover for blepharoconjunctivitis. Neomycin is a common cause of contact (type IV) dermatoblepharitis ; if a patient's lids worsen on the drop, suspect it.
Trimethoprim-sulfamethoxazole	Sequential folate pathway blockade	Oral option for MRSA lid disease and preseptal cellulitis; also an alternative regimen for ocular toxoplasmosis. Sulfa allergy and Stevens-Johnson risk.
Chloramphenicol	50S subunit	Broad, excellent penetration; widely used outside the United States. Rare but fatal aplastic anaemia limits systemic use.

ANTIVIRALS, ANTIFUNGALS, AND ANTIPARASITICS

Agent	Mechanism and use
Aciclovir / valaciclovir / famciclovir	Activated by viral thymidine kinase , then inhibit viral DNA polymerase: this selectivity is why they are so well tolerated. Oral aciclovir 400 mg five times daily for HSV epithelial keratitis, 400 mg twice daily for prophylaxis (halves recurrence, per HEDS). Zoster requires higher doses: aciclovir 800 mg five times daily or valaciclovir 1 g three times daily for 7 days, ideally started within 72 hours . Reduce dose in renal impairment.
Ganciclovir 0.15% gel; trifluridine 1%	Topical for HSV epithelial keratitis. Ganciclovir gel is far better tolerated; trifluridine is a thymidine analogue with substantial epithelial toxicity requiring nine-times-daily dosing.
Ganciclovir / valganciclovir / foscarnet / cidofovir	CMV retinitis. Ganciclovir causes neutropenia; foscarnet does not require viral kinase activation (so it works in thymidine-kinase-deficient strains) but is nephrotoxic and causes electrolyte disturbance; cidofovir causes nephrotoxicity and hypotony with uveitis.
Natamycin 5%	Polyene binding ergosterol. First-line for filamentous fungal keratitis , with better outcomes than voriconazole for <i>Fusarium</i> in the Mycotic Ulcer Treatment Trial. Poor penetration of intact epithelium, so debridement helps.

Agent	Mechanism and use
Amphotericin B 0.15%	Polyene; the choice for yeast (<i>Candida</i>) keratitis and used intravitreally. Nephrotoxic systemically.
Voriconazole / fluconazole	Triazoles inhibiting fungal cytochrome P450 (14- α -demethylase). Voriconazole penetrates the eye well orally and topically; watch hepatotoxicity, visual disturbance, and photosensitivity.
Polyhexamethylene biguanide, chlorhexidine, propamidine, hexamidine	<i>Acanthamoeba</i> : biguanides are cysticidal; combine with a diamidine and treat for months.
Pyrimethamine + sulfadiazine + folinic acid; or trimethoprim-sulfamethoxazole; or clindamycin	Ocular toxoplasmosis. Folinic acid is mandatory with pyrimethamine to prevent bone marrow suppression. Add oral steroid 24 to 48 hours after starting antimicrobials for sight-threatening lesions.
Ivermectin	Onchocerciasis and <i>Demodex</i> -associated disease; lotilaner 0.25% is now approved topically for <i>Demodex</i> blepharitis.

Anti-inflammatory and immunomodulatory agents

TOPICAL CORTICOSTEROIDS: POTENCY, PENETRATION, AND PRESSURE RISK

Agent	Relative strength	Notes
Difluprednate 0.05% emulsion	Very high	Excellent penetration; four times daily replaces prednisolone eight times daily in many uveitis protocols. Highest IOP-elevation risk , particularly in children.
Prednisolone acetate 1% (suspension)	High	The intraocular workhorse. Must be shaken : unshaken suspension delivers erratic doses. The <i>acetate</i> is lipophilic and penetrates the intact cornea; the <i>phosphate</i> salt is hydrophilic and penetrates poorly through intact epithelium but well through a defect.
Dexamethasone 0.1%	High	Potent; higher IOP risk than prednisolone in some series.
Loteprednol etabonate 0.2–0.5%	Moderate	Ester (not ketone) steroid : hydrolysed to inactive metabolites in the eye, giving a markedly lower IOP-elevation risk . Good for surface disease, allergy, and steroid responders.
Fluorometholone 0.1–0.25%	Moderate	Good surface penetration with limited intraocular penetration, so a lower pressure risk; a common choice for episcleritis and surface inflammation.
Rimexolone 1%	Moderate	Comparable efficacy to prednisolone in anterior uveitis with somewhat less pressure effect.
Hydrocortisone, medrysone	Low	Weak; largely historical.

STEROID MECHANISM AND THE FOUR CONSEQUENCES YOU MUST MONITOR

Mechanism: glucocorticoids induce **lipocortin (annexin A1)**, which inhibits **phospholipase A₂**, blocking arachidonic acid release and therefore *both* the cyclooxygenase (prostaglandins) and lipoxygenase (leukotrienes) pathways. NSAIDs block only cyclooxygenase, which is why steroids are broader and why NSAIDs never fully substitute.

1. **Raised IOP.** About **5%** of the general population are high responders (a rise of more than **15 mmHg**) and roughly a third are moderate responders; **the rate is much higher in patients with primary open-angle glaucoma, in first-degree relatives, in high myopes, in diabetics, and in children.** Onset is usually 2 to 6 weeks but can be days with potent agents. Check pressure at every steroid visit.
2. **Posterior subcapsular cataract**, dose- and duration-dependent, and not reversible.
3. **Infection:** potentiation of herpes simplex epithelial keratitis (a **contraindication** without antiviral cover), fungal keratitis, and *Acanthamoeba*; delayed epithelial and stromal healing with increased melting risk.
4. **Systemic effects** even from topical or periocular use in children: adrenal suppression and growth retardation with prolonged high-dose therapy.

Always taper after more than about two weeks of use, and never stop potent steroid abruptly in uveitis: rebound inflammation is common.

NSAIDS, IMMUNOMODULATORS, AND BIOLOGICS

Agent	Mechanism and indication
Ketorolac 0.4–0.5%, bromfenac 0.07–0.09%, nepafenac 0.1–0.3%, diclofenac 0.1%, flurbiprofen 0.03%	Cyclooxygenase inhibition. Uses: postoperative pain and inflammation, prevention and treatment of pseudophakic cystoid macular oedema , allergic itch (ketorolac), pain in corneal abrasion, and intraoperative miosis prevention (flurbiprofen). Risk: corneal melting and sterile ulceration, especially with generic diclofenac, in dry eye, in diabetics, and after surgery. Nepafenac is a prodrug with good posterior penetration.
Cyclosporine A 0.05%, 0.09%	Calcineurin inhibitor blocking T cell IL-2 transcription. Indicated for aqueous-deficient dry eye with an inflammatory component, vernal and atopic keratoconjunctivitis, and to reduce steroid dependence. Takes 3–6 months for full effect; burning is the main limiting side effect.
Lifitegrast 5%	LFA-1 antagonist blocking T cell adhesion to ICAM-1. Faster onset than cyclosporine for dry eye symptoms; dysgeusia (altered taste) is characteristic.
Varenicline nasal spray; perfluorohexyloctane	Trigeminal parasympathetic pathway stimulation of basal tear secretion; and an evaporation-limiting semifluorinated alkane for meibomian-driven dry eye.
Systemic immunomodulators	Methotrexate (dihydrofolate reductase inhibition; give folic acid; monitor liver and marrow), mycophenolate mofetil (inosine monophosphate dehydrogenase), azathioprine (purine synthesis; check TPMT), cyclosporine and tacrolimus (calcineurin; nephrotoxic, hypertensive), cyclophosphamide (alkylating; haemorrhagic cystitis and malignancy risk, reserved for ANCA vasculitis and necrotising scleritis).
Biologics	Adalimumab and infliximab (anti-TNF- α ; adalimumab is approved for non-infectious intermediate, posterior, and panuveitis, and is first-line biologic in Behçet and JIA-associated uveitis). Screen for latent tuberculosis and hepatitis B before starting. Etanercept is not effective for uveitis and can paradoxically induce it. Rituximab (anti-CD20) for granulomatosis with polyangiitis and ocular mucous membrane pemphigoid; tocilizumab (anti-IL-6) for giant cell arteritis, thyroid eye disease, and refractory uveitic macular oedema; teprotumumab (IGF-1R) for thyroid eye disease, with hearing loss and hyperglycaemia as key risks.

Agent	Mechanism and indication
Anti-VEGF intravitreal	Bevacizumab (full antibody, off-label), ranibizumab (Fab fragment), aflibercept (decoy receptor for VEGF-A, VEGF-B, and placental growth factor), brolucizumab (single-chain fragment; higher intraocular inflammation rate), faricimab (bispecific VEGF-A and angiopoietin-2). Indications: neovascular AMD, diabetic macular oedema, proliferative diabetic retinopathy, retinal vein occlusion macular oedema, myopic CNV, ROP. Risks: endophthalmitis (about 1 in 2000–5000 injections), IOP spikes, retinal tear, and a small theoretical thromboembolic signal. Pegcetacoplan and avacincaptad pegol are complement inhibitors for geographic atrophy.

ANTI-ALLERGY AGENTS

Class	Agents and role
Dual-acting antihistamine and mast cell stabiliser	Olopatadine, ketotifen, alcaftadine, bepotastine, epinastine, azelastine. First-line for allergic conjunctivitis: rapid itch relief plus prophylaxis. Once or twice daily.
Pure mast cell stabilisers	Cromolyn sodium, lodoxamide, nedocromil, pemirolast. Prophylactic only; require a loading period of days to weeks. Useful in vernal and atopic disease as maintenance.
Antihistamine plus vasoconstrictor	Pheniramine with naphazoline (over-the-counter). Effective short term; rebound hyperaemia (conjunctivitis medicamentosa) with chronic use.
Topical NSAID	Ketorolac for itch.
Topical steroid	Loteprednol for severe or vernal disease, in short pulses.
Topical cyclosporine or tacrolimus	Steroid-sparing therapy for vernal and atopic keratoconjunctivitis with shield ulcers.

Systemic drugs with ocular toxicity

THE TABLE TO MEMORISE: DRUG, FINDING, MONITORING

Drug	Ocular finding	Monitoring and management
Hydroxychloroquine / chloroquine	Bull's-eye maculopathy with parafoveal RPE and photoreceptor loss; in Asian patients the pattern is often pericentral rather than parafoveal. Also whorl keratopathy (chloroquine).	Risk rises sharply above 5.0 mg/kg/day of real body weight and after 5 years . Baseline examination within a year, then annual screening after five years (sooner with renal impairment, tamoxifen co-use, or high dose). Screen with automated visual fields plus SD-OCT , adding multifocal ERG and fundus autofluorescence where available. Damage is irreversible and can progress after cessation.
Ethambutol	Dose-dependent optic neuropathy : bilateral, symmetric, centrocaecal scotomas with red-green dyschromatopsia.	Risk above 15–25 mg/kg/day and with renal impairment. Baseline and regular acuity, colour vision, and fields. Stop the drug immediately ; recovery is possible but incomplete.

Drug	Ocular finding	Monitoring and management
Amiodarone	Cornea verticillata (near-universal, benign, dose-related) and rarely a distinct optic neuropathy with insidious bilateral disc swelling.	Verticillata needs no action. Optic neuropathy requires discussion with cardiology, since alternatives may be limited.
Tamoxifen	Refractile crystalline retinal deposits, cystoid macular oedema, and whorl keratopathy.	Usually with cumulative high dose; annual dilated examination.
Topiramate	Acute bilateral angle closure with myopic shift from ciliochoroidal effusion, typically within the first weeks. Iridotomy does not help.	Stop the drug, give cycloplegia and aqueous suppressants and topical steroid. Also occurs with sulfonamides, acetazolamide, and some antidepressants.
Tamsulosin and other α_1-blockers	Intraoperative floppy iris syndrome: iris billowing, prolapse, and progressive pupil constriction.	Ask every cataract surgery candidate. Effects persist for years after stopping, so cessation is not protective; inform the surgeon.
Corticosteroids (any route)	Raised IOP, posterior subcapsular cataract, and central serous chorioretinopathy.	Check IOP; taper; consider loteprednol or a steroid-sparing agent.
Sildenafil and other PDE5 inhibitors	Transient blue tinge and light sensitivity from PDE6 cross-inhibition; reported association with NAION.	Caution in retinitis pigmentosa; counsel about NAION symptoms.
Digoxin	Xanthopsia (yellow vision), snowy vision, blurred vision, and photopsia.	A sign of digoxin toxicity: check the level.
Phenothiazines: thioridazine, chlorpromazine	Thioridazine causes a pigmentary retinopathy (nummular then diffuse); chlorpromazine causes anterior lens and corneal endothelial pigment deposits .	Dose-dependent; thioridazine retinopathy can progress after stopping.

Drug	Ocular finding	Monitoring and management
Vigabatrin	Irreversible bilateral concentric visual field constriction in up to a third of users.	Mandatory field testing before and during treatment; a black box warning.
Pentosan polysulfate sodium	A distinctive pigmentary maculopathy with hyperpigmented spots and RPE changes.	Baseline and periodic macular imaging (OCT and autofluorescence) in long-term users for interstitial cystitis.
Isotretinoin	Severe dry eye, blepharoconjunctivitis, reduced dark adaptation, corneal opacities, and pseudotumour cerebri (especially with a tetracycline).	Never combine with a tetracycline. Warn contact lens wearers.
Tetracyclines and minocycline	Pseudotumour cerebri ; minocycline also causes blue-grey scleral and conjunctival pigmentation.	Examine discs if headache develops.
Bisphosphonates	Anterior uveitis, scleritis, and conjunctivitis.	Consider the drug in unexplained bilateral uveitis or scleritis.
Fingolimod	Macular oedema, typically in the first three to four months.	Baseline and 3 to 4 month OCT; higher risk in diabetics and with prior uveitis.
MEK inhibitors; taxanes; EGFR inhibitors	MEK: serous retinal detachments. Taxanes: cystoid macular oedema without leakage. EGFR (cetuximab, erlotinib): trichomegaly, dry eye, corneal erosions.	OCT surveillance and ocular surface support; coordinate with oncology.
Immune checkpoint inhibitors	Uveitis, dry eye, VKH-like panuveitis, thyroid eye disease, and myasthenia-like syndromes.	Emerging class effect; coordinate with oncology and treat aggressively.
Anticholinergics and antihistamines	Reduced accommodation, dry eye, mydriasis with rare angle closure.	Review the whole medication list in any presbyopia-onset or dry eye complaint.

Drug	Ocular finding	Monitoring and management
Ethanol, isoniazid, linezolid, methanol	Toxic optic neuropathy with centrocaecal scotomas; methanol causes acute blindness with formic acid accumulation.	B vitamins, remove the toxin; methanol needs urgent fomepizole or ethanol and dialysis.

PREGNANCY AND PAEDIATRIC PRESCRIBING

- The FDA letter categories have been replaced by narrative labelling, but the practical rules persist: **prefer topical over systemic, use punctal occlusion, and choose the oldest well-characterised agent.**
- **Avoid in pregnancy:** tetracyclines, fluoroquinolones where alternatives exist, sulfonamides near term, carbonic anhydrase inhibitors (teratogenic in animals), and brimonidine near delivery. Erythromycin, azithromycin, and β -lactams are the safer antibacterials.
- **Prostaglandin analogues** are theoretically capable of inducing uterine contraction and are usually avoided; β -blockers are commonly used with punctal occlusion but cross the placenta and enter breast milk.
- **Paediatrics:** **brimonidine is contraindicated under age 2** and avoided in young children (apnoea and CNS depression). Use cyclopentolate **0.5%** in infants, avoid phenylephrine **10%**, and remember that dose per kilogram from a standard drop is far higher in a neonate than in an adult.

NUMBERS TO KNOW: PHARMACOLOGY

Drop volume	35–50 μ L	Tear film capacity	7–9 μ L (max ~30)
Topical bioavailability	<5%	Punctal occlusion time	2–3 min
Prostaglandin IOP reduction	25–33%	β -blocker reduction	20–25%
Topical CAI reduction	15–20%	Oral CAI reduction	30–40%
Steroid high responders	~5%	Steroid response onset	2–6 weeks
Tropicamide duration	4–6 h	Atropine duration	7–14 days
Cyclopentolate cycloplegia peak	30–60 min	Proparacaine onset	13–30 s
Aciclovir HSV treatment dose	400 mg 5 $\times$ /day	HSV prophylaxis	400 mg 2 $\times$ /day
Zoster valaciclovir	1 g 3 $\times$ /day $\times$ 7 d	Antiviral window in zoster	$\leq$ 72 h
Hydroxychloroquine risk dose	>5.0 mg/kg/day	Screening after	5 years
Ethambutol risk dose	>15–25 mg/kg/day	Acetazolamide acute dose	500 mg
Intravitreal vancomycin	1 mg / 0.1 mL	Post-injection endophthalmitis	~1 in 2000–5000

Azithromycin for
chlamydia

1 g single oral dose

Ceftriaxone for
gonorrhoea

1 g IM/IV

RETRIEVAL PRACTICE

1. Name the receptor, agonist, and antagonist for the iris sphincter and for the dilator.
2. Why does a topical β -blocker cause bronchospasm, and how do you reduce that risk?
3. Which glaucoma drug raises trabecular outflow, and which lowers episcleral venous pressure?
4. Give three absolute or near-absolute cautions for oral acetazolamide.
5. Why is brimonidine contraindicated in infants?
6. State the sequence of management for acute angle closure and explain when pilocarpine is added.
7. Explain the mechanism of steroid anti-inflammatory action and why NSAIDs are narrower.
8. Which topical steroid has the lowest IOP risk and why?
9. Prednisolone acetate versus phosphate: which penetrates an intact cornea, and why?
10. Which antibiotic class inhibits matrix metalloproteinases, and name three ocular uses of that property.
11. A patient on topiramate presents with bilateral acute angle closure and a myopic shift. What is the mechanism and why is iridotomy unhelpful?
12. Give the hydroxychloroquine dose threshold, the screening interval, and the two core screening tests.
13. Which drug causes irreversible concentric field constriction, and what is required before prescribing?
14. A patient reports yellow vision and nausea. Which drug and what do you check?
15. Why must folinic acid accompany pyrimethamine?
16. Name the first-line antifungal for *Fusarium* keratitis and the trial supporting it.

KEY

1. Sphincter: M_3 ; agonist pilocarpine or carbachol; antagonist atropine, cyclopentolate, tropicamide. Dilator: α_1 ; agonist phenylephrine; antagonist dapiprazole or thymoxamine (and systemically tamsulosin).
2. Nasolacrimal drainage delivers drug to highly vascular nasal mucosa, bypassing hepatic first-pass metabolism. Reduce it with one drop only, punctal occlusion or gentle lid closure for 2 to 3 minutes, the lowest concentration, and a β_1 -selective agent such as betaxolol (or avoid the class entirely).
3. Pilocarpine (and laser trabeculoplasty and netarsudil) raise trabecular outflow; **netarsudil** lowers episcleral venous pressure.
4. Sick cell disease (acidosis promotes sickling), significant renal or hepatic failure, and true sulfonamide allergy; also avoid with high-dose salicylates.
5. It crosses an immature blood-brain barrier and causes apnoea, bradycardia, hypotension, hypothermia, and coma; contraindicated under age 2.
6. Topical β -blocker, α_2 -agonist, and CAI, plus oral or IV acetazolamide and topical steroid; corneal indentation and supine positioning; hyperosmotic if needed; then laser iridotomy in both eyes. Pilocarpine only after IOP falls below roughly 40 to 50 mmHg, because an ischaemic sphincter will not respond and it can worsen pupillary block by moving the lens forward.
7. Steroids induce lipocortin, which inhibits phospholipase A_2 , blocking arachidonic acid release and therefore both prostaglandin and leukotriene production. NSAIDs inhibit only cyclooxygenase, leaving leukotrienes intact.
8. Loteprednol etabonate, because it is an ester steroid that is rapidly hydrolysed to inactive metabolites within the eye.
9. The **acetate**, because it is lipophilic and can cross the lipophilic epithelium; the hydrophilic phosphate penetrates well only through an epithelial defect.
10. Tetracyclines, particularly doxycycline: meibomian gland dysfunction and ocular rosacea, recurrent corneal erosion, and corneal melting or peripheral ulcerative keratitis.
11. Ciliochoroidal effusion with forward rotation of the ciliary body and anterior lens displacement, producing secondary angle closure and a myopic shift. There is no pupillary block, so an iridotomy does

- not address the mechanism; stop the drug and give cycloplegia, aqueous suppressants, and steroid.
12. Above 5.0 mg/kg/day of real body weight; baseline within a year then annual screening after five years; automated visual fields and SD-OCT (adding multifocal ERG and fundus autofluorescence where available).
 13. Vigabatrin; documented baseline visual field testing and informed consent, with periodic monitoring, because the constriction is irreversible.
 14. Digoxin, causing xanthopsia; check the serum digoxin level and renal function, and consider toxicity.
 15. Pyrimethamine inhibits dihydrofolate reductase and causes bone marrow suppression; folinic acid (leucovorin) rescues host haematopoiesis without rescuing the parasite.
 16. Natamycin 5%; the Mycotic Ulcer Treatment Trial showed better outcomes than voriconazole, particularly for *Fusarium*.

CHAPTER RECAP

1. Less than 5% of a drop gets in; the rest drains to nasal mucosa and bypasses first-pass metabolism.
2. Build the autonomic table once and every mydriatic, miotic, and glaucoma item becomes derivable.
3. Tropicamide dilates but barely cyclopleges; cyclopentolate is the refraction agent; atropine lasts two weeks.
4. Prostaglandins raise U by 25 to 33%; β -blockers and α_2 -agonists and CAIs lower F; pilocarpine and netarsudil raise C.
5. Acute angle closure: suppress, steroid, indent, position, then pilocarpine, then iridotomy both eyes.
6. Steroids block phospholipase A₂, so both prostaglandins and leukotrienes; watch pressure, cataract, infection, and healing.
7. Loteprednol and fluorometholone for pressure-sensitive eyes; prednisolone acetate must be shaken.
8. Fourth-generation fluoroquinolone for keratitis; ceftriaxone for gonorrhoea; azithromycin 1 g for chlamydia; natamycin for *Fusarium*; biguanide plus diamidine for *Acanthamoeba*.
9. Doxycycline for MMP inhibition, not just for bacteria.
10. Learn the systemic toxicity table as drug plus finding plus monitoring; hydroxychloroquine, ethambutol, vigabatrin, and topiramate are the four that carry formal protocols.

Systemic disease and its ocular expression

Systemic health · 55–75 items: the largest single area

Emergencies / trauma (embedded)

Every ocular tissue area

ANSWER THESE COLD BEFORE READING

1. Give the diagnostic criteria for diabetes mellitus and the target HgbA1c, and summarise DCCT and UKPDS.
2. Classify diabetic retinopathy and diabetic macular oedema and give the treatment threshold for each stage.
3. Give the hypertensive retinopathy grades and the blood pressure definition of a hypertensive emergency.
4. List the ACR criteria features of giant cell arteritis and the immediate management.
5. Match each of the eight phakomatoses to its gene, inheritance, and ocular sign.
6. Define sensitivity, specificity, PPV, and likelihood ratio, and state which depend on prevalence.

BLUEPRINT WEIGHT

19%

55–75 scored items, roughly one in five. Candidates who under-prepare this area lose more points here than anywhere else.

Diabetes mellitus

DIAGNOSIS AND CONTROL TARGETS

Test	Normal	Prediabetes	Diabetes
Fasting plasma glucose	<100 mg/dL	100–125	≥126
2-hour oral glucose tolerance	<140 mg/dL	140–199	≥200
HgbA1c	<5.7%	5.7–6.4%	≥6.5%
Random glucose with symptoms	—	—	≥200

- **HgbA1c** reflects mean glycaemia over the preceding **2–3 months** (erythrocyte lifespan). General target **<7%**, individualised. Roughly, an A1c of **6%** corresponds to a mean glucose of **126 mg/dL**, and each **1%** rise adds about **29 mg/dL**. It is falsely low in haemolysis, recent transfusion, and pregnancy, and falsely high in iron deficiency and uraemia. **Part II PAM explicitly expects you to interpret an A1c without being given a reference range**, and the same expectation applies on Part I.
- **DCCT** (type 1) and **UKPDS** (type 2): intensive glycaemic control substantially reduces the incidence and progression of retinopathy, nephropathy, and neuropathy. UKPDS additionally showed that **tight blood pressure control independently reduces retinopathy progression and visual loss**. Note the **early worsening** phenomenon: a rapid improvement in control can transiently worsen retinopathy over 6 to 12 months, so patients with significant retinopathy who suddenly gain control need closer follow-up.

DIABETIC RETINOPATHY: CLASSIFICATION AND MANAGEMENT

Stage	Findings	Management and follow-up
No apparent retinopathy	None	Dilated examination annually (every 1 to 2 years if repeatedly normal in type 2 with good control).
Mild NPDR	Microaneurysms only	Annually; optimise systemic control.
Moderate NPDR	More than microaneurysms but less than severe	Every 6–12 months .
Severe NPDR	4-2-1 rule: severe haemorrhages in 4 quadrants, <i>or</i> venous beading in 2 quadrants, <i>or</i> IRMA in 1 quadrant	Every 3–4 months . Roughly half progress to proliferative disease within a year. Consider early panretinal photocoagulation or anti-VEGF in type 2 diabetes, in poor compliance, in pregnancy, or before cataract surgery.
PDR without high-risk characteristics	Neovascularisation of the disc or elsewhere	Every 2–3 months ; treat if progressing.
High-risk PDR	NVD greater than 1/3 disc area; <i>any</i> NVD with vitreous or preretinal haemorrhage; NVE greater than 1/2 disc area with haemorrhage	Treat now: panretinal photocoagulation (DRS showed a roughly 50% reduction in severe visual loss) and/or anti-VEGF. Vitrectomy for non-clearing haemorrhage or tractional detachment threatening the macula.
Diabetic macular oedema, non-centre-involving	Thickening not involving the central subfield	Observe or focal laser; monitor with OCT.
Centre-involving DMO	Central subfield thickening	Intravitreal anti-VEGF is first-line (DRCR Protocol T: aflibercept was superior at 1 year when baseline acuity was 20/50 or worse; all agents were similar with better baseline acuity). Steroid implants for pseudophakic or anti-VEGF-refractory eyes. Focal laser adjunctively.

DIABETES BEYOND THE RETINA

- **Refractive shifts** with glycaemic swings (index myopia with hyperglycaemia; hyperopic shift as control improves): **do not prescribe glasses during a period of unstable control.**
- **Reduced accommodation** and earlier presbyopia.
- **Cranial neuropathies:** pupil-sparing third nerve palsy, sixth and fourth nerve palsies, usually resolving in 8 to 12 weeks.
- **Corneal:** reduced sensation and neurotrophic keratopathy, recurrent erosion from abnormal basement membrane adhesion, delayed healing, increased endothelial polymegathism.
- **Iris:** poor dilation, pigment epithelial vacuolation (lacy vacuolisation), rubeosis.
- **Cataract:** earlier nuclear and cortical cataract, plus true snowflake cataract in young type 1 patients.
- **Glaucoma:** increased risk of primary open-angle glaucoma, and neovascular glaucoma from proliferative disease.
- **Infection:** increased risk of orbital cellulitis, and **mucormycosis in ketoacidosis:** a black necrotic eschar with orbital apex signs is a surgical emergency.
- **NAION** and papillopathy (diabetic papillopathy: bilateral disc swelling with surprisingly good acuity).
- **Screening rules:** type 1 at **5 years** after diagnosis or at puberty; type 2 **at diagnosis**; pregnancy before conception or in the first trimester and then each trimester, since **pregnancy accelerates retinopathy** (gestational diabetes alone does not require retinal screening).

Cardiovascular disease

HYPERTENSION: DEFINITIONS AND RETINAL GRADING

Category	Blood pressure	Retinal correlate and action
Normal	<120 / <80	–
Elevated	120–129 / <80	Lifestyle counselling.
Stage 1 hypertension	130–139 / 80–89	Referral for management.
Stage 2 hypertension	≥140 / ≥90	Referral; look for arteriolar narrowing and AV changes.
Hypertensive urgency	≥180 / ≥120 without target organ damage	Same-day medical evaluation.
Hypertensive emergency / malignant hypertension	≥180 / ≥120 with target organ damage	Emergency department immediately. Ocular target organ damage means grade 4 changes: disc oedema , or retinal haemorrhage and exudate with neurologic or renal signs.

KEITH-WAGENER-BARKER GRADING (AND THE MODERN MITCHELL-WONG SIMPLIFICATION)

Grade	Findings
1	Mild generalised arteriolar narrowing (A:V ratio falling below 2:3), increased tortuosity and wall opacification (copper wiring).
2	Definite focal narrowing and arteriovenous nicking (both vessels share an adventitial sheath at crossings).
3	Grade 2 plus retinal haemorrhages, cotton wool spots, hard exudates including a macular star, and Elschnig spots or Siegrist streaks from choroidal involvement.

Grade	Findings
4	Grade 3 plus optic disc oedema : this defines malignant hypertension and is an emergency.

VASCULAR OCULAR EMERGENCIES AND THEIR SYSTEMIC MEANING

Presentation	Ocular findings	Systemic implication and workup
Amaurosis fugax	Transient monocular vision loss, a "curtain", lasting minutes	A transient ischaemic attack of the eye. Same-day or urgent (within 24 hours) stroke workup: carotid imaging, ECG and rhythm monitoring, echocardiography, lipids, glucose, and in anyone over 50 an ESR, CRP, and platelet count for giant cell arteritis . Antiplatelet therapy and risk factor modification. Highest stroke risk is in the first days.
Central retinal artery occlusion	Sudden painless profound loss, whitened inner retina, cherry red spot , boxcarring, RAPD	Treat as an acute stroke : immediate referral to a stroke centre. Look for a Hollenhorst plaque (cholesterol embolus), calcific emboli (cardiac valves, higher mortality), and platelet-fibrin emboli. In anyone over 50 exclude giant cell arteritis urgently.
Branch retinal artery occlusion	Sectoral whitening and field loss	Same embolic workup.
Central retinal vein occlusion	Diffuse haemorrhage in all quadrants, dilated tortuous veins, disc oedema, cotton wool spots	Hypertension, hyperlipidaemia, diabetes, glaucoma, and hyperviscosity. In a patient under 50 , add a thrombophilia and inflammatory screen (antiphospholipid antibodies, homocysteine, protein C and S, factor V Leiden, vasculitis serology, and consider oral contraceptive use). Watch for iris and angle neovascularisation, most likely at 3 to 4 months in ischaemic CRVO.
Branch retinal vein occlusion	Sectoral haemorrhage respecting the horizontal raphe, at an arteriovenous crossing	Hypertension is the dominant risk factor.
Ocular ischaemic syndrome	Mid-peripheral dot-blot haemorrhages, dilated but not tortuous veins, narrowed arteries, neovascularisation, pain , low IOP despite rubeosis, and reduced retinal artery perfusion pressure	≥90% ipsilateral carotid stenosis. Carries about a 40% five-year mortality, mostly cardiac. Urgent vascular referral.

Presentation	Ocular findings	Systemic implication and workup
Giant cell arteritis	Arteritic AION with a chalky white swollen disc, or CRAO, or cilioretinal artery occlusion, or diplopia. Amaurosis fugax often precedes.	Features: age ≥ 50 (usually over 70), new headache, scalp tenderness, jaw claudication (the most specific symptom), temporal artery tenderness or reduced pulse, polymyalgia, weight loss, fever, and ESR typically >50 mm/h with a raised CRP and platelets. Start high-dose corticosteroid immediately, before biopsy: intravenous methylprednisolone for visual loss, oral prednisone 1 mg/kg otherwise. Temporal artery biopsy remains valid for up to about two weeks after starting steroid. Untreated, fellow eye involvement occurs in a large proportion within days to weeks. Tocilizumab is steroid-sparing.
Non-arteritic AION	Altitudinal field loss, segmental hyperaemic disc swelling, a crowded "disc at risk" fellow eye	Risk factors: hypertension, diabetes, hyperlipidaemia, nocturnal hypotension , obstructive sleep apnoea, and PDE5 inhibitors. No proven treatment; manage vascular risk and screen for sleep apnoea. About 15% fellow eye involvement over five years.
Infective endocarditis	Roth spots (white-centred haemorrhages), conjunctival petechiae, endogenous endophthalmitis	Blood cultures and echocardiography. Roth spots also occur in leukaemia, severe anaemia, diabetes, and hypertension, so they are not specific.
Carotid-cavernous fistula	Pulsatile proptosis, orbital bruit, dilated corkscrew episcleral vessels, raised IOP, chemosis, cranial neuropathies	High-flow (direct, traumatic) or low-flow (dural). Requires neuroimaging and neurointerventional referral.

Endocrine disease

THYROID EYE DISEASE

- An autoimmune orbitopathy driven by **TSH receptor antibodies** stimulating orbital fibroblasts to produce glycosaminoglycans and to differentiate into adipocytes, with lymphocytic infiltration of **extraocular muscle bellies (sparing the tendons)**: the imaging feature that distinguishes it from orbital myositis.
- **Muscle involvement order: "I'M SLow"**: Inferior rectus, Medial rectus, Superior rectus, Lateral rectus. Inferior rectus fibrosis restricts elevation and mimics a superior rectus palsy.
- **Signs**: lid retraction (Dalrymple), lid lag on downgaze (von Graefe), staring appearance (Kocher), infrequent blink (Stellwag), poor convergence (Möbius), proptosis, restrictive strabismus, chemosis, caruncular oedema, superior limbic keratoconjunctivitis, and **compressive optic neuropathy** at the crowded apex.
- **Activity versus severity**: activity is graded by the **clinical activity score** (pain, redness, swelling), severity by EUGOGO categories or the **NOSPECS** classification (No signs, Only signs, Soft tissue, Proptosis, Extraocular muscles, Corneal involvement, Sight loss). Treatment targets active disease.
- **Management**: **stop smoking** (the single strongest modifiable risk factor for progression and for failure of treatment), achieve euthyroidism, selenium in mild disease, lubrication and elevation of the head, intravenous methylprednisolone for moderate-to-severe active disease, **teprotumumab** (anti-IGF-1R; watch for hearing loss and hyperglycaemia), orbital radiotherapy, and surgical decompression urgently for optic neuropathy. Surgical order when needed: **decompression, then strabismus, then lid**.
- **Radioactive iodine** can worsen orbitopathy, so steroid cover is given in at-risk patients.

OTHER ENDOCRINE ASSOCIATIONS

Condition	Ocular findings
Pituitary adenoma	Bitemporal hemianopia beginning superotemporally; acromegaly or Cushing features; pituitary apoplexy presents with sudden headache, ophthalmoplegia, and visual loss and is a neurosurgical emergency.
Hypoparathyroidism	Cataract (punctate cortical), papilloedema, band keratopathy in hypercalcaemic states.
Cushing syndrome	Cataract, raised IOP, central serous chorioretinopathy, papilloedema.
Addison disease	Conjunctival and lid hyperpigmentation, papilloedema.
Pregnancy	Accelerated diabetic retinopathy, worsening of central serous chorioretinopathy, contact lens intolerance, transient refractive change, corneal thickening, reduced IOP, and rarely eclamptic serous detachments and cortical blindness. Avoid refractive prescribing changes and elective procedures.

Haematologic and oncologic disease

BLOOD DISORDERS WITH OCULAR SIGNS

Condition	Ocular findings and key facts
Sickle cell disease	SC and S-thalassaemia produce the most severe proliferative retinopathy , even though SS disease causes more systemic morbidity. Goldberg stages : 1 peripheral arteriolar occlusion, 2 arteriovenous anastomoses, 3 sea fan neovascularisation , 4 vitreous haemorrhage, 5 tractional retinal detachment. Also salmon patch haemorrhages, black sunbursts (RPE hyperplasia), iridescent spots, comma-shaped conjunctival capillaries, and angioid streaks . Hyphaema in sickle trait is dangerous: avoid carbonic anhydrase inhibitors, which cause acidosis and promote sickling in the anterior chamber .
Leukaemia	Roth spots, intraretinal haemorrhage, leukaemic infiltration of the choroid, iris, and optic nerve, hyperviscosity retinopathy, and opportunistic infection. Bilateral disc infiltration in a child requires urgent haematology involvement.
Multiple myeloma / Waldenström macroglobulinaemia	Hyperviscosity retinopathy : dilated tortuous "sausage-link" veins, haemorrhages, and CRVO-like appearances; also crystalline keratopathy and ciliary body cysts.
Anaemia	Retinal haemorrhages and cotton wool spots when haemoglobin falls below about 7–8 g/dL, particularly with concurrent thrombocytopenia; conjunctival pallor.
Thrombophilia and antiphospholipid syndrome	Retinal vein and artery occlusion in the young, arterial and venous thrombosis, recurrent miscarriage; look for lupus anticoagulant, anticardiolipin, and β_2 -glycoprotein antibodies.
Metastatic disease	Choroidal metastasis is the commonest intraocular malignancy: breast in women, lung in men. Orbital metastasis in adults from breast, lung, prostate; in children from neuroblastoma (with periorbital ecchymosis, "raccoon eyes") and Ewing sarcoma.
Paraneoplastic syndromes	CAR (cancer-associated retinopathy, anti-recoverin, small cell lung cancer), MAR (melanoma-associated retinopathy, with a negative ERG and shimmering photopsias), bilateral diffuse uveal melanocytic proliferation, and paraneoplastic optic neuropathy (anti-CRMP-5).

Rheumatologic and autoimmune disease

CONNECTIVE TISSUE AND VASCULITIC DISEASE

Disease	Serology	Ocular manifestations
Rheumatoid arthritis	Rheumatoid factor, anti-CCP (more specific)	Keratoconjunctivitis sicca (commonest), episcleritis, scleritis including necrotising scleritis and scleromalacia perforans , peripheral ulcerative keratitis , and corneal melting. Necrotising scleritis in RA signals systemic vasculitis and carries significant mortality: it requires systemic immunosuppression, not just topical treatment.
Juvenile idiopathic arthritis	ANA positive is the risk marker; RF negative	Chronic, bilateral, asymptomatic, non-granulomatous anterior uveitis with a white eye . Highest risk: oligoarticular, ANA positive, young girls, early onset . Systemic-onset (Still) disease essentially does not cause uveitis. Complications: band keratopathy, cataract, glaucoma, hypotony, amblyopia. Screening every 3 months in high-risk children, because there are no symptoms and no redness to prompt presentation.

Disease	Serology	Ocular manifestations
Spondyloarthropathies: ankylosing spondylitis, reactive arthritis, psoriatic arthritis, IBD-associated	HLA-B27	Recurrent, <i>acute</i> , unilateral or alternating, non-granulomatous anterior uveitis, often with fibrin and occasionally hypopyon. Reactive arthritis: " can't see, can't pee, can't climb a tree " (conjunctivitis or uveitis, urethritis, arthritis).
Systemic lupus erythematosus	ANA (sensitive), anti-dsDNA and anti-Smith (specific)	Keratoconjunctivitis sicca, retinal vasculitis with cotton wool spots, Purtscher-like retinopathy , choroidopathy with serous detachments, optic neuropathy, and hydroxychloroquine toxicity from treatment. Antiphospholipid antibodies drive vaso-occlusion.
Sjögren syndrome	Anti-SS-A (Ro) and anti-SS-B (La); Schirmer, ocular surface staining, salivary gland biopsy	Severe aqueous-deficient dry eye with filamentary keratitis and corneal melting; primary or secondary to RA or SLE. Increased lymphoma risk.
Sarcoidosis	Raised ACE and lysozyme, hypercalcaemia, chest imaging, gallium scan; biopsy shows non-caseating granulomas	Any uveitis pattern: granulomatous anterior uveitis with mutton-fat precipitates and Koeppe and Busacca nodules, candle-wax drippings (periphlebitis), snowballs, retinal and choroidal granulomas, optic nerve granuloma, lacrimal gland enlargement, conjunctival granulomas (the easiest biopsy site). Löfgren syndrome : fever, erythema nodosum, hilar adenopathy, arthritis, and uveitis. Heerfordt syndrome (uveoparotid fever): uveitis, parotid enlargement, fever, and facial nerve palsy.
Granulomatosis with polyangiitis	c-ANCA / anti-PR3	Orbital inflammatory mass with bone destruction and nasolacrimal obstruction, necrotising scleritis , peripheral ulcerative keratitis, retinal vasculitis. Classic triad of upper airway, lung, and kidney involvement. Requires cyclophosphamide or rituximab; topical treatment alone is negligent.
Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)	p-ANCA / anti-MPO , eosinophilia	Asthma, eosinophilia, vasculitis; retinal vascular occlusion, ischaemic optic neuropathy.
Polyarteritis nodosa	Hepatitis B association; angiography of medium vessels	Scleritis, peripheral ulcerative keratitis, retinal vasculitis, choroidal infarction (Elschnig spots).
Behçet disease	HLA-B51 ; clinical criteria	Recurrent oral and genital ulceration, occlusive retinal vasculitis affecting arteries and veins , recurrent non-granulomatous panuveitis with a shifting hypopyon , pathergy. Requires systemic immunosuppression, and anti-TNF therapy is well established.
Relapsing polychondritis	Anti-type II collagen	Scleritis, episcleritis, keratitis; saddle nose and floppy ear cartilage.

Disease	Serology	Ocular manifestations
Systemic sclerosis	Anti-centromere (limited), anti-topoisomerase I / Scl-70 (diffuse)	Tight lids with restricted closure, dry eye, retinopathy from hypertension and renal crisis.
Vogt-Koyanagi-Harada	HLA-DR4 / DRB1*0405	Bilateral granulomatous panuveitis with multifocal serous retinal detachments , disc oedema, and, later, sunset glow fundus and Dalen-Fuchs-like nodules, with meningismus, tinnitus, hearing loss, vitiligo, poliosis, and alopecia . Commonest in pigmented populations. Treat with prolonged high-dose systemic immunosuppression.
IgG4-related disease	Raised serum IgG4; biopsy with IgG4-positive plasma cells and storiform fibrosis	Bilateral lacrimal gland enlargement, orbital mass, extraocular muscle enlargement, and infraorbital nerve enlargement; markedly steroid-responsive.

Neurologic disease

NEUROLOGIC CONDITIONS WITH DEFINING OCULAR FEATURES

Condition	Ocular features and management points
Multiple sclerosis	Optic neuritis (pain on eye movement, central scotoma, RAPD, red desaturation, a normal or swollen disc, later temporal pallor), internuclear ophthalmoplegia especially bilateral, nystagmus, and Uhthoff phenomenon. ONTT : intravenous methylprednisolone speeds recovery but does not change final acuity; oral prednisone alone at 1 mg/kg is contraindicated because it increased recurrence . MRI with two or more typical white matter lesions raises the 15-year MS risk to about 72% versus 25% with a normal MRI. Consider MOG and aquaporin-4 (neuromyelitis optica) antibodies in severe, bilateral, or steroid-dependent cases, and in children.
Myasthenia gravis	Variable, fatigable ptosis and diplopia with a pupil that is always spared . Cogan lid twitch, enhancement of ptosis, improvement with rest or ice pack. Test acetylcholine receptor and MuSK antibodies, single-fibre EMG; image the chest for thymoma . Screen for thyroid disease. Aminoglycosides, fluoroquinolones, macrolides, and β-blockers can worsen it.
Idiopathic intracranial hypertension	Bilateral papilloedema, transient visual obscurations, pulsatile tinnitus, headache, and often a sixth nerve palsy; typically an obese young woman. Diagnosis by the modified Dandy criteria with normal imaging (except for empty sella, transverse sinus stenosis, or optic nerve sheath distension) and raised opening pressure. Manage with weight loss, acetazolamide (IIHTT), and surgery (shunt or optic nerve sheath fenestration) for progressive field loss. Perimetry, not acuity, tracks the disease.
Migraine	Visual aura with a scintillating, expanding fortification spectrum lasting 20–30 minutes , homonymous and usually followed by headache. Retinal migraine is monocular and requires exclusion of an embolic cause. Acephalgic migraine is common in older patients. A visual aura lasting under 5 or over 60 minutes, or one that is always on the same side, should prompt imaging.
Stroke	Homonymous field defects, cortical blindness, gaze palsies, INO, skew deviation, Horner syndrome with Wallenberg syndrome, and visual neglect. Any new homonymous defect is a stroke workup.

Condition	Ocular features and management points
Parkinson disease	Reduced blink and dry eye, convergence insufficiency, reduced contrast sensitivity and colour discrimination, apraxia of lid opening, visual hallucinations (also from dopaminergic therapy).
Progressive supranuclear palsy	Vertical (especially downgaze) supranuclear palsy, square-wave jerks, lid retraction, falls backwards.
Alzheimer disease	Reduced contrast sensitivity, impaired motion and complex visual processing, posterior cortical atrophy variant with visual agnosia and Balint syndrome.
Chiari malformation	Downbeat nystagmus , acute acquired comitant esotropia, papilloedema.
Mitochondrial disease	Leber hereditary optic neuropathy : mitochondrial (maternally inherited, mtDNA 11778, 3460, 14484), sequential bilateral painless central vision loss in young men, disc telangiectasia without leakage; idebenone and gene therapy are emerging. Chronic progressive external ophthalmoplegia : slowly progressive symmetric ptosis and ophthalmoplegia without diplopia; with pigmentary retinopathy and heart block it becomes Kearns-Sayre syndrome , requiring cardiology referral for pacing. MELAS : mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes with cortical visual loss and pigmentary retinopathy.

Phakomatoses and genetic syndromes

PHAKOMATOSES: GENE, INHERITANCE, OCULAR SIGN

Syndrome	Gene / locus	Ocular and systemic features
Neurofibromatosis 1 (von Recklinghausen)	<i>NF1</i> 17q11, AD	Lisch nodules (present in over 90% of adults, the most useful ocular sign), optic nerve glioma , plexiform neurofibroma of the lid with an S-shaped ptosis, congenital glaucoma with lid involvement, sphenoid dysplasia with pulsatile proptosis, choroidal nodules, prominent corneal nerves. Systemic: café-au-lait macules, axillary freckling, neurofibromas, bony dysplasia, pheochromocytoma.
Neurofibromatosis 2	<i>NF2</i> 22q12, AD	Bilateral vestibular schwannomas , posterior subcapsular or cortical cataract in the young , combined hamartoma of the retina and RPE, epiretinal membranes. Lisch nodules are <i>absent</i> .
Tuberous sclerosis (Bourneville)	<i>TSC1</i> 9q34 (hamartin), <i>TSC2</i> 16p13 (tuberin), AD	Retinal astrocytic hamartoma (mulberry, calcified) in about half, punched-out RPE depigmentation. Systemic: cortical tubers and seizures, facial angiofibroma (adenoma sebaceum), shagreen patch, ash-leaf hypopigmented macules, subungual fibromas, cardiac rhabdomyoma, renal angiomyolipoma. Everolimus for systemic disease.
Von Hippel-Lindau	<i>VHL</i> 3p25, AD	Retinal capillary haemangioblastoma with dilated feeder vessels and exudation, often multiple and bilateral. Systemic: cerebellar and spinal haemangioblastoma, renal cell carcinoma , pheochromocytoma, pancreatic and renal cysts. Lifelong surveillance for the patient and family.
Sturge-Weber (encephalotrigeminal angiomatosis)	<i>GNAQ</i> somatic mosaic, not inherited	Facial port-wine naevus in the V1 or V2 distribution, diffuse choroidal haemangioma ("tomato ketchup fundus"), episcleral vascular dilatation with raised episcleral venous pressure and glaucoma in 30 to 70%, especially when the naevus involves the upper lid. Leptomeningeal angioma with seizures.

Syndrome	Gene / locus	Ocular and systemic features
Ataxia-telangiectasia (Louis-Bar)	<i>ATM</i> 11q22, AR	Conjunctival telangiectasia, oculomotor apraxia, poor saccade initiation; cerebellar ataxia, immunodeficiency, lymphoma risk, radiation sensitivity.
Wyburn-Mason (racemose angiomatosis)	Sporadic	Retinal arteriovenous malformation with no intervening capillary bed; ipsilateral midbrain AVM.
Incontinentia pigmenti	<i>IKBKG</i> , X-linked dominant (lethal in males)	Peripheral retinal avascularity and neovascularisation resembling ROP, foveal hypoplasia; skin lesions in Blaschko lines.

CHROMOSOMAL AND CONNECTIVE TISSUE SYNDROMES

Syndrome	Ocular features
Down syndrome (trisomy 21)	Upslanting fissures, epicanthus, Brushfield spots , keratoconus , cataract, high refractive error and accommodative insufficiency, nystagmus, strabismus, blepharitis, and an increased risk of leukaemia. Accommodative lag is so common that bifocals are frequently indicated.
Turner syndrome (45,X)	Epicanthus, ptosis, strabismus, red-green colour defects (unmasked X), blue sclera, keratoconus.
Marfan syndrome	<i>FBN1</i> , AD. Superotemporal ectopia lentis , axial myopia, retinal detachment, flat cornea, hypoplastic iris and ciliary muscle with poor dilation, glaucoma. Systemic: aortic root dilatation and dissection (the cause of death), tall stature, arachnodactyly, pectus deformity, joint hypermobility. Requires cardiology surveillance.
Ehlers-Danlos syndrome	Blue sclera, thin cornea and keratoconus, angioid streaks, epicanthic folds, lens subluxation, retinal detachment; vascular type risks arterial rupture.
Osteogenesis imperfecta	Type I collagen defect: blue sclera , thin cornea, keratoconus, fractures, deafness.
Stickler syndrome	<i>COL2A1</i> , AD. Optically empty vitreous , high myopia, giant retinal tears and detachment (the leading inherited cause in children) , cataract, glaucoma; cleft palate, deafness, arthropathy, midface hypoplasia. Prophylactic retinal treatment is considered.
Pseudoxanthoma elasticum	<i>ABCC6</i> . Angioid streaks with peau d'orange, CNV; "plucked chicken skin" in flexures, gastrointestinal bleeding, vascular calcification.
Albinism	Oculocutaneous (tyrosinase-related, AR) and ocular (X-linked). Foveal hypoplasia, iris and fundus transillumination, nystagmus, reduced acuity, photophobia, and misrouting of the chiasm with excess crossed fibres on VEP. Hermansky-Pudlak adds a platelet storage pool defect with bleeding; Chediak-Higashi adds immunodeficiency.

Other systems in brief

RENAL, RESPIRATORY, GASTROINTESTINAL, DERMATOLOGIC

System / condition	Ocular relevance
Chronic kidney disease	Hypertensive and diabetic retinopathy, band keratopathy from calcium-phosphate deposition, red eye of dialysis, reduced clearance of renally excreted drugs (aciclovir, acetazolamide, fluconazole) requiring dose reduction.
Alport syndrome	Type IV collagen: anterior lenticonus, dot-and-fleck retinopathy, posterior polymorphous dystrophy; nephritis and sensorineural deafness.

System / condition	Ocular relevance
Asthma and COPD	Contraindication to non-selective topical β-blockers. Inhaled and systemic corticosteroids cause cataract and raised IOP.
Obstructive sleep apnoea	Floppy eyelid syndrome (a strong association: ask about snoring in any patient with an easily everted upper lid and chronic irritation), NAION, papilloedema, glaucoma, and central serous chorioretinopathy.
Inflammatory bowel disease	HLA-B27-associated anterior uveitis, episcleritis and scleritis, peripheral keratitis; uveitis activity does not track bowel activity.
Celiac disease	Fat-soluble vitamin deficiency with night blindness and dry eye.
Rosacea	Ocular rosacea often precedes or exceeds skin findings: meibomian gland dysfunction, telangiectatic lid margins, recurrent chalazia, marginal keratitis, and inferior corneal vascularisation. Treat with lid hygiene and oral doxycycline; consider topical ivermectin for <i>Demodex</i> .
Atopic dermatitis	Atopic keratoconjunctivitis, keratoconus, anterior subcapsular (shield) cataract, Dennie-Morgan folds, herpes simplex risk.
Psoriasis	Anterior uveitis (with psoriatic arthritis), blepharitis, dry eye.
Stevens-Johnson syndrome and toxic epidermal necrolysis	An ocular emergency. Acute pseudomembranous conjunctivitis with epithelial loss; late cicatrization, symblepharon, trichiasis, limbal stem cell deficiency, and keratinisation. Causes: sulfonamides, anticonvulsants, allopurinol, NSAIDs, <i>Mycoplasma</i> . Aggressive lubrication, symblepharon lysis, and amniotic membrane in the acute phase substantially reduce late scarring.
Mucous membrane (ocular cicatricial) pemphigoid	Type II hypersensitivity to basement membrane antigens; progressive conjunctival shrinkage, fornix foreshortening, symblepharon, keratinisation. Requires systemic immunosuppression (dapsone, mycophenolate, rituximab); surgery alone accelerates it.

Public health, epidemiology, and biostatistics

TEST PERFORMANCE: DEFINITIONS AND DEPENDENCIES

Measure	Formula	Property
Sensitivity	$TP / (TP + FN)$	Ability to detect disease. Independent of prevalence. A highly sensitive test, when <i>negative</i> , rules out disease (SnNout).
Specificity	$TN / (TN + FP)$	Ability to exclude the well. Independent of prevalence. A highly specific test, when <i>positive</i> , rules in disease (SpPin).
Positive predictive value	$TP / (TP + FP)$	Depends strongly on prevalence. Falls sharply in low-prevalence screening populations.
Negative predictive value	$TN / (TN + FN)$	Also prevalence-dependent, rising as prevalence falls.
Positive likelihood ratio	$\text{sensitivity} / (1 - \text{specificity})$	Prevalence-independent; combines with pre-test odds to give post-test odds. Values above 10 are strongly diagnostic.
Negative likelihood ratio	$(1 - \text{sensitivity}) / \text{specificity}$	Values below 0.1 strongly argue against disease.
Accuracy	$(TP + TN) / \text{total}$	Misleading when prevalence is very low or high.
Prevalence	$\text{existing cases} / \text{population}$	A snapshot; reflects both incidence and duration.

Measure	Formula	Property
Incidence	new cases / population at risk per time	Measures risk of developing disease.
Relative risk	risk exposed / risk unexposed	From cohort studies and trials.
Odds ratio	$(a/c) / (b/d) = ad / bc$	From case-control studies; approximates RR when the outcome is rare.
Absolute risk reduction	risk control - risk treated	The clinically meaningful measure.
Number needed to treat	$1 / ARR$	How many must be treated to prevent one event.

STUDY DESIGNS, STRONGEST TO WEAKEST FOR CAUSAL INFERENCE

Design	Strengths and limitations
Systematic review and meta-analysis of RCTs	Highest level; limited by the quality and heterogeneity of included trials and by publication bias.
Randomised controlled trial	Randomisation controls confounding, including unknown confounders; masking controls observer and placebo effects. Limited by cost, external validity, and ethics.
Cohort study	Prospective, gives incidence and relative risk; vulnerable to loss to follow-up and confounding.
Case-control study	Efficient for rare diseases; gives an odds ratio; vulnerable to recall and selection bias.
Cross-sectional study	Gives prevalence; cannot establish temporal sequence.
Case series and case report	Hypothesis-generating only.

- **Biases to name:** selection, recall, observer (Hawthorne), lead-time and length-time bias in screening, confounding, regression to the mean, attrition, and publication bias.
- **Statistics:** a **type I error (α)** is a false positive conclusion; a **type II error (β)** is a false negative. **Power** = $1 - \beta$, conventionally **0.80**, and it increases with sample size and effect size. A **95% confidence interval** that excludes the null value (0 for a difference, 1 for a ratio) corresponds to $p < 0.05$; the interval also conveys precision, which a p value does not. Use **parametric** tests (t test, ANOVA, Pearson) for normally distributed interval data and **non-parametric** tests (Mann-Whitney, Wilcoxon, Kruskal-Wallis, Spearman) otherwise; **chi-square** for categorical data.
- **Screening criteria (Wilson and Jungner):** the condition must be an important health problem with a recognisable latent stage, a suitable and acceptable test must exist, effective treatment must be available, and the whole programme must be cost-effective and continuous. This framework is why diabetic retinopathy screening is universally endorsed and population glaucoma screening is not.

NUMBERS TO KNOW: SYSTEMIC HEALTH

Diabetes: fasting glucose	≥ 126 mg/dL	HgbA1c	$\geq 6.5\%$
Prediabetes A1c	5.7–6.4%	General A1c target	$< 7\%$

Severe NPDR rule	4-2-1	High-risk PDR NVD	>1/3 disc area
Type 1 DR screening	5 yr after dx	Type 2 DR screening	at diagnosis
Hypertensive emergency	≥180/120 + organ damage	Normal A:V ratio	2:3
GCA typical ESR	>50 mm/h	GCA prednisone dose	1 mg/kg/day
Ocular ischaemic syndrome stenosis	≥90%	OIS 5-year mortality	~40%
CRVO neovascular glaucoma peak	3-4 months	NAION fellow eye risk (5 yr)	~15%
ONTT: MS risk with ≥2 lesions	~72% at 15 yr	with normal MRI	~25%
JIA uveitis screening interval	every 3 months	Sturge-Weber glaucoma rate	30-70%
Lisch nodules in adult NF1	>90%	Retinal hamartoma in TSC	~50%
Conventional statistical power	0.80	NNT	1 / ARR

RETRIEVAL PRACTICE

1. Give the three laboratory thresholds diagnostic of diabetes.
2. State the 4-2-1 rule and the follow-up interval it triggers.
3. Define high-risk proliferative diabetic retinopathy in three clauses.
4. A newly diagnosed type 2 diabetic asks when to have their first eye examination. Answer, and contrast with type 1.
5. Which retinal finding defines malignant hypertension, and what do you do?
6. A 74-year-old has sudden vision loss with a chalky white swollen disc and jaw pain on chewing. Give the diagnosis, the three initial tests, and the first action.
7. Mid-peripheral dot-blot haemorrhages, ocular pain, and rubeosis with a low IOP. Diagnosis, likely stenosis, and prognosis?
8. Which sickle genotype gives the worst retinopathy, and what drug must you avoid in sickle hyphaema?
9. Which JIA subtype and serology carries the highest uveitis risk, and how often do you screen?
10. Give the ANCA type for granulomatosis with polyangiitis and two ocular manifestations.
11. List the order of extraocular muscle involvement in thyroid eye disease and the single most important modifiable risk factor.
12. ONTT: what did intravenous steroid change, and what is contraindicated?
13. Match gene to syndrome: *VHL*, *TSC2*, *NF2*, *GNAQ*.
14. Which measures of test performance are independent of prevalence?
15. A trial reports an event rate of 12% in controls and 8% with treatment. Give ARR, RR, and NNT.
16. A patient with an easily everted upper lid and chronic irritation. What must you ask about, and why?

KEY

1. Fasting plasma glucose ≥ 126 mg/dL, 2-hour OGTT ≥ 200 mg/dL, or HgbA1c $\geq 6.5\%$ (or a random glucose ≥ 200 with symptoms).
2. Severe intraretinal haemorrhages in 4 quadrants, or venous beading in 2 quadrants, or IRMA in 1 quadrant defines severe NPDR; follow every 3 to 4 months.
3. NVD greater than one third of a disc area; any NVD with vitreous or preretinal haemorrhage; or NVE greater than half a disc area with haemorrhage.
4. Type 2: dilated examination **at diagnosis**, because retinopathy may already be present. Type 1: **5 years** after diagnosis or at puberty.
5. **Optic disc oedema** (grade 4 Keith-Wagener-Barker); send the patient to the emergency department the same day for blood pressure management and end-organ assessment.
6. Giant cell arteritis with arteritic anterior ischaemic optic neuropathy. Tests: ESR, CRP, and platelet count (then temporal artery biopsy). First action: **start high-dose corticosteroid immediately**, before the biopsy.
7. Ocular ischaemic syndrome; ipsilateral carotid stenosis of at least 90%; roughly 40% five-year mortality, mostly cardiovascular, so urgent vascular referral.
8. **SC and S-thalassaemia** genotypes; avoid **carbonic anhydrase inhibitors**, which cause systemic acidosis and promote sickling in the anterior chamber.
9. **Oligoarticular, ANA-positive** young girls with early onset; screen every 3 months.
10. **c-ANCA / anti-PR3**; necrotising scleritis and orbital inflammatory mass with bone destruction (also peripheral ulcerative keratitis and nasolacrimal obstruction).
11. Inferior rectus, medial rectus, superior rectus, lateral rectus ("I'M SLOW"); **smoking cessation**.
12. Intravenous methylprednisolone accelerated visual recovery but did not change final acuity; **oral prednisone alone at 1 mg/kg is contraindicated** because it increased the recurrence rate.
13. **VHL**: von Hippel-Lindau. **TSC2**: tuberous sclerosis. **NF2**: neurofibromatosis 2. **GNAQ**: Sturge-Weber (somatic mosaic).
14. **Sensitivity, specificity, and both likelihood ratios**. Predictive values depend on prevalence.
15. $ARR = 12\% - 8\% = 4\%$. $RR = 8/12 = 0.67$ (a 33% relative risk reduction). $NNT = 1/0.04 = 25$.
16. Ask about **snoring and daytime somnolence**: floppy eyelid syndrome is strongly associated with **obstructive sleep apnoea**, which itself carries cardiovascular, NAION, and glaucoma risk.

CHAPTER RECAP

1. Diabetes: three diagnostic thresholds, 4-2-1 for severe NPDR, high-risk PDR triggers treatment, anti-VEGF first for centre-involving oedema.
2. Type 2 screened at diagnosis, type 1 at five years, pregnancy each trimester.
3. Hypertensive grade 4 means disc oedema and an emergency department the same day.
4. Amaurosis fugax and CRAO are strokes; over 50, always exclude giant cell arteritis before anything else.
5. Thyroid eye disease spares tendons, follows I'M SLOW, and smoking is the modifiable driver.
6. SC sickle disease gives the worst retinopathy; never give a carbonic anhydrase inhibitor in sickle hyphaema.
7. ANA-positive oligoarticular JIA has a white, painless, sight-threatening uveitis screened three-monthly.
8. c-ANCA for GPA, HLA-B51 for Behçet, HLA-A29 for birdshot, DR4 for VKH.
9. Phakomatoses map one-to-one to genes and to a signature ocular lesion.
10. Sensitivity and specificity are prevalence-independent; predictive values are not. $NNT = 1/ARR$.

APPENDIX A

Master numbers table for last-day review

Every value the exam has a reasonable claim on, in one place. Cover the right-hand column of each pair, recite, then check. Work through it twice on the day before the exam, and once on the morning of.

Ocular dimensions

Structure	Value	Structure	Value
Axial length	23–24 mm	Globe volume	~6.5 mL
Orbital volume	~30 mL	Orbital depth	40–45 mm
Corneal horizontal diameter	11.5–12 mm	Corneal vertical diameter	10.5–11 mm
Central corneal thickness	520–540 μ m	Peripheral thickness	650–700 μ m
Corneal anterior radius	7.7–7.8 mm	Posterior radius	6.5–6.8 mm
Epithelium	50–52 μ m	Bowman layer	8–12 μ m
Descemet, adult	10–12 μ m	Endothelium	~5 μ m
Stromal lamellae	200–250	Collagen fibril diameter	22–35 nm
Sclera, posterior pole	1.0 mm	Sclera behind rectus insertion	0.3 mm
Scleral radius	~11.5 mm	Scleral fibril diameter	25–230 nm
Anterior chamber depth	3.0–3.5 mm	AC volume	~0.25 mL
Posterior chamber volume	~0.06 mL	Iris diameter	~12 mm
Pupil range	1–8 mm	Iris sphincter width	0.75–1 mm
Ciliary body width	~6 mm	Pars plicata / plana	2 / 4 mm
Ciliary processes	70–80	Vitrectomy port from limbus	3.5–4 mm
Lens diameter, adult	9–10 mm	Lens thickness, young	3.5–4 mm
Anterior lens capsule	~14–15.5 μ m	Posterior capsule	~4 μ m
Vitreous volume	~4 mL	Vitreous base pre/post ora	2 / 3–4 mm
Choroid, posterior pole	0.22–0.25 mm	Retina at parafovea	~0.56 mm
Retina at foveola	0.13–0.18 mm	Retina at ora	~0.1 mm
Macula	5.5–6.0 mm	Fovea	1.5 mm
Foveola	0.35 mm	Foveal avascular zone	0.4–0.5 mm
Optic disc, H $\times$ V	1.5 $\times$ 1.75–1.9 mm	1 disc diameter subtends	~5°
Ora serrata from limbus, nasal	6–7 mm	temporal	7–8 mm
Palpebral fissure H $\times$ W	8–11 $\times$ 28–30 mm	MRD1	4–5 mm
Upper / lower tarsus height	10–11 / 4–5 mm	Müller contribution	~2 mm
Punctum diameter	0.2–0.3 mm	Canaliculus length	8–10 mm

Structure	Value	Structure	Value
Lacrimal sac height	10–12 mm	Nasolacrimal duct	12–18 mm
Spiral of Tillaux MR / IR	5.5 / 6.5 mm	LR / SR	6.9 / 7.7 mm
Schlemm canal diameter	190–370 μ m	Collector channels	25–35

Cell counts and densities

Population	Value	Population	Value
Rods	110–125 million	Cones	6–7 million
Ganglion cells / ON axons	~1.2 million	RPE cells	~3.5 million
Peak cone density	~199,000/mm ²	Peak rod density	~160,000/mm ²
Endothelium at birth	5000–6000/mm ²	Adult endothelium	2000–3000/mm ²
Endothelial loss rate	~0.6% /yr	Decompensation density	~500/mm ²
Corneal stromal nerve trunks	70–80	Goblet cells, total	~1.5 million
Meibomian glands, upper / lower	30–40 / ~25	Lashes, upper / lower	100–150 / 50–75
Photoreceptors per RPE cell	30 macula, 20 periphery	Short ciliary nerves	6–10
Anterior ciliary arteries	7	Vortex veins	4–8

Optics and refraction

Quantity	Value	Quantity	Value
Total eye power	+58.6 to +60 D	Corneal power	~+43 D
Corneal anterior surface	+48.8 D	Corneal posterior surface	–5.8 D
Lens power in situ	+15 to +20 D	Keratometer index	1.3375
Corneal index	1.376	Aqueous / vitreous index	1.336
Lens index range	1.386–1.406	Critical angle, cornea to air	~46°
Anterior focal point of eye	~17 mm	Posterior focal length	~22.6 mm
Retinal image per degree	~0.29 mm	1 mm axial length	~3 D
1 prism dioptre	0.57°	1 degree	~1.75 Δ
Ocular chromatic aberration	~2 D	Optimum pupil size	2–3 mm
CR-39 index / Abbe	1.498 / 58	Polycarbonate index / Abbe	1.586 / 30
Trivex index / Abbe	1.53 / 43–45	Crown glass index / Abbe	1.523 / 59
CR-39 min thickness	2.0 mm	Glass min thickness	3.0 mm
Drop-ball test	15.9 g from 50 in	Pantoscopic tilt	8–12°
Sphere tolerance to 6.50 D	±0.13 D	Axis tolerance >1.50 D cyl	±2°
Vertical prism imbalance limit	±0.33 Δ	Flat-top seg OC below top	4–5 mm
Trifocal intermediate	50% of add	Aniseikonia per dioptre	1–2%
Aphakic spectacle magnification	20–35%	Knapp law plane	~15.7 mm

Quantity	Value	Quantity	Value
20/20 letter subtends	5 arcmin	Vernier acuity	2-5 arcsec
CSF peak	3-6 cpd	Photopic / scotopic peak	555 / 507 nm

Physiology and pharmacology

Quantity	Value	Quantity	Value
Mean IOP ± SD	15.5 ± 2.6 mmHg	Diurnal variation	3-5 mmHg
Aqueous formation	2.0-3.0 µL/min	Nocturnal reduction	~50%
Outflow facility C	0.2-0.3 µL/min/ mmHg	Episcleral venous pressure	8-10 mmHg
Trabecular / uveoscleral	80-90% / 10-20%	GAT applanation diameter	3.06 mm
GAT assumed CCT	~520 µm	Stromal swelling pressure	50-60 mmHg
Corneal hydration	78%	Overnight physiologic oedema	2-4%
Dk/t daily / extended wear	≥24 / ≥87	Harvitt-Bonanno targets	~35 / ~125
Tear film thickness	3-10 µm	Tear volume	6-9 µL
Tear turnover	~16% /min	Tear osmolarity, normal	295-310 mOsm/L
Dry eye osmolarity threshold	>308 mOsm/L	Normal TBUT	>10 s
Schirmer I normal	>10 mm / 5 min	Blink rate	12-15 /min
Drop volume	35-50 µL	Topical bioavailability	<5%
Prostaglandin IOP reduction	25-33%	β-blocker reduction	20-25%
Topical / oral CAI reduction	15-20% / 30-40%	Steroid high responders	~5%
Tropicamide duration	4-6 h	Atropine duration	7-14 days
Cone / rod dark adaptation	5-7 / 20-40 min	Rod outer segment turnover	~10 days
Dark / light membrane potential	-40 / -70 mV	Normal EOG Arden ratio	>1.8
Photopic CFF	50-60 Hz	Bloch critical duration	~100 ms
Lens glycolysis share	~85%	Lens protein content	~33%
Aqueous ascorbate vs plasma	~20×	Aqueous protein vs plasma	~1/200

Binocular vision norms

Finding	Value	Finding	Value
Distance phoria	1 exo ± 2	Near phoria	3 exo ± 3
Distance BO (b/br/rec)	9 / 19 / 10	Distance BI	x / 7 / 4
Near BO	17 / 21 / 11	Near BI	13 / 21 / 13
NRA / PRA	+2.00 / -2.37 D	MEM lag	+0.50 D
Accommodative facility	≥11 cpm	Vergence facility	~15 cpm
NPC break / recovery	5 / 7 cm	Calculated AC/A	4:1 ± 2
Hofstetter mean amplitude	18.5 - 0.30(age)	Hofstetter minimum	15 - 0.25(age)

Finding	Value	Finding	Value
Field: temporal / nasal	90–100° / 60°	superior / inferior	60° / 70–75°
Blind spot position	~15° temporal	Blind spot size	5 × 7°

Development and systemic thresholds

Item	Value	Item	Value
Axial length at birth	16–17 mm	Corneal power at birth	~51 D
Lens power at birth	~34 D	Newborn refraction	+2 to +3 D
Optic vesicle	day 25	Lens vesicle separation	day 33
Optic fissure closure	wk 5–7	Lid fusion / reopening	wk 8–9 / mo 5–7
Retinal vessels, nasal / temporal ora	36 / 40 wk	Foveal maturation	~4 mo
Acuity at birth / 1 yr	20/400 / 20/50	Acuity 20/20 by	3–5 yr
Stereopsis onset	3–6 mo	Amblyopia critical period	to ~7–9 yr
Diabetes fasting glucose	≥126 mg/dL	Diabetes HgbA1c	≥6.5%
Prediabetes A1c	5.7–6.4%	A1c target	<7%
Hypertensive emergency	≥180/120 + damage	Normal A:V ratio	2:3
GCA ESR	>50 mm/h	GCA prednisone	1 mg/kg/day
CD4 for CMV retinitis	<50 cells/μL	Hydroxychloroquine risk dose	>5.0 mg/kg/day
Legal blindness, acuity / field	20/200 / ≤20°	Kestenbaum add for 20/200	+10.00 D

APPENDIX B

Mnemonic index and mixed final drill

Mnemonic index

EVERY MNEMONIC IN THIS GUIDE, IN ONE PLACE

Mnemonic	What it encodes
Many Friendly Zebras Enjoyed Lazy Sunday Picnics	Orbital bones: Maxillary, Frontal, Zygomatic, Ethmoid, Lacrimal, Sphenoid, Palatine.
LR₆SO₄	Lateral rectus is CN VI, superior oblique is CN IV, all others CN III.
MILS 5.5 / 6.5 / 6.9 / 7.7	Spiral of Tillaux: Medial, Inferior, Lateral, Superior rectus insertion distances from the limbus.
SIN and RAID	Superiors Intort; Recti ADduct, obliques abduct.
Moll is Moist, Zeis is sebaceous, Meibomian is in the tarsus	Adnexal glands and their cysts: hidrocystoma, styte, chalazion.
Schwalbe, TM, Spur, Ciliary band, Iris	Gonioscopic angle structures from anterior to posterior.
Marilyn Monroe Always Gets Her Man in LA County	Macular-Mucopolysaccharide-Alcian blue; Granular-Hyaline-Masson; Lattice-Amyloid-Congo red.
ACID	Hypersensitivity types: Allergic (I), Cytotoxic (II), Immune complex (III), Delayed (IV).
Cranial Lacerations Prompt Many Serious Problems, And Might Send Doctors Nuts	Ophthalmic artery branches in order: Central retinal, Lacrimal, Posterior ciliary, Muscular, Supraorbital, Posterior ethmoidal, Anterior ethmoidal, Medial palpebral, Supratrochlear, Dorsal Nasal.
RAM-GAP	Duochrome: Red Add Minus, Green Add Plus.
SAM / FAP	Rigid lens tear lens: Steeper Add Minus, Flatter Add Plus.
LARS	Toric soft lens rotation: Left Add, Right Subtract.
I'M SLOW	Thyroid eye disease muscle order: Inferior, Medial, Superior, Lateral rectus.
4-2-1	Severe NPDR: haemorrhages in 4 quadrants, beading in 2, IRMA in 1.
ISNT	Normal neuroretinal rim thickness: Inferior > Superior > Nasal > Temporal.
Can't see, can't pee, can't climb a tree	Reactive arthritis: conjunctivitis or uveitis, urethritis, arthritis.
Hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter	Anticholinergic (atropine) toxicity.
SnNout / SpPin	A Sensitive test, when Negative, rules out; a Specific test, when Positive, rules in.
SAFE and FISTO	Trachoma: Surgery, Antibiotics, Facial cleanliness, Environmental improvement; and the grading Follicles, Intense inflammation, Scarring, Trichiasis, Opacity.
To Find Small Ocular Melanoma Doing IMaging	Risk factors distinguishing small melanoma from naevus: Thickness >2 mm, Fluid, Symptoms, Orange pigment, Margin at disc, ultrasound Hollow, Diameter, absence of Halo, absence of Drusen.

Look-alike pairs you must be able to separate

THE DISTINCTIONS THAT DECIDE CLOSE ITEMS

Pair	Deciding feature
Guttae vs Hassall-Henle bodies	Central (pathologic, Fuchs) vs peripheral (normal ageing).
Papilla vs follicle	Central vessel with fibrous septa (allergic, bacterial) vs avascular lymphoid centre with encircling vessels (viral, chlamydial, toxic).
Episcleritis vs scleritis	Blanches with phenylephrine, no pain vs does not blanch, boring night-waking pain, violaceous hue, possible systemic vasculitis.
HSV dendrite vs VZV pseudodendrite	Terminal bulbs and an ulcerated base vs elevated mucous plaque without terminal bulbs.
Compressive vs ischaemic CN III palsy	Pupil involved (urgent imaging) vs pupil spared (vascular, recovers in 8 to 12 weeks). Aberrant regeneration means compression.
Horner vs CN III ptosis	1 to 2 mm with miosis and reverse ptosis vs severe ptosis with ophthalmoplegia and possible mydriasis.
Superior oblique palsy vs skew deviation	Excyclotorsion of the higher eye, worse on ipsilateral tilt vs conjugate ocular tilt reaction with incyclotorsion of the lower eye and brainstem signs.
Restriction vs paresis	Positive forced duction, IOP rise on attempted gaze, normal saccadic velocity vs the opposite of all three.
Terrien vs pellucid marginal degeneration	Superior thinning with lipid line and intact epithelium vs inferior thinning with protrusion above the thinnest point and a crab-claw topography.
Phacolytic vs phacomorphic glaucoma	Open angle with macrophages engulfing leaked lens protein vs closed angle from an intumescent lens.
Arteritic vs non-arteritic AION	Chalky white swelling, over 70, jaw claudication, ESR >50, other eye at high risk vs segmental hyperaemic swelling, altitudinal defect, crowded fellow disc.
Optic disc drusen vs papilloedema	Lumpy margin, autofluorescent, spontaneous venous pulsation often present, no obscuration of vessels vs obscured vessels, hyperaemia, peripapillary haemorrhages, and Paton lines.
CRAO cherry red spot vs storage disease cherry red spot	Sudden painless monocular loss with inner retinal whitening vs bilateral, in an infant, with neurodegeneration.
Diffuse unilateral leukocoria differentials	Retinoblastoma (calcification), Coats disease (male, exudation, telangiectasia), PFV (microphthalmia, retrolental membrane), toxocariasis (granuloma, traction band), ROP (prematurity).
CLARE vs CLPU vs microbial keratitis	Intact epithelium with peripheral infiltrates vs small round full-thickness epithelial loss over a sterile infiltrate vs epithelial defect over an infiltrate with an anterior chamber reaction.
Jump vs displacement	Jump depends only on segment design and add; displacement adds the distance lens prism at the reading level.
Sensitivity/specificity vs predictive values	The first pair is prevalence-independent; the second is not.

Mixed final drill

Fifty items sampling all seventeen condition areas. Give yourself **77 seconds** each, which is the real exam pace, and log every error by condition area and discipline.

QUESTIONS 1 TO 25

No looking back. Write your answers, then check the key below.

1. Which wall of the orbit is thinnest, and which most commonly fractures in a blowout?
2. A patient has a 1.5 mm ptosis, miosis, and anhidrosis of the whole ipsilateral body. Which order of neuron is affected?
3. Trace lacrimal parasympathetic innervation from nucleus to gland.
4. Which corneal layer never regenerates, and what does injury to it produce?
5. Give the two collagen-related reasons the cornea is transparent and the sclera is not.
6. Name the structure housing limbal stem cells and one disease that destroys it.
7. Which trabecular layer creates most outflow resistance?
8. Write the Goldmann equation and identify the term altered by a carotid-cavernous fistula.
9. Explain how pilocarpine lowers IOP and what it does to uveoscleral outflow.
10. Which epithelium forms the blood-aqueous barrier, and name its two enzymes.
11. Anterior versus posterior lens capsule thickness, and the surgical significance.
12. Superotemporal versus inferonasal ectopia lentis: disease and molecular defect.
13. Rank the top three sites of vitreoretinal adhesion and the lesion each produces.
14. Which retinal layer marks the boundary between retinal and choroidal blood supply?
15. Name the two blood-retinal barriers and the cell type responsible for each.
16. What is the earliest histologic lesion of diabetic retinopathy?
17. Why does a glaucomatous arcuate defect stop abruptly at the horizontal midline nasally?
18. Which ganglion cell class contains melanopsin and where do its axons project?
19. Give the four segments of the optic nerve with approximate lengths.
20. A pituitary adenoma first produces which field defect, and why that quadrant?
21. Which LGN layers are magnocellular, and which receive crossed input?
22. Define internuclear ophthalmoplegia and give the commonest cause in a 28-year-old.
23. List the ciliary ganglion's three roots and state which synapses.
24. Give the three-drug sequence for a suspected Horner syndrome and what each proves.
25. Assign corneal epithelium, corneal stroma, iris stroma, and iris sphincter to embryonic sources.

KEY 1 TO 25

1. Medial (lamina papyracea) is thinnest; the posteromedial floor fractures most commonly because it is unsupported over the maxillary sinus.
2. First order (central), because whole-body anhidrosis implies a lesion above the sympathetic outflow.
3. Lacrimal (superior salivatory) nucleus, CN VII via nervus intermedius, greater superficial petrosal nerve, nerve of the pterygoid canal (vidian), pterygopalatine ganglion, zygomatic branch of V2, then the lacrimal nerve of V1.
4. Bowman layer; injury leaves permanent scarring.
5. Uniform small fibril diameter (22 to 35 nm) and regular short spacing giving destructive interference; the sclera has irregular diameters (25 to 230 nm) and irregular spacing.
6. Palisades of Vogt; chemical burn (also Stevens-Johnson, aniridia, contact lens overwear).
7. Juxtacanalicular (cribriform) tissue with the inner wall of Schlemm canal.
8. $P_o = (F - U)/C + P_v$; a carotid-cavernous fistula raises P_v .
9. Longitudinal ciliary muscle contraction pulls on the scleral spur, widening trabecular spaces and increasing conventional outflow; uveoscleral outflow decreases.
10. Non-pigmented ciliary epithelium; Na^+/K^+ ATPase and carbonic anhydrase II.
11. Anterior about 14 to 15.5 μm , posterior about 4 μm ; the posterior capsule is the fragile structure at risk in phacoemulsification.
12. Superotemporal: Marfan, *FBN1* fibrillin-1. Inferonasal: homocystinuria, cystathionine β -synthase.
13. Vitreous base (giant tear or dialysis), lattice margins (horseshoe tear), optic disc (Weiss ring on PVD).

14. The outer plexiform layer.
15. Inner: retinal capillary endothelial tight junctions with pericytes. Outer: RPE tight junctions.
16. Pericyte loss.
17. Arcuate nerve fibre bundles arch around the macula and do not cross the horizontal temporal raphe.
18. Intrinsically photosensitive retinal ganglion cells; to the suprachiasmatic nucleus and the olivary pretectal nucleus.
19. Intraocular 1 mm, intraorbital 25 to 30 mm, intracanalicular 4 to 10 mm, intracranial 10 to 16 mm.
20. Bitemporal superior quadrantanopia, because upward growth compresses the inferior chiasmal fibres, which carry the superior field.
21. Layers 1 and 2 are magnocellular; crossed fibres go to layers 1, 4, and 6.
22. A medial longitudinal fasciculus lesion causing ipsilateral adduction failure with contralateral abducting nystagmus; multiple sclerosis, especially if bilateral.
23. Parasympathetic (synapses), sensory from nasociliary (passes through), sympathetic from the carotid plexus (passes through).
24. Cocaine or apraclonidine confirms a Horner syndrome at any level; hydroxyamphetamine then localises, dilating in first- and second-order lesions and failing in third-order.
25. Corneal epithelium: surface ectoderm. Corneal stroma: neural crest. Iris stroma: neural crest. Iris sphincter: neuroectoderm.

QUESTIONS 26 TO 50

1. Give the vergence equation and use it: object at 20 cm before a **+10.00 D** lens. Where is the image?
2. Convert **-12.00 DS** at a 12 mm vertex to the corneal plane.
3. Transpose **+4.00 -2.50 × 110°**.
4. What is the power of **-2.00 -2.00 × 090°** in the 180 meridian?
5. Compute image jump for a 22 mm round-top with a **+3.00 add**.
6. Which lens material has the best impact resistance and the worst Abbe number?
7. Give the Dk/t thresholds for daily and extended wear and the oedema level they target.
8. K readings 43.00/43.00; rigid lens fitted at 43.50. What is the tear lens and which way do you adjust power?
9. Give the Kestenbaum add for 20/300 and the working distance it implies.
10. Which near low vision device gives the widest field, and what is its cost?
11. PD 62 mm, distance phoria 2 eso, near phoria 14 eso at 40 cm. Calculated AC/A and diagnosis?
12. Right hypertropia worse in left gaze and worse on right head tilt. Which muscle?
13. Give the PEDIG patching dose for moderate and severe amblyopia.
14. State the Köllner rule and two exceptions.
15. Which colour vision test classifies both red-green and blue-yellow defects and grades severity?
16. Which ERG component arises from ON bipolar cells, and what does a normal a-wave with a reduced b-wave mean?
17. Normal ERG with a grossly abnormal EOG: diagnosis and gene?
18. Delayed VEP latency with preserved amplitude: interpret.
19. Which organisms penetrate an intact corneal epithelium?
20. Name six mechanisms of ocular immune privilege.
21. Give the ANCA type and two ocular findings of granulomatosis with polyangiitis.
22. Give the 4-2-1 rule, the follow-up interval, and the definition of high-risk PDR.
23. Which topical steroid has the lowest IOP risk and why?
24. State the hydroxychloroquine dose threshold, screening interval, and two core tests.
25. A trial shows 15% events in controls and 10% with treatment. ARR, RR, and NNT?

KEY 26 TO 50

1. $L' = L + F$, $L = 1/(-0.20) = -5.00$ D; $L' = -5 + 10 = +5.00$ D, so the image is **20 cm to the right**, real and inverted, magnification -1.
2. $F_{\text{new}} = -12 / (1 - 0.012 \times -12) = -12/1.144 = \mathbf{-10.49}$, so **about -10.50 D**.
3. $+1.50 + 2.50 \times 0.20$.
4. θ from the cylinder axis (090) to 180 is 90° , so $F = -2.00 + (-2.00)(\sin^2 90) = \mathbf{-4.00$ D.
5. OC is 11 mm below the top, so jump = $1.1 \times 3.00 = \mathbf{3.30 \Delta}$.
6. Polycarbonate: best impact resistance, Abbe about 30.
7. At least 24 for daily wear and 87 for extended wear, targeting no more than about 4% oedema.
8. Fitted 0.50 D steeper, giving a **+0.50 D** tear lens; SAM, so **add 0.50 D of minus** to the lens power.
9. $300/20 = \mathbf{+15.00$ D, implying a working distance of about **6.7 cm**.
10. The spectacle-mounted high-plus microscope; the cost is a very short working distance and the need for base-in prism or monocular correction.
11. $AC/A = 6.2 + 0.4(14 - 2) = 6.2 + 4.8 = \mathbf{11:1}$: a high AC/A with near esophoria, so **convergence excess**, treated with a plus add.
12. **Right superior oblique**.
13. Two hours daily for moderate amblyopia, six hours for severe.
14. Outer retinal disease gives blue-yellow, optic nerve disease gives red-green. Exceptions: glaucoma and dominant optic atrophy (optic nerve but blue-yellow), Stargardt and cone dystrophy (outer retina but red-green).
15. The Hardy-Rand-Rittler (HRR) plates.
16. The b-wave, recorded through Müller cell potassium currents; a normal a-wave with a reduced b-wave (negative ERG) localises to the inner retina: congenital stationary night blindness, X-linked retinoschisis, CRAO, quinine toxicity.
17. Best vitelliform macular dystrophy; *BEST1* (bestrophin).
18. Demyelination, as in optic neuritis and multiple sclerosis.
19. *Neisseria gonorrhoeae* and *meningitidis*, *Corynebacterium diphtheriae*, *Shigella*, *Haemophilus* (aegyptius / influenzae biotype III), and *Listeria*; *Pseudomonas* invades rapidly with a lens biofilm.
20. No lymphatics or vessels in cornea, lens, and vitreous; blood-ocular barriers; immunosuppressive aqueous (TGF- β 2 and neuropeptides); Fas ligand expression; low MHC class II with few central antigen-presenting cells; and ACAID.
21. c-ANCA / anti-PR3; necrotising scleritis and an orbital inflammatory mass with bone destruction.
22. Severe haemorrhages in 4 quadrants, or venous beading in 2, or IRMA in 1 defines severe NPDR; review every 3 to 4 months. High-risk PDR is NVD greater than one third of a disc area, any NVD with vitreous or preretinal haemorrhage, or NVE greater than half a disc area with haemorrhage.
23. Loteprednol etabonate, because it is an ester steroid hydrolysed to inactive metabolites within the eye.
24. Above **5.0 mg/kg/day** of real body weight; annual screening after five years (baseline within a year); automated visual fields plus SD-OCT.
25. ARR = 5%; RR = $10/15 = \mathbf{0.67}$; NNT = $1/0.05 = \mathbf{20}$.

FINAL CHECKLIST BEFORE YOU SIT THE EXAM

1. Systemic health is 19% of the exam. Have you actually revised it, or only the ocular anatomy you enjoy?
2. Ametropia plus ophthalmic optics is 14%. Can you do vertex, transposition, oblique meridian, and prism calculations under time pressure without a calculator?
3. Conjunctiva, cornea, and refractive surgery is 12%. Do you know the six layers with numbers, and the dystrophy stain table?
4. Can you write the Goldmann equation and map every glaucoma drug onto it?
5. Can you recite Morgan norms and both AC/A formulae?
6. Have you memorised the visual field localisation table, including congruity and macular sparing?
7. Do you know which drugs need formal monitoring protocols: hydroxychloroquine, ethambutol, vigabatrin, pentosan polysulfate?
8. Can you assign every ocular structure to one of four embryonic sources?
9. Have you done at least one full 185-item timed block at **77 seconds** per item?
10. Multiple-response items are all-or-nothing and are 10 to 15% of the exam. Have you practised reading the "select 2" or "select 3" instruction *before* answering?